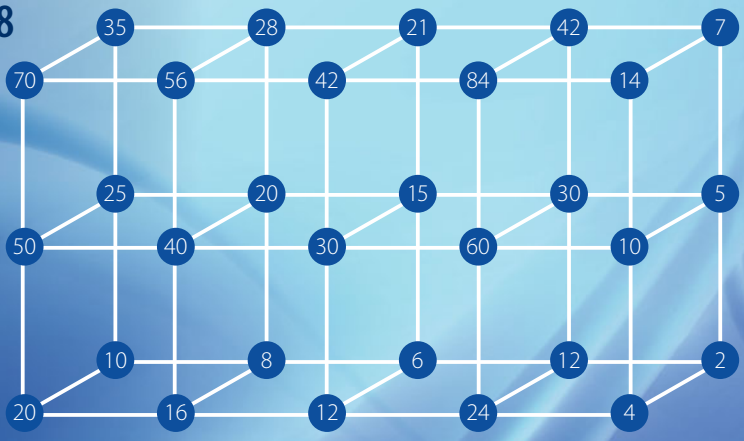
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Cristiano Bocci · Luca Chiantini

An Introduction to Algebraic Statistics with Tensors

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Cristiano Bocci · Luca Chiantini

An Introduction to Algebraic Statistics with Tensors

Springer

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Cover illustration (LaTeX): A decomposable 3-dimensional tensor of type $3\times 5\times 2$.

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We, the authors, dedicate this book to our great friend Tony Geramita. When the project started, Tony was one of the promoters and he should be among us in the list of authors of the text. Tony passed away when the book was at an early stage. We finished the book following the pattern traced in collaboration with him, and we always felt as if his encouragement to continue the project never faded.

Preface

Statistics and Algebraic Statistics

At the beginning of a book on Algebraic Statistics, it is undoubtedly a good idea to give the reader some idea of the goals of the discipline.

A reader who is already familiar with the basics of Statistics and Probability is probably curious about what the prefix "Algebraic" might mean. As we will see, Algebraic Statistics has its own way of approaching statistical problems, exploiting algebraic, geometric, or combinatorial properties. These problems are somewhat different from the one studied by Classical Statistics.

We will illustrate this point of view with some examples, which consider well-known statistical models and problems. At the same time, we will point out the difference between the two approaches to these examples.

The Treatment of Random Variables

The initial concern of Classical Statistics is the behavior of one random variable X. Usually X is identified with a function with values in the real numbers. This is clearly an approximation. For example, if one records the height of the members of a population, it is unlikely that the measure goes much further than the second decimal digit (assume that the unit is 1 m). So, the corresponding graph is a histogram, with a basic interval of 0.01 m. This is translated to a continuous variable, by sending the length of the basic interval to zero (and the size of the population increases).

For random variables of this type, the first natural distribution that one expects is the celebrated Gaussian distribution, which corresponds to the function

$$X(t) = \frac{1}{\sigma\sqrt{1\pi}} e^{\frac{-(t-\mu)^2}{2\sigma^2}}$$

where μ and σ are parameters which describe the *shape* of the curve (of course, other types of distributions are possible, in connection with special behaviors of the random variable $X(t)$).

The first goal of Classical Statistics is the study of the shape of the function $X(t)$, together with the determination of its numerical parameters.

When two or more variables are considered in the framework of Classical Statistics, their interplay can be studied with several techniques. For instance, if we consider both the heights and the weights of the members of a population and our goal is a proof of the (obvious) fact that the two variables are deeply connected, then we can consider the distribution over pairs (height, weight), which is represented by a bivariate Gaussian, in order to detect the existence of the connection.

The starting point of Algebraic Statistics is quite different. Instead of considering variables as continuous functions, Algebraic Statistics prefers to deal with a finite (and possibly small) range of values for the variable X. So, Algebraic Statistics emphasizes the discrete nature of the starting histogram, and tends to group together values in wider ranges, instead of splitting them. A distribution over the variable X is thus identified with a discrete function (to begin with, over the integers).

Algebraic Statistics is rarely interested in situations, where just one random variable is concerned.

Instead, networks containing several random variables are considered and some relevant questions raised in this perspective are

- Are there connections between the two or more random variables of the network?
- Which kind of connection is suggested by a set of data?
- Can one measure the complexity of the connections in a given network of interacting variables?

Since, from the new point of view, we are interested in determining the relations between discrete variables, in Algebraic Statistics a distribution over a set of variables is usually represented by matrices, when two variables are involved, or multidimensional matrices (i.e., tensors), as the number of variables increases.

It is a natural consequence of the previous discussion that while the main mathematical tools for Classical Statistics are based on multivariate analysis and measure theory, the underlying mathematical machinery for Algebraic Statistics is principally based on the Linear and Multi-linear Algebra of tensors (over the integers, at the start, but quickly one considers both real and complex tensors).

Relations Among Variables

Just to give an example, let us consider the behavior of a population after the introduction of a new medicine.

Assume that a population is affected by a disease, which dangerously alters the value of a glycemic indicator in the blood. This dangerous condition is partially treated with the new drug. Assume that the purpose of the experiment is to detect the existence of a substantial improvement in the health of the patients.

In Classical Statistics, one considers the distribution of the random variable $X_1 =$ the value of the glycemic indicator over a selected population of patients *before* the delivery of the drug, and the random variable $X_2 =$ the value of the glycemic indicator of patients *after* the delivery of the drug. Both distributions are likely to be represented by Gaussians, the first one centered at an abnormally high value of the glycemic indicator, the second one centered at a (hopefully) lower value. The comparison between the two distributions aims to detect if (and how far) the descent of the recorded values of the glycemic indicator is statistically meaningful, i.e., if it can be distinguished from the natural underlying ground noise. The celebrated *Student's t-test* is the world-accepted tool for comparing the means of two Gaussian distributions and for determining the existence of a statistically significant response.

In many experiments, the response variable is binary or categorical with k levels, leading to a 2×2, or a $2 \times k$, contingency table. Moreover, when there is more than one response variable and/or other control variables, the resulting data are summarized in a multiway contingency table, i.e., a tensor.

This structure may also come from the discretization of a continuous variable. As an example, consider a population divided into two subsets, one of which is treated with the drug while the other is treated with traditional methods. Then, the values of the glycemic indicator are divided into classes (in the roughest case just two classes, i.e., a threshold which separates two classes is established). After some passage of time, one records the distribution of the population in the four resulting categories (treated + under-threshold, treated + over-threshold ...) which determines a 2×2 matrix, whose properties encode the existence of a relation between the new treatment and an improved normalization of the value of the glycemic indicator (this is just to give an example: in the real world, a much more sophisticated analysis is recommended!).

Bernoulli Binary Models

Another celebrated model, which is different from the Gaussian distribution and is often introduced at the beginning of a course in Statistics, is the so-called Bernoulli model over one binary variable.

Assume we are given an object that can assume only two states. A coin, with the two traditional states H (heads) and T (tails), is a good representation. One has to

bear in mind, however, that in the real world, binary objects usually correspond to *biased* coins, i.e., coins for which the expected distribution over the two states is not even.

If p is the *probability* of obtaining a result (say H) by throwing the coin, then one can roughly estimate p by throwing the coin several times and determining the ratio

$$\frac{\text{number of throws giving } H}{\text{total number of throws}}$$

but this is usually considered too naïve. Instead, one divides the total set of throws into several packages, each consisting of r throws, and determines for how many packages, denoted $q(t)$, one obtained H exactly t times. The value of the constant p is thus determined by Bernoulli's formula:

$$q(t) = \binom{r}{t} p^t (1-p)^{r-t}.$$

By increasing the number of total throws (and thus increasing the number of packages and the number of throws r in each package), the function $q(t)$ tends to a real function, which can be treated with the usual analytic methods.

Notice that in this way, at the end of the process, the discrete variable *Coin* is substituted by a continuous variable $q(t)$. Usually one even goes one step further, by substituting the variable q with its logarithm, ending up with a linear description.

Algebraic Statistics is scarcely interested in knowing how a single given coin is biased. Instead, the main goal of Algebraic Statistics is to understand the connections between the behavior of *two* coins. Or, better, the connections between the behavior of a collection of coins.

Consequently, in Algebraic Statistics one defines a collection of variables, one for each coin, and defines a distribution by counting the records in which the variables $X_1, X_2, \ldots, X_n$ have a fixed combination of states. The distribution is transformed into a tensor of type $2 \times 2 \times \cdots \times 2$. All coins can be biased, with different loads: this does not matter too much. In fact, the main questions that one expects to solve are

- Are there connections between the outputs of two or more coins?
- Which kind of connection is suggested by the distribution?
- Can one divide the collection of coins into clusters, such that the behavior of coins of the same cluster is similar?

Answers are expected from an analysis of the associated tensor, i.e., in the framework of Multi-linear Algebra.

The importance of the last question can be better understood if one replaces coins with positions in a composite digital signal. Each position has, again, two possible states, 0 and 1. If the signal is the result of the superposition of many

elementary signals, coming from different sources, and digits coming from the same source behave similarly, then the division of the signal into clusters yields the reconstruction of the original message that each source issued.

Splitting into Types

Of course, the separation of several phenomena that are mixed together in a given distribution is also possible using methods of Classical Statistics.

In a famous analysis of 1894, the biologist Karl Pearson made a statistical study of the shape of a population of crabs (see [1]). He constructed the histogram for the ratio between the "forehead" breadth and the body length for 1000 crabs, sampled in Naples, Italy by W. F. R. Weldon. The resulting approximating curve was quite different from a Gaussian and presented a clear asymmetry around the average value. The shape of the function suggested the existence of two distinct types of crab, each determining its own Gaussian, that were mixed together in the observed histogram. Pearson succeeded in separating the two Gaussians with the method of *moments*. Roughly speaking, he introduced new statistical variables, induced by the same collection of data, and separated the types by studying the interactions between the Gaussians of these new variables.

This is the first instance of a computation which takes care of *several* parameters of the population under analysis, though the variables are derived from the same set of data. Understanding the interplay between the variables provides the fundamental step for a qualitative description of the population of crabs.

From the point of view of Algebraic Statistics, one could obtain the same description of the two types which compose the population by adding variables representing other ratios between lengths in the body of crabs, and analyzing the resulting tensor.

Mixture Models

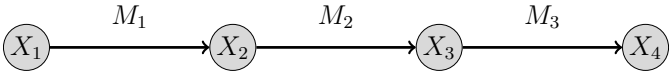
Summarizing, Algebraic Statistics becomes useful when the existence and the nature of the relations between several random variables are explored.

We stress that knowing the shape of the interaction between random variables is a central problem for the description of phenomena in Biology, Chemistry, Social Sciences, etc. Models for the description of the interactions are often referred to as *Mixture Models*. Thus, mixture models are a fundamental object of study in Algebraic Statistics.

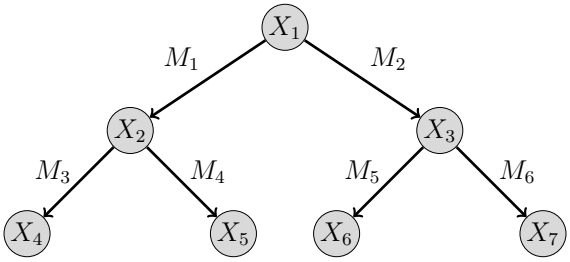
Perhaps, the most famous and easily described mixture models are the *Markov chains*, in which the set of variables is organized in a totally ordered chain, and the behavior of the variable X_i is only influenced by the behavior of the variable X_{i-1} (usually, this interaction depends on a given matrix).

Of course, much more complicated types of networks are expected when the complexity of the collection of variables under analysis increases. So, when one studies composite signals in the real world, or pieces of a DNA chain, or regions in a neural tissue, higher level models are likely to be necessary for an accurate description of the phenomenon.

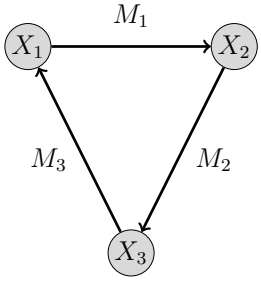
One thus moves from the study of Markov chains



to the study of Markov trees



and the study of neural nets



In Classical Statistics, the structure of the connections among variables is often a postulate. In Algebraic Statistics, determining the combinatorics and the topology of the network is a fundamental task. On the other hand, the time-dependent activating functions that transfer information from one variable to the next ones, deeply studied by Classical Statistics, are of no immediate interest for Algebraic Statistics which, at first, considers steady states of the configuration of variables.

The Multi-linear Algebra behind the aforementioned models is not completely understood. It requires a deep analysis of subsets of linear spaces described by parametric or implicit *polynomial* equations. This is the reason why, at a certain point, methods of Algebraic Geometry are invoked to push the analysis further.

Conclusion

The way we think about Algebraic Statistics focuses on aspects of the theory of random variables which are different from the targets of Classical Statistics. This is reflected in the point of view introduced in the book. Our general setting differs from the classical one and is closer to the one implicitly introduced in the books of Pachter and Sturmfels [2] and Sullivant [3]. Our aim is not to create a new formulation of the whole statistical theory, but only to present an algebraic natural way in which Statistics can handle problems related to mixture models.

The discipline is currently living in a rapidly expanding network of new insights and new areas of application. Our knowledge of what we can do in this area is constantly increasing and it is reasonable to hope that many of the problems introduced in this book will soon be solved or, if they cannot be solved completely, then they will at least be better understood. We feel that the time is right to provide a systematic foundation, with special attention to the application of tensor theory, for a field that promises to act as a stimulus for mathematical research in Statistics, and also as a source of suggestions for further developments in Multi-linear Algebra and Algebraic Geometry.

Siena, Italy — Cristiano Bocci
May 2019 — Luca Chiantini

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Part I
Algebraic Statistics

Chapter 1
Systems of Random Variables and Distributions



1.1 Systems of Random Variables

This section contains the basic definitions with which we will construct our statistical theory.

It is important to point out right away that in the field of Algebraic Statistics, a still rapidly developing area of study, the basic definitions are not yet standardized. Therefore, the definitions which we shall use in this text can differ significantly (more in form than in substance) from those of other texts.

Definition 1.1.1 A *random variable* is a variable x taking values in a finite non-empty set of symbols, denoted $A(x)$. The set $A(x)$ is called the *alphabet* of x or the *set of states* of x. We will say that every element of $A(x)$ is a *state* of the variable x.

A *system of random variables* (or random system) S is a finite set of random variables.

The condition of finiteness, required both for the alphabet of a random variable and the number of variables of a system, is typical of *Algebraic* Statistics. In other statistical situations this hypothesis is often not present.

Definition 1.1.2 A *subsystem* of a system S of random variables is a system defined by a subset $S' \subset S$.

Example 1.1.3 The simplest examples of a system of random variables are those containing a single random variable. A typical example is obtained by thinking of a die x as a random variable, i.e. as the unique element of S. Its alphabet is $A(x) = \{1, 2, 3, 4, 5, 6\}$.

Another familiar example comes by thinking of the only element of S as a coin c with alphabet $A(c) = \{H, T\}$ (heads and tails).

Example 1.1.4 On internet sites about soccer betting one finds systems in which each random variable has three states. More precisely the set S of random variables are (say) all the professional soccer games in a given country. For each random

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variable x, (i.e. game), its alphabet is $A(x) = \{1, 2, T\}$. The random variable takes value "1" if the game was won by the home team, value "2" if the game was won by the visiting team and value "T" if the game was a tie.

Example 1.1.5 (a) We can, similar to Example 1.1.3, construct a system S with two random variables, namely with two dice $\{x_1, x_2\}$, both having alphabet $A(x_i) = \{1, 2, 3, 4, 5, 6\}$.

(b) An example of another system of random variables T, closely related to the previous one but **different**, is given by taking a single random variable as the *ordered pair* of dice $x = (x_1, x_2)$ and, as alphabet $A(x)$, all possible values obtained by throwing the dice simultaneously: $\{(1, 1), \ldots (1, 6), \ldots, (6, 1), (6, 2), \ldots, (6, 6)\}$.

(c) Another example W, still related to the two above (but different), is given by taking as system the unique random variable the set consisting of two dice $z = \{x_1, x_2\}$ and as alphabet, $A(z)$, the sum of the values of the two dice after throwing them simultaneously: $A(z) = \{2, 3, 4, \ldots, 12\}$.

Remark 1.1.6 The random variables of the systems S, T and W might seem, at first glance, to be the same, but it is important to make clear that they are **very different**. In (a) there are two random variables while in (b) and (c) there is only one. Also notice that in T we have chosen an ordering of the two dice, while in W the random variable is an unordered set of two dice. With example (a) there is nothing stopping us from throwing the die x_1, say, twenty times and the die x_2 ten times. However, in both (b) and (c) the dice are each thrown the same number of times.

Example 1.1.7 There are many naturally occurring examples of systems with many random variables. In fact, some of the most significant ones come from applications in Economics and Biology and have an astronomical number of variables.

For example, in Economics and in market analysis, there are systems with one random variable for each company which trades in a particular market. It is easy to see that, in this case, we can have thousands, even tens of thousands, of variables.

In Biology, very important examples come from studying systems in which the random variables represent hundreds (or thousands) of positions in the DNA sequence of one or several species. The alphabet of each variable consists of the four basic ingredients of DNA: Adenine, Cytosine, Guanine and Thymine. As a shorthand notation, one usually denotes the alphabet of such random variables as $\{A, C, G, T\}$.

In this book, we will refer to the systems arising from DNA sequences, as *DNA-systems*.

Example 1.1.8 For cultural reasons (one of the authors was born and lives in Siena!), we will have several examples in the text of systems describing probabilistic events related to the famous and colourful Sienese horse race called the Palio di Siena. Horses which run in the Palio represent the various medieval neighbourhoods of the city (called *contrade*) and the Palio is a substitute for the deadly feuds which existed between the various sections of the city.

The names of the neighbourhoods are listed below with a shorthand letter abbreviation for each of them:

Aquila	(eagle)	(symbol: A)	Bruco	(caterpillar)	(symbol: B)
Chiocciola	(snail)	(symbol: H)	Civetta	(little owl)	(symbol: C)
Drago	(dragon)	(symbol: D)	Giraffa	(giraffe)	(symbol: G)
Istrice	(crested porcupine)	(symbol: I)	Leocorno	(unicorn)	(symbol: E)
Lupa	(she-wolf)	(symbol: L)	Nicchio	(conch)	(symbol: N)
Oca	(goose)	(symbol: O)	Onda	(wave)	(symbol: Q)
Pantera	(panther)	(symbol: P)	Selva	(forest)	(symbol: S)
Tartuca	(tortoise)	(symbol: R)	Torre	(tower)	(symbol: T)
Valdimontone	(valley of the ram)	(symbol: M).			

Definition 1.1.9 A random variable x of a system S is called a *boolean variable* if its alphabet has cardinality 2. A system is *boolean* if all its random variables are boolean.

Remark 1.1.10 The states of a boolean random variable can be thought of as the pair of conditions *(true, false)*. As a matter of fact the standard alphabet of a boolean random variable can be thought of as the elements of the finite field $\mathbb{Z}_2$, where $1 =$ true and $0 =$ false (this is our convention; be careful: in some texts this notation is reversed!). Other alphabets, such as heads-tails or even-odd, are also often used for the alphabets of boolean random variables.

Definition 1.1.11 A *map* or *morphism* between systems S and T of random variables is a pair $f = (F, G)$ where F is a function $F : S \to T$ and, for all $x \in S$, G defines a function between alphabets $G(x) : A(x) \to A(F(x))$.

The terminology used for functions can be transferred to maps of system of random variables. Thus we can have injective maps (in which case both F and each of the $G(x)$ are injective), surjective maps (in which case both F and each of the $G(x)$ are surjective), isomorphism (in which case both F and all the maps $G(x)$ are $1-1$ correspondences). With respect to these definitions, the systems of random variables form a category.

Example 1.1.12 If S' is a subsystem of S, the inclusion function $S' \to S$ defines, in a obvious way, an injective map of systems. In this case, the maps between alphabets are always represented by the identity map.

Example 1.1.13 Let $S = \{x\}$ be the system defined by a die as in Example 1.1.3 with alphabet $\{1, 2, 3, 4, 5, 6\}$. Let T be the system defined by $T = \{y\}$, with $A(y) = \{E, O\}$ ($E =$ even, $O =$ odd). The function $F : S \to T$, defined by $F(x) = y$, and the function $G : A(x) \to A(y)$ defined by $G(1) = G(3) = G(5) = O$ and $G(2) = G(4) = G(6) = E$, define a surjective map from S to T of systems of random variables.

The following definition will be of fundamental importance for the study of the relationship between systems of random variables.

Definition 1.1.14 The *(total) correlation* of a system S of random variables $\{x_1, \ldots, x_n\}$ is the system $\Pi S = \{x\}$, with a unique random variable $x = (x_1, \ldots, x_n)$ (the cartesian product of the elements $x_1, \ldots, x_n$ of S). Its alphabet is given by $A(x_1) \times \cdots \times A(x_n)$, the cartesian product of the alphabets of the individual random variables.

Remark 1.1.15 It is very important to notice that the definition of the total correlation uses the concept of cartesian product. Moreover the concept of cartesian product requires that we *fix an ordering of the variables in S*.

Thus, the total correlation of a system is not uniquely determined, but it changes as the chosen ordering of the random variables changes.

It is easy to see, however, that all the possible total correlations of the system S are isomorphic.

Example 1.1.16 If S is a system with *two* coins c_1, c_2, each having alphabet $\{H, T\}$, then the only random variable in its total correlation, has an alphabet with *four* elements $\{(T, T), (T, H), (H, T), (H, H)\}$, i.e. we have to distinguish between the states (H, T) and (T, H). This is how the ordering of the coins enters into the definition of the random variable (c_1, c_2) of the total correlation.

Example 1.1.17 Let S be the system of random variables consisting of two dice, D_1 and D_2 each having alphabet the set $\{1, 2, 3, 4, 5, 6\}$. The total correlation of this system, ΠS, is the system with a unique random variable $D = (D_1, D_2)$ and alphabet the set $\{(i, j) \mid 1 \leq i, j \leq 6\}$. So, the alphabet consists of 36 elements.

Now let T be the system whose unique random variable is the set $x = \{D_1, D_2\}$ and whose alphabet consists of the eleven numbers $\{2, 3, \ldots, 11, 12\}$.

We can consider the surjective morphism of systems $\phi : \Pi S \to T$ which takes the unique random variable of ΠS to the unique random variable of T and takes the element (i, j) of the alphabet of the unique variable of ΠS to $i + j$ in the alphabet of the unique variable of T.

Undoubtedly this morphism is familiar to us all!

Clearly if S is a system containing a single random variable, then X coincides with its total correlation.

Definition 1.1.18 Let $f : S \to T$ be a map of systems of random variables, defined by $F : S \to T$ and $G(x) : A(x) \to A(F(x))$ for all random variables $x \in S$, and suppose that F is bijective i.e. S and T have the same number of random variables.

Then f defines, in a natural way, a map $\Pi f : \Pi S \to \Pi T$ between the total correlations as follows: for each state $s = (s_1, \ldots, s_n)$ of the unique variable $(x_1, \ldots, x_n)$ of ΠS,

$$(\Pi f)(s) = (G(x_1)(s_1), \ldots, G(x_n)(s_n)).$$

1.2 Distributions

One of the basic notions in the study of systems of random variables is the idea of a *distribution*. Making the definition of a distribution precise will permit us to explain clearly the idea of an *observation* on the random variables of a system. This latter concept is extremely useful for the description of real phenomena.

Definition 1.2.1 Let K be any set. A *K-distribution* on a system S with random variables $x_1, \ldots, x_n$, is a set of functions $D = \{D_1, \ldots, D_n\}$, where for $1 \leq i \leq n$, D_i is a function from $A(x_i)$ to K.

Remark 1.2.2 In most concrete examples, K will be a numerical set, i.e. some subset of $\mathbb{C}$ (the complex numbers).

The usual use of the idea of a distribution is to associate to each state of a variable x_i in the system S, the number of times (or the percentage of times) such a state is observed in a sequence of observations.

Example 1.2.3 Let S be the system having as unique random variable a coin c, with alphabet $A(c) = \{T, H\}$ (the coin needs not be fair!).

Suppose we throw the coin n times and observe the state T exactly d_T times and the state H exactly d_H times ($d_T + d_H = n$). We can use those observations to get an $\mathbb{N}$-distribution ($\mathbb{N}$ is the set of natural numbers), denoted D_c, where $D_c : \{T, H\} \to \mathbb{N}$ by

$$D_c(T) = d_T, D_c(H) = d_H \ .$$

One can identify this distribution with the element $(d_T, d_H) \in \mathbb{N}^2$.

We can define a different distribution, D'_c on S (using the same series of observations) as follows:

$$D'_c : \{T, H\} \to \mathbb{Q} \text{ (the rational numbers)},$$

$$\text{where } D'_c(T) = d_T/n \text{ and } D'_c(H) = d_H/n.$$

Example 1.2.4 Now consider the system S with *two* coins c_1, c_2, and with alphabets $A(c_i) = \{T, H\}$.

Again, suppose we simultaneously throw both coins n times and observe that the first coin comes up with state T exactly d_1 times and with state H exactly e_1 times, while the second coin comes up T exactly d_2 times and comes up H exactly e_2 times.

From these observations we can define an $\mathbb{N}$-distribution, $D = (D_1, D_2)$, on S defined by the functions

$$D_1 : \{T, H\} \to \mathbb{N}, \quad D_1(T) = d_1, \ D_1(H) = e_1,$$
$$D_2 : \{T, H\} \to \mathbb{N}, \quad D_2(T) = d_2, \ D_2(H) = e_2$$

It is also possible to identify this distribution with the element

$$((d_1, e_1), (d_2, e_2)) \in \mathbb{N}^2 \times \mathbb{N}^2.$$

It is also possible to use this series of observations to define a distribution on the total correlation ΠS.

That system has a unique variable $c = c_1 \times c_2 = (c_1, c_2)$ with alphabet $A(c) = \{TT, TH, HT, HH\}$. The $\mathbb{N}$-distribution on ΠS we have in mind associates to each of the four states how often that state appeared in the series of throws.

Notice that if we only had the first distribution we could not calculate the second one since we would not have known (solely from the functions D_1 and D_2) how often each of the states in the second system were observed.

Definition 1.2.5 The set of K-distributions of a system S of random variables forms the *space of distributions* $\mathcal{D}_K(S)$.

Example 1.2.6 Consider the DNA-system S with random variables precisely 100 fixed positions (or sites) $p_1, \ldots, p_{100}$ on the DNA strand of a given organism. As usual, each variable has alphabet $\{A, C, G, T\}$. Since each alphabet has exactly four members, the space of $\mathbb{Z}$-distributions on S is $\mathcal{D}(S) = \mathbb{Z}^4 \times \cdots \times \mathbb{Z}^4(100\text{times}) = \mathbb{Z}^{400}$.

Suppose we now collect 1,000 organisms and observe which DNA component occurs in site i. With the data so obtained we can construct a $\mathbb{Z}$-distribution $D = \{D_1, \ldots, D_{100}\}$ on S where D_i associates to each of the members of the alphabet $A(p_i) = \{A, C, G, T\}$ the number of occurrences of the corresponding component in the $i-$th position. Note that for each D_i we have

$$D_i(A) + D_i(C) + D_i(G) + D_i(T) = 1{,}000.$$

Remark 1.2.7 Suppose that S is a system with random variables $x_1, \ldots, x_n$ and that the cardinality of each alphabet $A(x_i)$ is exactly a_i. As we have said before, a_i is simply the number of states that the random variable x_i can assume.

With this notation, the K-distributions on S can be seen as points in the space

$$K^{a_1} \times \cdots \times K^{a_n}.$$

We will often identify $\mathcal{D}_K(S)$ with this space.

It is also certainly true that $K^{a_1} \times \cdots \times K^{a_n} = K^{a_1 + \cdots + a_n}$, and so it might seem reasonable to say that this last is the set of distributions on S. However, since there are so many ways to make this last identification, we could easily lose track of what a particular distribution did on a member of the alphabet of one of the variables in S.

Remark 1.2.8 If S is a system with two variables x_1, x_2, whose alphabets have cardinality (respectively) a_1 and a_2, then the unique random variable in the total correlation

ΠS has $a_1 a_2$ states. Hence, as we said above, the space of K-distributions on ΠS could be identified with $K^{a_1 a_2}$.

Since we also wish to remember that the unique variable of ΠS arises as the cartesian product of the variables of S, it is even more convenient to think of $\mathcal{D}_K(\Pi S) = K^{a_1 a_2}$ as the set of $a_1 \times a_2$ matrices with coefficients in K.

Thus, for a distribution D on ΠS, we denote by D_{ij} the value associated to the state (i, j) of the unique variable, which corresponds to the states i of x_1 and j of x_2.

For system with a bigger number of variables, we need to use *multidimensional matrices*, commonly called *tensors* (see Definition 6.1.3).

The study of tensors is thus strongly connected to the study of systems of random variables when we want to fix relationships among the variables (i.e. look at distributions on the system). In fact, the algebra (and geometry) of spaces of tensors represents the point of connection between the study of statistics on discrete sets and other disciplines, such as Algebraic Geometry. The exploration of this connection is our main goal in this book. We will take up that connection in another chapter.

Definition 1.2.9 Let S and T be two systems of random variables and $f = (F, G) : S \to T$ a map of systems where F is a surjection. Let D be a distribution on S, and D' a distribution on T.

The *induced distribution* f_*^D on T (called *the image distribution*) is defined as follows: for t a state of the variable $y \in T$:

$$(f_*^D)_y(t) = \sum_{x \in F^{-1}(y), s \in G(x)^{-1}(t)} D_x(s).$$

The *induced distribution* $f_{D'}^*$ on S (called *the preimage distribution*) is defined as follows: for s a state of the variable x in S:

$$(f_{D'}^*)_x(s) = D'_{F(x)}(G(x)(s)).$$

We want to emphasize that distributions on a system of random variables should, from a certain point of view, be considered as data on a problem. Data from which one hopes to deduce other distributions or infer certain physical, biological or economic facts about the system. We illustrate this idea with the following example.

Example 1.2.10 In the city of Siena (Italy) two spectacular horse races have been run every year since the seventeenth century, with a few interruptions caused by the World Wars. Each race is called a *Palio*, and the Palio takes place in the main square of the city. In addition there have been some additional extraordinary Palios run from time to time. From the last interruption, which ended in 1945, up to now (2014), a total number of 152 Palios have taken place. Since the main square is large, but not enormous, not every contrada can participate in every Palio. There is a method, partly based on chance, that decides whether or not a contrada can participate in a particular Palio.

Table 1.1 Participation of the contrade at the 152 Palii (up to 2014)

x	*Name*	$D_x(1)$	$D_x(0)$	x	*Name*	$D_x(1)$	$D_x(0)$
A	*Aquila*	88	64	*B*	*Bruco*	92	60
H	*Chiocciola*	84	68	*C*	*Civetta*	90	62
D	*Drago*	95	57	*G*	*Giraffa*	89	63
I	*Istrice*	84	68	*E*	*Leocorno*	99	52
L	*Lupa*	89	63	*N*	*Nicchio*	84	68
O	*Oca*	87	65	*Q*	*Onda*	84	68
P	*Pantera*	96	56	*S*	*Selva*	89	63
R	*Tartuca*	91	61	*T*	*Torre*	90	62
M	*Valdimontone*	89	63				

Let's build a system with 17 boolean random variables, one for each contrada. For each variable we consider the alphabet $\{0, 1\}$. The space of $\mathbb{Z}$-distributions of this system is $\mathbb{Z}^2 \times \cdots \times \mathbb{Z}^2 = \mathbb{Z}^{34}$.

Let us define a distribution by indicating, for each contrada x, $D_x(1) =$ number of Palios where contrada x took part and $D_x(0) =$ number of Palios where contrada x did not participate. Thus we must always have $D_x(0) + D_x(1) = 152$. The data are given in Table 1.1.

We see that the Leocorno (unicorn) contrada participated in the most Palios while the contrada Istrice (crested porcupine), Nicchio (conch), Onda (wave), Chiocciola (snail) participated in the fewest.

On the same system, we can consider another distribution E, where $E_x(1) =$ number of Palios that contrada x won and $E_x(0) =$ number of Palios that contrada x lost (non-participation is considered a loss). The Win-Loss table is given in Table 1.2.

From the two tables we see that more participation in the Palios does not necessarily imply more victories.

Table 1.2 Win-Loss table of contrade at the 152 Palii (up to 2014)

x	*Name*	$E_x(0)$	$E_x(1)$	x	*Name*	$E_x(0)$	$E_x(1)$
A	*Aquila*	8	144	*B*	*Bruco*	5	147
H	*Chiocciola*	9	143	*C*	*Civetta*	8	144
D	*Drago*	11	141	*G*	*Giraffa*	12	140
I	*Istrice*	8	144	*E*	*Leocorno*	9	143
L	*Lupa*	5	147	*N*	*Nicchio*	9	143
O	*Oca*	14	138	*Q*	*Onda*	9	143
P	*Pantera*	8	144	*S*	*Selva*	15	137
R	*Tartuca*	10	142	*T*	*Torre*	3	149
M	*Valdimontone*	9	143				

1.3 Measurements on a Distribution

We now introduce the concepts of *sampling* and *scaling* on a distribution for a system of random variables.

Definition 1.3.1 Let K be a numerical set and let $D = (D_1, \ldots, D_n)$ be a distribution on the system of random variables $S = \{x_1, \ldots, x_n\}$. The number

$$c_D(x_i) = \sum_{s \in A(x_i)} D_i(s).$$

is called the *sampling* of the variable x_i in D. We will say that D has *constant sampling* if all variables in S have the same sampling in D.

A K-distribution D on S is called *probabilistic* if each $x_i \in S$ has sampling equal to 1.

Remark 1.3.2 Let S be a system with random variables $\{x_1, \ldots, x_n\}$ and let $D = (D_1, \ldots, D_n)$ be a K-distribution on S, where K is a numerical field.

If every variable x_i has sampling $c_D(x_i) \neq 0$, we can obtain from D an *associated probabilistic distribution* $\tilde{D} = (\tilde{D}_1, \ldots \tilde{D}_n)$ defined as follows:

$$\text{for all } i \text{ and for all states } s \in A(x_i) \text{ set } \tilde{D}_i(s) = \frac{D_i(s)}{c_D(x_i)}.$$

Remark 1.3.3 In Example 1.2.3, the distribution D' is exactly the probabilistic distribution associated to D (seen as a $\mathbb{Q}$-distribution).

Convention. To simplify the notation in what follows and since we will always be thinking of the set K as some set of numbers, usually clear from the context, we won't mention K again but will speak simply of a *distribution* on a system S of random variables.

Warning. We want to remind the reader again that the basic notation in Algebraic Statistics is far from being standardized. In particular, the notation for a distribution is quite varied in the literature and in other texts.

E.g. if s_{ij} is the $j-$th state of the $i-$th variable x_i of the system S, and D is a distribution on S, we will denote this by writing $D_i(s_{ij})$ as the value of D on that state.

You will also find this number $D_i(s_{ij})$ denoted by $D_{x_i = s_{ij}}$.

Example 1.3.4 Suppose we have a tennis tournament with 8 players where a player is eliminated as soon as that player loses a match. So, in the first set of matches four players are eliminated and in the second two more are eliminated and then we have the final match between the remaining two players.

We can associate to this tournament a system with 8 boolean random variables, one variable for each player. We denote by D the distribution that, for each player x_i, is defined as:

$$D_i(0) = \text{ number of matches lost;}$$

$$D_i(1) = \text{ number of matches won.}$$

Clearly the sampling $c(x_i)$ of every player x_i represents the number of matches played. For example, $c(x_i) = 3$ if and only if x_i is a finalist, while $c(x_i) = 1$ for the four players eliminated at the end of the first match. Hence D is not a distribution with constant sampling.

Notice that this distribution doesn't have any variable with sampling equal to 0 and hence there is an associated probabilistic distribution $\tilde{D}$, which represents the statistics of victories. For example, for the winner x_k, one has

$$\tilde{D}_k(0) = 0, \quad \tilde{D}_k(1) = 1.$$

Instead, for a player x_j eliminated in the semi-final,

$$\tilde{D}_j(0) = \tilde{D}_j(1) = \frac{1}{2}.$$

While for a player x_i eliminated after the first round we have

$$\tilde{D}_i(0) = 1, \quad \tilde{D}_i(1) = 0.$$

The concept of an *associated probabilistic distribution* to a distribution D is quite important in texts concerned with the analytic Theory of Probability. This is true to such an extent that those texts work directly only with probabilistic distributions.

This is not the path we have chosen in this text. For us the concept that will be more important than a probabilistic distribution is the concept of *scaling*. This latter idea is more useful in connecting the space of distributions with the usual spaces in which Algebraic Geometry is done.

Definition 1.3.5 Let $D = (D_1, \ldots, D_n)$ be a distribution on a system S with random variables $\{x_1, \ldots, x_n\}$. A distribution $D' = (D'_1, \ldots, D'_n)$ is a *scaling* of D if, for any $x = x_i \in S$, there exists a constant $\lambda_x \in K \setminus \{0\}$ such that, for all states $s \in A(x)$, $D'_x(s) = \lambda_x D_x(s)$.

Remark 1.3.6 Notice that the probabilistic distribution, D', associated to a distribution D is an example of a scaling of D, where $\lambda_x = 1/c(x)$.

Note moreover that, given a scaling D' of D, if D, D' have the same sampling, then they must coincide.

Remark 1.3.7 In the next chapters we will see that scaling doesn't substantially change a distribution. Using a projectivization method, we will consider two distributions "equal" if they differ only by a scaling.

Proposition 1.3.8 *Let $f : S \to T$ be a map of systems which is a bijection on the sets of variables. Let D be a distribution on S and D' a scaling of D. Then $f_*^{D'}$ is a scaling of f_*^D.*

Proof Let y be a variable of T and let $t \in A(y)$. Since f is a bijection there is a unique $x \in S$ for which $f(x) = y$. Then by definition we have

$$(f_*^{D'})_y(t) = \sum_{s \in A(x)} D'(s) = \sum_{s \in A(x)} \lambda_x D(s) = \lambda_x (f_*^D)_y(t). \qquad \square$$

1.4 Exercises

Exercise 1 Let us consider the random system associated with the tennis tournament, see Example 1.3.4.

Compute the probabilistic distribution for the finalist who did not win the tournament.

Compute the probabilistic distribution for a tournament with 16 participants.

Exercise 2 Let S be a random system with variables $x_1, \ldots, x_n$ and assume that all the variables have the same alphabet $A = \{s_1, \ldots, s_m\}$. Then one can create the *dual* system S' by taking $s_1, \ldots, s_m$ as variables, each s_i with alphabet $X = \{x_1, \ldots, x_n\}$.

Determine the relation between the dimension of the spaces of K-distributions of S and S'.

Exercise 3 Let S be a random system and let S' be a subsystem of S.

Determine the relation between the spaces of K-distributions of the correlations of S and S'.

Exercise 4 Let $f : S \to S'$ be a surjective map of random systems.

Prove that if a distribution D on S' has constant sampling, then the same is true for f_D^*.

Exercise 5 One can define a *partial correlation* over a system S, by connecting only *some* of the variables.

For instance, if S has variables $x_1, \ldots, x_n$ and $m < n$, one can consider the partial correlation on the variables $x_1, \ldots, x_m$ as a system T whose variables are $Y, x_{m+1}, \ldots x_n$, where Y stands for the variable $x_1 \times \cdots \times x_m$, with alphabet the product $A(x_1) \times \cdots \times A(x_m)$.

If S has variables c_1, c_2, c_3, all of them with alphabet $\{T, H\}$ (see Example 1.1.3), determine the space of K-distributions of the partial correlation T with random variables $c_1 \times c_2$ and c_3.

Chapter 2
Basic Statistics



In this chapter, we focus on some basic examples in Probability and Statistics. We phrase these concepts using the language and definitions we have given in the previous chapter.

2.1 Basic Probability

Given a system S of random variables, we introduce the concept of the *probability* of each of the states $s_1, \ldots, s_n$ of a random variable x in S.

The first thing to keep in mind is that if we have n states then, in general, there is no reason to believe, a priori, that each of the states has probability $1/n$. Usually, in fact, we don't know what the probability for each of the states is when we begin to study a system of random variable.

Consider, for example, the case in which S has one random variable x which represents one running of the Palio di Siena (see Example 1.1.8). Then x has ten states, one for each of the ten contrade which will participate in that Palio. The probability that a particular contrada will win is not equal for each of the ten contrade participating in the Palio, but will depend on (among other things) the strength of the contrada's horse, the ability of the jockey, the historical deviousness of the contrada, etc.

Example 2.1.1 Let's consider the data associated with the Italian Series A Soccer games of 2005/2006. There were 380 games played that season with 176 games won by the home team, 108 games which resulted in a tie and the remaining 96 games won by the visiting team.

Suppose now we want to construct a random system S with a unique random variable x as in Example 1.1.4. Recall that the alphabet for x is $\mathcal{A}(x) = \{1, 2, T\}$ where 1 means the game was won by the home team, 2 means the game was won by the visiting team and T means the game ended in a tie. The 2005/2006 Season offers us a distribution D such that $D_s(1) = 176$, $D_s(T) = 108$, $D_s(2) = 96$.

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The normalization of D, which we will denote $\tilde{D}$, gives us some reasonable probabilities for the new season games. Namely:

$$\tilde{D}_s(1) = \frac{176}{380} \simeq 46.3\% \quad \tilde{D}_s(T) = \frac{108}{380} \simeq 28.4\% \quad \tilde{D}_s(2) = \frac{96}{380} \simeq 25.3\%.$$

So, making reference to a previous season, we have gathered some insight into the probabilities to assign the various states.

Before we continue our discussion of the probability of the states of a random variable in a random system, we want to introduce some more terminology for distributions.

Example 2.1.2 Let S be a system of random variables.

The *equidistribution*, denoted E, on S is the distribution that associates to every state of every random variable in the system, the value 1.

Now suppose the random variables of S are denoted by $x_1, \ldots, x_r$ and that the states of the variable x_i are $\{s_{i1}, \ldots, s_{in_i}\}$, then the *associated probabilistic distribution* of E, denoted $\tilde{E}$ (see Remark 1.3.2) is defined by

$$p^E_{x_i}(s_{ij}) = \frac{1}{n_i} \text{ for every } j,\ 1 \leq j \leq n_i.$$

Notice that this distribution is the one which gives us equal probability for all states in the alphabet of a fixed random variable.

This same probability distribution is clearly obtained if we start with the distribution cE, $c \in \mathbb{R}$, which associates to each state of each variable in the system, the value c and then take $c\tilde{E}_{x_i}(s_{ij}) = \frac{c}{cn_i}$.

Clearly, the equidistribution has constant sampling if and only if all variables have the same number of states.

Remark 2.1.3 There is a famous, but imprecise, formula for calculating the probability for something to occur. It is usually written as

$$\frac{\text{positive cases}}{\text{possible cases}}.$$

The problem with this formula is best illustrated by the following example. Assume that a person likes to play tennis very much. If he were asked: If you had a chance to play Roger Federer for one point what is your probability of winning that point? Using the formula above he would offer as a reply that there are two possible outcomes and one of those has the person winning the point, so the probability is 0.5 (but maybe who has seen this person playing tennis would appreciate the absurdity of that reply!).

The problem with the formula is that it is only a reasonable formula if all outcomes are equally probable, i.e. if we have an equidistribution on the possible states—something far from the case in the example above.

Example 2.1.4 The Palio of Siena usually takes place twice a year, in July and August. Only 10 of the 17 contrade can participate in a Palio. How are the ten chosen?

First, the 7 contrade which did not participate in the previous year's Palio are automatically chosen for this year's Palio. The remaining three places are chosen, by a draw, from among the 10 contrade which did participate in the previous year's Palio.

What is the probability that a contrada x, which participated in the previous year's Palio, will participate in the current year's Palio?

To answer this question we construct two systems of random variables. The first system, T, will have only one random variable, which we will denote by e (for *extraction*). What are the elements in the alphabet $A(e)$ of e? In particular, how many states does e have ? We need to choose 3 contrade from a set of 10 contrade. We have 10 choices for the first extracted contrada, then 9 choices for the second extracted contrada and finally 8 choices for the third extracted contrada. The total is $10 \cdot 9 \cdot 8 = 720$ states, where each state is an ordered triplet which tells us which contrade was chosen first, which was chosen second and which was chosen third.

Since we will assume that the extractions will be made honestly, we can consider the equidistribution on $A(e)$ whose probability distribution associates to each triplet in the alphabet the value $1/720$.

Also the second system S has only one random variable corresponding to exactly one of the ten contrade from which we will make the extraction. Such a variable, which we will call c (where c really is the name of one of the contrade), is boolean. Its alphabet can be considered as $\mathbb{Z}_2$, where 1 signifies that c participates in the Palio and 0 signifies that it does not participate.

Consider the map $f_c : T \to S$, sending e to c and each triplet in $t \in A(e)$ to 1 or 0, depending on whether or not c is in t or c is not in t.

The probability that c will participate in the Palio of next July is defined by the probability distribution associated to $D = (f_c)^E_*$ on S.

$D(1)$ is equal to the number of triplets containing c. How many of these triplets are there?

The triplets where c is the first element are obtained by choosing the second element among the other 9 contrade and the third element among the remaining 8, hence we have 72 such triplets. Similarly for the number of triplets where c is in the second or in the third position, for a total of $72 \cdot 3 = 216$ triplets. Hence $D(1) = 216$ and, obviously, $D(0) = 720 - 216 = 504$. Thus, the probability for a contrada c to participate in the Palio of next July, assuming it already run in the Palio of the previous July, is:

$$p = \frac{D(1)}{D(0) + D(1)} = \frac{216}{720} = \frac{3}{10} \simeq 30\%.$$

In this example we could use the intuitive formula since we assumed the 720 possible states for the random variable e were equally distributed. With this in mind, the previous procedure yields a general construction which, for elementary situations, is enough to compute the *probability* that a given event occurs.

We introduce the definition of *booleanization* of a system of random variables. As a matter of fact, in the previous example, we passed from a system T of random variables to a boolean system associated to S. Practically speaking, we divide all possible states of every random variables in *good* states and *bad* states, sending the first one to 1 and the others to 0 (accordingly with the fact that a fixed contrada c is extracted or not).

Definition 2.1.5 We call *booleanization* or *dicotomia* of a system T, with variables $\{x_1, \ldots, x_n\}$, any pair (S, f) where S is a boolean system, with variables $\{y_1, \ldots, y_n\}$, and f is a map $f : T \to S$ which determines a bijection on the sets of variables.

Thus, the output of Example 2.1.4 is a consequence of the following result.

Proposition 2.1.6 *Let T be a system with a unique variable x and let E be the equidistribution on T. Consider a booleanization $f : T \to S$ of T and let D be the image distribution of E under f. The probability induced by D on S corresponds to the ratio where the denominator is the number of all states of the variable in T (the* possibile cases*) and the numerator is the sum of all states of the variable of T sent to* 1 *through f (the* positive cases*).*

Remark 2.1.7 It is obvious that, given a *rational* distribution D on a system S with a unique variable, we can always find a system S and a map $f : S \to T$ in such a way D is the image, through f, of the equidistribution on S.

Example 2.1.8 Let us give another example to stress how the formula

$$\frac{\text{positive cases}}{\text{possible cases}}$$

is rephrased in our setting.

Let T be a random system with one variable x representing a (fair) die. Thus x has 6 states. Which is the probability that launching the die one gets an even number?

Yes, everybody expects that the answer is $1/2 = 50\%$, but let us get the result in our setting.

We construct a boolean system S with one variable y and states $\{0, 1\}$ and construct the map $f : T \to S$ which sends 1, 3, 5 to 0 (bad cases) and 2, 4, 6 to 1 (good cases). Then the image in f of the equidistribution over T is the distribution D such that $D(0) = D(1) = 3$. Thus we get the expected answer $D(1)/(D(0) + D(1)) = 3/6 = 1/2$.

Remark 2.1.9 Notice that the previous construction of *probability*, in practice, is strongly influenced by the choice of the equidistribution on T. It is reliable only when we have good reasons (which, often, cannot be *mathematical* reasons) to believe that the states of any variable of T are *equiprobable*.

We cannot emphasize enough that in many concrete cases it is absolutely impossible to know, ahead of time, if the choice of the equidistribution on T makes sense.

It's enough to consider the example of a coin, about which we know nothing except that it has two states, H = heads or T = tails. What is the probability that after

tossing the coin we obtain one of the two possibilities? A priori, no one can affirm with certainty that the probability of each state is $1/2$ since the coin may not be a fair coin.

In a certain sense, physical, biological and economic worlds are filled with "unfair" coins. For example, on examining the first place in the DNA chain of large numbers of organisms, one observes that the four basic elements do not appear equally often; the element T appears much more frequently than the element A. In like manner, if one were attempting to understand the possible results of a soccer match, there is no way one would say that the three possibilities $1, 2, T$ are equally probable.

2.2 Booleanization and Logic Connectors

The elementary construction of probability, obtained in the previous section, becomes more involved when one wants to consider a series of events. Let us start with an example, taken again from the Palio.

Example 2.2.1 The procedure for the choice of the ten contradas which run in the Palio of July is repeated verbatim for the Palio of August. In August the race is open for the seven contrade that did not participate to the Palio of August of the previous year, plus three contrade, chosen by chance.

The procedures for the participation to the two Palios ran every year are independent. Hence it is possible that a contrada, one year, can participate to both Palio, or does not participate at all.

Assume that one contrada c did not participate to both Palio in 2014. Then it will participate to both Palio in 2015. What is the probability that it participates to both palio in 2016? What is the probability that it participates to at least one palio in 2016?

Observe that the problem can be easily solved by applying the results of Example 2.1.4. obtaining the expected solutions $\left(\frac{3}{10}\right)^2 = \frac{9}{100}$ and $1 - \left(\frac{7}{10}\right)^2 = 51/100$.

However, to give a more formal answer to these questions, we define a system T with two random variables $J =$ July and $A =$ August. For each variable, the alphabet is the set of possible triplets of extracted contrade. Hence each variable has 720 states (see Example 2.1.4).

Consider now $T' = \Pi S$, the total correlation on T. This furnishes all possible results of the extractions for the two Palii in 2008. S' has only one variable, with $(720)^2 = 518400$ states.

Consider the boolean system S with unique variable c and alphabet $\{0, 1\}$.

To compute the probability that c participates to both Palii, we build the map $\epsilon : T' \to S$, where ϵ sends (obviously) the variable $y \in T'$ to c and sends each state s of y, corresponding to a pair of triplets, to 1 or 0, depending if c appears in both the triplets or not.

How many states s are sent to 1? There are 216 triplets, among the 720 possible ones, where c appears. Then the pairs of triplets, in both of which c appears, are $216 \cdot 216 = 46656$.

The equidistribution E on T' induces on S the distribution $D = \epsilon_*^E$ such that $D(1) = 46656$ and $D(0) = 518400 - 46656 = 471744$. Hence, the probability for the contrada c to participate to both Palii, in 2016, is

$$\tilde{D}(1) = \frac{D(1)}{D(0) + D(1)} = \frac{46656}{518400} = \frac{9}{100} = 9\%.$$

To compute the probability that c participates to at least one Palio, we build the map $u : T' \to S$ where this time u sends each state s of y, corresponding to a pair of triplets, to 1 or 0, depending if c appears or not in at least one of the triplets in the pair.

How many states s are sent to 1 now? There are 216 triplets, among the 720 possible ones, where c appears in the first element of s. Among the $720 - 216 = 504$ remaining ones, there are $504 \cdot 216$ cases where c appears in the second element of s. Then, the pairs of triplets having c in at least one triplet are $216 \cdot 720 + 504 \cdot 216 = 264384$.

The equidistribution E on T' induces on T the distribution $R = u_*^E$ such that $R(1) = 264384$ and $R(0) = 518400 - 264384 = 254016$. Hence, the probability for the contrada c to participate to at least one Palio, in 2016, is

$$\tilde{R}(1) = \frac{R(1)}{R(0) + R(1)} = \frac{264384}{518400} = \frac{51}{100} = 51\%.$$

A different approach to this problem can be found in Example 2.3.19.

Notice that the example suggests a way to define, in general, what happens for the composition of two events. It suggests, however, that there are several ways in which the composition can be constructed, depending of which kind of logical operator one uses to combine the two events.

In the example, we used the two operators "for any" and "at least one", but other choices are possible (see Exercise 6).

Let us review, briefly, the setting of logic connectors.

Definition 2.2.2 Define an *elementary logic connector* simply as a map $\mathbb{Z}_2 \times \mathbb{Z}_2 \to \mathbb{Z}_2$. In other words, an elementary logic connector is an operation on $\mathbb{Z}_2$. Clearly there are exactly 16 elementary logic connectors.

Example 2.2.3 The most famous elementary logic connectors are AND, OR and AUT (which means "one of them, but not both"), described respectively by parts (a), (b) and (c) of Table 2.1.

Definition 2.2.4 Let S be a systems with two random variables x_1, x_2, and consider a booleanization E of S. Let ℓ be a logic connector. Then ℓ defines a booleanization E_ℓ of the total correlation ΠS by setting, for each pair s_1, s_2 of states of x_1, x_2:

$$E_\ell(s, t) = \ell(E(s), E(t)).$$

Table 2.1 Truth tables of AND (a), OR (b) and AUT (c)

AND	0	1
0	0	0
1	0	1

(a)

OR	0	1
0	0	1
1	1	1

(b)

AUT	0	1
0	0	1
1	1	0

(c)

The reader can easily rephrase Example 2.2.1 in terms of the booleanization induced by a connector ℓ.

Remark 2.2.5 One can extend the notion of logic connectors on n-tuples, by taking n-ary operations on $\mathbb{Z}_2$, i.e. maps $(\mathbb{Z}_2)^n \to \mathbb{Z}_2$. This yields an extension of a booleanization on a system S, with an arbitrary number of random variables, to a booleanization of ΠS.

The theory of logic connectors has useful applications. Just to mention one, it introduces a theoretical study of queries on databases.

However, since a systematic study of logic connectors goes far beyond the scopes of this book, we will not continue the discussion here.

2.3 Independence Connections and Marginalization

The numbers appearing in Example 2.2.1 are quite large and, in similar but slightly more complicate situations, become rapidly untreatable.

Fortunately, there is a simpler setting to reproduce the computations of probability performed in Example 2.2.1. It is based on the notion of "independence".

In order to make this definition we first show a connection between tensors and distributions, which extends the connection between distribution on a system with two variables and matrices, introduced in Remark 1.2.8.

Tensors are fundamental objects in Multi-linear Algebra, and they are introduced in Chap. 6 of the related part of the book. We introduce here the way tensors enter into the study of random systems, and this will provide the fundamental connection between Algebraic Statistics and Multi-linear Algebra.

Remark 2.3.1 Let S be a system of random variables, say $x_1, \ldots, x_n$ and suppose that x_i has a_i states and let $T = \Pi(S)$ be the total correlation of S. Recall that a distribution D on T assigns to every state of the unique variable $x = (x_1, \ldots, x_n)$, of T, a number. But the states of x correspond to ordered n-tuples $(s_{1j_1}, \ldots, s_{nj_n})$ where s_{ij_i} is a state of the variable x_i, i.e. a state of T corresponds to a particular location in a multidimensional array, and a distribution on T puts a number in that location. I.e. a distribution on T can be identified with an n-dimensional tensor of type $a_1 \times \cdots \times a_n$ and conversely.

The entry $T_{i_1\ldots,i_n}$ corresponds to the states i_1 of the variable x_1, i_2 of the variable $x_2, \ldots, i_n$ of the variable x_n.

Remark 2.3.2 When the original system S has only two variables (*random dipole*), then the distributions in the total correlation ΠS are represented as 2-dimensional tensors, i.e. usual matrices.

Example 2.3.3 Assume we have a system S of three coins, that we toss 15 times and record their outputs. Assume that we get the following result:

- 1 time we get HHH;
- 2 times we get HHT and TTH;
- 3 times we get HTH and HTT;
- 4 times we get TTT;
- we never get THH and THT

Then the corresponding distribution, in the space of $\mathbb{R}$-distributions of S which is $\mathcal{D}(S) = \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R}^2$, is $((9, 6), (3, 12), (6, 9))$, where $(9, 6)$ corresponds to 9 Heads and 6 Tails for the first coin, $(3, 12)$ corresponds to 3 Heads and 12 Tails for the second coin, and $(6, 9)$ corresponds to 6 Heads and 9 Tails for the third coin. The distribution over $T = \Pi S$ that corresponds to our data is the tensor of Example 6.4.15, namely:

$$D = \begin{array}{ccc} & 2 & 3 \\ 1 & & 3 \\ & 0 & 4 \\ 0 & & 2 \end{array}$$

(we identify H = 1 and T = 2).

In general, as we can immediately see from the previous Example, knowing the distribution on a system S of three coins is not enough to reconstruct the corresponding distribution on the total correlation $T = \Pi S$.

On the other hand, there is a special class of distributions in ΠS where the reconstruction is possible. It is the class of independence distributions defined below.

Definition 2.3.4 Let S be a system of random variables, $x_1, \ldots, x_n$ where each x_i has a_i states. Then $\mathcal{D}_K(S)$ is identified with $K^{a_1} \times \cdots \times K^{a_n}$. Set $T = \Pi S$ as the total correlation of S.

Define a function

$$\Pi : \mathcal{D}_K(S) \to \mathcal{D}_K(T)$$

called the *independence connection* (or *Segre connection*), in the following way: Π sends the distribution

$$D = ((d_{11}, \ldots, d_{1a_1}), \ldots, (d_{n1}, \ldots, d_{na_n}))$$

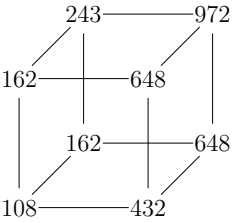
to the distribution $D' = \Pi(D)$ on ΠS (thought of as a tensor) such that

$$D'_{i_1,\dots,i_n} = d_{1i_1} \cdots d_{ni_n}.$$

Coherently with the notation of Chap. 6 we will denote the image of D as $v_1 \otimes \cdots \otimes v_n$, where $v_i = (d_{i1}, \dots, d_{ia_i})$.

Example 2.3.5 The distribution on ΠS in Example 2.3.3 above is not the image of $D = ((9, 6), (3, 12), (6, 9))$ under the independence connection Π, even after scaling. Namely the image of D under Π is

$D' =$



where $162 = 9 \cdot 3 \cdot 6$, $648 = 6 \cdot 12 \cdot 9$ and so on.

Remark 2.3.6 The image of the independence connection is, by definition, exactly the set of tensors of rank 1, as introduced in Definition 6.3.3.

The elements of the image will be called *distributions of independence*.

Proposition 2.3.7 *If D is a probabilistic distribution, then $\Pi(D)$ is a probabilistic distribution.*

Proof Let S be a system, with random variables $X = \{x_1, \dots, x_n\}$ and let $\Pi S = Y$ be a total correlation of S with alphabet B. Let D be a probabilistic distribution on S. Denote by y the unique random variable of ΠS. We need to prove that

$$\sum_{s_{1j_1} \in A(x_1), \dots, s_{nj_n} = A(x_n)} \Pi(D)_y(s_{1j_1}, \dots, s_{nj_n}) = 1.$$

We can observe that

$$\sum_{s_{1j_1} \in A(x_1), \dots, s_{nj_n} \in A(x_n)} \Pi(D)_y(s_{1j_1}, \dots, s_{nj_n}) =$$

$$= \sum_{s_{1j_1} \in A(x_1), \dots, s_{nj_n}} D_{x_1}(s_{1j_1}) \cdot D_{x_2}(s_{2j_2}) \cdot \cdots \cdot D_{x_n}(s_{nj_n}) =$$

$$= \Big(\sum_{s_{1j_1} \in A(x_1)} D_{x_1}(s_{1j_1}) \Big) \Big(\sum_{s_{2j_2} \in A(x_2), \dots, s_{nj_n} \in A(x_n)} D_{x_2}(s_{2j_2}) \cdot \cdots \cdot D_{x_n}(s_{nj_n}) \Big).$$

Since $\sum_{s_{1j_1} \in A(x_1)} D_{x_1}(s_{1j_1}) = 1$, the claim follow by induction on the number of random variables of S. □

Proposition 2.3.8 *Let $f = (F, G) : S \to T$ be a map between systems and D a distribution on S. Let $D' = f_*^D$ be the induced distribution on T. Then the distribution $\Pi(D)$ induced in the total correlation of S has, as image in Πf, the distribution $\Pi(D')$.*

Proof Denote by $x_1, \ldots, x_n$ the random variables in S and by $y_1, \ldots, y_n$ the random variables in T, with $y_i = F(x_i)$. For each state $t = (t_1, \ldots t_n)$ of the variable $(y_1 \times \cdots \times y_n)$ of ΠT, one has $\Pi(D')(t) = \Pi_{i=1,\ldots,n} D'(t_i)$ and

$$D'(t_i) = \sum_{s_{ij} \in A(x_i), G(x_i)(s_{ij}) = t_i} D(s_{ij}).$$

We know that $\Pi(D)(s_{1j_1}, \ldots, s_{nj_n}) = D(s_{1j_1}) \cdot \cdots \cdot D(s_{nj_n})$, hence

$$(\Pi f)_*^{\Pi(D)}(t_1, \ldots, t_n) = \sum_{(s_{1j_1}, \ldots, s_{nj_n}) \to (t_1, \ldots, t_n)} \Pi(D)(s_{1j_1}, \ldots, s_{nj_n})$$

coincides with $\Pi(D')(t_1, \ldots, t_n)$. □

Example 2.3.9 Let S be a boolean system with two random variables x, y, both with alphabet $\{0, 1\}$. Let D be the distribution defined as

$$D_x(0) = \frac{1}{6}, D_x(1) = \frac{5}{6}, D_y(0) = \frac{1}{6}, D_y(1) = \frac{5}{6}.$$

(which is clearly a probabilistic distribution).

Its product distribution on the total correlation with variable $z = x \times y$ is defined as

$$\begin{aligned}
\Pi(D)_z(0, 0) &= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \\
\Pi(D)_z(0, 1) &= \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{36} \\
\Pi(D)_z(1, 0) &= \frac{5}{6} \cdot \frac{1}{6} = \frac{5}{36} \\
\Pi(D)_z(1, 1) &= \frac{5}{6} \cdot \frac{5}{6} = \frac{25}{36}
\end{aligned}$$

which is still a probabilistic distribution since $\frac{1}{36} + \frac{5}{36} + \frac{5}{36} + \frac{25}{36} = 1$.

The independence connection can be, in some sense, inverted. For this aim, we need the definition of *(total) marginalization*.

Definition 2.3.10 Let S be a system of random variables and $T = \Pi S$ its total correlation. Define a function $M : \mathcal{D}_K(T) \to \mathcal{D}_K(S)$, called (total) marginalization in the following way. Given a tensor (thought as distribution on ΠS) D', $M(D')$ is

the total marginalization of D': $M(D')$ send the jth state of the variable x_i of S to the number $\sum D'_{a_i,\ldots,a_n}$, where the sum is taken over all elements in the tensor whose $i-$th index is equal to j.

We say that a distribution D on S is *coherent* with the distribution D' on ΠS if $M(D') = D$, i.e. $D' \in M^{-1}(D)$.

For a distribution D on S, we denote:

$$Co(D) = \{ \text{ distributions } D' \text{ on } \Pi S, \text{ coherent with } D\} = M^{-1}(D).$$

Assume that S is a system, with random variables $x_1, \ldots, x_n$, and let D be a distribution on S and D' be a distribution on ΠS. Then D' is coherent with D if, for all $i = 1, \ldots, n$ and for each state s_{ij} of x_i, one has

$$D(s_{ij_i}) = \sum D'(s_{1j_1}, \ldots, s_{ij_i}, \ldots, s_{nj_n})$$

where the sum ranges for all choices of states s_{kj_k} of x_k, with $k \neq i$.

Proposition 2.3.11 *If there exists is a distribution D' on ΠS coherent with the distribution D on S, then D has constant sampling.*

Proof It is sufficient to show that the sampling of D on no matter variable x_i is equal to the sampling of D' on the unique random variable of ΠS. One has:

$$\begin{aligned} c(x_i) &= \sum_{s_{ij_i} \in A(x_i)} D_{x_i}(s_{ij_i}) = \\ &= \sum_{s_{ij_i} \in A(x_i)} \left(\sum_{k \neq i, s_{kj_k} \in A(x_k)} D'(s_{1j_1}, \ldots, s_{ij_i}, \ldots, s_{nj_n}) \right) = \\ &= \sum_{s_{kj_k} \in A(x_k)} D'(s_{1j_1}, \ldots, s_{nj_n}) = \sum D'(s_{1j_1}, \ldots, s_{nj_n}) \end{aligned} \tag{2.3.1}$$

where the last sum runs on the states of the unique random variable of ΠS. □

Remark 2.3.12 From the previous proof it follows that if a distribution D on S is probabilistic, then every distribution D' su ΠS, coherent with D, is still probabilistic.

Example 2.3.13 Let us see now how independence connection and marginalization work together.

To this aim we introduce an example (honestly, rather harsh) on the efficiency of a medicine, that we will use also in the chapter of statistical models.

Consider a pharmaceutical company who want to verify if a specific product is efficient against a given pathology.

The company will try to verify such efficiency hiring volunteers (the *population*) affected by the pathology and giving the drug to some of them, a placebo to others. From the registration of the number of healings, the conclusions must be drawn.

The situation is described by a boolean system S, whose two variables x, y represent respectively the administration of the drug and the healing (as usual $1 =$ yes $0 =$ no).

On this system we introduce the following distribution D (with constant sampling)

$$D_x(0) = 20, \quad D_x(1) = 80, \quad D_y(0) = 30, \quad D_y(1) = 70.$$

This corresponds to an experiment with 100 persons, affected by the pathology: 80 persons receive the drug, while the other 20 receive the placebo. At the end of the observation, 30 subjects are still sick, while the remaining 70 are healed.

If Π is the independence connection, then $\Pi(D)$ is the 2×2 tensor (matrix):

$$\begin{pmatrix} 600 & 1400 \\ 2400 & 5600 \end{pmatrix}$$

We easily observe that $\Pi(D)$ is a matrix of rank 1 and this says that the medicine has not effects (is independent) on the healing of the patients.

The marginalization of $\Pi(D)$ gives the distribution D'' on X:

$$D''_x(0) = 600 + 1400 = 2000, \quad D''_x(1) = 2400 + 5600 = 8000,$$

$$D''_y(0) = 600 + 2400 = 3000, \quad D''_y(1) = 1400 + 5600 = 7000$$

We notice that D'' is a scaling of D, with scaling factor equal to 100, which is the sampling.

Already from the previous Example 2.3.13, it is clear that the marginalization is in general not injective, in other words $Co(D)$ needs not to be a singleton.

Indeed in the Example, both distributions D' and D'' are coherent with $\tilde{D}$, as one can immediately compute.

Example 2.3.14 Consider a boolean system with two variables representing to coins c_1, c_2, each of them has alphabet $\{H, T\}$. We throw, separately, 100 times the coins obtaining, for the first coins c_1, 30 times H and 70 times T, and for the second coin c_2, 60 times H and 40 times T. Thus one has a distribution D given by $((30, 70), (60, 40))$.

Using the independence connection Π, we get, on the unique variable of the total correlation T of S, the distribution

$$\begin{aligned} \Pi(D)(H, H) &= 1800 \\ \Pi(D)(H, T) &= 1200 \\ \Pi(D)(T, H) &= 4200 \\ \Pi(D)(T, T) &= 2800 \end{aligned}$$

Normalizing this distribution, we notice that the probability to have (H, T) is $1200/10000 = 12\%$. The marginalization is the distribution $M(\Pi D)$ with values $((3000, 7000), (6000, 4000))$.

Clearly $M(\Pi D)$ is a scaling of D.

The previous examples can be generalized.

Proposition 2.3.15 *Let S be a system and T its total correlation. Denote, as usual, by Π the independence connection from S to T and by M the marginalization from T to S.*

If D is a distribution on S, then $M(\Pi(D))$ is a scaling of D.

Proof If s_{11} is the first state of the first variable of S, then

$$\begin{aligned} M(\Pi(D(s_{11}))) &= \sum \Pi(D(s_{11}, s_{2,j_2}, \ldots, s_{nj_n})) \\ &= \sum D(s_{11})D(s_{2,j_2}) \ldots D(s_{nj_n}) \\ &= D(s_{11})(\sum D(s_{2,j_2}) \ldots D(s_{nj_n})) \end{aligned}$$

and the same equality holds for all other states.

Hence, given $c_1 = \sum D(s_{11})D(s_{2,j_2}) \ldots D(s_{nj_n})$, for each state s_{1i} of the first variable of S, we get

$$M(\Pi(D(s_{1i}))) = D(s_{1i})c_1.$$

Similar equalities hold for the other variables of S, thus $M(\Pi(D))$ is a scaling of D. □

Example 2.3.16 The viceversa of the previous proposition is not valid in general.

As a matter of fact, in Example 2.3.14, consider a distribution D' on T defined as

$$\begin{aligned} D'(H, H) &= 6 \\ D'(H, T) &= 1 \\ D'(T, H) &= 3 \\ D'(T, T) &= 1 \end{aligned}$$

obtained throwing 11 times the two coins.

The marginalization M gives the distribution $((7, 4), (9, 2))$ on S. If we apply the independence connection Π, we get on T

$$\begin{aligned} \Pi(M(D'))(H, H) &= 63 \\ \Pi(M(D'))(H, T) &= 14 \\ \Pi(M(D'))(T, H) &= 36 \\ \Pi(M(D'))(T, T) &= 8 \end{aligned}$$

which is not a scaling of D'.

In the previous Example, the initial distribution D' does not lie in the image of the independence connection, hence $\Gamma(M(D'))$, which obviously lie in the image, cannot be equal to D'.

When we start from a distribution D' which lies in the image of the independence connection, then the marginalization works (up to scaling) as the inverse of the independence connection.

Proposition 2.3.17 *Let S be a system and T its total correlation. Denote, as usual, by Π the independence connection from S to T and by M the (total) marginalization from T to S.*

If $D' = \Pi(D)$ then $\Pi((M(D'))$ is a scaling of D'.

Proof Suppose S has n variables $x_1, \ldots, x_n$, such that the variable x_i has a_i states. By our assumption on D', there exist vectors $v_i = (v_{i1}, \ldots, v_{ia_i}) \in K^{a_i}$ such that, as tensor, $D' = v_1 \otimes \cdots \otimes v_n$.

Then $M(D')$ associates the vector

$$M(D')_j = (\sum_{i_j=1} v_{1i_1} \ldots v_{ni_n}), \ldots, \sum_{i_j=a_j} v_{1i_1} \ldots v_{ni_n})$$

to the state of the variable x_j. Hence, given $c_i =$ sampling of $v_1 \otimes \cdots \otimes \hat{v}_i \otimes \cdots \otimes v_n$, $M(D')$ associates the value $v_{ij}c_i$ to the $j-$th state of x_i. Thus

$$\Pi(M(D')) = (c_1 \cdots c_n)D'.$$

If one of the c_i's is 0, we get the empty distribution, otherwise we get a scaling of D'. □

Corollary 2.3.18 *For every distribution D, with non-empty constant sampling, on S, there exists one and only one distribution of independence D' on $\Pi S = T$ such that D is the marginalization of D'. In other words, $Co(D)$ intersects the image of the independence correlation in a singleton.*

Proof Let D', D'' be two distributions of independence with the same marginalization $D = (v_1, \ldots, v_n)$. Then D', D'' have the same sampling c, which is equal to the sampling of the variables in D. By the previous proposition, D', D'' are both equal to a scaling of $\Pi(D)$. Since they have the same sampling, they must coincide.

For the existence, it is enough to consider $(1/c)\Pi D$. □

Let us go back to see how the use of the independence connection can simplify our computations when we know that two systems are independent.

Example 2.3.19 Consider again Example 2.2.1 and let us use the previous constructions to simplify the computations.

We have a distribution D on the system T whose unique variable is the contrada c, with alphabet $\mathbb{Z}_2$. We know that the normalization Δ of D sends 1 to $3/10$ and 0 to $7/10$.

This defines, on ΠT, the distribution $\Pi(\Delta)$:

$$\Pi\Delta(1,1) = \frac{3}{10}\cdot\frac{3}{10} = \frac{9}{100}, \quad \Pi\Delta(1,0) = \frac{3}{10}\cdot\frac{7}{10} = \frac{21}{100},$$
$$\Pi\Delta(0,1) = \frac{7}{10}\cdot\frac{3}{10} = \frac{21}{100}, \quad \Pi\Delta(0,0) = \frac{7}{10}\cdot\frac{7}{10} = \frac{49}{100}. \tag{2.3.2}$$

If we consider the logic connector AND, this sends $(1, 1)$ to 1 and the other pairs to 0. Thus, the distribution induced by $\Pi(\Delta)$ sends 1 to $9/100$ and 0 to $(21 + 21 + 49)/100 = 91/100$.

If we consider the logic connector OR, this sends $(0, 0)$ to 0 and the other pairs to 1. Thus, the distribution induced by $\Pi(\Delta)$ sends 1 to $(9 + 21 + 21)/100 = 51/100$ and 0 to $49/100$.

The logic connector AUT sends $(0, 0)$ and $(1, 1)$ to 0 and the others pairs to 1. Thus, the distribution induced by $\Pi(\Delta)$ sends 1 to $(21 + 21)9/100 = 42/100$ and 0 to $(9 + 49)/100 = 58/100$.

And so on.

It is important to observe that the results are consistent with the ones of Example 2.2.1, but the computations are simpler.

Let us finish by showing some properties of the space $Co(D)$ of distributions coherent with a given distribution D on a system S.

Theorem 2.3.20 *For every distribution D, with constant sampling, on a system S, $Co(D)$ is an affine subspace (i.e. a translate of a vector subspace) in $\mathcal{D}(\Pi S)$.*

Proof We provide the proof only in the case when S is a random dipole, i.e. it has only two variables, leaving to the reader the straightforward extension to the general case.

Let x, y be the random variables of S, with states $(s_1, \ldots, s_m)$ and $(t_1, \ldots, t_n)$ respectively. We will prove that $Co(D)$ is an affine subspace of dimension $mn - m - n + 1$. Let D' be a distribution on ΠS, identified with the matrix $D' = (d_{ij}) \in \mathbb{R}^{m,n}$. Since $D' \in Co(D)$, then the row sum of the matrix of D' must give values $D(s_1), \ldots, D(s_m)$ of D on the states of x, while similarly the column sum must give the values $D(t_1), \ldots, D(t_n)$ of D on the states of y. Thus D' is in $Co(D)$ if and only if it is solution of the linear system with $n + m$ equations and nm unknowns:

$$\begin{cases} d_{11} + \cdots + d_{1n} & = D_x(s_1) \\ \quad\vdots & = \quad\vdots \\ d_{m1} + \cdots + d_{mn} & = D_x(s_m) \\ d_{11} + \cdots + d_{m1} & = D_y(t_1) \\ \quad\vdots & = \quad\vdots \\ d_{1n} + \cdots + d_{mn} & = D_y(t_n) \end{cases}$$

It follows that $Co(D)$ is an affine subspace of $\mathcal{D}(\Pi S) = \mathbb{R}^{m,n}$. The matrix associated to the previous linear system has a block structure given by

$$\left(\begin{array}{c|c|c|c|c} M_1 & M_2 & \dots & M_m & D_x \\ I & I & \dots & I & D_y \end{array}\right)$$

where I is the $n \times n$ identity matrix, M_i is the $m \times n$ matrix whose entries are 1 in the i-th row and 0 otherwise, D_x and D_y represent respectively the column vectors $(D_x(s_1), \dots, D_x(s_m))$ and $(D_y(t_1), \dots, D_y(t_n))$.

Denote by H the matrix

$$\left(\begin{array}{c|c|c|c} M_1 & M_2 & \dots & M_m \\ I & I & \dots & I \end{array}\right)$$

We observe that the $m + n$ rows of H are not independent since the vector $(1, 1, \dots, 1)$ can be obtained as sum of the first m row and as sum of the last n rows. Hence the rank of H is at most $n + m - 1$.

In particular, the system has solution if and only if the constant terms satisfy

$$D_x(s_1) + \cdots + D_x(s_m) = D_y(t_1) + \cdots + D_y(t_n),$$

which is equivalent to the hypothesis that D has constant sampling.

To conclude the proof, it is enough to check that H has rank at least $n + m - 1$, that is H has a $(n + m - 1) \times (n + m - 1)$ submatrix of maximal rank.

Observe that the $n \times n$ block in the left bottom corner is an identity matrix, with rank n. Deleting the last n rows and the first n columns of H, we get the $m \times (mn - n)$ matrix $H' = (M_2|M_3|\dots|M_m)$ that has null first row, but rank equal to $m - 1$, since the columns in position $1, n + 1, 2n + 1, \dots, (m - 2)n + 1$ contain an identity matrix of size $(m - 1) \times (m - 1)$. □

Example 2.3.21 Let us show the trueness of the previous theorem, verifying that H has rank $m + n - 1$, in some numerical case.

For example, if $m = 2, n = 3$, the matrix H is:

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

If $m = 3, n = 2$, the matrix H is:

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

We can say a little more on the Geometry of $Co(D)$.

Definition 2.3.22 In $\mathcal{D}(\Pi S)$ define the *unitary simplex* $\mathscr{U}$ as the subspace formed by tensors whose coefficients sum to 1.

$\mathscr{U}$ is an affine subspace of codimension 1, which contains all the probabilistic distributions on ΠS.

We previously said that if D' is coherent with D and D has constant sampling k, then also the sampling of D', on the unique variable of ΠS, is k. In other terms, the matrix (d_{ij}) associated to D' satisfies $\sum_{i,j} d_{ij} = k$. From this fact we get the following

Proposition 2.3.23 *For every distribution D on S with constant sampling, the affine space $Co(D)$ is parallel to $\mathscr{U}$.*

We finish by showing that, for a fixed distribution D over S, one can find in $Co(D)$ distributions with rather different properties.

Example 2.3.24 Assume one needs to find properties of a distribution D' on ΠS, but knows only the marginalisation $D = M(D')$. Even if $Co(D)$ is in general infinite, sometimes only few further information is needed to determine some property of D'.

For instance, let S be a random dipole, whose variables x, y represent one position in the DNA chain of a population of cells in two different moments, so that both variables have alphabet $\{A, C, G, T\}$. After treating the cells with some procedure, a researcher wants to know if some systematic change occurred in the DNA sequence (which can change also randomly). The total correlation of S has one variable with 16 states. A distribution D' on ΠS corresponds to a 4×4 matrix.

If the researcher cannot trace the evolution of every single cell, but can only determine the total number of DNA basis that occur in the given position before and after the treatment, i.e. if the researcher can only know the distribution D on S, any conclusion on the dependence of the final basis in terms of the initial one is impossible.

But assume that one can trace the behaviour of cells having a A in the fixed position. So one can record the value of the distribution D' on the state (A, A). Since there exists only one distribution of independence D'' which is coherent with D if the observed value of $D'(A, A)$ does not coincide with $D''(A, A)$, then the researcher has an evidence towards the non-independence of the variables, in the distribution D'.

Notice that, on the contrary, if $D'(A, A) = D''(A, A)$, one cannot conclude that D' is a distribution of independence, since infinitely many distributions in $Co(D)$ assume on (A, A) a fixed value.

Example 2.3.25 Consider the following situation. In a bridge game, two assiduous players A, B follow this rule: they play alternately one day a match in pairs together, the other day a match in opposing pairs. After 100 days, the situation is as follows: A won 30 games and lost 70, while B won 40 and lost 60. Can one determine analytically the total number of wins and defeats? Can one check whether the win or the defeat of each of them dependent on or not playing in pairs with each other?

We can give an affermative answer to both questions. Here we have a boolean dipole S, with two variables A, B and state $1 =$ victory, $0 =$ defeat. The distribution D on S is defined by

$$D_A(1) = 30, \quad D_A(0) = 70, \quad D_B(1) = 40, \quad D_B(0) = 60.$$

Clearly D has constant sampling equal to 100. From these data, we want to determine a distribution D' on ΠS, coherent with D, which explains the situation. We already know that we have infinitely many distributions on ΠS, coherent with D. By Theorem 2.3.20, these distributions D' fill an affine subspace of $\mathbb{R}^{2,2}$ having dimension $2 \cdot 2 - 2 - 2 + 1 = 1$.

The extra datum with respect to Example 2.3.13 is to know that the players played alternately in pairs and opposed. Then among the 100 played matches, 50 times they played together, then the observed result could be only (0, 0) o (1, 1), while 50 time were opposed, and the observed result could be only (0, 1) or (1, 0). Hence, the matrix (d_{ij}) of the required distribution D' must satisfy the condition:

$$d_{11} + d_{22} = d_{12} + d_{21} \quad (= 50).$$

The matrix of distribution of independence D'', coherent with D, is given by (30, 70)(40, 60), divided by the sampling of D, i.e. 100. One has

$$D'' = \begin{pmatrix} 12 & 18 \\ 28 & 42 \end{pmatrix}.$$

All others distributions, coherent with D'', are obtained summing to D'' the solution of the homogeneous system

$$\begin{cases} d_{11} + d_{12} = 0 \\ d_{11} + d_{21} = 0 \\ d_{22} + d_{12} = 0 \\ d_{22} + d_{21} = 0 \end{cases}$$

which are multiple of

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then a generic distribution, coherent with D, has matrix:

$$D' = \begin{pmatrix} 12 - z & 18 + z \\ 28 + z & 42 - z \end{pmatrix}.$$

Requiring $d_{11} + d_{22} = d_{12} + d_{21}$, we get $z = 2$, then the only possible matrix is:

$$D' = \begin{pmatrix} 10 & 20 \\ 30 & 40 \end{pmatrix}.$$

Then playing in pairs A and B won 10 times and lost 40, while playing against A has won 20 times, B has won 30 times.

Finally, the percentage of wins depends on the players A and B, because the determinant of D' is $-200 \neq 0$ (both have advantage to not play in pairs with each other).

Example 2.3.26 Sometimes even extra information is not enough to determine a unique $D' \in Co(D)$.

Consider the following situation. There are two sections A, B in a school. The Director can assign, every year, some grants for some students, depending on the financial situation of the school. Every year, the Director assigns at most two grants, but when the balance of the school is poor, he can assign only one grant, or no grants at all.

No section can have a privilege with respect to the other: if two grants are assigned, they are divided one for each section. If one year only one grant is assigned to a section, then the next year in which only one grant is assigned, the grant will be reserved for the other section.

After 25 years the situation is: section A obtained 15 grants, and also section B obtained 15 grants.

Can one deduce, from these data, in how many years no grants were assigned?

Can one deduce that the fact that section A gets a grant is an advantage or not for section B?

Both answers to these questions are negative.

Create a random system S, which is a boolean dipole, with variables A, B whose states are $1 =$ obtains a grant, $0 =$ obtains no grants. We know that the distribution of grants that we observe in S is $D = ((10, 15), (10, 15))$. We would like to know the distribution D' on ΠS such that $D'(1, 0) =$ number of years in which A got a grant and did not, and so on. Since both sections obtained 15 grants, the number of years in which only one grant was assigned is even. Moreover the number of years in which section A obtained a grant and section B did not is equal to the number of years in which section B obtained a grant and section A did not. In other words $D'(1, 0) = D'(0, 1)$, i.e. the matrix representing D' is symmetric.

But this is still not enough to determine D'!

Indeed all distributions in $Co(D)$ are symmetric, because the values of D on the states of A and B coincide, so that if $D' = (d_{ij})$ then

$$d_{11} + d_{12} = d_{11} + d_{21}, \qquad d_{21} + d_{22} = d_{12} + d_{22},$$

thus $d_{12} = d_{21}$.

For instance, the unique distribution of independence coherent with D is $d_{11} = 4$, $d_{12} = 6, d_{21} = 6, d_{22} = 9$, which is symmetric. In geometric terms, $Co(D)$ is parallel to the space of symmetric matrices.

In the example, $Co(D)$ contains the matrices:

$$D_1 = \begin{pmatrix} 3 & 7 \\ 7 & 8 \end{pmatrix} \qquad D_2 = \begin{pmatrix} 5 & 5 \\ 5 & 10 \end{pmatrix} \qquad D_3 = \begin{pmatrix} 2 & 8 \\ 8 & 7 \end{pmatrix}.$$

In D_1 and D_2 the fact that A gets a grant is a vantage for section B, while in D_3 it represents a disadvantage.

2.4 Exercises

Exercise 6 Following Example 2.2.1 compute the probability that a contrada c participates to one Palio in 2016, but not both, under the assumption that c didn't participate to any Palio in 2015.

Chapter 3
Statistical Models



3.1 Models

In this chapter, we introduce the concept of *model*, essential point of statistical inference. The concept is reviewed here by our algebraic interpretation. The general definition is very simple:

Definition 3.1.1 A model on a system S of random variables is a subset M of the space of distributions $\mathcal{D}(S)$.

Of course, in its total generality, the previous definition it is not so significant.

The Algebraic Statistics consists of practice in focus only on certain particular types of models.

Definition 3.1.2 A model M on a system S is called *algebraic model* if, in the coordinates of $\mathcal{D}(S)$, M corresponds to the set of solutions of a finite system of polynomial equations. If the polynomials are homogeneous, then M si called a *homogeneous algebraic model*.

It occurs that the algebraic models are those mainly studied by the proper methods of Algebra and Algebraic Geometry.

In the statistical reality, it occurs that many models, which are important for the study of stochastic systems (discrete), are actually algebraic models.

Example 3.1.3 On a system S, the set of distributions with constant sampling is a model M. Such a model is a homogeneous algebraic one.

As a matter of fact, if $x_1, \ldots, x_n$ are the random variables in S and we identify $\mathcal{D}_{\mathbb{R}}(S)$ with $\mathbb{R}^{a_1} \times \cdots \times \mathbb{R}^{a_n}$, where the coordinates are $y_{11}, \ldots, y_{1a_1}, y_{21}, \ldots, y_{2a_2}, \ldots, y_{n1}, \ldots, y_{na_n}$, then M is defined by the homogeneous equations:

$$y_{11} + \cdots + y_{1a_1} = y_{21} + \cdots + y_{2a_2} = \cdots = y_{n1} + \cdots + y_{na_n}.$$

The probabilistic distributions form a submodel of the previous model, which is still algebraic, but not homogeneous!

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3.2 Independence Models

The most famous class of algebraic models is the one given by independence models. Given a system S, the independence model on S is, in effect, a subset of the space of distributions of the total correlation $T = \Pi S$, containing the distributions in which the variables are independent among them. The basic example is Example 2.3.13, where we consider a Boolean system S, whose two variables x, y represent, respectively, the administration of the drug and the healing.

This example justifies the definition of a model of independence, for the random systems with two variables (dipoles), already in fact introduced in the previous chapters.

Definition 3.2.1 Let S be a system with two random variables x_1, x_2 and let $T = \Pi S$. The space of $K-$distributions on T is identified with the space of matrices K^{a_1,a_2}, where a_i is the number of states of the variable x_i.

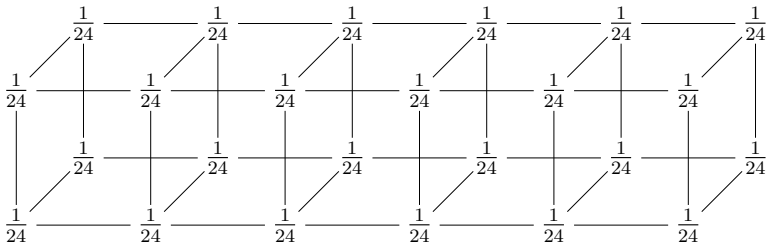
We recall that a distribution $D \in \mathcal{D}_K(T)$ is a distribution of independence if D, as matrix, has rank ≤ 1.

The independence model on S is the subset of $\mathcal{D}_K(T)$ of distributions of rank ≤ 1.

To extend the definition of independence to systems with more variables, consider the following example.

Example 3.2.2 Let S be a random system, having three random variables x_1, x_2, x_3 representing, respectively, a die and two coins (this time not loaded). Let $T = \Pi S$ and consider the $\mathbb{R}$-distribution D on T defined by the tensor

$D =$



It is clear that D is a distribution of independence and probabilistic. You can read it like the fact that the probability that a d comes out from the die at the same time of the faces (for example, T and H) from the two coins, is the product of the probability $\frac{1}{6}$ that d comes out of the die times the probability $\frac{1}{2}$ that T comes out of the coin times the probability $\frac{1}{2}$ that H comes out of the other coin.

Hence, we can use Definition 6.3.3 to define the *independence model*.

Definition 3.2.3 Let S be a system with random variables $x_1, \ldots, x_n$ and let $T = \Pi S$. The space of K-distributions on T is identified with the space of tensors $K^{a_1,\ldots,a_n}$, where a_i is the number of states of the variable x_i.

A distribution $D \in \mathcal{D}_K(T)$ is a *distribution of independence* if D lies in the image of independence connection (see Definition 2.3.4), i.e., as a tensor, it has rank 1 (see Definition 6.3.3).

The *independence model* on X is the subset of $\mathcal{D}_K(T)$ consisting of all distributions of independence (that is, of all tensors of rank 1).

The model of independence, therefore, corresponds to the subset of simple (or decomposable) tensors in a tensor space (see Remark 6.3.4).

We have seen, in Theorem 6.4.13 of the chapter on Tensorial Algebra, how such subset can be described. Since all relations (6.4.1) correspond to the vanishing of one (quadratic) polynomial expression in the coefficients of the tensor, we have

Corollary 3.2.4 *The model of independence is an algebraic model.*

Note that for the tensors $2 \times 2 \times 2$, the independence model is defined by 12 quadratic equations (6 faces + 6 diagonal).

The equations corresponding to the equalities (6.4.1) describe a set of equations for the model of independence. However such a set, in general, it is not minimal.

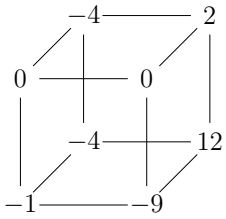
The distributions of independence represent situations in which there is no link between the behavior of the various random variables of S, which are, therefore, independent.

There are, of course, intermediate cases between a total link and a null link, as seen in the following:

Example 3.2.5 Let S be a system with 3 random variables. The space of distribution $\mathcal{D}(\Pi S)$ consists of tensors of dimension 3 and type (d_1, d_2, d_3). We say that a distribution $D \in \mathcal{D}(\Pi S)$ is *without triple correlation* if there exist three matrices $A \in R^{d_1,d_2}$, $B \in R^{d_1,d_3}$, $C \in R^{d_2,d_3}$ such that for all i, j, k:

$$D(i, j, k) = A(i, j)B(i, k)C(j, k).$$

An example, when S is Boolean, is given by the tensor



which is obtained by the matrices

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix} \quad C = \begin{pmatrix} 1 & -2 \\ 3 & 2 \end{pmatrix}$$

3.3 Connections and Parametric Models

Another important example of models in Algebraic Statistics is provided by the *parametric models*. They are models whose elements have coefficients that vary according to certain parameters. To be able to define parametric models, it is necessary first to fix the concept of *connection* between two random systems.

Definition 3.3.1 Let S, T be system of random variables. We call K*-connection* between S and T any function Γ from the space of K-distributions $\mathcal{D}_K(S)$ to the space of K-distributions $\mathcal{D}_K(T)$.

As usual, when the field K is understood, we will omit it in the notation.

After all, therefore, connections are nothing more than functions between a space K^s and a space K^t. The name we gave, in reference to the fact that these are two spaces connected to random systems, serves to emphasize the use we will make of connections: *to transport* distributions from the system S to the system T.
In this regard, if T has n random variables $y_1, \ldots, y_n$, and the alphabet of each variable y_i has d_i elements, then $\mathcal{D}_K(T)$ can be identified with $K^{d_1} \times \cdots \times K^{d_n}$. In this case, it is sometimes useful to think of a connection Γ as a set of functions $\Gamma_i : \mathcal{D}(S) \to K^{d_i}$.

If $s_1, \ldots, s_a$ are all possible states of the variables in S, and $t_{i1}, \ldots, t_{id_i}$ are the possible states of the variable y_i, then we will also write

$$\begin{cases} t_{i1} & = & \Gamma_{i1}(s_1, \ldots, s_a) \\ \ldots & = & \ldots \\ t_{id_i} & = & \Gamma_{id_i}(s_1, \ldots, s_a) \end{cases}$$

The definition of connection given here, in principle, it is extremely general: no particular properties are required for the Γ_i functions; not even continuity. Of course, in concrete cases, we will study in particular certain connections having well-defined properties.

It is clear that, in the absence of any property, we cannot hope that the more general connections satisfy many properties.

Let us look at some significant examples of connections.

Example 3.3.2 Let S be a random system and S' a subsystem of S. You get a connection from S to S', called *projection* simply by forgetting the components of the distributions which correspond to random variables not contained in S'.

Example 3.3.3 Let S be a random system and $T = \Pi S$ its total correlation. Assume that S has random variables $x_1, \ldots, x_n$, and each variable x_i has a_i states, then $\mathcal{D}_K(S)$ is identified with $K^{a_1} \times \cdots \times K^{a_n}$. In Definition 2.3.4, we defined a connection $\Gamma : \mathcal{D}_K(S) \to \mathcal{D}_K(T)$, called *connection of independence* or *Segre connection*, in the following way: Γ sends the distribution

$$D = ((d_{11}, \ldots, d_{1a_1}), \ldots, (d_{n1}, \ldots, d_{na_n}))$$

to the tensor (thought as distribution on ΠS) $D' = \Gamma(D)$ such that

$$D'_{i_1,\dots,i_n} = d_{1i_1} \cdots d_{ni_n}.$$

It is clear, by construction, that the image of the connection is formed exactly by the distributions of independence on ΠS.

Clearly there are other interesting types of connection. A practical example is the following:

Example 3.3.4 Consider a population of microorganisms in which we have elements of two types, A, B, that can pair together randomly. In the end of the couplings, we will have microorganisms with genera of type AA or BB, or mixed type $AB = BA$.

The initial situation corresponds to a Boolean system with a variable (the initial type t_0) which assumes the values A, B. At the end, we still have a system with only one variable (the final type t) that can assume the 3 values AA, AB, BB.

If we initially insert a distribution with $a = D(A)$ elements of type A and $b = D(B)$ elements of type B, which distribution we can expect on the final variable t?

An individual has a chance to meet another individual of type A or B which is proportional to (a, b), then the final distribution on t will be D' given by $D'(AA) = a^2$, $D'(AB) = 2ab$, $D'(BB) = b^2$. This procedure corresponds to the connection $\Gamma : \mathbb{R}^2 \to \mathbb{R}^3$ $\Gamma(a, b) = (a^2, 2ab, b^2)$.

Definition 3.3.5 We say that a model $V \subset \mathcal{D}(T)$ is *parametric* if there exists a random system S and a connection Γ from S to T such that V is the image of S under Γ in $\mathcal{D}(T)$, i.e., $V = \Gamma(\mathcal{D}(S))$.

A model is *polynomial parametric* if Γ is defined by polynomials.

A model is *toric* if Γ is defined by monomials.

The motivation for defining *parametric models* should be clear from the representation of a connection. If $s_1, \dots, s_a$ are all possible states of the variables in S, and $t_{i1}, \dots, t_{id_i}$ are the possible states of the variables y_i of T, then in the parametric model defined by the connection Γ we have

$$\begin{cases} t_{i1} & = & \Gamma_{i1}(s_1, \dots, s_a) \\ \dots & = & \dots \\ t_{id_i} & = & \Gamma_{id_i}(s_1, \dots, s_a) \end{cases}$$

where the Γ_{ij}'s represent the components of Γ.

The model definition we initially gave is so vast to be generally poorly usable. In reality, the models we will use in the following will always be algebraic models or polynomial parametric.

Example 3.3.6 It is clear from the Example 3.3.3 that the model of independence is given by the image of the independence connection, defined by the Segre map (see the Definition 10.5.9), so it is a parametric model.

The tensors T of the independence model have in fact coefficients that satisfy parametric equations

$$\begin{cases} \dots \\ T_{i_1 \dots i_n} = v_{1i_1} v_{2i_2} \cdots v_{ni_n} \\ \dots \end{cases} \tag{3.3.1}$$

From its parametric equations, (3.3.1), we quickly realize that the independence model is a toric model.

Example 3.3.7 The model of Example 3.3.4 is a toric model, since it is defined by the equations

$$\begin{cases} x &= a^2 \\ y &= 2ab \\ z &= b^2 \end{cases}.$$

Remark 3.3.8 It is evident, but it is good to underline it, that for the definitions we gave, being an algebraic or polynomial parametric model is independent from changes in coordinates. Being a toric model instead it can depend on the choice of coordinates.

Definition 3.3.9 The term *linear model* denotes, in general, a model on S defined in $\mathcal{D}(S)$ by linear equations.

Obviously, every linear model is algebraic and also polynomial parametric, because you can always parametrize a linear space.

Example 3.3.10 Even if a connection Γ, between the K-distributions of two random systems S and T, is defined by polynomials, the polynomial parametric model that Γ defines it is not necessarily algebraic!.

In fact, if we consider $K = \mathbb{R}$ and two random systems S and T each having one single random variable with a single state, the connection $\Gamma : \mathbb{R} \to \mathbb{R}$, $\Gamma(s) = s^2$ certainly determines a polynomial parametric model (even toric) which corresponds to $\mathbb{R}_{\geq 0} \subset \mathbb{R}$, so it can not be defined in $\mathbb{R}$ as vanishing of polynomials.

We will see, however, that by widening the field of definition of distributions, as we will do in the next chapter switching to distributions on $\mathbb{C}$, under a certain point of view all polynomial parametric models will, in fact, be algebraic models.

The following counterexample is a milestone in the development of so much part of the Modern mathematics. Unlike Example 3.3.10, it cannot be recovered by enlarging our field of action.

Example 3.3.11 Not all algebraic models are polynomial parametric.

We consider in fact a random system S with only one variable having three states. In the distribution space $\mathcal{D}(S) = \mathbb{R}^3$, we consider the algebraic model V defined by the unique equation $x^3 + y^3 - z^3 = 0$.

There cannot be a polynomial connection Γ from a system S' to S whose image is V.

In fact, suppose the existence of three polynomials p, q, r, such that $x = p, y = q, z = r$. Obviously, the three polynomials must satisfy identically the equation $p^3 + q^3 - r^3 = 0$. It is enough to verify that there are no three polynomials satisfying the previous relationship. Provided to set values for the other variables, we can assume that p, q, r are polynomials in a single variable t. We can also suppose that the three polynomials do not have common factors. Let us say that $\deg(p) \geq \deg(q) \geq \deg(r)$. Deriving the equation

$$p(t)^3 + q(t)^3 - r(t)^3 = 0,$$

with respect to t, we get

$$p^2(t)p'(t) + q^2(t)q'(t) - r^2(t)r'(t) = 0.$$

The two previous equations give a homogeneous linear system

$$\begin{pmatrix} p(t) & q(t) & -r(t) \\ p'(t) & q'(t) & -r'(t) \end{pmatrix}.$$

The solution $p^2(t), q^2(t), r^2(t)$ must be proportional to the 2×2-minors of the matrix, hence $p^2(t)$ is proportional to $q(t)r'(t) - q'(t)r(t)$, and so on. Considering the equality $p^2(t)(p(t)r'(t) - p'(t)r(t)) = q^2(t)(q(t)r'(t) - q'(t)r(t))$, we get that $p^2(t)$ divides $q(t)r'(t) - q'(t)r(t)$, hence $2\deg(p(t)) \leq \deg(q) + deg(r) - 1$ which contradicts the fact that $\deg(p) \geq \deg(q) \geq \deg(r)$.

Naturally, there are examples of models that arise from connections that they do not relate a system and its total correlation.

Example 3.3.12 Let us say we have a bacterial culture in which we insert bacteria corresponding to two types of genome, which we will call A, B.

Suppose, according to the genetic makeup, the bacteria can develop characteristics concerning the thickness of the membrane and of the core. To simplify, let us say that in this example, cells can develop nucleus and membrane large or small.

According to the theory to be verified, the cells of type A develop, in the descent, a thick membrane in 20% of cases and develop large core in 40% of cases. Cells of type B develop thick membrane in the 25% of cases and a large core in one-third of the cases. Moreover, the theory expects that the two phenomena are independent, in the intuitive sense that developing a thick membrane is not influenced by, nor influences, the development of a large core.

We build two random systems. The first S, which is Boolean, has only one variable random c (= cell) with A, B states. The second T with two boolean variables, m (= membrane) and n (= core). We denote for both with 0 the status "big" and with 1 the status "small".

The theory induces a connection Γ between S and T. In the four states of the two variables of T, which we will indicate with x_0, x_1, y_0, y_1, this connection is defined by

$$\begin{cases} x_0 &= \frac{1}{5}a + \frac{1}{4}b \\ x_1 &= \frac{4}{5}a + \frac{3}{4}b \\ y_0 &= \frac{2}{5}a + \frac{1}{3}b \\ y_1 &= \frac{3}{5}a + \frac{2}{3}b \end{cases}$$

where a, b correspond to the two states of S. As a matter of fact, suppose we introduce 160 cells, 100 of type A, and 60 of type B. This leads to consider a distribution D on S given by $D = (100, 60) \in \mathbb{R}^2$.

The distribution on T defined by the connection is given by

$$\Gamma(D) = ((35, 125), (60, 100)) \in (\mathbb{R}^2) \times (\mathbb{R}^2).$$

This reflects the fact that in the cell population (reported at 160) we expect to eventually observe 35 cells with large membrane and 60 cells with a large core.

If the experiment, more realistically, manages to capture the percentage of cells with the two characteristics (shuffled), then we can consider a connection that links S with the total correlation ΠT: indicating with $x_{00}, x_{01}, x_{10}, x_{11}$ the variables corresponding to the four states of the only variable of ΠT, then such connection Γ' is defined by

$$\begin{cases} x_{00} &= \frac{(\frac{1}{5}a+\frac{1}{4}b)(\frac{2}{5}a+\frac{1}{3}b)}{(a+b)^2} \\ x_{01} &= \frac{(\frac{1}{5}a+\frac{1}{4}b)(\frac{3}{5}a+\frac{2}{3}b)}{(a+b)^2} \\ x_{10} &= \frac{(\frac{4}{5}a+\frac{3}{4}b)(\frac{2}{5}a+\frac{1}{3}b)}{(a+b)^2} \\ x_{11} &= \frac{(\frac{4}{5}a+\frac{3}{4}b)(\frac{3}{5}a+\frac{2}{3}b)}{(a+b)^2} \end{cases}$$

This connection, starting at D, determines the (approximate) probabilistic distribution on ΠT:

$$\Gamma'(D) = (0.082, 0.137, 0.293, 0.488) \in \mathbb{R}^4.$$

From an algebraic point of view, an experiment will be in agreement with the model if the percentages observed will be exactly those described by the latter connection: 8.2% of cells with a thick membrane, nucleus, etc.

In the real world, of course, some tolerance should be expected from experimental data error. The control of this experimental tolerance will be not addressed in this book, as it is a part of standard statistical theories.

3.4 Toric Models and Exponential Matrices

Recall that a toric model is a parametric model on a system T corresponding to a connection from S to T which is defined by monomials.

Definition 3.4.1 Let W be a toric model defined by a connection Γ from S to T. Let $s_1, \ldots, s_q$ be all possible states of all the variables of S and let $t_1, \ldots, t_p$ the states of all the variables of T. One has, for every i, $t_i = \Gamma_i(s_1, \ldots, s_q)$, where each Γ_i is a monomial in the s_j.

We will call *exponential matrix* of W the matrix $E = (e_{ij})$, where e_{ij} is the exponent of s_j in t_i.

E is, therefore, a $p \times q$ array of nonnegative integers. We will call it complex associated with W the subset of $\mathbb{Z}^q$ formed by the points corresponding to the rows of E.

Proposition 3.4.2 *Let W be a toric model defined by a monomial connection Γ from S to T and let E be its exponential matrix.*

Each linear relationship $\sum a_i R_i = 0$ among the R_i rows of E corresponds to implicit polynomial equations that are satisfied by all points in W.

Proof Consider a relation $\sum a_i R_i = 0$ among the rows of E. We associate to this relation a polynomial equation

$$\prod_{a_i \geq 0} t_i^{a_i} - z \prod_{a_j < 0} t_j^{a_j} = 0$$

where, by indicating with $c(\Gamma_i)$ the monomial coefficient,

$$z = \frac{\prod_{a_i \geq 0} c(\Gamma_i)^{a_i}}{\prod_{a_j < 0} c(\Gamma_j)^{a_j}}$$

We verify that this polynomial relationship is satisfied by all points in W.

In effect, by replacing $t_1, \ldots, t_p$ with their expressions in terms of Γ, we get two monomials with equal exponents and opposite coefficients, which are canceled. □

Note that the polynomial equations obtained previously, are in fact binomial.

Definition 3.4.3 The polynomial equations associated with the linear relations between the rows of the exponential matrix of a toric model W define an algebraic model containing W. This model takes the name of *algebraic model generated by W*.

It is clear from Example 3.3.10 that the algebraic model generated by a toric model W always contains W, but it does not always coincide with W. Let us see a couple of examples about it.

Example 3.4.4 We still consider the example of the model of independence on a dipole S.

Resuming the terminology of the previous section, we indicate with $t_1, \ldots, t_n$ the states of the first variable T and with $u_1, \ldots, u_m$ the states of the second variable. The resulting model is parametrically defined, on ΠS, by $y_{(t_i,u_j)} = t_i u_j$. It is, therefore, a toric model, whose exponential matrix is given by

$$\begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_m \\ R_{m+1} \\ R_{m+2} \\ \vdots \\ R_{mn} \end{pmatrix} = \left(\begin{array}{c|c} 1\,0\,\ldots\,0 & 1\,0\,\ldots\,0 \\ 1\,0\,\ldots\,0 & 0\,1\,\ldots\,0 \\ \vdots\,\vdots\,\ldots\,\vdots & \vdots\,\vdots\,\ldots\,\vdots \\ 1\,0\,\ldots\,0 & 0\,0\,\ldots\,1 \\ 0\,1\,\ldots\,0 & 1\,0\,\ldots\,0 \\ 0\,1\,\ldots\,0 & 0\,1\,\ldots\,0 \\ \vdots\,\vdots\,\ldots\,\vdots & \vdots\,\vdots\,\ldots\,\vdots \\ 0\,0\,\ldots\,1 & 0\,0\,\ldots\,1 \end{array}\right)$$

from which we can see all relations, between rows, of the form

$$R_{qm+h} + R_{pm+k} = R_{qm+k} + R_{pm+h}$$

which define, as equations in $\mathbb{R}^{mn} = \mathbb{R}^{m,n}$, exactly the 2×2-minors of the matrices in the model.

It follows that the algebraic model associated to this connection coincides with the space of matrices of rank ≤ 1, which is exactly the image of the connection of independence.

Example 3.4.5 Come back to the connection of Example 3.3.4. It defines a polynomial parametric model W on $\mathbb{R}^3$ given by the parametric equations

$$\begin{cases} x &= a^2 \\ y &= 2ab \\ z &= b^2 \end{cases}$$

The associated exponential matrix is

$$\begin{pmatrix} 2\,0 \\ 1\,1 \\ 0\,2 \end{pmatrix}$$

which, as unique relation between rows, satisfies $R_1 + R_3 = 2R_2$. Using the formula for the coefficients, we get the equation in $\mathbb{R}^3$:

$$4xz = y^2.$$

The algebraic model W' defined by this equation does not coincide with W. As a matter of fact, it is clear that the points in W have nonnegative x, z, while the point $(-1, 2, -1)$ is in W'.

However, one has $W = W' \cap B$ where B is the subset of the points in $\mathbb{R}^3$ with nonnegative coordinates. In fact, if (x, y, z) is a point in B which satisfies the equation, then, posing $a = \sqrt{x}$, $b = \sqrt{z}$, one has $y = ab$.

Remark 3.4.6 **The scientific method.**
Given a parametric model, defined by a connection Γ from S to T, if we know an "initial" distribution D over S (the experiment data) and we measure the distribution $D' = \Gamma(D)$ that we derive on T (the result of the experiment), we can easily deduce if the hypothesized model fits or not with reality.

But if we have no way of knowing the D distribution and we can only measure the D' distribution, as happens in many real cases, then it would be a great help to know some polynomial F that vanishes on the model, that is, to know its implicit equations. In fact, in this case the simple check of the fact that $F(D') = 0$ can give us many indications: if the vanishing does not occur, our model is clearly inadequate; instead if it occurs, it gives a clue in favor of the validity of the model.

If we then knew that the model is also algebraic and we knew the equations, their control over many distributions, results of experiments, would give a good scientific evidence on the validity of the model itself.

Chapter 4
Complex Projective Algebraic Statistics



No, we are not exaggerating. We are instead simplifying.

In fact, many of the phenomena associated with the main statistic models are better understood if studied, at least in the first measure, from the projective point of view and on an algebraically closed numerical field.

The main link between Algebraic Statistics and Projective Algebraic Geometry is based on the constructions of this chapter.

4.1 Motivations

We have seen as many models of interest in the fields of statistics are defined, in the space of the distributions of a system S, by (polynomials) algebraic equations of degree greater than one. To understand such models, a mathematical approach is to initially study the subsets of a space defined by the vanishing of polynomials. This is equivalent to studying the theory of solutions of systems of polynomial equations of arbitrary degree, which goes under the name of Algebraic Geometry.

The methods of Algebraic Geometry are based on various theories: certainly, on Linear and Multi-linear Algebra, but also on rings theory (in particular, on the theory of rings of polynomials) and on Complex Analysis. We will repeat here only a part of the main results that can be applied to statistical problems. It must be kept in mind, however, that the theory in question is rather developed, and topics that will not be introduced here, they could be important in the future also from a statistical point of view.

The first step to take is to define the ambient space in which we move. Being to study solutions of nonlinear equations, from an algebraic point of view, it is natural to pass from the real field, fundamental for applications but without some algebraic elements, to the complex field which, being algebraically closed, allows a complete understanding of the solutions of polynomial equations.

We will then have to expand, from a theoretical point of view, the space of distributions, to admit points with complex coordinates. These points, corresponding

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to distributions with complex numbers, will allow us a more immediate characterization of the sets of solutions of algebraic systems. Naturally, when we have to reread the results in the common statistical theory, we will have to return to models exclusively defined on the real field, that intersects with the real space contained in every complex space. This final step, which in general poses technical problems that are absolutely nontrivial, can, however, be overlooked in a first reading, where the indications that we will get on the complexes will still help us to understand of real phenomena.

Once this first enlargement is accepted, to arrive at an understanding even more in-depth of algebraic phenomena, it is opportune to operate a second step, perhaps apparently even more challenging: the passage from affine spaces $\mathbb{C}^n$ to the associated projective spaces.

The reasons for this second enlargement are justified, from a geometric point of view, with the need to work in compact ambients. Compactness is indeed an essential property for our geometric understanding. In very descriptive terms, thanks to the introduction of *points to infinity*, we will avoid to lose the solutions when, passing to the limit, we should have phenomena of parallelism or however asymptotic phenomena. The possibility of following the reasonings *passing to the limit* is one of the winning cards that geometry in projective setting offers, compared to that in similar environments.

Of course, projective compactification, to make sense in statistical problems, it must be carried out properly, differentiating, for example, the passage to the limit of the various random variables.

For those who find too cumbersome the procedure to use homogeneous coordinates to describe distributions on random systems, perhaps it is worth remembering that a similar procedure, in statistics, is always present: the normalization. In practice, if we have a distribution D on a random variable x having states $s_1, \ldots, s_n$, then it is natural to replace D with the $\bar{D}$ distribution obtained dividing each $D_x(s_i)$ by the sampling $\sum D_x(s_j)$ (in the event that x is not neutral with respect to D). Note that, in doing so, in the space of distributions that concerns the variable x, we get to replace the point $(D_x(s_1), \ldots, D_x(s_n))$ with the point $(\bar{D}_x(s_1), \ldots, \bar{D}_x(s_n))$. If, in the affine space, the point is changed, in the projective space, where the n -tuple represents homogeneous coordinates, passing to normalization, the point does not change! In effect, every point in the projective space can always be represented by homogeneous coordinates $(a_1, \ldots, a_n)$ such that $\sum a_j = 1$.

From another point of view, the classical statistical theory, in the space $\mathbb{R}^n$ of the distributions of a random variable x as above, tended to shrink to the hyperplane defined by the equation $\sum a_j = 1$, as we have seen, for example, in the statement of Varcenko's Theorem. Instead, projective statistical theory works on distributions *up to scaling*, then does not need this restriction, since from a projective point of view, normalization, like any other scaling, is irrelevant.

Therefore, it is quite simple to convince ourselves that, in the end, these are two equivalent approaches. The difficulty of passing from one to another basically resides in habit. The advantage of using the projective language consists in being able to access directly to the vast literature on Algebraic Geometry which, in many ways, mainly uses this terminology.

During this chapter, we will use many topics from the Part III. We strongly recommend to the reader which is not expert in Algebraic Geometry to study such chapters before to start the treatment of projective algebraic models.

4.2 Projective Algebraic Models

What is the link of all of this with Algebraic Statistics?

When we meet a distribution $D \in \mathcal{D}(S)$, where S is a system of random variables, we are, in practice, collecting a set of observed data. To the aim of our interpretation of data, is usually irrelevant (within certain reasonable limits) the sampling of the variables.

If, for example, we are evaluating the efficiency of a medicine, to administer the medicine to 100 sick people and having 50 healed suggests that the medicine is efficient in about the 50% of cases. From this point of view, we obtain the same conclusion if we administer the medicine to 120 sick and record 60 healed.

So, in our setting, we will consider that a distribution gives the same information on the analyzed phenomenon than any of its scaling.

In Statistics, the problem is solved by choosing, among all possible scaling of a distribution, the *associated probabilistic distribution*, introduced in Definition 1.3.2. Such distribution is uniquely determined, given a distribution D, but it exists only *when all variables have sampling different from* 0 *in* D.

The associated probabilistic distribution lies in a linear subspace of $\mathcal{D}(S) = K^{a_1} \times \cdots \times K^{a_n}$, defined by the vectors $v = ((v_{11}, \ldots, v_{1a_1}), \ldots, (v_{n1}, \ldots, v_{na_n}))$ satisfying the equations

$$v_{11} + \cdots + v_{1a_1} = \cdots = v_{n1} + \cdots + v_{na_n} = 1.$$

Instead, in our setting, we prefer to consider the *multi-projective space of distributions*.

Definition 4.2.1 Given a system S, with random variables $x_1, \ldots, x_n$, we call *multi-projective space of distributions* the multi-projective space

$$\mathbb{P}(\mathcal{D}(S)) = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$$

where x_i has $a_i + 1$ states. (note the increasing by 1!).

The elements of $\mathbb{P}(\mathcal{D}(S))$ are thus equivalence classes, each of them containing a distribution D and all its scaling. In such a way we can link more easily (Projective) Algebraic Geometry with the study of significative statistical models.

In this new view, only the statistical models which are independent of the scaling (*multiconi*) in the space of distributions, are significative.

Because the overwhelming majority of important models (*if correctly interpreted*) are independent of the scaling, so they are cones, the depth investigation of our theory will not be affected.

Definition 4.2.2 A *K-projective model* on S is a subset of $\mathbb{P}(\mathcal{D}_K(S))$. A *K-projective algebraic model* on S is a model corresponding to a (multi)projective variety, i.e., defined by the vanishing of multi-homogeneous polynomials.

There is a natural surjection from $\mathcal{D}(S) \setminus \{0\}$ to $\mathbb{P}(\mathcal{D}(S))$. The preimage of a projective model on S, in such projection, is a model M on S with the following property:

$$\text{if } D \in M \text{ and } D' \text{ is a scaling of } D, \text{ with } D' \neq 0, \text{ then } D' \in M.$$

Example 4.2.3 The independence model can be thought as a projective algebraic model, because it is defined by many multi-homogeneous equations (see Theorem 6.4.13).

A linear model is a projective algebraic model when its defining linear polynomials do not have a constant term.

Example 4.2.4 The projective algebraic models on a system S with a single variable (as in the case of the total correlation) are strictly related with the cones of the vector space $\mathcal{D}(S)$. Every cone defines a projective model on S. Vice versa, given a projective model on S, its preimage in the projection $\mathcal{D}(S) \to \mathbb{P}(S)$ is a cone.

Example 4.2.5 Consider the system S formed by two ordinary dice. The projective space of distributions is $\mathbb{P}^5 \times \mathbb{P}^5$. Instead, the projective space of distribution of ΠS is $\mathbb{P}^{35}$, corresponding to the space of 6×6 matrices.

If S is a system where the two variables represent a die and a coin, the projective space of distributions is $\mathbb{P}^5 \times \mathbb{P}^1$. In this case, it is easy to observe that the only variable in the total correlation ΠS has 12 states, hence the projective space of distributions of ΠS is $\mathbb{P}^{11}$.

If a system S has n Boolean variables, then its projective space of distributions is a product of n copies of $\mathbb{P}^1$, and the space of distributions of ΠS is $\mathbb{P}^{2^n-1}$.

4.3 Parametric Models

We are now able to define projective parametric models on A. For this aim, we will use Definition 9.3.1 of projective map, in Sect. 9.3.

Definition 4.3.1 If S, T are random systems, we call *projective connection* any projective map $\Gamma : \mathbb{P}(\mathcal{D}(S)) \to \mathbb{P}(\mathcal{D}(T))$. Notice, in particular, that if Γ is a projective connection, then the image of any scaling D' of a distribution D is a scaling of $\Gamma(D)$.

We say that a model M is *projective parametric* if it is the image $\mathbb{P}(\mathcal{D}(S))$ of a projective connection Γ.

Many interesting parametric models have a counterpart projective parametric.

Example 4.3.2 The independence model is projective parametric. As a matter of fact let S be a system with random variables $x_1, \ldots, x_n$ and let $a_i + 1$ be the number of states of the variable x_i. Hence, the total correlation ΠS has a unique variable, with $\Pi(a_i + 1)$ states.

The model of independence of S corresponds to the map

$$\mathbb{P}(\mathcal{D}(S)) = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n} \to \mathbb{P}(\mathcal{D}(\Pi S)) = \mathbb{P}^M$$

($M = -1 + \Pi(a_i + 1)$) defined by

$$\begin{cases} \vdots & = & \vdots \\ t_{i_1,\ldots,i_n} & = & s_{1i_1} s_{2i_2} \cdots s_{ni_n} \\ \vdots & = & \vdots \end{cases}$$

where we have numbered the coordinates of an element of $\mathbb{P}(\mathcal{D}(\Pi X))$, as usual, identifying this element as a tensor.

It is evident from the same definition that the model of independence corresponds to a *Segre variety* (compare with Definition 10.5.9).

Notice that, in general, M is quite bigger than a_i. For example, if $n = 3$ and $a_1 = a_2 = a_3 = 3$, then $M = 63$ and the model corresponds to the Segre variety of $\mathbb{P}^3 \times \mathbb{P}^3 \times \mathbb{P}^3$ embedded in $\mathbb{P}^{63}$.

We recall that the product $\mathbb{P}^1 \times \mathbb{P}^1$ *is not* isomorphic to $\mathbb{P}^2$. Through the Segre map, $\mathbb{P}^1 \times \mathbb{P}^1$ corresponds to a surface in $\mathbb{P}^3$, image of the map

$$((x_1, x_2), (y_1, y_2)) \mapsto (x_1 y_1, x_1 y_2, x_2 y_1, x_2 y_2)$$

that is, in parametric form

$$\begin{cases} a_{11} & = & x_1 y_1 \\ a_{12} & = & x_1 y_2 \\ a_{21} & = & x_2 y_1 \\ a_{22} & = & x_2 y_2 \end{cases}$$

This surface, which represents the (projective) model of independence of a Boolean system with two variables, is defined by a single equation (determinant of the corresponding 2×2 matrix) $a_{11} a_{22} = a_{12} a_{21}$.

Example 4.3.3 On a random system with three variables x_1, x_2, x_3, the model *without triple correlation* of Example 3.2.5 it is not, strictly speaking, parametric projective.

In fact, taking up the terminology of the example, this model is defined by considering the model S' given by the union of the total correlations of the three subsystems of S that are obtained by canceling one of the variables in turn. S' also has

three variables, corresponding to (x_1, x_2), (x_1, x_3), and (x_2, x_3). The models without triple correlation are obtained from the connection from S' to S, which sends every triplet of matrices $(A, B, C) \in \mathcal{D}(S')$, with $A \in \mathbb{C}^{d_1,d_2}$, $B \in \mathbb{C}^{d_1,d_3}$, $C \in \mathbb{C}^{d_2,d_3}$, to the tensor $D \in \mathcal{D}(\prod S)$ defined by

$$D(i, j, k) = A(i, j)B(i, k)C(j, k).$$

It is clear that all components of this map are multi-homogeneous of the same degree, but, in general, they do not define a map

$$\mathbb{P}^{d_1d_2-1} \times \mathbb{P}^{d_1d_3-1} \times \mathbb{P}^{d_2d_3-1} \to \mathbb{P}^{d_1d_2d_3-1}.$$

because even if A, B, C are all different from zero, however it is not clear that their image is not zero.

In order to obtain our model as the image of a well-defined projective map, we must restrict to a subvariety of $\mathbb{P}^{d_1d_2-1} \times \mathbb{P}^{d_1d_3-1} \times \mathbb{P}^{d_2d_3-1}$.

For instance, when the three variables are Boolean, we must restrict the model to a suitable model X of distributions on S', so that we get a well-defined map from a variety $X \subset \mathbb{P}^3 \times \mathbb{P}^3 \times \mathbb{P}^3$ to $\mathbb{P}^7$. This map is obtained by composing the Segre map $\mathbb{P}^3 \times \mathbb{P}^3 \times \mathbb{P}^3 \to \mathbb{P}^{63}$ with a suitable projection $\mathbb{P}^{63} \to \mathbb{P}^7$. It is a matter of computations that the subvariety X should not intersect a configuration of products of linear spaces, containing, for instance,

$$\begin{array}{ll} L_1 \times \mathbb{P}^3 \times L_1'' & \text{where } L_1 = \{x_1 = x_3 = 0\}\ L_1'' = \{z_3 = z_4 = 0\} \\ L_2 \times L_2' \times \mathbb{P}^3 & \text{where } L_2 = \{x_1 = x_2 = 0\}\ L_1'' = \{y_3 = y_4 = 0\} \\ \quad \ldots & \end{array}$$

The fact that the image of a Segre map can be interpreted as (projective) model of independence of an random system, by Theorem 6.4.13, guarantees us that *the Segre varieties are all projective varieties*.

Let us see how, in general, there are projective parametric models which are not algebraic models.

Example 4.3.4 Consider two systems S, S', both with a single random variable.

We identify both the projective spaces of distribution over $\mathbb{R}$, $\mathbb{P}(\mathcal{D}(S))$ and $\mathbb{P}(\mathcal{D}(S'))$, with $\mathbb{P}^1_{\mathbb{R}}$. We can define a projective connection $\Gamma : \mathbb{P}(\mathcal{D}(S)) \to \mathbb{P}(\mathcal{D}(S'))$ by posing $\Gamma(x_0, x_1) = (x_0^2, x_1^2)$.

It is easy to check that the image W of Γ contains infinitely many points of $\mathbb{P}^1_R$. However, it does not contain all points: as a matter of fact, the point with homogenous coordinates $(1, -1)$ is not in the image.

On the other hand, each projective variety in $\mathbb{P}^1_{\mathbb{R}}$, being defined by the vanishing of a homogeneous polynomial in two variables, or coincides with $\mathbb{P}^1_{\mathbb{R}}$, or it can only contain a finite number of points.

So W can not be a projective variety.

Example 4.3.5 Let us go back to the situation represented in Example 3.3.4. Recall that the initial situation corresponds to a Boolean system S with a variable (of states A, B) while the final situation corresponded to a system S' with only one variable that could take the 3 values AA, AB, BB.

The connection Γ, defined by $\Gamma(a, b) = (a^2, 2ab, b^2)$, is clearly a projective map between $\mathbb{P}(\mathcal{D}(S)) = \mathbb{P}^1_{\mathbb{R}}$ and $\mathbb{P}(\mathcal{D}(S')) = \mathbb{P}^2_{\mathbb{R}}$. The image corresponds to the subset $W \subset \mathbb{P}^2_{\mathbb{R}}$ defined by the points satisfying the equation $y^2 = 4xz$.

It should be noted, however, that not all the homogeneous coordinates of these points can be obtained in the map. In fact the point P of coordinates $(1, 2, 1)$ is in the image (we get it for $(a, b) = (1, 1)$, but no pair of $\mathbb{R}^2$ gives $(-1, -2, -1)$, which are also coordinates of P.

From Chow's Theorem (see Theorem 10.6.3), it immediately follows:

Theorem 4.3.6 *Each projective parametric model is a projective algebraic model.*

This theorem generalizes the situation already seen for the model of independence and explains how each projective parametric model can be defined by homogeneous polynomial equations.

The proof of Chow's Theorem also explains *theoretically* how the homogeneous equations of a projective parametric model can be found.

In practice, as one can imagine, when the number of variables grows, it is not easy to follow step by step the directions, and find an effective set of equations, even with the help of computational tools. The use of *Groebner bases*, which we will introduce later, allows an optimization of the procedure (see Chap. 13).

The advantage of presenting a model with homogeneous equations (*implicit equations*), rather than through parametric equations, should instead be clear in the daily practice of Algebraic Statistics: to test whether a given phenomenon, i.e., a given distribution, falls within the model imagined by a theory (in more imaginative words: if an experiment confirms or less a theory), once the implicit equations are known, it is sufficient to check if the distribution satisfies them. A similar computation is elementary, for every single equation. In everyday practice, the complication derives only from the fact that normally each model is described by an astronomical number of equations, sometimes with approximate coefficients. However, these problems can be managed by methods of sample searching and error checking.

Instead, having to show that a given distribution belongs to a model of which only parametric equations are known, the problem moves to show the existence of parameters for which the parameterization returns the starting distribution. Such a problem of existence is extremely difficult to control, even in the presence of a few, precise equations. Imagine when the equations are thousands, with approximate coefficients!

Chapter 5
Conditional Independence



An intermediate case between total independence and generic situations of the dependence of random variables concerns the so-called conditional independence.

To understand the practical meaning of models of conditional independence, let us start with two examples. In the examples, we will use the fact that the naïve notion of independence between two variables corresponds to distributions whose associated matrices have rank 1 (see Definition 3.2.1 and the thereby discussion), while the conditional independence corresponds to have rank 1 in the slices of the associated tensor.

Example 5.0.1 Let us take again an example, presented by B. Sturmfels in a conference, and cited as *urban legend*.

In England, a magazine, specialized in curious statistics, commissioned a study on the following problem: being football fans increase hair loss?

The authors of the study interviewed many people, reporting the answers to two questions:

(A) Are you a football fan? (possible replies 1 = no, 2 = a little, 3 = a lot).
(B) Do you lose your hair? (possible replies 1 = no, 2 = a little, 3 = a lot).

The results were then listed in the following 3×3-matrix

$$D = \begin{array}{c|ccc} B\backslash A & 1 & 2 & 3 \\ \hline 1 & 72 & 41 & 15 \\ 2 & 60 & 55 & 45 \\ 3 & 40 & 70 & 82 \end{array}$$

As we can easily see, the matrix has not rank 1.

We interpret the fact in terms of Algebraic Statistics.

The starting random system S includes two variables (A = football fan, B = hair loss), each with three states. D is the distribution on ΠS that arises from the

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C. Bocci and L. Chiantini, *An Introduction to Algebraic Statistics with Tensors*, UNITEXT 118, https://doi.org/10.1007/978-3-030-24624-2_5

investigation. Since D has not rank 1, that is, it does not belong to model of independence on ΠS, *the two variables are not independent*.

In other words, being football fans *affects* hair loss.

The result is surprising, although the unequivocal consequence of the collected data and the magazine launched into a series of interpretations of the case.

In reality, the true interpretation was very simple. A clue to the solution of the mystery was contained in the fact that *the matrix D has rank* 2.

In fact, the magazine had mixed, in the result of the investigation, the data related to two distinct groups: *M*en and *W*omen. The M group is more prone to being a football fan and losing hair compared to the W group. The lack of homogeneity of the sample led to a false result, in fact dividing the results of the investigation with respect to an additional Boolean variable (the *gender G* of the sample) you get a tensor $3 \times 3 \times 2$ whose scan along the third index (front–back) is made of two matrices of rank 1.

$$T = \begin{array}{ccccccc}
 & 2 & & 6 & & 8 & \\
70 & & 35 & & 7 & & \\
 & 10 & & 30 & & 40 & \\
50 & & 25 & & 5 & & \\
 & 20 & & 60 & & 80 & \\
20 & & 10 & & 2 & &
\end{array}$$

The previous matrix D represents the marginalization relative to the first index. So D is the sum of two matrices of rank 1, and in fact has rank 2.

Note that the two starting variables A, B *are* really dependent on each other, as follows: if a person is subject to hair loss, he is more likely to be a man, so he is more likely to be a football fan (in the example cited, in fact a bit dated, it was taken as hypothesis that men are more susceptible than women to hair loss, and they are also more likely to follow football).

The fact that D has rank smaller than the maximum, though it does not imply independence of the two variables, should suggest to researchers a connection between the two variables, mediated by a *hidden variable* G.

A similar example is the following.

Example 5.0.2 This example is another classic of the algebraic statistical study: the example of a scientific research that leads to a result only apparently significant.

Osteoporosis is a bone disease that mainly affects elderly people. Let us face the problem: having a driving license affects the vulnerability to osteoporosis? The question is apparently idiotic: how can the sensitivity to a bone disease be influenced by the possession of the license? Yet paradoxically the results would seem to state the opposite.

In fact, a researcher builds a system formed by two Boolean variables to study the phenomenon: possession of the license and the state of illness. Then he considers a population of older people, say 100 individuals, examines them with respect to the possession of the license and the state of the bones, and builds a D distribution on the total correlation. The result is expressed by the matrix:

$$\begin{pmatrix} 13 & 37 \\ 22 & 28 \end{pmatrix}$$

The matrix expresses the fact that 13 people have at the same time driving license and osteoporosis, 37 have a license but not osteoporosis, etc.

The result is incontrovertible! The matrix of D has determinant -450, so it is a far away from having rank 1. Therefore, *there is* correlation between having a driving license and contracting osteoporosis. In the specific case it is clear, from the examination of the results, that having the driving license makes *less likely* the manifestation of osteoporosis. Great unexpected discovery. Such research risks ending up in some scientific journal serious (let's hope not!) and to be picked up by media of half the world. You could create unfounded expectations of healing, with crows of old men and old ladies to the assault of the driving schools. There would be some clinicians ready to explain that driving vehicles, cause the movement of the pedals and steering wheel, is a beneficial workout that tones the bones and makes them more resistant to osteoporosis.

Unfortunately, we have to switch off easy enthusiasms, because the reality is a little different.

The weak point of the statistical experiment lies in the fact that the chosen sample is not homogeneous. In fact among the selected individuals there are mixed elderly men and women. If the sample selection is random, it is likely that there is an equal split: 50 men and 50 women. However, osteoporosis does not affect the two sexes in a homogeneous way. Women are a lot more susceptible to the disease than men. On the other hand, especially in the elderly population, for a man it is much more usual to have a driving license compared to a woman.

The situation is clarified if the chosen random system has 3 variables: at the possession of the license x_1 and at the osteoporosis x_2 we add the Boolean variable x_3 indicating sex ($0 =$ man, $1 =$ woman). In the total correlation of this system, which is a tensor of size 3 and type $(2, 2, 2)$, the real distribution is:

$$D' = \begin{array}{ccc} 8 & & 32 \\ & 5 \quad 5 & \\ & 2 \quad\quad 8 & \\ 20 & & 20 \end{array}$$

which has not rank 1, as there are submatrices with determinant different from 0. The tensor tells us that x_1 is independent from x_2 given x_3 ($x_1 \sqcup x_2 | \{x_3\}$), because the front and back matrices have both rank 1, i.e., fixing the male or female population, in both we see that the possession of the license does not affect on the likelihood of contracting osteoporosis, as was widely expected.

Note that D represents the marginalization of D' along x_3, therefore it is not true that $x_1 \sqcup x_2$. In other words, x_1 and x_2 are actually dependent on each other. What is the meaning of this statement? It must be read this way. We charge a subject z who has a driving license. As the percentage of licensed persons who are men rather than women is higher, it is more likely that z is a man. As such, it is less likely to develop osteoporosis. Conversely, if a subject has osteoporosis, it is more likely to be female, so it is less likely to have a driving license.

Our perception still remains a bit perplexed. The reason lies in the psychological fact that the property of being a man or a woman, for an individual, it is obviously seen as far more fundamental than having a driving license or even to the development of osteoporosis.

The previous examples explain the usefulness of introducing the concepts of *conditional independence* of random variables and also the concept of *hidden variables*.

5.1 Models of Conditional Independence

In this section, we introduce the concept of conditional independence and we show its basic properties.

Remark 5.1.1 As usual, we are more interested to show the geometric or algebraic structure of probability distributions, hence the explicit computation of *conditional probability* is out of our scopes. Consequently we will not introduce, in a formal way, the celebrated Bayes Formula (but see Example 5.1.13 for an instance of how the formula can be recovered in our setting).

We will refer to the concepts of Tensorial Algebra contained in Chap. 8 about scan and contraction. In particular, see Definitions 8.2.1 and 8.3.1.

Definition 5.1.2 Let S be a random system with variables $x_1, \ldots, x_n$.

Let $A \subset \{1, \ldots, n\}$. A distribution D on the total correlation of S *satisfies the condition* $\sqcup A$ (or even that A is independent) if the contraction of D along $\{1, \ldots, n\} \setminus A$ has rank 1.

Given $B = \{1, \ldots, n\} \setminus A$, we will say that D *satisfies the condition* $\sqcup A|B$ (which means A *is independent, given* B) if all the elements of the scan D along B have rank 1.

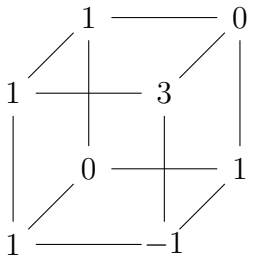
The previous definitions can be generalized and cumulated in the following.

Definition 5.1.3 If A, B are disjoint subsets of $\{1, \ldots, n\}$, we will say that D *satisfies the condition* $\sqcup A|B$ (A *is independent, given* B) if, given $C = \{1, \ldots, n\} \setminus (A \cup B)$, the contraction D' of D along C satisfies the $\sqcup A|B$ condition, that is, all the elements of the scan of D' along B have rank 1.

When $A = \{x_i, x_j\}$ has two elements, we will also write $x_i \sqcup x_j$ and $x_i \sqcup x_j|B$ instead of $\sqcup A$ and $\sqcup A|B$, respectively. It is clear that $\sqcup A$ is equivalent to $\sqcup A|B$ with $B = \emptyset$.

Example 5.1.4 The tensor

$D =$



describes a distribution on the total correlation of a Boolean system with three variables x_1, x_2, x_3.

In D, one has $x_2 \sqcup x_3$, since the contraction along x_1 gives

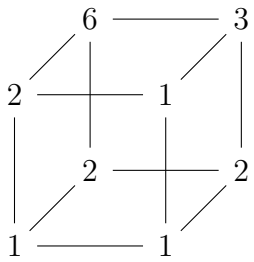
$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}.$$

However, in D, $x_1 \sqcup x_3$ *does not* hold, since the contraction along x_2 gives

$$\begin{pmatrix} 4 & 1 \\ 0 & 1 \end{pmatrix}.$$

Example 5.1.5 The tensor

$D' =$



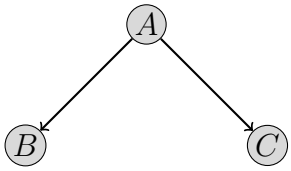
on the total correlation of a Boolean system with three variables x_1, x_2, x_3, describes a distribution D for which one has $(x_2 \sqcup x_3 | \{x_1\})$, since the two submatrices obtained by taking $x_3 = 0$ and $x_3 = 1$ have rank 1. Notice that in D, $x_1 \sqcup x_2$ *does not* hold, since the contraction along x_3 gives

$$\begin{pmatrix} 8 & 4 \\ 3 & 3 \end{pmatrix}$$

of rank 2.

Examples 5.0.1 and 5.0.2 represent situations, where two initial variables are independent, given the third one (the gender).

Example 5.1.6 Consider the transmission chain of a Boolean signal, with a headquarter A and two locations B, C non-connected, represented by the oriented graph

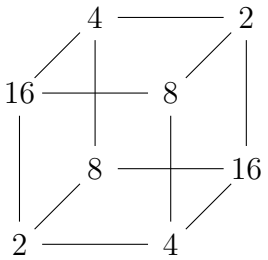


Suppose that the following matrices are associated with the edges AB, AC

$$M_{AB} = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix} \qquad M_{AC} = \begin{pmatrix} \frac{4}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{4}{5} \end{pmatrix}$$

These matrices represent the transmission of the signal, in the sense that if A transmits 30 times the signal 0, B transcribes the distribution $M_{AB} \cdot (30, 0) = (20, 10)$, that is, it transcripts 20 times the signal 0 and 10 times the signal 1. Similarly, C transcribes the distribution $M_{AC} \cdot (30, 0) = (24, 6)$.

If A sends a signal of 30 bits 0 and 30 bits 1, the distribution resulting from the graphic model, in the Boolean system with three variables A, B, C, is given by the tensor:



Observe that the tensor has not rank 1, in fact the three variables are not independent. On the other hand, the two submatrices that they get by fixing $A = 0$ and $A = 1$ both have null determinant, so that $(B \sqcup C|A)$ holds. Instead, the marginalization of the tensor in the direction of A gives the matrix:

$$\begin{pmatrix} 12 & 18 \\ 18 & 12 \end{pmatrix}$$

that has rank 2, so $B \sqcup C$ *does not* hold.

In fact, if one does not consider the contribution of A, the fact that B receives 0 makes it reliable that the bit emitted was really 0, so it is more likely that C also receives a 0. So, if we do not fix the status of A, then B and C are actually dependent. On the other hand, if we know the status of A, then B and C can receive correctly or erratically, independently of each other.

Definition 5.1.7 The matrices used in the previous example are of a type widely used in applications of Algebraic Statistics, especially for the theory of strings of symbols (digital signals, DNA, etc.). They are called *Jukes–Cantor's matrices*.

In general, a Jukes–Cantor matrix is a square matrix $n \times n$ whose diagonal elements are all equal to a value a, while all the other elements are equal to a value b.

These matrices represent the fact that, for example, in the transmission of a signal, if the transmitter A issues a value x_i, the probability of the station B to receive correctly x_i is proportional to a, independently from x_i, while the probability to receive a wrong digit (proportional to $(n-1)b$) is distributed uniformly on all the other values x_j, $j \neq i$.

Proposition 5.1.8 *Let M be a real-valued Jukes–Cantor matrix of order $n \times n$, having a on the main diagonal and b elsewhere, with $a > b > 1$. Then M has rank n.*

Proof We prove the statement by induction on the rank. The cases $n = 1, 2$ are trivial. For n generic, notice that deleting the last row and column we get a Jukes–Cantor matrix of order $(n-1) \times (n-1)$. Hence, we can suppose, by induction that the first $n-1$ rows of M are linearly independent.

If the last row R_n is linear combination of the previous ones, that is, there exists a relation

$$R_n = a_1 R_1 + \cdots + a_{n-1} R_{n-1},$$

then comparing the last entries we get $a = (a_1 + \cdots + a_{n-1})b$, from which $(a_1 + \cdots + a_{n-1}) > 1$. Thus, at least one of the a_i's is positive. Suppose, that $a_1 > 0$. Comparing the first entry of the rows, we get

$$b = a_1 a + (a_2 + \cdots + a_{n-1})b > (a_1 + a_2 + \cdots + a_{n-1})b > b,$$

giving a contradiction.

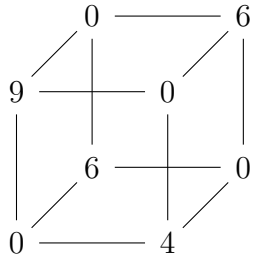
Example 5.1.9 Let us go back to the Example 2.3.26 of the school with two sections A, B, where scholarships are distributed. Let us say the situation after 25 years is given by

$$D = \begin{pmatrix} 9 & 6 \\ 6 & 4 \end{pmatrix}.$$

The matrix defines a distribution on the total correlation of the Boolean system which has two variables A, B, corresponding to the two sections. As the matrix has rank 1, this distribution indicates independence among the possibilities of A, B to have a scholarship.

We introduce a third random variable N, which is 0 if the year is normal, i.e., one scholarship is assigned, and 1 if the year is exceptional, that is, or 2 scholarships are distributed or no scholarships are distributed at all. In the total correlation of the new system, we necessarily obtain the distribution defined by the tensor

$D' =$



since in the normal years only one of the two sections has the scholarship, something that cannot happen in exceptional years.

The tensor D' clearly does not have rank 1. Also note that the elements of the scan of D' along N *do not* have rank 1. As a matter of fact, both in exceptional years

and in normal years, knowing if the A section has had or not the scholarship even determines whether or not B has had it.

On the other hand, $A \sqcup B$, because the marginalization of D' along the variable N gives the matrix D, which is a matrix of independence.

Definition 5.1.10 Given a set of conditions $(\sqcup A_i | B_i)$ as above, the distributions that satisfy all of them form a model in $\mathcal{D}(\Pi S)$. These models are called *models of conditional independence*.

Proposition 5.1.11 *All the models of conditional independence are homogeneous algebraic models, defined by equations of degree* ≤ 2.

All the models of conditional independence are polynomial parametric models.

Each model defined by a single condition $(\sqcup A|B)$ *is a toric model, provided a homogeneous change of coordinates.*

Proof By Theorem 6.4.13, we know that imposing a tensor to have rank 1 corresponds to the vanishing of certain 2×2 determinants. The equations that are obtained are homogeneous polynomial (of degree two). Therefore, every condition $(\sqcup A|B)$ is defined by the composition of quadratic equations with a marginalization, hence from the composition of quadratic and linear equations. Therefore, the resulting model is algebraic. To prove the second statement, we note that if D satisfies a condition $(\sqcup A|B)$ with $A \cup B = \{1, \dots n\}$ (i.e., no marginalization) then for each element D' of the scan of D along B there must exist $v_1 \dots, v_a$, where a is the cardinality of A, such that $D' = v_1 \otimes \cdots \otimes v_a$. It is clear that such condition is polynomial parametric, in fact toric. When $A \cup B \neq \{1, \dots n\}$, the same fact holds on the coefficients that are obtained from the marginalization that depend linearly on the coefficients of D. □

Example 5.1.12 Consider a Boolean system S with three variables $\{x_1, x_2, x_3\}$, so that the distribution space $\mathcal{D}(\Pi S)$ corresponds to the space of tensors of type $(2, 2, 2)$.

The model determined by $(x_1 \sqcup x_2 | x_3)$ contains all distributions D satisfying

$$\begin{cases} D(1,1,1)D(1,2,2) - D(1,2,1)D(1,1,2) &= 0 \\ D(2,1,1)D(2,2,2) - D(2,2,1)D(2,1,2) &= 0 \end{cases}.$$

The same model can be described in parametric form by

$$\begin{cases} D(1,1,1) &= ac \\ D(1,2,1) &= ad \\ D(1,1,2) &= bc \\ D(1,2,2) &= bd \\ D(2,1,1) &= a'c' \\ D(2,2,1) &= a'd' \\ D(2,1,2) &= b'c' \\ D(2,2,2) &= b'd' \end{cases}$$

showing that it is toric.

The model determined by $x_1 \sqcup x_2$ contains all distributions D satisfying

$$(D(1,1,1)+D(2,1,1))(D(1,2,2)+D(2,2,2))- \\ -(D(1,2,1)+D(2,2,1))(D(1,1,2)+D(2,1,2))=0 \quad (5.1.1)$$

that is, all the distributions defined by

$$\begin{cases} (D(1,1,1)+D(2,1,1)) &= ac \\ (D(1,2,2)+D(2,2,2)) &= bd \\ (D(1,2,1)+D(2,2,1)) &= ad \\ (D(1,1,2)+D(2,1,2)) &= bc \end{cases}$$

hence it corresponds to the polynomial parametric model

$$\begin{cases} D(1,1,1) &= x \\ D(2,1,1) &= ac-x \\ D(1,2,2) &= y \\ D(2,2,2) &= bd-y \\ D(1,2,1) &= z \\ D(2,2,1) &= ad-z \\ D(1,1,2) &= t \\ D(2,1,2) &= bc-t \end{cases}$$

This last parameterization, in the new coordinates $D'(i,j,k)$, with $D'(1,j,k) = D(1,j,k)$, $D'(2,j,k) = D(1,j,k)+D(2,j,k)$, becomes

$$\begin{cases} D'(1,1,1) &= x \\ D'(2,1,1) &= ac \\ D'(1,2,2) &= y \\ D'(2,2,2) &= bd \\ D'(1,2,1) &= z \\ D'(2,2,1) &= ad \\ D'(1,1,2) &= t \\ D'(2,1,2) &= bc \end{cases}$$

which represents a toric model.

Example 5.1.13 (Bayes formula) We illustrate in this setting a special instance of the Bayes' formula.

Consider a system in which A sends a digital signal to B. The connection between A and B is not perfect, so that what B receives is modified by the multiplication by a Jukes–Cantor matrix

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix}.$$

We assume that A sends a string of $\{0, 1\}$, containing α occurrences of 0 and β occurrences of 1.

Write $\alpha' = \alpha/(\alpha + \beta)$ and $\beta' = \beta/(\alpha + \beta)$, so that $\alpha' + \beta' = 1$. Also, after scaling, we may assume that the Jukes–Cantor matrix satisfies $a + b = 1$.

It follows from the setting that B receives a string with about $\alpha a + \beta b$ 0's and $\alpha b + \beta a$ 1's. Notice that $(\alpha' a + \beta' b) + (\alpha' b + \beta' a) = 1$.

The probability that A sends 0 is thus $p(A) = \alpha'$, while the probability that B receives 0, assuming that A sends 0, is $p(A|B) = a$. The probability that B receives 0 is

$$p(B) = \frac{\alpha a + \beta b}{(\alpha a + \beta b) + (\alpha b + \beta a)} = \frac{\alpha a + \beta b}{(\alpha + \beta)(a + b)} = \alpha' a + \beta' b$$

while the probability that A did send 0 when B receives 0 is

$$p(A|B) = \frac{\alpha a}{\alpha a + \beta b} = (\alpha' a)\frac{\alpha + \beta}{(\alpha a + \beta b)} = \frac{\alpha' a}{\alpha' a + \beta' b}.$$

It follows that

$$p(A)p(B|A) = p(B)p(A|B).$$

5.2 Markov Chains and Trees

Among all the situations concerning conditional independence, a important case apart is represented by *Markov chains*.

In common practice, a *Markov chain* is a random system if the variables are strictly ordered and the state assumed by each variable is determined *exclusively* from the state assumed by the previous variable.

If exclusivity is intended strictly, other conditions such as time or external factors to the system do not influence the passage from a variable to the consecutive one.

Therefore, if in a distribution D, with sampling c, in which the variable x_i is always in the state ϵ and the variable x_{i+1} is d times in the state ξ, then in a distribution of sampling $2c$, in which the variable x_i is always in the state ϵ, the variable x_{i+1} must assume $2d$ times the state ξ. And if in another distribution D', with sampling c', in which the variable x_i is always in the state ϵ' and the variable x_{i+1} is d' in the state ξ, then in a distribution with sampling $c + c'$, in which the variable x_i is c times in

the state ϵ and c' times in the state ϵ', the variable x_{i+1} must assume $d + d'$ times the state ξ.

This motivates the following:

Definition 5.2.1 Let S be a random system with variables $x_1, \ldots, x_n$ (thus, the variables have a natural ordering). Let a_i be the number of states of variable x_i. Consider matrices $M_1, \ldots, M_{n-1}$, where each matrix M_i has a_i columns and a_{i+1} rows.

We call *Markov model* with matrices $M_1, \ldots, M_{n-1}$ the model on the total correlation of S, formed by distributions D, whose total marginalization $(v_1, \ldots, v_n)$, $v_i \in K^{a_i}$, satisfy all the following conditions

$$v_{i+1} = M_i v_i, \qquad i = 1, \ldots, n-1.$$

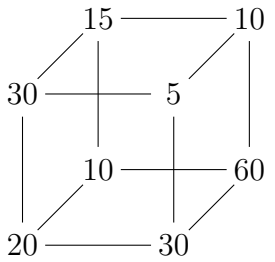
We simply call *Markov model* the model on the total correlation of S, formed by distributions D, satisfying a Markov model, for some choice of the matrices.

Example 5.2.2 Consider a system formed by three headquarters A, B, C transmitting a Boolean signal. A transmits the signal to B, which in turn retransmitted it to C. The signal is disturbed according to the matrices of Jukes–Cantor

$$M = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix} \qquad N = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

If A transmits 60 times 0 and 120 times 1, the distribution observed on the total correlation is

$D' =$



The total marginalization of D is given by (60, 120), (75, 105), (85, 95). Since one has

$$M \begin{pmatrix} 60 \\ 120 \end{pmatrix} = \begin{pmatrix} 75 \\ 105 \end{pmatrix} \qquad N \begin{pmatrix} 75 \\ 105 \end{pmatrix} = \begin{pmatrix} 85 \\ 95 \end{pmatrix}$$

it follows that D is the distribution of the Markov model associated to the matrices M, N.

From the previous example it is clear that, when we have three variables, the Markov model with matrices M, N is formed by distributions $D = (D_{ijk})$ such that $D_{ijk} = d_j M_{ij} N_{jk}$, where $(d_1, \ldots, d_n)$ represents the marginalization of D over the variables x_2, x_3.

Proposition 5.2.3 *The distributions of the Markov model are exactly the distributions satisfying all conditional independences*

$$x_i \sqcup x_k | x_j$$

for any choice of i, j, k *with* $i < j < k$.

Proof We give only a sketch of the proof.

We prove one direction for $n = 3$. If D satisfies a Markov model, relatively to the matrices M, N, then, denoted by (v_1, v_2, v_3) the total marginalization of D, consider $R = \{2\}$ and $Q : R \to J_{n_2}$, $Q(2) = j$. For what has been said following the Example 5.2.2, the element R, Q of D is given by a multiple of $C_j \otimes R_j$, where C_j is the jth column of M while R_j is the jth column of N. Therefore, all these elements have rank 1.

The general case is solved by marginalizing the distribution D in such a way to restrict it to the variables x_i, x_j, x_k.

For the other direction, we describe what happens for a system of three Boolean variables. Given the distribution $D_{ijk} \neq 0$ that satisfies $x_1 \sqcup x_3 | x_2$, unless renumbering the states, we can assume $D_{222} \neq 0$. Consider the submatrices of D

$$M' = \begin{pmatrix} D_{112} & D_{122} \\ D_{212} & D_{222} \end{pmatrix} \qquad N' = \begin{pmatrix} D_{211} & D_{212} \\ D_{221} & D_{222} \end{pmatrix}.$$

Given two numbers h, k such that $hk = D_{212}/D_{222}$, we multiply the second column of M' by h and the second row of N' by k.

The two matrices thus obtained, appropriately scaled, determine matrices M, N describing the Markov model satisfied by D.

In the general case, the procedure is similar but more complicated. □

For a more complete discussion, the reader is referred to the article by Eriksson, Ranestad, Sturmfels, and Sullivant in [1].

Corollary 5.2.4 *Markov's chain models are algebraic models and also polynomial parametric models (since, in general, many conditional independences are involved, these models are not generally toric).*

Remark 5.2.5 Consider a system consisting of three variables x_1, x_2, x_3 with the same number of states.

In practice, the Markov chain model is almost always associated to matrices M, N that are invertible.

In this case, the obtained distributions are the same that we obtain by considering the Markov model on the same system, ordered so that $x_3 \to x_2 \to x_1$, with matrices N^{-1}, M^{-1}.

Thus the Markov chains, when the transition matrices are invertible, *cannot distinguish* who transmits the signal or receives it. From the point of view of distributions, the two chains



are, in fact, indistinguishable.

Markov chains can be generalized to models which are defined over trees.

Definition 5.2.6 Let G be a directed tree.

Consider a random system S whose variables $x_1, \ldots, x_n$ are the vertices of G (which will be partially ordered by the direction on the tree). Let a_i be the number of states of the variable x_i. For any (directed) edge x_i, x_j, we consider a matrix M_{ij} with a_i columns and a_j rows.

We call *Tree Markov model* on G, with matrices $\{M_{ij}\}$ the model on the total correlation of S, formed by distributions D, whose total marginalization $(v_1, \ldots, v_n)$, $v_i \in K^{a_i}$, satisfies the following conditions:

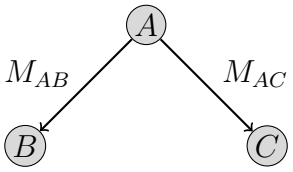
$$v_j = M_{ij} v_i, \ \forall i, j.$$

We simply call *Markov model on* G the model on the total correlation of S, formed by distributions D satisfying the condition of a Tree Markov model on G for some choice of the matrices M_{ij}.

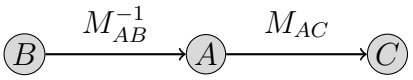
Example 5.2.7 The Markov chains models are obviously examples of tree Markov models.

The simplest example of tree Markov model, beyond chains, is the one shown in the Example 5.1.6.

Remaining in the Example 5.1.6, it is immediate to understand that, for the same motivations expressed in the Remark 5.2.5, when the matrix M_{AB} is invertible, the model associated with the scheme



is indistinguishable from the following Markov chain model



The previous argument suggests that the tree Markov models are described by models of conditional independence. In fact, the hint is valid because in a tree, given two vertices x_i, x_j, there exists at most a minimal path that connects them.

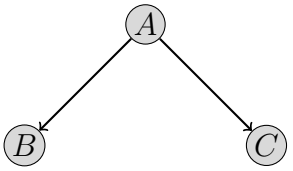
Theorem 5.2.8 *Given a tree G and a random system S whose variables are the vertices $x_1, \ldots, x_n$ of G, then a distribution D on the total correlation of S is in the tree Markov model associated with G, for some choice of matrices M_{ij}, if and only if D satisfies all conditional independencies*

$$x_i \sqcup x_j | x_k$$

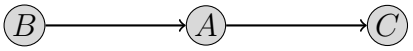
whenever x_k is in the minimal path joining x_i, x_j.

For the proof, refer to the book [2] or to the aforementioned article by Eriksson, Ranestad, Sturmfels, and Sullivant in [1].

Example 5.2.9 Both the tree Markov model associated to the following tree



and the Markov chain model

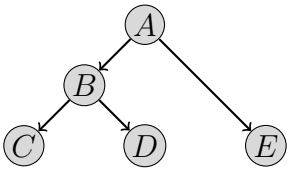


are equivalent to the conditional independence model

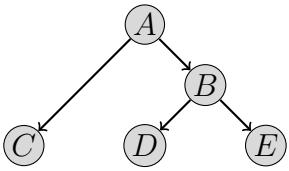
$$B \sqcup C | A.$$

Example 5.2.10 An interesting example of application of the tree Markov models is in the study of phylogenetic, where one tries to reconstruct the genealogical tree of an evolution (which can be biological, but also chemistry or linguistics, etc.). See, for example, [3–6].

For example, suppose we have to fix the evolutionary situation of five species, A, B, C, D, E, starting from the ancestor A. We can hypothesize two different evolutionary situations, represented by graphs G_1, G_2, where

$G_1 =$ 

that is, B, E directly descend from A while C, D descend from B; or

$G_2 =$ 

that is, B, C directly descend from A while E, D descend from B.

We build a random system on the variables A, B, C, D, E, which we can also consider Boolean. If the situation concerns biological evolution, the two states could represent the presence of purine or pyrimidine bases in the positions of the DNA chain of the species. In this case, a distribution is represented by a tensor of type $2 \times 2 \times 2 \times 2 \times 2$.

The models associated with the two graphs G_1, G_2 can be distinguished as, for example, in the first one you have $A \sqcup C|B$, which does not happen in the second case.

5.3 Hidden Variables

Let us go back to the initial Examples 5.0.1 and 5.0.2 of this chapter.

The situation presented in these examples foresees the presence of *hidden variables*, that is, variables whose presence was not known at the beginning, but which condition the dependence between the *observable* variables.

Also in the Example 5.2.10 a similar situation can occur. If the species A, B from which the others derive are only hypothesized in the past, it is clear that one cannot hope to observe the DNA, then the distributions on the variables A, B are not known, so what we observe is not the *true* original tensor, but only its marginalization along the variables A, B.

How can we hope to determine the presence of hidden variables?

One way is suggested by the Example 5.0.1 and uses the concept of rank (see the Definition 6.3.3). In that situation, the distributions on the two observable variables ($A =$ football fan, $B =$ hair loss) were represented by 3×3-matrices. The existence of the hidden variable ($G =$ gender) implied that the matrix of the distribution D was the marginalization of a tensor T of type $3 \times 3 \times 2$, whose scan along the hidden

variable is formed by two matrices M_1, M_2 of rank 1. Therefore, $M = M_1 + M_2$ had rank ≤ 2.

Remark 5.3.1 Let S be a system of random variables $y, x_1, \ldots, x_n$, where y has r states while every x_i has a_i states. A distribution D on ΠS in the conditional independence model $\sqcup\{x_1, \ldots, x_n\}|y$ is represented by a tensor of type $r \times a_1 \times \cdots \times a_n$ whose scan along the first variable is formed by elements of rank 1. Therefore, the marginalization of D along the variable y has rank $\leq r$.

Vice versa, consider a system S' with random variables $x_1, \ldots, x_n$ as above and let D' be a distribution of rank $\leq r$ on $\Pi S'$. Then there is a distribution D on S (not necessarily unique!) whose marginalization along y is D'. In fact we can write

$$D' = D_1 + \cdots + D_r,$$

with each D_i of rank ≤ 1, hence the tensor whose elements along the first direction are $D_1, \ldots, D_r$ represents the distribution D we looked for.

The previous remark justifies the definition of *hidden variable model*.

Definition 5.3.2 On the total correlation of a random system S we will call *hidden variable model* with r states the subset of $\mathbb{P}(\mathcal{D}(\Pi S))$ formed by points corresponding to tensors of rank $\leq r$.

Because the rank of a tensor T is invariant when you multiply T for a constant $\neq 0$, the definition is well placed in the world of projective distributions.

The model of independence is a particular (and degenerate) case of hidden variable models.

Example 5.3.3 Consider a random dipole S, consisting of the variables A, B, having a, b states, respectively. The distributions on ΠS are represented by matrices M of type $a \times b$.

When $r < \min\{a, b\}$, the hidden variable model with r states is equal to the subset of the matrices of rank $\leq r$. It is clear that this model is (projective) algebraic, because it is described by the vanishing of all subdeterminants $(r + 1) \times (r + 1)$, which are homogeneous polynomials of degree $r + 1$ in the coefficients of the matrix M.

When $r \geq \min\{a, b\}$, the hidden variable model with r states can still be defined, but it becomes trivial: all $a \times b$-matrices have rank $\leq r$.

The previous example can be generalized. The hidden variable models with r states become trivial, i.e., they coincide with the entire space of distributions, for r big enough. Furthermore, they are all projective parametric models, therefore also projective algebraic, for Chow's Theorem. The hidden variables models are in fact linked to the geometric concept of *secant variety* of a subset of a projective space. Here, we recall the basic definitions, but a more general treatment will be given in Chap. 12.

Definition 5.3.4 Let Y be a subset of a projective space $\mathbb{P}^n$. We will say that $P \in \mathbb{P}^n$ *belongs to a space r-secant to Y* if there are points $P_1, \dots, P_r \in Y$ (not necessarily distinct) such that the homogeneous coordinates of P are linear combination of the homogeneous coordinates of $P_1, \dots, P_n$. It is clear that the definition is well placed, because it is invariant when multiplying the coordinates of P for a nonzero constant.

We will call *r-secant variety of Y*, denoted by $S_r^0(Y)$, the subset of $\mathbb{P}^n$ formed by the points belonging to a space r-secant to Y.

Remark 5.3.5 It is clear that $S_1^0(Y) = Y$. Moreover, $S_i^0(Y) \subseteq S_{i+1}^0(Y)$ (and equality could hold).

When the cone over Y span the vector space K^{n+1}, then Y contains $n+1$ points whose coordinates are linearly independent, hence $S_{n+1}^0(Y) = \mathbb{P}^n$. In fact, it is clear that $S_{n+1}^0(Y) \neq \mathbb{P}^n$ if and only if the cone over Y is contained in a proper subspace of K^{n+1}, that is if and only if Y is contained in (at least) one hyperplane in $\mathbb{P}^n$.

Notice that we can have $S_r^0(Y) = \mathbb{P}^n$ also for r much smaller than $n+1$.

Proposition 5.3.6 *In the space of tensors $\mathbb{P} = \mathbb{P}(K^{a_1 \cdots a_n})$, a tensor has rank $\leq r$ if and only if it is in the r-secant variety of the Segre variety X.*

It follows that the model with a hidden variable with r state corresponds to the secant variety $S_r^0(X)$ of the Segre variety.

Proof By definition (see Proposition 10.5.12), the Segre variety X is exactly the set of tensors of rank 1.

In general, if a tensor has rank $\leq r$, then it is sum of r tensors of rank 1, hence it lies in the r-secant variety of X.

Vice versa, if T lies in the r-secant variety of X, then there exist tensors $T_1, \dots, T_r$ of X (hence tensors or rank 1) such that

$$T = \alpha_1 T_1 + \cdots + \alpha_r T_r.$$

Hence, since the rank of $\alpha_i T_i$ is 1, at least if α_i is 0, we get that T is sum of $\leq r$ tensors of rank 1. □

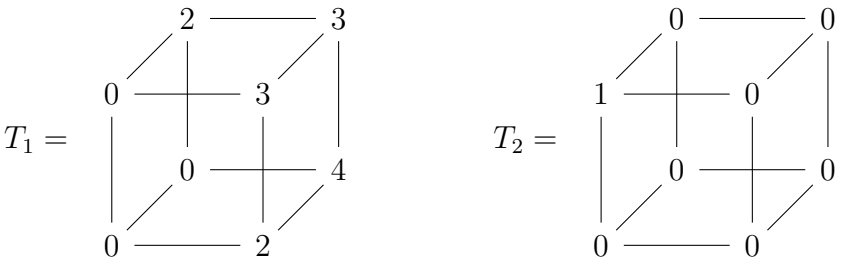
Secant varieties have long been studied in Projective Geometry for their applications to the study of projections of algebraic varieties. Their use in hidden variable models is one of the largest points of contact between Algebraic Statistics and Algebraic Geometry.

An important fact in the study of hidden variable models is that (unfortunately) such models *are not* algebraic models (and therefore not even projective parametric).

Proposition 5.3.7 *In the projective space $\mathbb{P} = \mathbb{P}^7$ of tensors of type $2 \times 2 \times 2$ over $\mathbb{C}$, the subset Y of tensors or rank ≤ 2 is not an algebraic variety.*

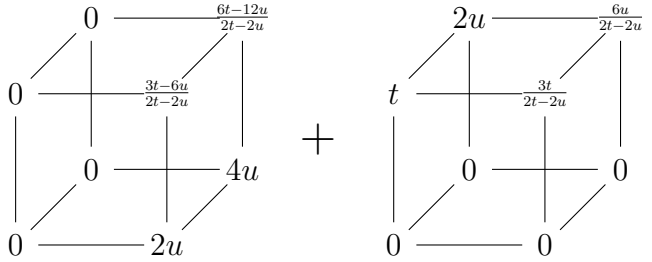
Proof We will use the tensor of rank 3 of Example 6.4.15, which proves, moreover, that Y do not coincide with $\mathbb{P}$.

Consider the tensors of the form $D = uT_1 + tT_2$, where



Such tensors span a space of dimension 2 in the vector space of tensors, hence a line $L \subset \mathbb{P}$. For $(u, t) = (1, 1)$ we get the tensor D of Example 6.4.15, that has rank 3. Hence $L \not\subset Y$.

We check now that all other points in L, different from D, are in Y. In fact if $D' \in L \setminus \{D\}$, then $D' = uT_1 + tT_2$, where (u, t) is not proportional to $(1, 1)$, that is $u \neq t$. Then D' can be decomposed in the sum of two tensors of rank 1 as follows:



If Y were an algebraic model, there would be at least one homogeneous polynomial which vanishes on all points of L, except D. But by restricting this polynomial to L, one would obtain a homogeneous polynomial $p \in \mathbb{C}[u, t]$ that vanishes everywhere, except in the coordinates of D, that is, in the pairs (u, u). On the other hand, every nonzero homogeneous polynomial in $\mathbb{C}[u, t]$ decomposes into the product of a finite number of homogeneous linear factors, so p, which cannot be null because it does not vanishes in D, can vanish only in a finite number of points of the projective line of coordinates u, t, that is, in L. □

To overcome this problem, we define the algebraic secant variety and consequently the algebraic model of hidden variable.

Definition 5.3.8 Let Y be a subset of a projective space $\mathbb{P}^n$. We will call *algebraic r-secant variety of* Y, denoted by $S_r(Y)$, the closure, in Zariski's topology, of $S_r^0(Y)$. This closure corresponds to the *smallest* algebraic variety containing $S_r^0(Y)$.

On the total correlation of a random system S we will call *algebraic model of hidden variable* with r states the subset of $\mathbb{P}(\mathcal{D}(\Pi S))$ formed by the algebraic r-secant variety of the Segre variety corresponding to the tensors of rank 1.

Example 5.3.9 In the projective space $\mathbb{P} = \mathbb{P}^7$ of tensors of type $2 \times 2 \times 2$ over $\mathbb{C}$, let X be the Segre variety given by the embedding of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$.

The algebraic 2-secant variety of S coincides with $\mathbb{P}^7$. It occurs indeed that every tensor of rank > 2 is limit of tensors of rank 2.

Remark 5.3.10 We can try to characterize hidden variable models as parametric models. Consider, for example, the product $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ and his embedding X in $\mathbb{P}^7$. The 2-secant variety can, at first glance, be obtained as a parametric variety defined by the equations

$$Q = \alpha P_1 + \beta P_2 \qquad a, b \in \mathbb{C} \quad P_1, P_2 \in X$$

which combined with the parametric equations of X, leads to overall parametric equations

$$\begin{cases} x_{111} = & \alpha a_1 b_1 c_1 + \beta a'_1 b'_1 c'_1 \\ x_{112} = & \alpha a_1 b_1 c_2 + \beta a'_1 b'_1 c'_2 \\ \dots & \dots \\ x_{222} = & \alpha a_2 b_2 c_2 + \beta a'_2 b'_2 c'_2 \end{cases}$$

where $P_1 = (a_1, a_2) \otimes (b_1, b_2) \otimes (c_1, c_2)$ and $P_2 = (a'_1, a'_2) \otimes (b'_1, b'_2) \otimes (c'_1, c'_2)$.

Unluckily this parameterization cannot be defined globally.

As a matter of fact, moving freely the parameters, we must consider also the cases when $P_1 = P_2$. In this situation, for some choice of α, β,the image would be the point $(0, \dots, 0)$, which does not exist in the projective setting. Hence, the parameterization is only partial.

If we exclude the parameter values for which the image would give $(0, \dots, 0)$, we get a well-defined function on a Zariski open set of $(\mathbb{P}^1)^7$. The image Y of that open, however, is not a Zariski closed set of $\mathbb{P}^7$. Zariski's closure of Y in $\mathbb{P}^7$ coincides with the whole $\mathbb{P}^7$.

Part of the study of secant varieties is based on the calculation of the dimension. From what has just been said in the previous remark, a limitation of the dimension of algebraic secant varieties is always possible.

Proposition 5.3.11 *The algebraic r-secant variety of the Segre variety X, image of the Segre map of the product $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$, has dimension bounded by*

$$\dim(S_r(X)) \leq \min\{N, ar + r - 1\} \tag{5.3.1}$$

where $N = (a_1 + 1) \cdots (a_m + 1) - 1$ is the dimension where X is embedded and $a = a_1 + \cdots + a_n$ is the dimension of X.

Proof That the dimension of $S_r(X)$ is at most N depends on the fact that the dimension of an algebraic variety in $\mathbb{P}^N$ can not exceed the dimension of the ambient space (see Proposition 11.2.14).

The second limitation $\dim(S_r(X)) \leq ar + r - 1$ follows from Theorem 11.2.24, since, generalizing the previous example, on an open Zariski, $\dim(S_r(X))$ is the image of a polynomial map from $(Y)^s \times \mathbb{P}^{r-1}$ to $\mathbb{P}^N$. □

If instead of the Segre map we take the Veronese map, a similar situation is obtained.

Proposition 5.3.12 *The algebraic r-secant variety of the Veronese variety X, image of the Veronese map of degree d on $\mathbb{P}^n$, has dimension bounded by*

$$\dim(S_r(X)) \leq \min\{N, nr + r - 1\} \tag{5.3.2}$$

where $N = \binom{n+d}{d} - 1$ is the dimension of the space where X is embedded.

In both situations, we will call *expected r-secant dimension* of the Segre variety (respectively, of the Veronese variety) the second member of the inequality (5.3.1) (respectively of the inequality (5.3.2)).

Definition 5.3.13 We call *generic rank* of tensors of type $(a_1 + 1) \times \cdots \times (a_n + 1)$ the minimum r such that, if X is the Segre variety of $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$ in $\mathbb{P}^N$, one has $S_r(X) = \mathbb{P}^N$.

We call *generic symmetric rank* of symmetric tensors of type $n \times \cdots \times n$ (d times) the minimum r such that, if X is the Veronese variety of degree d of $\mathbb{P}^n$ in $\mathbb{P}^N$, one has $S_r(X) = \mathbb{P}^N$.

Example 5.3.14 The generic rank of $n \times n$ matrices is n.

The generic rank of $2 \times 2 \times 2$ tensors is 2.

The generic rank of $3 \times 3 \times 3$ tensors *cannot be* 3. As a matter of fact, such tensors of rank 1 correspond to Segre embedding X of $\mathbb{P}^2 \times \mathbb{P}^2 \times \mathbb{P}^2$ in $\mathbb{P}^{26}$.The dimension of the algebraic 3-secant variety $S_3(X)$, by Proposition 5.3.11, is bounded by $6 \cdot 3 + 3 - 1 = 20$, hence $\dim(S_3(X)) \neq 26$.

The last part of the previous example provides a general principle.

Proposition 5.3.15 *Given $a = a_1 + \cdots + a_n$ and $N = (a_1 + 1) \cdots (a_n + 1) - 1$, the generic rank r_g of tensors of type $(a_1 + 1) \times \cdots \times (a_n + 1)$ satisfies*

$$r_g \geq \frac{N+1}{a+1}.$$

Given $N = \binom{n+d}{d} - 1$, the generic symmetric rank r_g^s of symmetric tensors of type $(n + 1) \times \cdots \times (n + 1)$ (d times) satisfies

$$r_g^s \geq \frac{N+1}{n+1}.$$

Note that, in general, there are tensors whose rank is lower than the generic rank, but there may also be tensors whose rank is *greater* than the generic rank (this cannot happen in the case of matrices). See the Example 6.4.15.

Example 5.3.16 In general, we could expect that the generic rank r_g is exactly equal to the smaller integer $\geq (N+1)/(n+1)$. This does not always occur. This is already obvious, in the case of matrix spaces.

For tensors of higher dimension, consider the case of $3 \times 3 \times 3$ tensors, for which $N = 26$ and $n = 6$. The minimum integer $\geq (N+1)/(n+1)$ is 4, but the generic rank is 5.

The tensors for which the generic rank is larger than the minimum integer greater than or equal to $(N+1)/(n+1)$ are called *defective*.

We know a few examples of defective tensors, but a complete classification of them is not known. A discussion of the defectiveness (as a proof of the statement on $3 \times 3 \times 3$ tensors) is beyond the scope of this Introduction and we refer to the text of Landsberg [7].

The importance of the generic rank in the study of hidden variables is evident. Given a random system S with variables $x_1, \ldots, x_n$, where x_i has $a_i + 1$ states, the algebraic model of hidden variable with r states, on the total correlation of S, is equivalent to the algebraic secant variety $S_r(X)$ where X is the Segre variety of $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$. The distributions that are in this model should suggest that the phenomenon under observation it is actually driven by a variable (in fact: *hidden*) with r-states.

However, if $r \geq r_g$, this suggestion is null. In fact, in this case, $S_r(X)$ is equal to the whole space of the distributions, then practically all of distributions suggest the presence of such a variable. This, from the practical side, simply means that the information given the additional hidden variable is null. In practice, therefore, the existence or nonexistence of the hidden variable does not add any useful information to the understanding of the phenomenon.

Example 5.3.17 Consider the study of DNA strings. If we observe the distribution of the bases on 3 positions of the string, we get distributions described by $4 \times 4 \times 4$ tensors. The tensors of this type are not defective, so being $n = 9$, $N = 63$, the generic rank is 7.

The observation of a rank 6 distribution then suggests the presence of a hidden variable with 6 states (as the subdivision of our sample into 6 different species).

The observation of a rank 7 distribution does not, therefore, give us any practical evidence of the real existence of a hidden variable with 7 states.

If we really suspect the existence of a hidden variable (the species) with 7 or more states, how can we verify this?

The answer is that such an observation is not possible considering only three positions of DNA. However, if we go on to observe *four* positions, we get a $4 \times 4 \times 4 \times 4$ tensor. The tensors of this type (which are not even them defective) have

generic rank equal to $\lceil 256/13 \rceil = 20$. If in this case we still get distributions of rank 7, which is much less than 20, our assumption received a formidable experimental evidence.

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Part II
Multi-linear Algebra

Chapter 6
Tensors



6.1 Basic Definitions

The main objects of multi-linear algebra that we will use in the study of Algebraic Statistics are multidimensional matrices, that we will call *tensors*.

One begins by observing that matrices are very versatile objects! One can use them for keeping track of information in a systematic way. In this case, the entries in the matrix are "place holders" for the information. Any elementary book on Matrix Theory will be filled with examples (ranging from uses in Accounting, Biology, and Combinatorics to uses in Zoology) which illustrate how thinking of matrices in this way gives a very important perspective for certain types of applied problems.

On the other hand, from the first course on Linear Algebra, we know that matrices can be used to describe important mathematical objects. For example, one can use matrices to describe linear transformations between vector spaces or to represent quadratic forms. Coupled with the calculus these ideas form the backbone of much of mathematical thinking.

We want to now mention yet another way that matrices can be used: namely to describe bilinear forms. To see this let M be an $m \times n$ matrix with entries from the field K. Consider the two vector spaces K^m and K^n and suppose they have the standard basis. If $v \in K^m$ and $w \in K^n$ we will represent them as $1 \times m$ and $1 \times n$ matrices, respectively, where the entries in the matrices are the coordinates of v and w with respect to the chosen basis. So, let

$$v = \begin{pmatrix} \alpha_1 & \cdots & \alpha_m \end{pmatrix}$$

and

$$w = \begin{pmatrix} \beta_1 & \cdots & \beta_n \end{pmatrix} .$$

The matrix M above can be used to define a function

$$K^m \times K^n \to K$$

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described by

$$(v, w) \to vMw^t$$

where the expression on the right is simply the multiplication of three matrices (^{t} denoting matrix transpose). Notice that this function is linear both in K^m and in K^n, and hence is called a *bilinear form*.

On the other hand, given any bilinear form $B : K^m \times K^n \to K$, i.e., a function which is linear in both arguments, and choosing a basis for both K^m and K^n, we can associate to that bilinear form an $m \times n$, matrix, N, as follows: if $\{v_1, \ldots, v_m\}$ is the basis chosen for K^m and $\{w_1, \ldots, w_n\}$ is the basis chosen for K^n then we form the $m \times n$ matrix $N = (n_{i,j})$ where $n_{i,j} := B(v_i, w_j)$.

It is easy to see that if $v \in K^m$, $v = \sum_{i=1}^m \alpha_i v_i$ and $w \in K^n$, $w = \sum_{j=1}^n \beta_j w_j$ then

$$B(v, w) = \begin{pmatrix} \alpha_1 & \cdots & \alpha_m \end{pmatrix} N \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}.$$

Thus, bilinear forms mapping $K^m \times K^n \to K$ (and a choice of basis for both K^m and for K^n) are in 1-1 correspondence with $m \times n$ matrices with entries from K.

Remark 6.1.1 One should note that although $K^m \times K^n$ is a vector space of dimension $m + n$, the bilinear map defined above from that vector space to K is **not** a linear map. In fact, any vector in the cartesian product of the form $(v, \overline{0})$ or $(\overline{0}, w)$ (where $\overline{0}$ is the zero vector) is sent to 0 under the bilinear form, but the sum of those two vectors is (v, w) which does not necessarily get sent to 0 by the bilinear form.

Example 6.1.2 Recall that if S is a system with two random variables, say x and y, where $A(x)$ contains m elements and $A(y)$ contains n elements, then we used an $m \times n$ matrix M to encode all the information of a distribution on the total correlation ΠS. The (i, j) entry in M was the value of the distribution on the (i, j)th element in the alphabet of the unique random variable (x, y) of the system ΠS (see Definition 1.1.14). This is an example where we used a matrix as a convenient place to store the information of a distribution on ΠS.

However, if we consider the ith element of the alphabet of the random variable x as corresponding to the matrix

$$v = \begin{pmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{pmatrix}$$

(where the 1 occurs in the ith place in this $1 \times m$ matrix) and

$$w = \begin{pmatrix} 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{pmatrix}$$

(where this time the 1 occurs in the jth place in this $1 \times n$ matrix) then the product vMw^t is precisely the (i, j) entry in the matrix M. But, as we noted above, this is the value of the distribution on the (i, j) element in the alphabet of the unique random variable in the total correlation we described above.

So, although the matrix M started out being considered simply as a place holder for information, we see that considering it as a bilinear form on an appropriate pair of vector spaces it can also be used to give us information about the original distribution.

Tensors will give us a way to generalize what we have just seen for two random variables to any finite number of random variables. So, tensors will encode information about the connections between distinct variables in a random system. As the study of the properties of such connections is a fundamental goal in Algebraic Statistics, it is clear that the role of tensors is ubiquitous in this book.

From the discussion above concerning bilinear forms and matrices, we see that we have a choice as to how to proceed. We can define tensors as multidimensional arrays or we can define tensors as multi-linear functions on a cartesian product of a finite number of vector spaces. Both points of view are equally valid and will eventually bring us to the same place. The two ways are equivalent, as we saw above for bilinear forms, although sometimes one point of view is preferable to the other. We will continue with both points of view but, for low dimensional tensors, we will usually prefer to deal with the multidimensional arrays.

Before we get too involved in studying tensors, this is probably a good time to forewarn the reader that although matrices are very familiar objects for which there are well-understood tools to aid in their study, that is far from the case for multidimensional matrices, i.e., tensors. The search for appropriate tools to study tensors is part of ongoing research. The abundance of research on tensors (research being carried out by mathematicians, computer scientists, statisticians, and engineers as well as by people in other scientific fields) attests to the importance that these objects have nowadays in real-life applications.

Notation. For every positive integer i, we will denote by $[i]$ the set $\{1, \ldots, i\}$.

For the rest of the section, K can indicate any set, but in practice, K will always be a set of numbers (like $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$).

Definition 6.1.3 A *tensor* T over K, of dimension n and type

$$a_1 \times \cdots \times a_n$$

is a multidimensional table of elements of K, in which any element is determined by a multi-index $(i_1, \ldots, i_n)$, where i_j ranges between 1 and a_j.

In more formal terms, a tensor T as above is a map:

$$T : [a_1] \times \cdots \times [a_n] \to K.$$

Equivalently (when K is a field) such a tensor T is a multi-linear map

$$T : K^{a_1} \times \cdots \times K^{a_n} \to K$$

where we consider the standard basis for each of the K^{a_i}.

Remark 6.1.4 If we think of T as a multi-linear map and suppose that for each $1 \leq i \leq n$, $\{e^i_j \mid 1 \leq j \leq a_i\}$ is the standard basis for K^{a_i} then the entry in the multidimensional array representation of T corresponding to the multi-index $(i_1, \ldots, i_n)$ is

$$T(e^1_{i_1}, e^2_{i_2}, \ldots, e^n_{i_n}) .$$

Tensors are a natural generalization of matrices. Indeed matrices of real numbers and of type $m \times n$ correspond exactly to tensors over $\mathbb{R}$ of dimension 2 and type $m \times n$.

Example 6.1.5 An example of a tensor over $\mathbb{R}$, of dimension 3 and type $2 \times 2 \times 2$ is:

$$T = \begin{array}{ccc} 2 & \text{———} & 1 \\ | & & | \\ 4 & \text{———} & 0 \\ | & & | \\ -1 & \text{———} & 3 \\ | & & | \\ 4 & \text{———} & 7 \end{array}$$

Notation. Although we have written a $2 \times 2 \times 2$ tensor above, we have not made clear which place in that array corresponds to $T(e^1_{i_1}, e^2_{i_2}, e^3_{i_3})$. We will have to make a convention about that. Again, the conventions in the case of three-dimensional tensors are not uniform across all books on Multi-linear Algebra, but we will attempt to motivate the notation that we use, and is most common, by looking at the cases in which there is widespread agreement, i.e., the cases of one-dimensional and two-dimensional tensors.

Let's start by recalling the conventions for how to represent a one-dimensional tensor, i.e., a linear function

$$T : K^n \to K .$$

Recall that such a tensor can be represented by a $1 \times n$ matrix as follows: let $e_1, \ldots, e_n$ be the standard basis for K^n and suppose that $T(e_i) = \alpha_i$, then the matrix for this linear map is

$$\begin{pmatrix} \alpha_1 & \cdots & \alpha_n \end{pmatrix} .$$

So, if $v = \sum_{i=1}^n \gamma_i e_i$ is any vector in K^n then

$$T(v) = \begin{pmatrix} \alpha_1 & \cdots & \alpha_n \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_n \end{pmatrix}$$

Now suppose that we have a two-dimensional tensor T of type $m \times n$, i.e., a bilinear form

$$T : K^m \times K^n \to K \ .$$

Recall that such a tensor is represented by an $m \times n$ matrix, $\mathcal{A}$, as follows: let

$$\{e_j^1 \mid 1 \le j \le m\} \text{ be the standard basis for } K^m;$$

$$\{e_j^2 \mid 1 \le j \le n\} \text{ be the standard basis for } K^n$$

then

$$\mathcal{A} = (\alpha_{i,j}) \text{ where } \alpha_{i,j} := T(e_i^1, e_j^2) \ .$$

So

$$\mathcal{A} = \begin{pmatrix} \alpha_{1,1} & \alpha_{1,2} & \cdots & \alpha_{1,n} \\ \vdots & \vdots & \dots & \vdots \\ \alpha_{m,1} & \alpha_{m,2} & \cdots & \alpha_{m,n} \end{pmatrix}$$

Now suppose we have a three-dimensional tensor T of type $m \times n \times r$, i.e., a trilinear form

$$T : K^m \times K^n \times K^r \to K \ .$$

This tensor is represented by an $m \times n \times r$ box, $\mathcal{A}$, as follows: let

$$\{e_j^1 \mid 1 \le j \le m\} \text{ and } \{e_j^2 \mid 1 \le j \le n\}$$

be the standard basis for K^m and K^n respectively (as above) and let

$$\{e_j^3 \mid 1 \le j \le r\} \text{ be the standard basis for } K^r \ .$$

Then

$$\mathcal{A} = (\alpha_{i,j,k}) \text{ where } \alpha_{(i,j,k)} := T(e_i^1, e_j^2, e_k^3) \ .$$

How will we arrange these values in a rectangular box? We let the front (or *first*) face of the box be the $m \times n$ matrix whose (i, j) entry is $T(e_i^1, e_j^2, e_1^3)$. The second face, parallel to the first face, is the $m \times n$ matrix whose (i, j) entry is $T(e_i^1, e_j^2, e_2^3)$. We continue in this way so that the back face (the rth face), parallel to the first face is the $m \times n$ matrix whose (i, j) entry is $T(e_i^1, e_j^2, e_r^3)$.

Example 6.1.6 Let T be the three-dimensional tensor of type $3 \times 2 \times 2$, whose (i, j, k) entry is equal to ijk (the product of the three numbers). Then the $3 \times 2 \times 2$ rectangle has first face a 3×2 matrix whose (i, j) entry is $(ij) \cdot 1$. The second (or back) face is a 3×2 matrix whose (i, j) entry is $(ij) \cdot 2$. We put this all together to get our $3 \times 2 \times 2$ tensor.

$$T_1 = \begin{array}{ccc} 2 & & 4 \\ & 1 \quad 2 & \\ 4 & & 8 \\ & 2 \quad 4 & \\ 6 & & 12 \\ & 3 \quad 6 & \end{array}$$

To be assured that you have the conventions straight for trilinear forms, verify that the three-dimensional tensor of type $3 \times 2 \times 2$ whose multidimensional matrix representation has entries $(i, j, k) = i + j + k$, looks like

$$T_2 = \begin{array}{ccc} 4 & & 5 \\ & 3 \quad 4 & \\ 5 & & 6 \\ & 4 \quad 5 & \\ 6 & & 7 \\ & 5 \quad 6 & \end{array}$$

Remark 6.1.7 We saw above that elements of K^n can be considered as tensors of dimension 1 and type n. Notice that they can also be considered as tensors of dimension 2 and type $1 \times n$, or tensors of dimension 3 and type $1 \times 1 \times n$, etc.

Similarly, $n \times m$ matrices are tensors of dimension 2 but they can also be seen as tensors of dimension 3 and type $1 \times n \times m$, etc.

Elements of K can be seen as tensors of dimension 0.

As a generalization of what we can do with matrices, we mention the following easy fact.

Proposition 6.1.8 *When K is a field, the set of all tensors of fixed dimension n and type $a_1 \times \cdots \times a_n$ is a vector space where the operations are defined over elements with corresponding multi-indices.*

This space, whose dimension is the product $a_1 \cdots a_n$, will be denoted by $K^{a_1,\ldots,a_n}$. One basis for this vector space is obtained by considering all the multidimensional

matrices with a 1 in precisely one place and a zero in every other place. If that unique 1 is in the position $(i_1, \ldots, i_n)$*, we refer to that basis vector as* $e_{(i_1,\ldots,i_n)}$.

The null element of a space of tensors is the tensor having all entries equal to 0.

Now that we have established our convention about how the entries in a multidimensional array can be thought of, it remains to be precise about how a multidimensional array gives us a multi-linear map.

So, suppose we have a tensor T which is a tensor of dimension n and type $a_1 \times \cdots \times a_n$. Let $\mathcal{A} = (\alpha_{i_1,i_2,\ldots,i_n})$, where $1 \leq i_j \leq a_j$, be the multidimensional array which represents this tensor. We want to use $\mathcal{A}$ to define a multi-linear map

$$\mathcal{T} : K^{a_1} \times \cdots \times K^{a_n} \to K$$

whose multidimensional matrix representation is precisely $\mathcal{A}$. Let $v_j \in K^{a_j}$, where v_j has coordinates $(v_{j,1}, \ldots, v_{j,a_j})$ with respect to the standard basis for K^{a_j}. Then define

$$\mathcal{T}(v_1, v_2, \ldots, v_n) = \sum (\alpha_{i_1,i_2,\ldots,i_n})(v_{1,i_1} \cdot v_{2,i_2} \cdots v_{n,i_n}) \; .$$

Now if $\{e_i^{[j]} \mid 1 \leq i \leq a_j, \; 1 \leq j \leq n\}$ is the standard basis for K^{a_j} then it is easy to see that

$$\mathcal{T}(e_{i_1}^{[1]}, \ldots, e_{i_n}^{[n]}) = \alpha_{i_1,i_2,\ldots,i_n}$$

Since the $(e_{i_1}^{[1]}, \ldots, e_{i_n}^{[n]})$ form a basis for the space $K^{a_1} \times \ldots K^{a_n}$ and $\mathcal{T}$ is the unique multi-linear map with values equal to the entries in the multidimensional matrix $\mathcal{A}$ we are done.

6.2 The Tensor Product

Besides the natural operations (addition and scalar multiplication) between tensors of the same type, there is another operation, the *tensor product*, which combines tensors of any type. This tensor product is fundamental for our analysis of the properties of tensors.

The simplest way to define the tensor product is to think of tensors as multi-linear maps. With that in mind, we make the following definition.

Definition 6.2.1 Let $T \in K^{a_1,\ldots,a_n}$, $U \in K^{a'_1,\ldots,a'_m}$ be tensors. We define the *tensor product* $T \otimes U$ as the tensor $W \in K^{a_1,\ldots,a_n,a'_1,\ldots,a'_m}$ such that

if $v_i \in K^{a_i}$, $w_j \in K^{a'_j}$ then

$$W(v_1, \ldots, v_n, w_1, \ldots, w_m) = T(v_1, \ldots, v_n)U(w_1, \ldots, w_m).$$

We extend this definition to consider more factors. So, for any finite collection of tensors $T_j \in K^{a_{j1},\ldots,a_{jn_j}}$, $j = 1, \ldots, m$, one can define their tensor product as the tensor

$$W = T_1 \otimes \cdots \otimes T_m \in K^{a_{11},\ldots,a_{1n_1},\ldots,a_{m1},\ldots,a_{mn_m}}$$

such that

$$W(i_{11}, \ldots, i_{1n_1}, \ldots, i_{m1}, \ldots, i_{mn_m}) = T_1(i_{11}, \ldots, i_{1n_1}) \cdots T_m(i_{m1}, \ldots, i_{mn_m}).$$

This innocent looking definition actually contains some new and wonderful ideas. The following examples will illustrate some of the things that come from the definition. The reader should keep in mind how different this multiplication is from the usual multiplication that we know for matrices.

Example 6.2.2 Given 2 one-dimensional tensors v and w of type m and n, respectively, we write $v = (\alpha_1, \ldots, \alpha_m) \in K^m$ and $w = (\beta_1, \ldots, \beta_n) \in K^n$. Then v defines a linear map (which we'll also call v)

$$v : K^m \to K \text{ defined by: } \quad v(x_1, \ldots, x_m) = \sum_{i=1}^{m} \alpha_i x_i$$

and w a linear map (again abusively denoted w)

$$w : K^n \to K \text{ defined by: } \quad w(y_1, \ldots, y_n) = \sum_{i=1}^{n} \beta_i y_i \ .$$

By definition, the tensor product $v \otimes w$ is the bilinear map:

$$v \otimes w : K^m \times K^n \to K$$

defined by

$$v \otimes w : ((x_1, \ldots, x_m), (y_1, \ldots, y_n)) \to (\sum_{i=1}^{m} \alpha_i x_i)(\sum_{i=1}^{n} \beta_i y_i).$$

If we let $\{e_1, \ldots, e_m\}$ be the standard basis for K^m and $\{e'_1, \ldots, e'_n\}$ be the standard basis for K^n then

$$v \otimes w : (e_i, e'_j) \to \alpha_i \beta_j$$

and so the matrix for this bilinear form is $v^t w$.

To give a very specific example of this let $v = (1, 2) \in \mathbb{R}^2$ and $w = (2, -1, 3) \in \mathbb{R}^3$. Then:

$$v \otimes w = v^t w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 2 & -1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 & 3 \\ 4 & -2 & 6 \end{pmatrix}$$

We could just as well have considered the tensor $w \otimes v$. In the specific example we just considered, notice that

$$w \otimes v = w^t v = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ -1 & -2 \\ 3 & 6 \end{pmatrix} = (v^t w)^t.$$

We see here that the tensor product is not commutative. In fact, the two multiplications did not even give us tensors of the same type.

Example 6.2.3 Let's now consider a slightly more complicated example. This time we will take the tensor product of v, a one-dimensional tensor of type 2, and multiply it by w, a two-dimensional tensor of type 2×2. We can represent v by a 1×2 matrix and w by a 2×2 matrix. So, let

$$v = (2, -3) \in \mathbb{R}^2 \text{ and } w = \begin{pmatrix} 2 & -1 \\ 4 & 3. \end{pmatrix}$$

Then v defines a linear map

$$v : K^2 \to K \text{ given by } v(x_1, x_2) = 2x_1 - 3x_2$$

and w defines a bilinear map

$$w : K^2 \times K^2 \to K \text{ given by } w : ((y_1, y_2), (z_1, z_2)) = \begin{pmatrix} y_1 & y_2 \end{pmatrix} w \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} =$$

$$= 2y_1z_1 + 4y_2z_1 - y_1z_2 + 3y_2z_2.$$

Putting these all together we have a trilinear form,

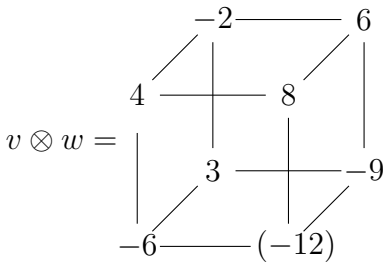
$$v \otimes w : (K^2) \times (K^2 \times K^2) \to K$$

defined by

$$v \otimes w((x_1, x_2), (y_1, y_2), (z_1, z_2)) = (2x_1 - 3x_2)(2y_1z_1 + 4y_2z_1 - y_1z_2 + 3y_2z_2) =$$

$$= 4x_1y_1z_1 + 8x_1y_2z_1 - 2x_1y_1z_2 + 6x_1y_2z_2 - 6x_2y_1z_1 - 12x_2y_2z_1 + 3x_2y_1z_2 - 9x_2y_2z_2.$$

From this, we express $v \otimes w$ as a $2 \times 2 \times 2$ multidimensional matrix, namely,



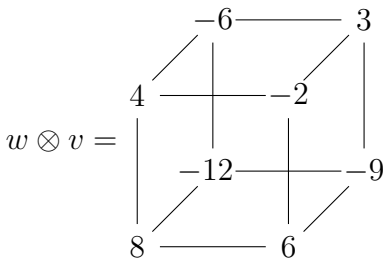
On the other hand,

$$w \otimes v((x_1, x_2), (y_1, y_2), (z_1, z_2)) = w((x_1, x_2), (y_1, y_2))v(z_1, z_2) =$$

$$(2x_1y_1 + 4x_2y_1 - x_1y_2 + 3x_2y_2)(2z_1 - 3z_2) =$$

$$= 4x_1y_1z_1 + 8x_2y_1z_1 - 2x_1y_2z_1 + 6x_2y_2z_1 - 6x_1y_1z_2 - 12x_2y_1z_2 + 3x_1y_2z_2 - 9x_2y_2z_2 \,.$$

So, the multidimensional array for $w \otimes v$ is



Example 6.2.4 Observe that if T, U are $n \times n$ matrices, the tensor product $T \otimes U$ *does not* coincide with their row-by-column product. The tensor product of these two matrices is a tensor of dimension 4, of type $n \times n \times n \times n$.

As we just noted, the tensor product does not define an internal operation in the spaces of tensors of the same dimension and same type. It is possible, however, to define something called the *tensor algebra* on which the tensor product behaves like a product. We will just give the definition of the tensor algebra, but won't have occasion to use it in this text.

Definition 6.2.5 Let K be a field. The *tensor algebra over the space* K^n is the direct sum

$$\mathscr{T}(n) = K \oplus K^n \oplus K^{n,n} \oplus \cdots \oplus K^{n,\ldots,n} \oplus \cdots$$

The tensor product defines a homogeneous operation inside $\mathscr{T}(n)$.

Remark 6.2.6 It is an easy (but messy) consequence of our definition that the tensor product is an associative product, i.e., if T, U, V are tensors, then

$$T \otimes (U \otimes V) = (T \otimes U) \otimes V.$$

Notice that the tensor product is *not*, in general, a commutative product (see Example 6.2.3 above). Indeed, in that example we saw that even the spaces in which $T \otimes U$ and $U \otimes T$ lie can be different.

Remark 6.2.7 The tensor product of tensors has the following properties: for any $T, T' \in K^{a_1,\dots,a_n}$, $U, U' \in K^{a'_1,\dots,a'_m}$ and $\lambda \in K$, one has

- $T \otimes (U + U') = T \otimes U + T \otimes U'$;
- $(T + T') \otimes U = T \otimes U + T' \otimes U$;
- $(\lambda T) \otimes U = T \otimes (\lambda U) = \lambda(T \otimes U)$.

This can be expressed by saying that the tensor product is linear over the two factors.

More generally, the tensor product defines a map

$$K^{a_{11},\dots,a_{1n_1}} \times \cdots \times K^{a_{m1},\dots,a_{mn_m}} \longrightarrow K^{a_{11},\dots,a_{1n_1},\dots,a_{m1},\dots,a_{mn_m}}$$

which is linear in any factor. For this reason, we say that the tensor product is a multi-linear product in its factors.

The following useful proposition holds for the tensor product.

Proposition 6.2.8 (Vanishing Law) *Let T, U be tensors. Then:*
- If $T = 0$ or $U = 0$, then $T \otimes U = 0$.
- Conversely, if $T \otimes U = 0$ then either $T = 0$ or $U = 0$.

Proof Assume $T' \in K^{a_1,\dots,a_n}$, $U \in K^{a'_1,\dots,a'_m}$.

If $T = 0$ then for any choice of the indices $i_1, \dots, i_n$, $j_1, \dots, j_m$ one has

$$(T \otimes U)_{i_1,\dots,i_n,j_1,\dots,j_m} = T_{i_1,\dots,i_n} \cdot U_{j_1,\dots,j_m} = 0 \cdot U_{j_1,\dots,j_m} = 0.$$

A similar computation holds when $U = 0$.

Conversely, if $T \neq 0$ and $U \neq 0$, then there exist two sets of indices, $i_1, \dots, i_n$ and $j_1, \dots, j_m$, such that $T_{i_1,\dots,i_n} \neq 0$ and $U_{j_1,\dots,j_m} \neq 0$. Thus

$$(T \otimes U)_{i_1,\dots,i_n,j_1,\dots,j_m} = T_{i_1,\dots,i_n} \cdot U_{j_1,\dots,j_m} \neq 0.$$

□

The bilinear map

$$K^{a_1,\dots,a_n} \times K^{a'_1,\dots,a'_m} \to K^{a_1,\dots,a_n,a'_1,\dots,a'_m}$$

determined by the tensor product is not injective (as the Vanishing Law clearly shows). However, we can characterize tensors $T, T' \in K^{a_1,\ldots,a_n}$ and $U, U' \in K^{a'_1,\ldots,a'_m}$ such that $T \otimes U = T' \otimes U'$.

Proposition 6.2.9 *Let $T, T' \in K^{a_1,\ldots,a_n}$ and $U, U' \in K^{a'_1,\ldots,a'_m}$ satisfy*

$$T \otimes U = T' \otimes U' \neq 0.$$

Then there exists a nonzero scalar $\alpha \in K$ such that $T' = \alpha T$ and $U' = \frac{1}{\alpha} U$.

In particular, if $U = U'$ then $T = T'$ (and conversely).

Proof Put $Z = T \otimes U = T' \otimes U'$. Since $Z \neq 0$, there exists a choice of indices such that

$$Z_{i_1,\ldots,i_n,j_1,\ldots,j_m} = T_{i_1,\ldots,i_n} \cdot U_{j_1,\ldots,j_m} = T'_{i_1,\ldots,i_n} \cdot U'_{j_1,\ldots,j_m} \neq 0.$$

Thus $T_{i_1,\ldots,i_n} \neq 0$.

Let

$$\beta = T'_{i_1,\ldots,i_n} / T_{i_1,\ldots,i_n} .$$

Since $\beta \neq 0$, it is easy to show that

$$U'_{k_1,\ldots,k_m} = \frac{T'_{i_1,\ldots,i_n}}{T_{i_1,\ldots,i_n}} U_{k_1,\ldots,k_m} = \beta U_{k_1,\ldots,k_m},$$

for all $k_1, \ldots, k_m$, i.e. $U' = \beta U$.

Similarly, since $U_{j_1,\ldots,j_m} \neq 0$, we can let $\alpha \neq 0$ be the quotient $U'_{j_1,\ldots,j_m} / U_{j_1,\ldots,j_m}$. As above one shows that $T' = \alpha T$.

Finally, by multi-linearity, we get $Z = T' \otimes U' = (\alpha\beta)(T \otimes U)$. Hence, $\alpha\beta = 1$, i.e., $\beta = \frac{1}{\alpha}$.

The final statement of the proposition follows from the previous one, since α is 1. □

By using the associativity of the tensor product and slightly modifying the proof of the preceding proposition one can prove, by induction on the number of factors, the following result:

Proposition 6.2.10 *Let $T_1, U_1 \in K^{a_{11},\ldots,a_{1n_1}}, \ldots, T_s, U_s \in K^{a_{s1},\ldots,a_{sn_s}}$ satisfy*

$$T_1 \otimes T_2 \otimes \cdots \otimes T_s = U_1 \otimes U_2 \otimes \cdots \otimes U_s \neq 0.$$

Then there exist nonzero scalars $\alpha_1, \ldots, \alpha_s \in K$ such that $U_i = \alpha_i T_i$ for all i, and moreover $\alpha_1 \cdots \alpha_s = 1$.

Remark 6.2.11 We mentioned above that the tensor product of two bilinear forms, represented by matrices M and N, respectively, doesn't correspond to the product of

the two matrices M and N. Indeed, in most cases, we cannot even take the product of the two matrices!

However, when M is an $n \times m$ matrix and N is an $m \times s$ matrix we can form their product as matrices and also form their tensor product. It turns out that there is a relation between these two objects.

The tensor product is an element of the vector space $K^n \times K^m \times K^m \times K^s$ while the matrix product can be considered as an element of $K^n \times K^s$. How can we recover the regular product from the tensor product?

Now the tensor product is the tensor Q of dimension 4 and type (n, m, m, s), such that $Q(i, j, k, l) = M(i, j)N(k, l)$. The row-by-column product of M, N is obtained by sending Q to the matrix $Z \in K^{n,s}$ defined by

$$Z(i, l) = \sum_j Q(i, j, j, l).$$

So, the ordinary matrix product is obtained, in this case, by taking the tensor product and following that by a projection onto the space $K^n \times K^s = K^{n,s}$.

6.3 Rank of Tensors

In the next two sections we generalize, to tensors of any dimension, a definition which is basic in the theory of matrices, namely the notion of the *rank* of a matrix.

To find the appropriate generalization to tensors, we will have to choose among the many equivalent ways one can define the rank of a matrix. It turns out that it is not convenient to choose, as the definition of rank, its characterization as the dimension of either the row space or the column space of a matrix. We will use, instead, a characterization of the rank of a matrix which is probably less familiar to the reader, but which turns out to be perfect for a generalization to arbitrary tensors. The starting point is a simple characterization of matrices of rank 1.

Proposition 6.3.1 *Let $M = (m_{ij})$ be a nonzero $m \times n$ matrix with coefficients in a field K. M has rank 1 if and only if there are nonzero vectors $v \in K^m$, $w \in K^n$ such that,*

$$M = v \otimes w = v^t w.$$

Proof Assume that v and w exist. Since $M = v^t w$ every row of M is a multiple of w and so the row space of M has dimension 1 and hence the rank of M is 1.

Conversely, if the rank of M is 1 then every row of M is a multiple of some nonzero vector, which we will call w. I.e. the ith row of M is $c_i w$. If we set $v = (c_1, \ldots, c_m)$ then clearly $M = v^t w = v \otimes w$. □

Thus, one can define matrices of rank 1 in terms of the tensor product of vectors.

Although the rank of a matrix M is usually defined as the dimension of either the row space or column space of M, we now give a neat characterization of rank(M) in terms of matrices of rank 1.

Proposition 6.3.2 *Let $M \neq 0$ be an $m \times n$ matrix. Then the rank of M is equal to the smallest integer r such that M is a sum of r matrices of rank* 1.

Proof Assume $M = M_1 + \cdots + M_r$, where every M_i has rank 1. Then we may write $M_i = (v_i)^t w_i$ where $v_i \in K^m$, $w_i \in K^n$. Form the matrix A whose columns are the vectors v_i^t, and the matrix B whose rows are the vectors w_i. It is easy to see that

$$M = AB$$

and so the rows of M are linear combinations of the rows of B. Since B has only r rows we obtain that $rank(M) \leq r$.

Conversely, assume that M has rank r. Then we can find r linearly independent vectors in K^n which generate the row space of M. Call those vectors $w_1, \ldots, w_r$. Suppose that the ith row of M is $c_{i,1}w_1 + \cdots + c_{i,r}w_r$. Form the vector $v_i = (c_{i,1}, \ldots, c_{i,r})$ and construct a matrix A whose ith column is v_i^t. If B is the matrix whose jth row is w_j then $M = AB = \sum_{i=1}^{r} v_i^t w_i$ is a sum of r matrices of rank 1 and we are done. □

The two previous results on matrices allow us to extend the definition of *rank* to tensors of any type.

Definition 6.3.3 A nonzero tensor $T \in K^{a_1,\ldots,a_n}$ has *rank* 1 if there are vectors $v_i \in K^{a_i}$ such that $T = v_1 \otimes \cdots \otimes v_n$. (since the tensor product is associative, there is no need to specify the order in which the tensor products in the formula are performed).

We define the *rank* of a nonzero tensor T to be the minimum r such that there exist r tensors $T_1, \ldots, T_r$ of rank 1 with

$$T = T_1 + \cdots + T_r. \tag{6.3.1}$$

Remark 6.3.4 A tensor of rank 1 is also called a *simple* or *decomposable* tensor.

For any tensor T of rank r, the expression (6.3.1) is called a *(decomposable) decomposition* of T.

We will sometimes just refer to the decomposable decomposition of T as a *decomposition* of T or a *rank decomposition* of T.

By convention we say that null tensors, i.e., tensors whose entries are all 0, have rank 0.

Remark 6.3.5 Let T be a tensor of rank 1 and let $\alpha \neq 0$, $\alpha \in K$. Then, using the multi-linearity of the tensor product, we see that αT also has rank 1. More generally, if T has rank r then αT also has rank r. Then (exactly as for matrices), the union of the null tensor with all the tensors in $K^{a_1,\ldots,a_n}$ of rank r is closed under scalar multiplication.

Subsets of vector spaces that are closed under scalar multiplication are called *cones*. Thus the set of tensors in $K^{a_1,\ldots,a_n}$ of fixed rank (plus 0) is a cone.

On the other hand (again exactly as happens for matrices), in general the sum of two tensors in $K^{a_1,\ldots,a_n}$ of rank r need not have rank r. Thus, the set of tensors in $K^{a_1,\ldots,a_n}$ having fixed rank (union the null tensor) is *not* a subspace of $K^{a_1,\ldots,a_n}$.

6.4 Tensors of Rank 1

In this section, we give a useful characterization of tensors of rank 1. There exists a generalization for *matrices* of higher rank but, unfortunately, there does not exist a similar characterization for tensors of higher rank and having dimension ≥ 3.

Recall that we are using the notation $[i] = \{1, 2, \ldots, i-1, i\}$.

Definition 6.4.1 Let $0 < m \leq n$ be integers. An injective *nondecreasing* function

$$f : [m] \to [n]$$

is a function with the property that

$$\text{whenever } a, b \in [m] \text{ and } a < b \text{ then } f(a) < f(b) .$$

With this technical definition made we are now able to define the notion of a *subtensor* of a given tensor.

Definition 6.4.2 Let T be a tensor in $K^{a_1,\ldots,a_n}$. We consider T as a map

$$T : [a_1] \times \cdots \times [a_n] \to K .$$

For any choice of positive integers $a'_j \leq a_j$ $(1 \leq j \leq n)$ and for any choice of injective, nondecreasing maps $f_j : [a'_j] \to [a_j]$ we define the tensor $T' \in K^{a'_1,\ldots,a'_n}$ as follows:

$$T' : [a'_1] \times \cdots \times [a'_n] \to K$$

where

$$T'_{i_1 \ldots i_n} = T_{i_{f_1(i_1)} \ldots i_{f_n(i_n)}} .$$

Remark 6.4.3 This is a formal (and perhaps a bit odd) way to say that we are fixing a few values for the indices $i_1, \ldots, i_n$ and forgetting the elements of T whose kth index is not in the range of the map f_k.

Since we usually think of a tensor of type $1 \times a_2 \times \cdots \times a_n$ as a tensor of type $a_2 \times \cdots \times a_n$, when a $a'_k = 1$ we simply *forget* the kth index in T. In this case, the dimension of T' is $n - m$, where m is the number of indices for which $a'_k = 1$.

Example 6.4.4 A $3 \times 2 \times 2$ tensors T can be denoted as follows:

$$T = \begin{array}{ccc} T_{112} & - & T_{122} \\ | & \diagdown & | \\ T_{111} & - & T_{121} \\ | & & | \\ & T_{212} & - & T_{222} \\ & | & \diagdown & | \\ & T_{211} & - & T_{221} \\ & | & & | \\ & & T_{312} & - & T_{322} \\ & & | & \diagdown & | \\ & & T_{311} & - & T_{321} \end{array}$$

and an instance is

$$T = \begin{array}{ccc} 1 & - & 0 \\ | & \diagdown & | \\ -2 & - & 4 \\ | & & | \\ & 2 & - & 3 \\ & | & \diagdown & | \\ & 0 & - & 1 \\ & | & & | \\ & & -3 & - & 4 \\ & & | & \diagdown & | \\ & & 2 & - & 1 \end{array}$$

If one takes the maps $f_2 = f_3 =$ identity, $f_1 : [2] \to [3]$ defined as $f_1(1) = 1$, $f_1(2) = 3$, then the corresponding subtensor is

$$T' = \begin{array}{ccc} 1 & - & 0 \\ | & \diagdown & | \\ -2 & - & 4 \\ | & & | \\ & -3 & - & 4 \\ & | & \diagdown & | \\ & 2 & - & 1 \end{array}$$

i.e. one just cancels the layer corresponding to the elements whose first index is 2.

If, instead, one takes $f_2 = f_3 =$ identity, $f_1 : [1] \to [3]$ defined as $f_1(1) = 1$, then one gets the matrix in the top face:

$$T' = \begin{pmatrix} -2 & 4 \\ 1 & 0 \end{pmatrix}$$

Definition 6.4.5 A subtensor of T of dimension 2 is called a *submatrix* of T. Note that any submatrix of T is a 2×2 matrix inside T (considered as a multidimensional array) *which is parallel to one of the faces of the array.* So, for instance, in the Example 6.4.4, the array

$$\begin{pmatrix} T_{112} & T_{122} \\ T_{211} & T_{221} \end{pmatrix}$$

is not a submatrix of T.

Proposition 6.4.6 *If T has rank 1, and T' is a subtensor of T then either T' is the null tensor or T' has rank 1.*

In particular, if T has rank 1, then the determinant of any 2×2 submatrix of T vanishes.

Proof Assume that $T \in K^{a_1,\dots,a_n}$ has rank 1. Then there exist vectors $v_i \in K^{a_i}$ such that $T = v_1 \otimes \cdots \otimes v_n$. Eliminating from T the elements whose kth index has some value q corresponds to eliminating the qth component in the vector v_k. Thus, the corresponding subtensor T' is the tensor product of the vectors $v'_1, \dots, v'_n$, where $v'_i = v_i$ if $i \neq k$, and v'_k is the vector obtained from v_k by eliminating the qth component. Thus T' has rank ≤ 1 (it has rank 0 if $v'_k = 0$). For a general subtensor $T' \in K^{a'_1,\dots,a'_n}$ of T, we obtain the result arguing step by step, by deleting each time one value for one index of T, i.e., arguing by induction on $(a_1 + \cdots + a_n) - (a'_1 + \cdots + a'_n)$.

The second claim in the statement of the theorem is immediate from what we have just said and the fact that a 2×2 matrix of rank 1 has determinant 0. □

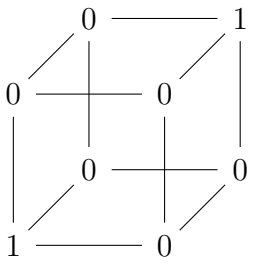
Corollary 6.4.7 *The rank of a subtensor of T cannot be bigger than the rank of T.*

Proof If T has rank 1, the claim follows from Proposition 6.4.6. For tensors T of higher rank r, the claim follows since if $T = T_1 + \cdots + T_k$, with T_i of rank 1, then a subtensor T' of T is equal to $T'_1 + \cdots + T'_k$, where T'_i is the subtensor of T_i obtained by eliminating all the elements corresponding to elements of T eliminated in the passage from T to T'. Thus, by Proposition 6.4.6 each T'_i is either 0 or it has rank 1, and the claim follows. □

Example 6.4.8 Recall that a nonzero matrix has rank 1 if and only if all of its 2×2 submatrices have determinant equal to zero. This is not true for tensors of dimension greater than 2, as the following example shows. Recall our earlier warning about the subtle differences between matrices and tensors of dimension greater than 2.

Consider the $2 \times 2 \times 2$ tensor T, defined by

$$\begin{aligned} &T_{1,1,1} = 0 \; T_{1,2,1} = 0 \; T_{2,1,1} = 1 \; T_{2,2,1} = 0 \quad \text{(front face)} \\ &T_{1,1,2} = 0 \; T_{1,2,2} = 1 \; T_{2,1,2} = 0 \; T_{2,2,2} = 0 \quad \text{(back face)} \, . \end{aligned}$$

$T =$ 

It is clear that all the 2×2 faces of T have determinant equal to 0. On the other hand, if T has rank 1, i.e., $T = (\alpha_1, \alpha_2) \otimes (\beta_1, \beta_2) \otimes (\gamma_1, \gamma_2)$, then $T_{2,1,1} = \alpha_2\beta_1\gamma_1 \neq 0$ which implies $\alpha_2, \beta_1, \gamma_1 \neq 0$. However, $T_{2,1,2} = T_{1,1,1} = T_{2,2,1} = 0$ implies $\alpha_1 = \beta_2 = \gamma_2 = 0$. But then $T_{1,2,2} = \alpha_1\beta_2\gamma_2 = 1 \neq 0$ yields a contradiction.

We want to find a set of conditions which describe the set of all tensors of rank 1. To this aim, we need to introduce some new piece of notation.

Notation. Recall that we denote by $[n]$ the set $\{1, \ldots, n\}$.

Fix a subset $J \subset [n]$. Then for any fixed pair of multi-indices $I_1 = (k_1, \ldots, k_n)$ and $I_2 = (l_1, \ldots, l_n)$, we denote by $J(I_1, I_2)$ the multi-index $(m_1, \ldots, m_n)$ where

$$m_j = \begin{cases} k_j & \text{if } j \in J, \\ l_j & \text{otherwise.} \end{cases}$$

Example 6.4.9 Let $n = 4$ and set $J = \{2, 3\} \subset [4]$. Consider the two multi-indices $I_1 = (1, 3, 3, 2)$ and $I_2 = (2, 1, 3, 4)$. Then $J(I_1, I_2) = (2, 3, 3, 4)$. Notice that if $J' = [n] \setminus J = \{1, 4\}$ then $J'(I_1, I_2) = (1, 1, 3, 2)$.

Remark 6.4.10 If T has rank 1, then for any pair of multi-indices $I_1 = (k_1, \ldots, k_n)$ and $I_2 = (l_1, \ldots, l_n)$ and for any subset $J \subset [n]$, the entries of T satisfy:

$$T_{I_1} T_{I_2} = T_{J(I_1,I_2)} T_{J'(I_1,I_2)} \tag{6.4.1}$$

where $J' = [n] \setminus J$.

To see why this is so recall that since T has rank 1 we can write $T = v_1 \otimes \cdots \otimes v_n$, with $v_i = (v_{i1}, v_{i2}, \ldots)$. In this case both of the products in (6.4.1) are equal

$$v_{1k_1} v_{1l_1} \cdots v_{nk_n} v_{nl_n}$$

and so the result is obvious.

Remark 6.4.11 When I_1, I_2 differ only in two indices, the equality (6.4.1) simply says that the determinant of a 2×2 submatrix of T is 0.

Example 6.4.12 Look back to Example 6.4.8, and notice that if one takes $I_1 = (1, 1, 1), I_2 = (2, 2, 2)$ and $J = \{1\} \subset [3]$, then $J(I_1, I_2) = (1, 2, 2)$ and $J'(I_1, I_2) = (2, 1, 1)$ so that formula (6.4.1) does not hold, since

$$T_{I_1} T_{I_2} = 0 \neq 1 = T_{J(I_1,I_2)} T_{J'(I_1,I_2)}.$$

Theorem 6.4.13 *A tensor $T \neq 0$ of dimension n has rank 1 if and only if it satisfies all the equalities (6.4.1), for any choice of multi-indices I_1, I_2, and $J \subset [n]$.*

Proof Thanks to Remark 6.4.10, we need only prove that if all the equalities (6.4.1) hold, then T has rank 1.

Let us argue by induction on the dimension n of $T \in K^{a_1,\ldots,a_n}$. The case $n = 2$ is well known: a matrix has rank 1 if and only if all its 2×2 minors vanish.

For $n > 2$, pick an entry $T_{I_1} = T_{k_1,\ldots,k_n} \neq 0$ in T.

Let $J_1 \subset [a_1]$ where $J_1 = \{1\}$ and let $f_1 : J_1 \to [a_i]$ be defined by $f_1(1) = k_1$. For $2 \leq i \leq n$, let $f_i = identity$. Let T' be the subtensor corresponding to these data. T' is a tensor of dimension $n - 1$ and hence satisfies the equalities (6.4.1). By induction, we obtain that $\text{rank}(T') = 1$, so there are vectors $v_2, \ldots v_n$ such that, for any choice of $i_2, \ldots, i_n$, one gets

$$T_{k_1,i_2,\ldots,i_n} = T'_{i_2,\ldots,i_n} = v_{2i_2} \cdots v_{ni_n}. \tag{a}$$

For all $m \in [a_1]$ define the number

$$p_m = \frac{T_{m,k_2,\ldots,k_n}}{T_{k_1,k_2,\ldots,k_n}}. \tag{b}$$

We use those numbers to define the vector $v_1 = (p_1, \ldots, p_{a_1})$.

We now claim that $T = v_1 \otimes v_2 \otimes \cdots \otimes v_n$.

Indeed for any $I_2 = (l_1, \ldots, l_n)$, by setting $J = \{1\}$, and hence $J' = \{2, \ldots, n\}$, one obtains from the equalities (6.4.1) that

$$T_{I_1} T_{I_2} = T_{J(I_1,I_2)} T_{J'(I_1,I_2)} = T_{k_1,l_2,\ldots,l_n} T_{l_1,k_2,\ldots,k_n} = T_{k_1,l_2,\ldots,l_n} \cdot p_{l_1} T_{k_1,k_2,\ldots,k_n}.$$

Using the terms at the beginning and end of this string of equalities and also taking into account (a) and (b) above, we obtain

$$v_{2k_2} \cdots v_{nk_n} T_{I_2} = v_{2l_2} \cdots v_{nl_n} \cdot v_{1l_1} \cdot v_{2k_2} \cdots v_{nk_n}.$$

Since $T_{I_1} \neq 0$, and hence $v_{2k_2}, \ldots, v_{nk_n} \neq 0$, we can divide both sides of this equality by $v_{2k_2}, \ldots, v_{nk_n}$ and finally get

$$T_{I_2} = v_{2l_2} \cdots v_{nl_n} \cdot v_{1l_1},$$

which proves the claim. □

Observe that the rank 1 analysis of a tensor reduces to compute if finitely many flattening matrices have rank one (see Definition 8.3.4 and Proposition 8.3.7), and this can be accomplished with Gaussian elimination as well, without the need to compute all 2×2 minors.

The equations corresponding to the equalities (6.4.1) determine a set of polynomial (quadratic) equations, in the space of tensors $K^{a_1,\dots,a_n}$, which describe the locus of decomposable tensors (interestingly enough, it turns out that in many cases this set of equations is not minimal).

In any event, Theorem 6.4.13 provides a finite procedure which allows us to decide if a given tensor has rank 1 or not. We simply plug the coordinates of the given tensor into the equations we just described and see if all the equations vanish or not.

Moreover, in Theorem 6.4.13 it is enough to take subsets J given by one single element, and even are sufficient $n - 1$ of them. Unfortunately, as the dimension grows, the number of operations required in the algorithm rapidly becomes quite large!

Recall that for matrices there is a much simpler method for calculating the rank of the matrix: one uses Gaussian reduction to find out how many nonzero rows that reduction has. That number is the rank. We really don't have to calculate the determinants of all the 2×2 submatrices of the original matrix.

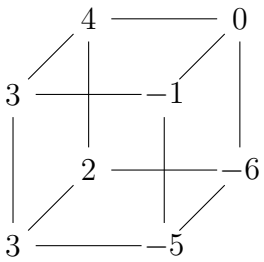
There is nothing like the simple and well-known Gaussian reduction algorithm (which incidentally calculates the rank for a tensor of dimension 2) for calculating the rank of tensors of dimension greater than 2. All known procedures for calculating the rank of such a tensor quickly become not effective.

There are many other ways in which the behavior of rank for tensors having dimension greater than 2 differs considerably from the behavior of rank for matrices (tensors of dimension exactly 2). For example, although a matrix of size $m \times n$ (a two-dimensional tensor of type (m, n)) cannot have rank which exceeds the minimum of m and n, tensors of type $a_1 \times \cdots \times a_n$ (for $n > 2$) may have rank bigger than $\max\{a_i\}$. Although the *general* matrix of size $m \times n$ has rank $= \min\{m, n\}$ (the maximum possible rank) there are often special tensors of a given dimension and type whose rank is bigger than the rank of a general tensor of that dimension and type.

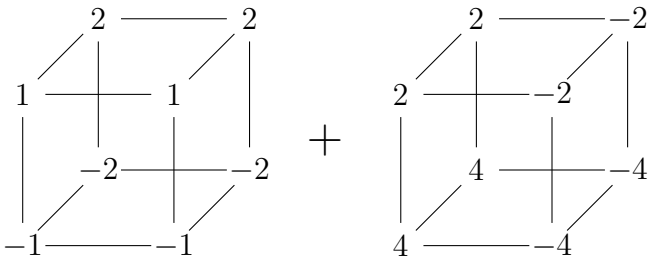
The attempt to get a clearer picture of how rank behaves for tensors of a given dimension and type has many difficult problems associated to it. Is there some nice geometric structure for the set of tensors having a given rank? Is the set of tensors of prescribed rank not empty? What is the maximum rank for a tensor of given dimension and type? These questions, and several variants of them, are the subject of research for many mathematicians and other scientists today.

We conclude this section with some examples which illustrate that although there is no algorithm for finding the rank of a given tensor, one can sometimes decide, using ad hoc methods, exactly what is the rank of the tensor.

Example 6.4.14 The following tensor of type $2 \times 2 \times 2$ has rank 2:



Indeed it cannot have rank 1, because some of its 2×2 submatrices have determinant different from 0. T has rank 2 because it is the sum of two tensors of rank 1 (one can check, using the algorithm, that the summands have rank 1):



Example 6.4.15 The tensor

$$D = \begin{array}{ccc} & 2 & 3 \\ 1 & & 3 \\ & 0 & 4 \\ 0 & & 2 \end{array}$$

has rank 3, i.e., one cannot write D as a sum of two tensors of rank 1. Let us see why.

Let's assume that D is the sum of two tensors $T = (T_{ijk})$ e $T' = (T'_{ijk})$ of rank 1 and let's try to derive a contradiction from that assumption.

Notice that the vector $(D_{211}, D_{212}) = (0, 0)$ would have to be equal to the sum of the vectors $(T_{211}, T_{212}) + (T'_{211}, T'_{212})$, Consequently, the two vectors (T_{211}, T_{212}) and (T'_{211}, T'_{212}) are opposite of each other and hence span a subspace $W \subset K^2$ of dimension ≤ 1.

If one (hence both) of these vectors is nonzero, then also the vectors (T_{111}, T_{112}), (T_{221}, T_{222}), and (T'_{111}, T'_{112}), (T'_{221}, T'_{222}), would also have to belong to W because all the 2×2 determinants of T and T' vanish. But notice that (T_{121}, T_{122}) and (T'_{121}, T'_{122}) must also belong to W by Remark 6.4.10 (take $J = \{3\} \subset [3]$).

It follows that both vectors $(D_{111}, D_{112}) = (1, 2)$ and $(D_{121}, D_{122}) = (3, 3)$, must belong to W. This is a contradiction, since $\dim(W) = 1$ and $(1, 2)$, $(3, 3)$ are linearly independent.

So, we are forced to the conclusion that $(T_{211}, T_{212}) = (T'_{211}, T'_{212}) = (0, 0)$. Since the sum of (T_{111}, T_{112}) and (T'_{111}, T'_{112}) is $(1, 2) \neq (0, 0)$ we may assume that one of them, say (T_{111}, T_{112}), is nonzero. As T has rank 1, there exists $a \in K$ such that $(T_{221}, T_{222}) = a(T_{111}, T_{112})$ (we are again using Remark 6.4.10).

Now, the determinant of the front face of the tensor T is 0, i.e.,

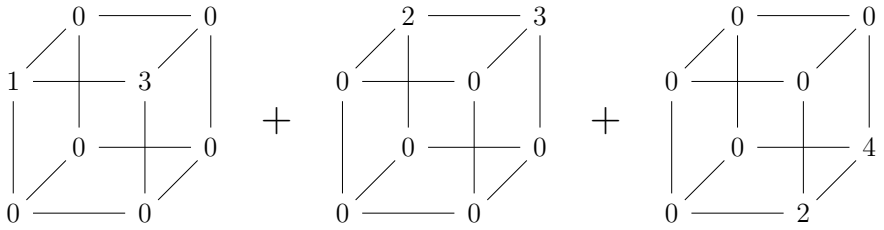
$$0 = T_{111}T_{221} - T_{121}T_{211}.$$

Since $T_{211} = 0$ and $T_{221} = aT_{111}$ we get $0 = aT_{111}^2$. Doing the same argument on the back face of the tensor T we get $0 = aT_{112}^2$. It follows that $a = 0$ and so the bottom face of the tensor T consists only of zeroes.

It follows that $(T'_{221}, T'_{222}) = (2, 4)$. Since the tensor T' has rank 1, it follows that the vector (T'_{111}, T'_{112}) is also a multiple of $(2, 4)$, as is the vector (T'_{121}, T'_{122}).

Since $(T_{111}, T_{112}) = (1, 2) - (T'_{111}, T'_{112})$, and both $(1, 2)$ and (T'_{111}, T'_{112}) are multiples of $(2, 4)$, it follows that the vector (T_{111}, T_{112}) (which we assumed was not 0) is also a multiple of $(2, 4)$. Thus, since the tensor T has rank 1, the vector (T_{121}, T_{122}) is also a multiple of $(2, 4)$. Since we already noted that (T'_{121}, T'_{122}) is a multiple of $(2, 4)$ it follows that the vector $(3, 3)$ is a multiple of $(2, 4)$, which is the final contradiction.

Notice that a decomposition of the tensor is given by



6.5 Exercises

Exercise 7 Give a graphical representation of the following tensors:

(a) T is the three-dimensional tensor of type $3 \times 2 \times 2$ whose (i, j, k) entry is equal to $i \cdot (j + k)$;

(b) T is the three-dimensional tensor of type $2 \times 3 \times 2$ whose (i, j, k) entry is equal to $i \cdot (j + k)$;
(c) T is the three-dimensional tensor of type $2 \times 2 \times 3$ whose (i, j, k) entry is equal to $i \cdot (j + k)$.

Exercise 8 Find two tensors of rank 1 whose sum is the $2 \times 2 \times 2$ tensor $v \otimes w$ of Example 6.2.3.

Exercise 9 Show that the tensor:

$$T = \begin{array}{ccc} 3 & - & 3 \\ | & & | \\ 2 & - & 2 \\ | & & | \\ 6 & - & 3 \\ | & & | \\ 4 & - & 2 \\ | & & | \\ 9 & - & 3 \\ | & & | \\ 6 & - & 2 \end{array}$$

is the tensor product of a matrix times the vector $(2, 3)$.

Exercise 10 Prove that the tensor T of Exercise 9 has rank 2.

Chapter 7
Symmetric Tensors



In this chapter, we make a specific analysis of the behavior of symmetric tensors, with respect to the rank and the decomposition.

We will see, indeed, that besides their utility to understand some models of random systems, symmetric tensors have a relevant role in the study of the algebra and the computational complexity of polynomials.

7.1 Generalities and Examples

Definition 7.1.1 A *cubic* tensor is a tensor of type $a_1 \times \cdots \times a_n$ where all the a_i's are equal, i.e., a tensor of type $d \times \cdots \times d$ (n times).

We say that a *cubic* tensor T is *symmetric* if for any multi-index $(i_1, \ldots, i_n)$ and for any permutation σ on the set $\{i_1, \ldots, i_n\}$, it satisfies

$$T_{i_1,\ldots,i_n} = T_{i_{\sigma(1)},\ldots,i_{\sigma(n)}}.$$

Example 7.1.2 When T is a square matrix, then the condition for the symmetry of T simply requires that $T_{i,j} = T_{j,i}$ for any choice of the indices. In other words, our definition of *symmetric tensor* coincides with the plain old definition of symmetric matrix, when T has dimension 2.

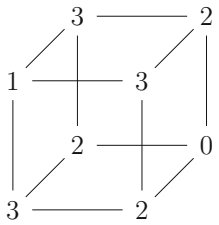
If T is a cubic tensor of type $2 \times 2 \times 2$, then T is symmetric if and only if the following equalities hold:

$$\begin{cases} T_{1,1,2} &= T_{1,2,1} = T_{2,1,1} \\ T_{2,2,1} &= T_{2,1,2} = T_{1,2,2} \end{cases}$$

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An example of a $2 \times 2 \times 2$ symmetric tensor is the following:



Remark 7.1.3 The set of symmetric tensors is a linear subspace of $K^{d,\dots,d}$. Namely, it is defined by a set of linear equations:

$$T_{i_1,\dots,i_n} = T_{\sigma(i_1),\dots,\sigma(i_n)}$$

in the coordinates of $K^{d,\dots,d}$.

As a vector space itself, the space of symmetric tensors of type $d \times \cdots \times d$, n times, is usually denoted by $Sym^n(K^d)$.

Of course, from the point of view of multi-linear forms, $Sym^n(K^d)$ coincides with the space of *symmetric* multi-linear maps $(K^d)^n \to K$.

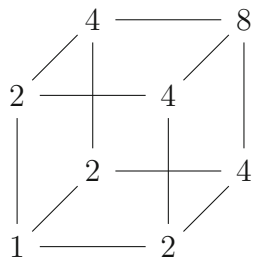
As we will see later (see Proposition 7.3.8), the dimension of $Sym^n(K^d)$ is

$$\dim(Sym^n(K^d)) = \binom{n+d-1}{n} = \binom{n+d-1}{d-1}.$$

7.2 The Rank of a Symmetric Tensor

Next step is the study of the behavior of symmetric tensors with respect to the rank. It is easy to realize that there are *symmetric tensors of rank* 1, i.e., the space $Sym^n(K^d)$ intersects the set of decomposable tensors. Just to give an instance, look at:

$D =$



The following proposition determines how one construct decomposable, symmetric tensors.

Proposition 7.2.1 *Let T be a cubic tensor of type $d \times \cdots \times d$, n times. Then T is symmetric, of rank* 1*, if and only if there exist a nonzero scalar $\lambda \in K$ and a nonzero vector $v \in K^d$ with:*

$$T = \lambda(v \otimes v \otimes \cdots \otimes v).$$

If moreover K is an algebraically closed field (as the complex field $\mathbb{C}$), then we may assume $\lambda = 1$.

Proof If $T = \lambda(v \otimes v \otimes \cdots \otimes v)$, $v \neq 0$, then T cannot be zero by Proposition 6.2.8, thus it has rank 1. Moreover if $v = (\alpha_1, \ldots, \alpha_d)$, then for any multi-index $(i_1, \ldots, i_n)$ and for any permutation σ:

$$T_{i_1,\ldots,i_n} = \alpha_{i_1} \cdots \alpha_{i_n} = T_{\sigma(i_1),\ldots,\sigma(i_n)}$$

thus T is symmetric.

Conversely, assume that T is symmetric of rank 1, say $T = v_1 \otimes \cdots \otimes v_n$, where no $v_i \in K^d$ can be 0, by Proposition 6.2.8. Write $v_i = (v_{i,1}, \ldots, v_{i,d})$ and fix a multi-index $(i_1, \ldots, i_n)$ such that $v_{1,i_1} \neq 0, \ldots, v_{n,i_n} \neq 0$. Then $T_{i_1,\ldots,i_n} = v_{1,i_1} \cdots v_{n,i_n}$ cannot vanish. Define $b_2 = v_{2,i_1}/v_{1,i_1}$. Then we claim that $v_2 = b_2 v_1$. Namely, for all j we have, by symmetry:

$$v_{1,i_1} v_{2,j} v_{3,i_3} \cdots v_{n,i_n} = T_{i_1,j,i_3,\ldots,i_n} = T_{j,i_1,i_3,\ldots,i_n} = v_{1,j} v_{2,i_1} v_{3,i_3} \cdots v_{n,i_n},$$

which means that $v_{1,i_1} v_{2,j} = v_{1,j} v_{2,i_1}$, so that $v_{2,j} = b_2 v_{1,j}$. Similarly we can define $b_3 = v_{3,i_1}/v_{1,i_1}, \ldots, b_d = v_{d,i_1}/v_{1,i_1}$, and obtain that $v_3 = b_3 v_1, \ldots, v_d = b_d v_1$. Thus, if $\lambda = b_2 \cdot b_3 \cdot \cdots \cdot b_d$, then

$$T = v_1 \otimes v_2 \otimes \cdots \otimes v_n = v_1 \otimes (b_2 v_1) \otimes \cdots \otimes (b_n v_1) = \alpha(v_1 \otimes v_1 \otimes \cdots \otimes v_1).$$

When K is algebraically close, then take a dth root β of $\lambda \in K$ and define $v = \beta v_1$. Then $T = \beta^d(v_1 \otimes v_1 \otimes \cdots \otimes v_1) = v \otimes v \otimes \cdots \otimes v$. □

Notice that purely algebraic properties of K can be relevant in determining the shape of a decomposition of a tensor.

Remark 7.2.2 In the sequel, we will often write $v^{\otimes d}$ for $v \otimes v \otimes \cdots \otimes v$, d times.

If K is algebraically closed, then a symmetric tensor $T \in Sym^n(K^d)$ of rank 1 has a finite number (exactly: d) decompositions as a product $T = v^{\otimes d}$.

Namely if $w \otimes \cdots \otimes w = v \otimes \cdots \otimes v$, then by Proposition 6.2.9 there exists a scalar β such that $w = \beta v$ and moreover $\beta^d = 1$, thus w is equal to v multiplied by a dth root of unity.

Passing from rank 1 to higher ranks, the situation becomes suddenly more involved.

The definition itself of *rank* of a symmetric tensors is not completely trivial, as we have two natural choices for it:

- **First choice**. The rank of a symmetric tensor $T \in Sym^n(K^d)$ is simply its rank as a tensor in $K^{d,\dots,d}$, i.e., it is the minimum r for which one has r decomposable tensors $T_1,\dots,T_r$ with
$$T = T_1 + \cdots + T_r.$$

- **Second choice**. The rank of a symmetric tensor $T \in Sym^n(K^d)$ is the minimum r for which one has r *symmetric* decomposable tensors $T_1,\dots,T_r$ with
$$T = T_1 + \cdots + T_r.$$

Then, the natural question is about which choice gives the *correct definition*. Here, *correct* definition means the definition which proves to be most useful, for the applications to Multi-linear Algebra and random systems.

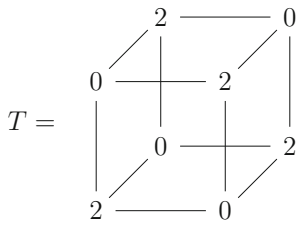
The reader could be disappointed in knowing that there is no clear preference between the two options: each can be preferable, depending on the point of view.

Thus, we will leave the word *rank* for the minimum r for which one has a decomposition $T = T_1 + \cdots + T_r$, with the T_i's not necessarily symmetric (i.e., the first choice above).

Then, we give the following:

Definition 7.2.3 The *symmetric rank* $\mathrm{srank}(T)$ of a symmetric tensor $T \in Sym^n(K^d)$ is the minimum r for which one has r *symmetric* decomposable tensors $T_1,\dots,T_r$ with
$$T = T_1 + \cdots + T_r.$$

Example 7.2.4 The symmetric tensor

$T =$ 

has not rank 1, as one can compute by taking the determinant of some face.

T has rank 2, because it is expressible as the sum of two decomposable tensors $T = T_1 + T_2$, where

$$T_1 = \begin{array}{ccc} 1 & \text{———} & 1 \\ \text{\textbackslash} & & \text{\textbackslash} \\ & 1 \text{———} 1 & \\ & \vdots \quad \vdots & \\ & 1 \text{———} 1 & \\ \text{\textbackslash} & & \text{\textbackslash} \\ 1 & \text{———} & 1 \end{array}$$

$$T_2 = \begin{array}{ccc} 1 & \text{———} & -1 \\ \text{\textbackslash} & & \text{\textbackslash} \\ & -1 \text{———} 1 & \\ & \vdots \quad \vdots & \\ & -1 \text{———} 1 & \\ \text{\textbackslash} & & \text{\textbackslash} \\ 1 & \text{———} & -1 \end{array}$$

and $T_1 = (1, 1)^{\otimes 3}$, $T_2 = (-1, 1)^{\otimes 3}$.

Example 7.2.5 The tensor (over $\mathbb{C}$):

$$T = \begin{array}{ccc} 0 & \text{———} & 7 \\ \text{\textbackslash} & & \text{\textbackslash} \\ & 1 \text{———} 0 & \\ & \vdots \quad \vdots & \\ & 7 \text{———} 8 & \\ \text{\textbackslash} & & \text{\textbackslash} \\ 0 & \text{———} & 7 \end{array}$$

is not decomposable. Let us prove that the symmetric rank is bigger than 2.

Assume that $T = (a, b)^{\otimes 3} + (c, d)^{\otimes 3}$. Then we have

$$\begin{cases} a^3c^3 & = 1 \\ a^2b + c^2d & = 0 \\ ab^2 + cd^2 & = 7 \\ b^3d^3 & = 8. \end{cases}$$

Notices that none of a, b, c, d can be 0. Moreover we have $ac = \epsilon$ and $bd = 2\epsilon'$, where ϵ, ϵ' are two cubic roots of unit, not necessarily equal. But then $c = \epsilon/a$ and $d = \epsilon'/b$, so that $a^2b + c^2d = 0$ yields $1 + \epsilon^2\epsilon' = 0$, which cannot hold, because -1 is not a cubic root of unit.

Remark 7.2.6 Proposition 7.2.1 shows in particular that any symmetric tensor of (general) rank 1 has also the symmetric rank equal to 1.

The relations between the rank and the symmetric rank of a tensor are not obvious at all, when the ranks are bigger than 1. It is clear that

$$\operatorname{srank}(T) \geq \operatorname{rank}(T).$$

Very recently, Shitov found an example where the strict inequality holds (see [1]).

Shitov's example is quite peculiar: the tensor has dimension 3 and type $800 \times 800 \times 800$, whose rank is very high with respect of general tensors of same dimension and type.

The difficulty in finding examples where the two ranks are different, despite the large number of concrete tensors tested, suggested to the French mathematician Pierre Comon to launch the following:

Problem 7.2.7 (*Comon 2000*) Find conditions such that the symmetric rank and rank of a symmetric tensor coincide.

In other words, find conditions for $T \in Sym^n(\mathbb{C}^d)$ such that if there exists a decomposition $T = T_1 + \cdots + T_r$ in terms of tensors of rank 1, then there exists also a decomposition with the same number of summands, in which each T_i is *symmetric*, of rank 1.

The condition is known for some types of tensors. For instance, it is easy to prove that the Comon Problem holds for any symmetric matrix T (and this is left as an exercise at the end of the chapter).

The reader could wonder that such a question, which seems rather elementary in its formulation, could yield a problem which is still open, after being studied by many mathematicians, with modern techniques.

This explains a reason why, at the beginning of the chapter, we warned the reader that problems that are simple for Linear Algebra and matrices can suddenly become prohibitive, as the dimension of the tensors grows.

7.3 Symmetric Tensors and Polynomials

Homogeneous polynomials and symmetric tensors are two apparently rather different mathematical objects, that indeed have a strict interaction, so that one can skip from each other, translating properties of tensors to properties of polynomials, and vice versa.

The main construction behind this interaction is probably well known to the reader, for the case of polynomials of degree 2. It is a standard fact that one can associate a symmetric matrix to quadratic homogeneous polynomial, in a one-to-one correspondence, so that properties of the quadratic form (as well as properties of quadratic hypersurfaces) can be read as properties of the associated matrix.

The aim of this section is to point out that a similar correspondence holds, more generally, between homogeneous forms of any degree and symmetric tensors of higher dimension.

Definition 7.3.1 There is a natural map between a space $K^{n,\dots,n}$ of cubic tensors of dimension d and the space of homogeneous polynomials of degree d in n variables (i.e., the dth graded piece R_d of the ring of polynomials $R = K[x_1, \dots, x_n]$), defined by sending a tensor T to the polynomial F_T such that

$$F_T = \sum_{i_1,\ldots,i_n} T_{i_1,\ldots,i_n} x_{i_1} \cdots x_{i_n}.$$

It is clear that the previous correspondence is not one to one, as soon as general tensors are considered. Namely, for the case $n, d = 2$, one immediately sees that the two matrices

$$\begin{pmatrix} 2 & 3 \\ -1 & 1 \end{pmatrix} \qquad \begin{pmatrix} 2 & 0 \\ 2 & 1 \end{pmatrix}$$

define the same polynomial of degree 2 in two variables $F = 2x_1^2 + 2x_1x_2 + x_2^2$.

The correspondence becomes one to one (and onto) when restricted to symmetric tensors. To see this, we need to introduce a piece of notation.

Definition 7.3.2 For any multi-index $(i_1, \ldots, i_d)$, we will define the multiplicity $m(i_1, \ldots, i_d)$ as the number of different permutations of the multi-index.

Definition 7.3.3 Let $R = K[x_1, \ldots, x_n]$ be the ring of polynomials, with coefficients in K, with n variables. Then there are linear isomorphisms

$$p : Sym^d(K^n) \to R_d \qquad t : R_d \to Sym^d(K^n)$$

defined as follows. The map p is the restriction to $Sym^d(K^n)$ of the previous map

$$p(T) = \sum_{i_1,\ldots,i_d} T_{i_1,\ldots,i_d} x_{i_1} \cdots x_{i_d}.$$

The map t is defined by sending the polynomial F to the tensor $t(F)$ such that

$$t(F)_{i_1,\ldots,i_d} = \frac{1}{m(i_1, \ldots, i_d)} (\text{the coefficient of } x_{i_1} \cdots x_{i_d} \text{ in } F).$$

Example 7.3.4 If G is a quadratic homogeneous polynomial in 3 variables $G = Ax^2 + Bxy + Cy^2 + Dxz + Eyz + Fz^2$, then $t(G)$ is the symmetric matrix

$$t(G) = \begin{pmatrix} A & B/2 & D/2 \\ B/2 & C & E/2 \\ D/2 & E/2 & F, \end{pmatrix}$$

which the usual matrix of the bilinear form associated to G.

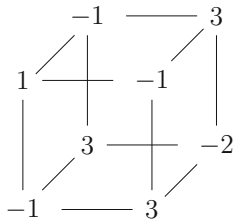
Example 7.3.5 Consider the homogeneous cubic polynomial in two variables

$$F(x_1, x_2) = x_1^3 - 3x_1^2x_2 + 9x_1x_2^2 - 2x_2^3.$$

Since one easily computes that

$$m(1, 1, 1) = 1, \quad m(1, 1, 2) = m(2, 1, 1) = 3, \quad m(2, 2, 2) = 1,$$

then the symmetric tensor $t(F)$ is:

$T =$ 

It is an easy exercise to prove that the two maps p and t defined above are inverse to each other.

Once the correspondence is settled, one can easily speak about the rank or the the symmetric rank of a polynomial.

Definition 7.3.6 For any homogeneous polynomial $G \in K[x_1, \ldots, x_n]$, we define the *rank* (respectively, the *symmetric rank*) of G as the *rank* (respectively, the *symmetric rank*) of the associated tensor $t(G)$.

Example 7.3.7 The polynomial $G = x_1^3 + 21x_1x_2^2 + 8x_2^3$ has rank 3, since the associated tensor $t(G)$ is exactly the $2 \times 2 \times 2$ symmetric tensor of Example 7.2.5.

Proposition 7.3.8 *The linear space $Sym^d(K^n)$ has dimension*

$$\dim(Sym^d(K^n)) = \binom{n+d-1}{d} = \binom{n+d-1}{n-1}.$$

Proof This is obvious once one knows that $\binom{n+d-1}{d}$ is the dimension of the space of homogeneous polynomials R_d of degree d in n variables. We prove it for the sake of completeness.

Since monomials of degree d in n variables are a basis for R_d, it is enough to count the number of such monomials.

The proof goes by induction on n. For $n = 2$ the statement is easy: we have $d + 1$ monomials, namely $x_1^d, x_1^{d-1}x_2, \ldots, x_2^d$.

Assume the formula holds for $n - 1$ variables $x_2, \ldots, x_n$. Every monomial of degree d in n variables is obtained by multiplying x_1^a by any monomial of degree $d - a$ in $x_2, \ldots, x_n$. Thus, we have 1 monomial with x_1^d, n monomials with x_1^{d-1},..., $\binom{n+d-a-2}{d-a}$ monomials with x_1^a, and so on. Summing up

$$\dim(Sym^d(K^n)) = \sum_{a=0}^{d} \binom{n+d-a-2}{d-a},$$

and the sum is $\binom{n+d-1}{d}$, by standard facts on binomial coefficients. □

7.4 The Complexity of Polynomials

In this section, we rephrase the results on the rank of symmetric tensors in terms of the associated polynomials.

It will turn out that the rank decomposition of a polynomial is the analogue of a long-standing series of problems in Number Theory, for the expression of integers as a sum of powers.

In principle, from the point of view of Algebraic Statistic, the complexity of a polynomial is the complexity of the associated symmetric tensor. So, the most elementary case of polynomials corresponds to symmetric tensor of rank 1. We start with a description of polynomials of this type.

Remark 7.4.1 Before we proceed, we need to come back to the *multiplicity* of a multi-index $J = (i_1, \dots, i_d)$, introduced in Definition 7.3.2.

In the correspondence between polynomials and tensors, the element $T_{i_1,\dots,i_d}$ is linked with the coefficient of the monomial $x_{i_1} \cdots x_{i_d}$. Notice that $i_1, \dots, i_d$ need not be distinct, so the monomial $x_{i_1} \cdots x_{i_d}$ could be written unproperly. The usual way in which $x_{i_1} \cdots x_{i_d}$ is written is:

$$x_{i_1} \cdots x_{i_d} = x_1^{m_J(1)} x_2^{m_J(2)} \cdots x_n^{m_J(n)},$$

where $m_J(i)$ indicates the times in which i occurs in the multi-index J.

With the notation just introduced, one can describe the multiplicity $m(i_1, \dots, i_d)$. Indeed a permutation changes the multi-index, unless it simply switches indices i_a, i_b which are equal. Since the number of permutations over a set with m elements is $m!$, then one finds that

$$m(J) = m(i_1, \dots, i_d) = \frac{d!}{m_J(1)! \cdots m_J(n)!}.$$

Proposition 7.4.2 *Let G be a homogeneous polynomial of degree d in n variables, so that $t(G) \in Sym^d(K^n)$.*

Then $t(G)$ has rank 1 *if and only if there exists a homogeneous* linear *polynomial $L \in K[x_1, \dots, x_n]$, such that $G = L^d$.*

Proof It is sufficient to prove that $t(G) = v^{\otimes d}$, where $v = (\alpha_1, \dots, \alpha_n) \in K^n$, if and only if $G = (\alpha_1 x_1 + \cdots + \alpha_n x_n)^d$.

To this aim, just notice that the coefficient of the monomial $x_1^{m_1} \cdots x_n^{m_n}$ in $p(v^{\otimes d})$ is the sum of the entries of the tensors $v^{\otimes d}$ whose multi-index J satisfies $m_J(1) = m_1, \dots, m_J(n) = m_n$. These entries are all equal to $\alpha_1^{m_1} \cdots \alpha_n^{m_n}$ and their number is $m(J)$. On the other hand, by the well-known Newton formula, $m(J)(\alpha_1^{m_1} \cdots \alpha_n^{m_n})$ is exactly the coefficient of the monomial $x_1^{m_1} \cdots x_n^{m_n}$ in the power $(\alpha_1 x_1 + \cdots + \alpha_n x_n)^d$.

Corollary 7.4.3 *The symmetric rank of a homogeneous polynomial $G \in K[x_1, \dots, x_n]_d$ is the minimum r for which there are r linear homogeneous*

forms $L_1, \ldots, L_r \in K[x_1, \ldots, x_n]$, *with*

$$G = L_1^d + \cdots + L_r^d. \tag{7.1}$$

The symmetric rank is the number that computes the complexity of symmetric tensors, hence the complexity of homogeneous polynomials, from the point of view of Algebraic Statistics. Hence, it turns out that the simplest polynomials, in this sense, are powers of linear forms. We guess that nobody will object to the statement that powers are rather simple!

We should mention, however, that sometimes the behavior of polynomials with respect to the complexity can be much less intuitive.

For instance, the rank of monomials is usually very high, so that the complexity of general monomials is over the average (and we expect that most people will be surprised). Even worse, efficient formulas for the rank of monomials were obtained only very recently by Enrico Carlini, Maria Virginia Catalisano, and Anthony V. Geramita (see [2]). For other famous polynomials, as the determinant of a matrix of indeterminates, we do not even know the rank. All we have are lower and upper bounds, not matching.

We finish the chapter by mentioning that the problem of finding the rank of polynomials reflects a well-known problem in Number Theory. Solving a question posed by Diophantus, the Italian mathematician Giuseppe Lagrange proved that any positive integer N can be written as a sum of four squares, i.e., for any positive integer G, there are integers L_1, L_2, L_3, L_4 such that $G = L_1^2 + L_2^2 + L_3^2 + L_4^2$. The problem has been generalized by the English mathematician Edward Waring, who asked in 1770 for the minimum integer $r(k)$ such that any positive integer G can be written as a sum of $r(k)$ powers L_i^k. In other words, find the minimum $r(k)$ such that any positive integers are of the form

$$G = L_1^k + \cdots L_{r(k)}^k.$$

The analogy with the decomposition (7.1) that computes the symmetric rank of a polynomial is evident.

The determination of $r(k)$ is called, from then, the *Waring problem* for integers. Because of the analogy, the symmetric rank of a polynomial is also called the *Waring rank*.

For integers, few values of $r(k)$ are known, e.g., $r(2) = 4$, $r(3) = 9$, $r(4) = 19$. There are also variations on the Waring problem, as asking for the minimum $r'(k)$ such that all positive integers, except for a finite subset, are the sum of $r'(k)$ kth powers (the *little* Waring problem).

Going back to the polynomial case, as for integers, a complete description of the maximal complexity that a homogeneous polynomial of the given degree in a given number of variables can have, is not known. We have only upper bounds for the maximal rank. On the other hand, we know the solution of an analogue to the little Waring problem, for polynomials over the complex field.

Theorem 7.4.4 (Alexander-Hirschowitz 1995) *Over the complex field, the symmetric rank of a* general *homogeneous polynomial of degree d in n variables (here* general *means: all polynomials outside a set of measure* 0 *in* $\mathbb{C}[x_1, \ldots, x_n]_d$*; or also: all polynomials outside a Zariski closed subset of the space* $\mathbb{C}[x_1, \ldots, x_n]_d$*, see Remark 9.1.10) is*

$$r = \lceil \frac{\binom{n+d-1}{d}}{n} \rceil$$

except for the following cases:

- $d = 2$, *any* n, *where* $r = n$;
- $d = 3$, $n = 5$, *where* $r = 8$;
- $d = 4$, $n = 3$, *where* $r = 6$.
- $d = 4$, $n = 4$, *where* $r = 10$.
- $d = 4$, $n = 5$, *where* $r = 15$.

The original proof of this theorem requires the *Horace method*. It is long and difficult and occupies a whole series of papers [3–7].

For specific tensors, an efficient way to compute the rank requires the use of *inverse systems*, which will be explained in the next chapter.

7.5 Exercises

Exercise 11 Prove that the two maps p and t introduced in Definition 7.3.3 are linear and inverse to each other.

Exercise 12 Prove Comon's Problem for matrices: a symmetric matrix M has rank r if and only if there are r *symmetric* matrices of rank 1, M_1,…, M_r, such that $M = M_1 + \cdots + M_r$.

Exercise 13 Prove that the tensor T of Example 7.2.5 cannot have rank 2.

Exercise 14 Prove that the tensor T of Example 7.2.5 has symmetric rank $\text{srank}(T) = 3$ (so, after Exercise 13, also the rank is 3).

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Chapter 8
Marginalization and Flattenings



We collect in this chapter some of the most useful operations on tensors, in view of the applications to Algebraic Statistics.

8.1 Marginalization

The concept of marginalization goes back to the beginnings of a statistical analysis of discrete random variables, when, mainly, only a pair of variables were compared and correlated. In this case, distributions in the total correlation corresponded to matrices, and it was natural to annotate the sums of rows and columns at the edges (margins) of the matrices.

Definition 8.1.1 For matrices of given type $n \times m$ over a field K, which can be seen as points of the vector space $K^{n,m}$, the *marginalization* is the linear map $\mu : K^{n,m} \to K^n \times K^m$ which sends the matrix $A = (a_{ij})$ to the pair $((v_1, \ldots, v_n), (w_1, \ldots, w_m)) \in K^n \times K^m$, where

$$v_i = \sum_{j=1}^{m} a_{ij}, \qquad w_j = \sum_{i=1}^{n} a_{ij}.$$

In practice, the v_i's correspond to the sums of the columns, the w_j's correspond to the sums of the rows.

Notice that we can define as well the marginalization of A by $((1, \ldots, 1)A, (1, \ldots, 1)A^t)$.

Below there is an example of marginalization of a 3×3 matrix.

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$$M = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 2 & 4 & 7 \end{pmatrix} \quad \rightarrow \quad \begin{matrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 2 & 4 & 7 \end{pmatrix} & \begin{matrix} 6 \\ 1 \\ 13 \end{matrix} \\ 3 \; 8 \; 9 & \end{matrix}$$

$$\mu(M) = ((6, 1, 13), (3, 8, 9)).$$

The notion can be extended (with some complication only for the notation) to tensors of any dimension.

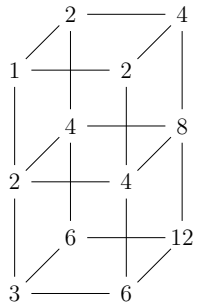
Definition 8.1.2 For tensors of given type $a_1 \times \cdots \times a_n$ over a field K, which can be seen as points of the vector space $K^{a_1,\ldots,a_n}$, the *marginalization* is the linear map $\mu : K^{a_1,\ldots,a_n} \rightarrow K^{a_1} \times \cdots \times K^{a_n}$ which sends the tensor $A = (\alpha_{q_1 \ldots q_n})$ to the n-uple $((v_{11}, \ldots, v_{1a_1}), \ldots, (v_{n1}, \ldots, v_{na_n})) \in K^{a_1} \times \cdots \times K^{a_n}$, where

$$v_{ij} = \sum_{q_i = j} \alpha_{q_1 \ldots q_n},$$

i.e., in each sum we fix the ith index and take the sum over all the elements of the tensor in which the ith index is equal to j.

Example 8.1.3 The marginalization of the $3 \times 2 \times 2$ tensor of Example 6.1.6

$T =$



is $((18, 36), (9, 18, 27), (18, 36))$.

Since the marginalization μ is a linear map, we can analyze its linear properties. It is immediate to realize that, except for trivial cases, μ is not injective.

Example 8.1.4 Even for 2×2 matrices, the matrix

$$M = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

belongs to the Kernel of μ. Indeed, M generates the Kernel of μ.

Even the surjectivity of μ fails in general. This is obvious for 2×2 matrices, since μ is a noninjective linear map between K^4 and itself.

Indeed, if $((v_1, \dots, v_n), (w_1, \dots, w_m))$ is the marginalization of the matrix $A = (a_{ij})$, then clearly $v_1 + \cdots + v_n$ is equal to the sum of all the entries of A, thus it is also equal to $w_1 + \cdots + w_m$. We prove that this is the only restriction for a vector in $K^n \times K^m$ to belong to the image of the marginalization.

Proposition 8.1.5 *A vector* $((v_1, \dots, v_n), (w_1, \dots, w_m)) \in K^n \times K^m$ *is the marginalization of a matrix* $A = (a_{ij}) \in K^{n,m}$ *if and only if*

$$v_1 + \cdots + v_n = w_1 + \cdots + w_m.$$

Proof The fact that the condition is necessary follows from the few lines before the proposition. We need to prove that the condition is sufficient. So, assume that $\sum v_i = \sum w_j$. A way to prove the claim, which can be easily extended even to higher dimensional tensors, is the following. Write e_i for the element in K^n with 1 in the ith position and 0 elsewhere, so that $e_1, \dots, e_n$ is the canonical basis of K^n. Define similarly $e'_1, \dots, e'_m \in K^m$. It is clear that any pair (e_i, e'_j) belongs to the image of μ: just take the marginalization of the matrix having $a_{ij} = 1$ and all the remaining entries equal to 0. So, it is sufficient to prove that if $v_1 + \cdots + v_n = w_1 + \cdots + w_m$, then $(v, w) = ((v_1, \dots, v_n), (w_1, \dots, w_m))$ belongs to the subspace generated by the (e_i, e'_j)'s. Assume that $n \leq m$ (if the converse holds, just take the transpose). Then notice that

$$(v, w) = v_1(e_1, e'_1) + \sum_{i=2}^{n}(v_1 + \cdots + v_i - w_1 - \cdots - w_{i-1})(e_i, e'_i) + \sum_{i=1}^{n-1}(w_1 + \cdots + w_i - v_1 - \cdots - v_i)(e_{i+1}, e'_i). \quad \square$$

Corollary 8.1.6 *The image of* μ *has dimension* $n + m - 1$. *Thus the kernel of* μ *has dimension* $nm - n - m + 1$.

We can extend the previous computation to tensors of arbitrary dimension.

Proposition 8.1.7 *A vector*

$$((v_{11}, \dots, v_{1a_1}), \dots, (v_{n1}, \dots, v_{na_n})) \in K^{a_1} \times \cdots \times K^{a_n}$$

is the marginalization of a tensor of type $a_1 \times \cdots \times a_n$ *if and only if*

$$v_{11} + \cdots + v_{1a_1} = v_{21} + \cdots + v_{2a_2} = \cdots = v_{n1} + \cdots + v_{na_n}.$$

Definition 8.1.8 If the marginalization $\mu(T)$ of a tensor T is the element $((v_{11}, \dots, v_{1a_1}), \dots, (v_{n1}, \dots, v_{na_n}))$ of $K^{a_1} \times \cdots \times K^{a_n}$, then the vector $\mu(T)_i = (v_{i1}, \dots, v_{ia_i}) \in K^{a_i}$ is called the *i-contraction* of T.

Next, we will study the connections between the marginalization of a tensor T and the tensor product of its contractions.

Proposition 8.1.9 *Let $T \in K^{a_1,\dots,a_n}$ be a tensor of rank* 1. *Let $u_i \in K^{a_i}$ be the i-contraction of T. Then T is a scaling of $u_1 \otimes \cdots \otimes u_n$.*

Proof Assume $T = w_1 \otimes \cdots \otimes w_n$ where $w_i = (w_{i1}, \dots, w_{ia_i})$. Then $T_{i_1,\dots,i_n} = w_{1i_1} \cdots w_{ni_n}$, hence $u_i = (u_{i1}, \dots, u_{ia_i})$ where

$$u_{ij} = \sum_{j_i = j} w_{1j_1} \cdots w_{nj_n} = w_{ij_i} \sum w_{1j_1} \cdots \hat{w}_{ij_i} \cdots w_{nj_n},$$

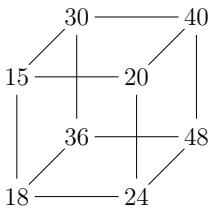
hence u_i is a multiple of w_i (by $\sum w_{1j_1} \cdots \hat{w}_{ij_i} \cdots w_{dj_d}$). □

Remark 8.1.10 We can be even more precise about the scaling factor of the previous proposition. Namely, assume $T = w_1 \otimes \cdots \otimes w_n$ where $w_i = (w_{i1}, \dots, w_{ia_i})$, and set $W_i = \sum w_{i1} + \cdots + w_{ia_i}$. Then $u_1 \otimes \cdots \otimes u_n = WT$, where

$$W = \Pi_{i=1}^{n} W_1 \cdots \hat{W}_i \cdots W_n.$$

Example 8.1.11 Let $T \in K^{2,2,2}$ be the rank 1 tensor, product of $(1, 2) \otimes (3, 4) \otimes (5, 6)$. Then,

$T =$



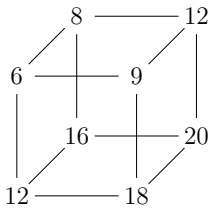
so that the the marginalization of T is (u_1, u_2, u_3)=((77, 154), (99, 132), (105, 136)). Clearly, $(77, 154) = 77(1, 2)$, $(99, 132) = 33(3, 4)$ and $(105, 136) = 21(5, 6)$. Here $W_1 = 3$, $W_2 = 7$, $W_3 = 11$, so that

$$u_1 \otimes u_2 \otimes u_3 = (3 \cdot 7)(3 \cdot 11)(7 \cdot 11)T = (53361)T.$$

When T has rank > 1, then clearly it cannot coincide with a multiple of $u_1 \otimes \cdots \otimes u_n$. For general T, the product of its contractions $u_1 \otimes \cdots \otimes u_n$ determines a good rank 1 approximation of T.

Example 8.1.12 Consider the tensor

$T =$



which is an approximation of $(1,2) \otimes (2,3) \otimes (3,4)$ (only one entry has been changed). The marginalization of T is $(u_1, u_2, u_3) = ((35, 66), (42, 59), (45, 56))$. The product $u_1 \otimes u_2 \otimes u_3$, divided by $35 \cdot 42 \cdot 45 = 66150$ gives the rank 1 tensor (approximate):

$$T' = \begin{array}{ccc} 7.5 & \text{———} & 10.5 \\ | \backslash & & | \backslash \\ 6 & \text{———} & 8.4 \\ | & & | \\ & 14.1 \text{———} 19.8 & \\ | & & | \\ 11.3 & \text{———} & 15.9 \end{array}$$

not really far from T.

Indeed, for some purposes, the product of the contractions can be considered as a good rank 1 approximation of a given tensor. We warn the reader that, on the other hand, there are other methods for the rank 1 approximation of tensors which, in many cases, produce a result much closer to the best possible rank 1 approximation. See, for instance, Remark 8.3.12.

8.2 Contractions

The concept of i-contraction of a tensor can be extended to any subset of indices. In this case, we will talk about *partial contraction*. The definition is straightforward, though it can look odd, at a first sight, since it requires many levels of indices.

Definition 8.2.1 For a tensor T of type $a_1 \times \cdots \times a_n$ and for any subset J of the set $[n] = \{1, \ldots, n\}$, of cardinality $n - q$, define the *J-contraction* T^J as follows. Set $\{1, \ldots, n\} \setminus J = \{j_1, \ldots, j_q\}$. For any choice of $k_1, \ldots, k_q$ with $1 \leq k_s \leq a_{j_s}$, put

$$T^J_{k_1,\ldots,k_q} = \sum T_{i_1,\ldots,i_n},$$

where the sum ranges on all the entries $T_{i_1,\ldots,i_n}$ in which $i_{j_s} = k_s$ for $s = 1, \ldots, q$.

Some example is needed.

Example 8.2.2 Consider the tensor:

$$T = \begin{array}{ccc} 18 & \text{———} & 20 \\ | \backslash & & | \backslash \\ 12 & \text{———} & 16 \\ | & & | \\ & 9 \text{———} 12 & \\ | & & | \\ 6 & \text{———} & 8 \end{array}$$

The contraction of T along $J = \{2\}$ means that we take the sum of the left face with the right face, so that, e.g., $T^J_{11} = T_{111} + T_{121}$, and so on. We get

$$T^J = \begin{pmatrix} 28 & 38 \\ 14 & 21 \end{pmatrix}.$$

The contraction of T along $J' = \{3\}$ means that we take the sum of the front face with the rear face, so that, e.g., $T^{J'}_{11} = T_{111} + T_{112}$, and so on. We get

$$T^{J'} = \begin{pmatrix} 30 & 36 \\ 15 & 20 \end{pmatrix}.$$

Instead, if we take the contraction along $J'' = \{2, 3\}$, we get a vector $T^{J''} \in K^2$, whose entries are the sums of the upper and the lower faces. Indeed $T^{J''}_1 = T_{111} + T_{112} + T_{121} + T_{122}$, so that

$$T^{J''} = (66, 35).$$

In this case, $T^{J''}$ is the 1-st contraction of T.

The last observation of the previous example generalizes.

Proposition 8.2.3 *If $J = \{1, \ldots, n\} \setminus \{i\}$, then T^J is the ith contraction of T.*

Proof Just compare the two definitions.

Remark 8.2.4 The contraction along $J \subset \{1, \ldots, n\}$ determines a linear map between spaces of tensors.

If $J' \subset J \subset \{1, \ldots, n\}$, then the contraction $T^{J'}$ is a contraction of T^J.

The relation on the ranks of a tensor and its contractions is expressed as follows:

Proposition 8.2.5 *If T has rank 1, then every contraction of T is either 0 or it has rank 1. As a consequence, for every $J \subset \{1, \ldots, n\}$,*

$$\mathrm{rank}(\mathrm{T}^{\mathrm{J}}) \leq \mathrm{rank}\mathrm{T}.$$

Proof In view of Remark 8.2.4, it is enough to prove the first statement when J is a singleton. Assume, for simplicity, that $J = \{n\}$. Then by definition

$$T^J_{i_1 \ldots i_{n-1}} = \sum_{j=1}^{a_n} T_{1_1 \ldots i_{n-1} j}.$$

If $T = v_1 \otimes \cdots \otimes v_n$, where $v_i = (v_{i1}, \ldots, v_{ia_i})$, then $T_{i_1 \ldots i_{n-1} j} = v_{1i_1} \cdots v_{n-1\ i_{n-1}} v_{nj}$. It follows that, setting $w = v_{n1} + \cdots + v_{na_n}$, then

$$T^J_{i_1 \ldots i_{n-1}} = w v_{1i_1} \cdots v_{n-1\ i_{n-1}},$$

so that $T^J = wv_1 \otimes \cdots \otimes v_{n-1}$.

The second statement follows immediately from the first one and from the linearity of the contraction. Indeed, if $T = T_1 + \cdots + T_r$, where each T_i has rank 1, then $T^J = T_1^J + \cdots + T_r^J$.

The following proposition is indeed an extension of Proposition 8.1.9.

Proposition 8.2.6 *Let $T \in K^{a_1,\ldots,a_n}$ be a tensor of rank 1. Let $Q_1, \ldots, Q_m$ be a partition of $\{1, \ldots, n\}$, with the property that every element of Q_i is smaller than every element of Q_{i+1}, for all i. Define $J_i = \{1, \ldots, n\} \setminus Q_i$.*

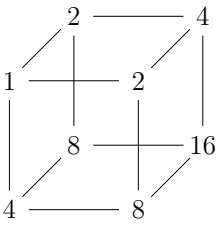
Then the tensor product $T^{J_1} \otimes \cdots \otimes T^{J_m}$ is a scalar multiple of T.

Proof The proof is straightforward when we take the (maximal) partition where $m = n$ and $Q_i = \{i\}$ for all i. Indeed in this case T^{J_i} is the ith contraction of T, and we can apply Proposition 8.1.9.

For the general case, we can use Proposition 8.1.9 again and induction on n. Indeed, by assumption, if $j_i = \min\{Q_i\} - 1$, then the kth contraction of T^{J_i} is equal to the $(j_i + k)$th contraction of T. □

Example 8.2.7 Consider the rank 1 tensor:

$T =$

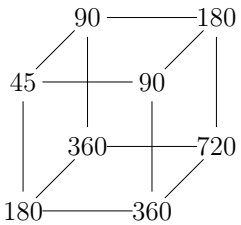


and consider the partition $Q_1 = \{1, 2\}$, $Q_2 = \{3\}$, so that $J_1 = \{3\}$ and $J_2 = \{1, 2\}$. Then

$$T^{J_1} = \begin{pmatrix} 3 & 6 \\ 12 & 24 \end{pmatrix},$$

while $T^{J_2} = (15, 30)$. We get:

$T^{J_1} \otimes T^{J^2} =$



hence $T^{J_1} \otimes T^{J^2} = 45T$.

8.3 Scan and Flattening

The following operations is natural for tensors and often allow a direct computation of the rank.

Definition 8.3.1 Let $T \in K^{a_1,\dots,a_n}$ be a tensor. For any $j = 1, \dots, n$ we denote with $Scan(T)_j$ the set of a_j tensors obtained by fixing the index j in all the possible ways.

Technically, $Scan(T)_j$ is the set of tensors $T_1, \dots T_{a_j}$ in $K^{a_1,\dots,\hat{a}_j,\dots,a_n}$ such that

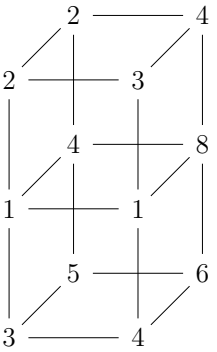
$$(T_q)_{i_1\dots i_{n-1}} = T_{i_1\dots q\dots i_{n-1}},$$

where the index q occurs in the jth place.

By applying the definition recursively, we obtain the definition of scan of a tensor T along a subset $J \subset \{1, \dots, n\}$ of two, three, etc., indices.

Example 8.3.2 Consider the tensor:

$T =$



The scan $Scan(T)_1$ is given by the three matrices

$$\begin{pmatrix} 2 & 3 \\ 2 & 4 \end{pmatrix} \quad \begin{pmatrix} 1 & 1 \\ 4 & 8 \end{pmatrix} \quad \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix},$$

while $Scan(T)_2$ is the couple of matrices

$$\begin{pmatrix} 2 & 2 \\ 1 & 4 \\ 3 & 5 \end{pmatrix} \quad \begin{pmatrix} 3 & 4 \\ 1 & 8 \\ 4 & 6 \end{pmatrix}.$$

Remark 8.3.3 There is an obvious relation between the scan and the contraction of a tensor T. If $J \subset \{1, \dots, n\}$ is any subset of the set of indices and $J' = \{1, \dots, n\} \setminus J$ then the $J-$contraction of T equals the sum of the tensors of the scan of the tensor T along J'.

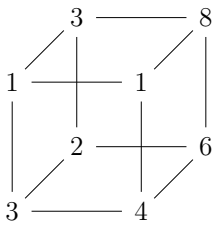
We define the flattening of a tensor by taking the scan along one index, and arranging the resulting tensors in one big tensor.

Definition 8.3.4 Let $T \in K^{a_1,\dots,a_n}$ be a tensor. The *flattening* of T along the last index is defined as follows. For any positive $q \leq a_{n-1}a_n$, one finds uniquely defined integers α, β such that $q - 1 = (\beta - 1)a_{n-1} + (\alpha - 1)$, with $1 \leq \beta \leq a_n$, $1 \leq \alpha \leq a_{n-1}$. Then the flattening of T along the last index is the tensor $FT \in K^{a_1,\dots,a_{n-2},a_{n-1}a_n}$ with:

$$FT_{i_1\dots i_{n-2}q} = T_{i_1\dots i_{n-2}\alpha\beta}.$$

Example 8.3.5 Consider the tensor:

$T =$

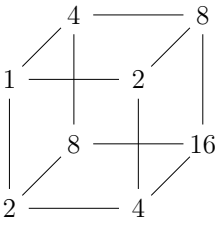


Its flattening is the 2×4 matrix:

$$\begin{pmatrix} 1 & 1 \\ 3 & 4 \\ 3 & 8 \\ 2 & 6 \end{pmatrix}.$$

The flattening of the rank 1 tensor:

$T =$



is the 2×4 matrix of rank 1:

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 4 & 8 \\ 8 & 16 \end{pmatrix}.$$

Remark 8.3.6 One can apply the flattening procedure after a permutation of the indices. In this way, in fact, one can define the flattening along any of the indices. We leave the details to the reader.

Moreover, one can take an ordered series of indices and perform a sequence of flattening procedures, in order to reduce the dimension of the tensor.

The final target which is usually the most useful for applications is the flattening reduction of a tensor to a (usually rather huge) matrix, by performing $n-2$ flattenings of an n-dimensional tensor. If we do not use permutations, the final output is a matrix of size $a_1 \times (a_2 \cdots a_n)$.

The reason why the flattenings are useful, for the analysis of tensors, is based on the following property.

Proposition 8.3.7 *A tensor T has rank 1, if and only if all its flattening has rank 1.*

Proof Let $T \in K^{a_1,\ldots,a_n}$ be a tensor of rank 1, $T = v_1 \otimes \cdots \otimes v_n$ where $v_i = (v_{i1}, \ldots, v_{ia_i})$. Since $T \neq 0$, then also $FT \neq 0$. Then one computes directly that the flattening FT is equal to $v_1 \otimes \cdots \otimes v_{n-2} \otimes w$, where

$$w = (v_{n-1\,1}v_{n1}, \ldots, v_{n-1a_{n-1}}v_{n1}, v_{n-1\,1}v_{n2}, \ldots, v_{n-1n_{a-1}}v_{na_n}).$$

Conversely, recall that, from Theorem 6.4.13, a tensor T has rank 1 when, for all choices $p, q = 1, \ldots, n$ and numbers $1 \leq \alpha, \gamma \leq a_p$, $1 \leq \beta, \delta \leq a_q$ one has

$$\begin{aligned} &T_{i_1\cdots i_{p-1}\alpha i_{p+1}\ldots i_{q-1}\beta i_{q+1}\ldots i_n} \cdot T_{i_1\cdots i_{p-1}\gamma i_{p+1}\ldots i_{q-1}\delta i_{q+1}\ldots i_n} - \\ &-T_{i_1\cdots i_{p-1}\alpha i_{p+1}\ldots i_{q-1}\delta i_{q+1}\ldots i_n} \cdot T_{i_1\cdots i_{p-1}\gamma i_{p+1}\ldots i_{q-1}\beta i_{q+1}\ldots i_n} = 0 \end{aligned} \tag{8.1}$$

If we take the flattening over the two indices p and q, the left term of previous equation is a 2×2 minor of the flattening.

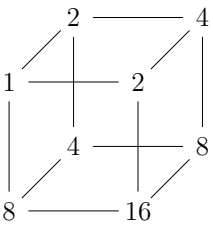
The second tensor in Example 8.3.5 shows the flattening of a $2 \times 2 \times 2$ tensor of rank 1.

Notice that one implication for the Proposition works indeed for any rank. Namely, if T has rank r, then all of its flattenings have rank $\leq r$.

On the other hand, the converse does not hold at least when the rank is big. For instance, there are tensors of type $2 \times 2 \times 2$ and rank 3 (see Example 6.4.15) for which, of course, the 2×4 flattenings cannot have rank 3.

Example 8.3.8 The tensor

$T =$



has rank > 1, because some determinants of its faces are not 0. On the other hand, its flattening is the 2×4 matrix

$$\begin{pmatrix} 1 & 2 \\ 8 & 16 \\ 2 & 4 \\ 4 & 8 \end{pmatrix}$$

which has rank 1.

We have the following, straightforward:

Proposition 8.3.9 *Let FT have rank 1, $FT = v_1 \otimes \cdots \otimes v_{n-2} \otimes w$, with $w \in K^{a_{n-1},a_n}$. Assume that we can split w in a_{n-1} pieces of length a_n which are proportional, i.e., assume that there is $v_n = (\alpha_1, \ldots, \alpha_{a_n})$ with*

$$w = (\beta_1\alpha_1, \ldots, \beta_1\alpha_{a_n}, \beta_2\alpha_1, \ldots, \beta_{a_{n-1}}\alpha_{a_n}).$$

Then T has rank 1, and setting $v_{n-1} = (\beta_1, \ldots, \beta_{a_{n-1}})$, then:

$$T = v_1 \otimes \cdots \otimes v_{n-2} \otimes v_{n-1} \otimes v_n.$$

As a corollary of Proposition 8.3.7, we get:

Proposition 8.3.10 *If a tensor T has rank r, then its flattening has rank $\leq r$.*

Proof The flattening is a linear operation in the space of tensors. Thus, if $T = T_1 + \cdots + T_r$ with each T_i of rank 1, then also $FT = FT_1 + \cdots + FT_r$, and each FT_i has rank 1. The claim follows.

Of course, the rank of the flattening FT can be strictly smaller than the rank of T. For instance, we know from Example 6.4.15 that there are $2 \times 2 \times 2$ tensors T of rank 3. The flattening FT, which is a 2×4 matrix, cannot have rank bigger than 2.

Remark 8.3.11 Here is one application of the flattening procedure to the computation of the rank.

Assume we are given a tensor $T \in K^{a_1,\ldots,a_n}$ and assume we would like to know if the rank of T is r. If $r < a_1$, then we can perform a series of flattenings along the last indices, obtaining a matrix F^*T of size $a_1 \times (a_2 \cdots a_n)$. Then, we can compute the rank of the matrix (and we have plenty of fast procedures to do this). If F^*T has rank $> r$, then there is no chance that the original tensor T has rank r. If F^*T has rank r, then this can be considered as a cue toward the fact that $rank(T) = r$.

Of course, a similar process is possible, by using permutations on the indices, when $r \geq a_1$ but $r < a_i$ for some i.

The flattening process is clearly invertible, so that one can reconstruct the original tensor T from the flattening FT, thus also from the matrix F^*T resulting from a process of $n - 2$ flattenings.

On the other hand, since a matrix of rank $r > 1$ has infinitely many decompositions as a sum of r matrices of rank 1, then by taking one decomposition of F^*T as a

sum of r matrices of rank 1 one cannot hope to reconstruct automatically from that decomposition of T as a sum of r tensors of rank 1.

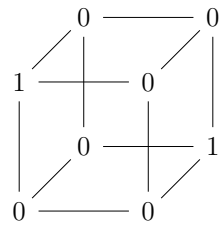
Indeed, the existence of a decomposition for T is subject to the existence of a decomposition for F^*T in which every summand satisfies the condition of Proposition 8.3.9.

Remark 8.3.12 One can try to find an approximation of a given tensor T with a tensor of prescribed, small rank $r < a_1$ by taking the matrix F^*T, resulting from a process of $n-2$ flattenings, and considering the rank r approximation for F^*T obtained by the standard SVD approximation process for matrices (see [1]).

For instance, one can find in this way a rank 1 approximation for a tensor, which in principle is not equal to the rank 1 approximation obtained by the marginalization (see Example 8.1.12).

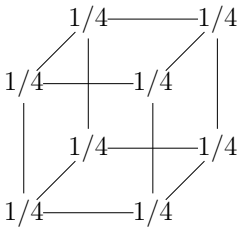
Example 8.3.13 Consider the tensor:

$T =$



of Example 8.1.12. The contractions of T are $(1,1),(1,1),(1,1)$, whose tensor product, divided by $4 = (1+1)+(1+1)$ determines a rank 1 approximation:

$T_1 =$



The flattening of T is the matrix:

$$FT = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The rank 1 approximation of FT, by the SVD process, is:

$$FT = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

This matrix defines the tensor:

$$T_2 = \begin{array}{ccc} & 0 & \\ 1 & & 0 \\ & 0 & \\ 0 & & 0 \end{array} \quad \begin{array}{ccc} 0 & & 0 \\ & 0 & \\ 0 & & 0 \\ & 0 & \end{array}$$

which is another rank 1 approximation of T.

If we consider the tensors as vectors in K^8, then the natural distance $d(T, T_1)$ is $\sqrt{3/2}$, while the natural distance $d(T, T_2)$ is 1. So T_2 is "closer" to T than T_1. On the other hand, for some purposes, one could consider T_1 as a better rank 1 approximation of T than T_2 (e.g., it preserves marginalization).

8.4 Exercises

Exercise 15 Prove the assertion in Example 8.1.4: the matrix M defined there generates the Kernel of the marginalization.

Exercise 16 Find generators for the Kernel of the marginalization map of 3×3 matrices.

Exercise 17 Find generators for the image of the marginalization map of tensors and prove Proposition 8.1.7.

Exercise 18 Prove the statements of Remark 8.2.4.

Reference

1. Markovsky, I.: Low-Rank Approximation: Algorithms, Implementation, Applications. Springer, Berlin (2012)

Part III
Commutative Algebra and Algebraic Geometry

Chapter 9
Elements of Projective Algebraic Geometry



The scope of this part of the book is to provide a quick introduction to the main tools of the Algebraic Geometry of projective spaces that are necessary to understand some aspects of algebraic models in Statistics.

The material collected here is not self-contained. For many technical results, as the Nullstellensatz or the theory of fields extensions, we will refer to specific texts on the subject.

We assume in the sequel that the reader knows the basic definitions of algebraic structures, as rings, ideals, homomorphisms, etc., as well as the main properties of polynomial rings.

This part of the book could also be used for a short course or a cutway through the main results of algebraic and projective geometry which are relevant in the study of Statistics.

9.1 Projective Varieties

The first step is a definition of the ambient space.

Since we need to deal with subsets defined by polynomial equations, the starting point is the polynomial rings over the *complex* field, the field where solutions of polynomial equations live properly. Several claims that we are going to illustrate also work on any algebraically closed field. We deal only with the complex field, in order to avoid details on the structure of fields of arbitrary characteristic.

The main feature of Projective Geometry is that the coordinates of a point are defined only *up to scaling*.

Definition 9.1.1 Let V be a linear space over $\mathbb{C}$. Define on $V^* = V \setminus \{0\}$ an equivalence relation $\sim$ which associates v, v' if and only if there exists $\alpha \in \mathbb{C}$ with $v' = \alpha v$. The quotient $\mathbb{P}(V) = V^*/\sim$ is the *projective space associated to* V.

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The *projective dimension* of $\mathbb{P}(V)$ is the number $\dim(V) - 1$ (it is a constant fact that the dimension of a projective space is always 1 less than its linear dimension).

When $V = \mathbb{C}^{n+1}$, we will denote the projective space $\mathbb{P}(V)$ also with $\mathbb{P}^n$.

Points of the projective space are thus equivalent classes of vectors, in the relation $\sim$, hence are formed by a vector $v \neq 0$ together with all its multiples.

In particular, $P \in \mathbb{P}^n$ is an equivalence class of $(n+1)$-tuples of complex numbers. We will denote with *homogeneous coordinates* of P any representative of the equivalence class.

Notice that the coordinates, in a projective space, are no longer uniquely defined, but only defined modulo scalar multiplication. We will also write $P = [p_0 : \cdots : p_n]$ when $(p_0, \ldots, p_n)$ is a representative for the homogeneous coordinates of P.

Remark 9.1.2 $\mathbb{P}^n$ contains several subsets in natural one-to-one correspondence with $\mathbb{C}^n$.

Indeed, take the subset U_i of points with homogeneous coordinates $[x_0 : \cdots : x_n]$ whose i-th coordinate x_i is nonzero. The condition is clearly independent from the representative of the class that we choose. There is a one-to-one correspondence $U_i \leftrightarrow \mathbb{C}^n$, obtained as follows:

$$[x_0 : \cdots : x_n] \mapsto (\frac{x_0}{x_i}, \frac{x_1}{x_i}, \ldots, \frac{\hat{x_i}}{x_i}, \ldots, \frac{x_n}{x_i})$$

U_i is called the *i-th affine subspace*.

Notice that when $P = [p_0 : \cdots : p_n] \in U_i$, hence $p_i \neq 0$, then there exists a unique representative of P with $p_i = 1$. The previous process identifies $P \in U_i$ with the point of $\mathbb{C}^n$ whose coordinates correspond to such representative of P, excluding the i-th coordinate.

Definition 9.1.3 A subset C of a linear space V is a *cone* if for any $v \in C$ and $a \in \mathbb{C}$ then $av \in C$.

Remark 9.1.4 Cones in the linear space $\mathbb{C}^{n+1}$ define subsets in the associated projective space $\mathbb{P}^n$.

Indeed there is an obvious map $\pi : \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}^n$ that sends $(n+1)$-tuples to their equivalence classes. If $W \subset \mathbb{P}^n$ is any subset, then $\pi^{-1}(W) \cup \{0\}$ is a cone in $\mathbb{C}^{n+1}$.

Conversely, every cone in $\mathbb{C}^{n+1}$ is the inverse image in π of a subset of $\mathbb{P}^n$ (one must add 0).

The same argument applies if we substitute $\mathbb{C}^{n+1}$ and $\mathbb{P}^n$ with, respectively, a generic linear space V and its associated projective space $\mathbb{P}(V)$.

In general, one cannot expect that a polynomial $f \in \mathbb{C}[x_0, \ldots, x_n]$ defines a cone in $\mathbb{C}^{n+1}$. This turns out to be true when f is *homogeneous*. Indeed, if f is homogeneous of degree d and $a \in \mathbb{C}$ is any scalar, then

$$f(ax_1, \ldots, ax_n) = a^d f(x_1, \ldots, x_n),$$

thus for $a \neq 0$ the vanishing of $f(ax_1, \ldots, ax_n)$ is equivalent to the vanishing of $f(x_1, \ldots, x_n)$.

The observation can be reversed, as follows.

Lemma 9.1.5 *Let f be a polynomial in $\mathbb{C}[x_0, \ldots, x_n]$, of degree bigger than 0. Then there exist a point $P = (p_0, \ldots, p_n) \in \mathbb{C}^{n+1}$ with $f(P) = 0$ and infinitely many points $Q = (q_0, \ldots, q_n) \in \mathbb{C}^{n+1}$ with $f(Q) \neq 0$.*

Proof Make induction on the number of variables.

When f has only one variable, then the first claim is exactly the definition of *algebraically closed field*. The second claim holds because every nonzero polynomial of degree d has at most d roots.

If we know both claims for n variables, then write $f \in \mathbb{C}[x_0, \ldots, x_n]$ as a polynomial in x_0, with coefficients in $\mathbb{C}[x_1, \ldots, x_n]$:

$$f = f_d x_0^d + f_{d-1} x_0^{d-1} + \cdots + f_0$$

where each f_i is a polynomial in $x_1, \ldots, x_n$. We may assume that $f_d \neq 0$, otherwise f has only $n-1$ variables, and the claim holds by induction. By induction, there are infinitely many points $(q_1, \ldots, q_n) \in \mathbb{C}^n$ which are not a solution of f_d (notice that such points exist trivially if $f_d \neq 0$ is constant). Then for any $(q_1, \ldots, q_n)$ the polynomial $g = f(x_0, q_1, \ldots, q_n)$ has just one variable and degree $d > 0$, hence there are infinitely many $q_0 \in \mathbb{C}$ with $g(q_0) = f(q_0, \ldots, q_n) \neq 0$ and at least one $p_0 \in \mathbb{C}$ such that $g(p_0) = f(p_0, q_1, \ldots, q_n) = 0$. □

We recall that any polynomial $f(x) \in \mathbb{C}[x_0, \ldots, x_n]$ can be written uniquely as a sum of homogeneous polynomials

$$f(x) = f_d + f_{d-1} + \cdots + f_0,$$

with f_i homogeneous of degree i for all i. The previous sum is called the *homogeneous decomposition of $f(x)$*.

Proposition 9.1.6 *Let $f = f(t)$ be a polynomial in $\mathbb{C}[x_0, \ldots, x_n]$ of degree $d > 0$. Assume that $f(t)$ is not homogeneous.*

Then there exists $P = (p_0, \ldots, p_n) \in \mathbb{C}^{n+1}$ and a scalar $\alpha \in \mathbb{C} \setminus \{0\}$ such that $f(P) = 0$ and $f(\alpha P) \neq 0$.

Proof Take the homogeneous decomposition of $f(x)$

$$f(x) = f_d + f_{d-1} + \cdots + f_0.$$

Since $f(x)$ is not homogeneous, we may assume $f_d, f_i \neq 0$ for some $i < d$. Take the minimal i with $f_i \neq 0$. Choose $y = (y_0, \ldots, y_n) \in \mathbb{C}^{n+1}$ with $f_d(y) \neq 0$ (it exists by Lemma 9.1.5). Then $f(ay) = a^d f_d(y) + a^{d-1} f_{d-1}(y) + \cdots + a^i f_i(y)$ is a polynomial of degree $d > 0$ in the unique variable a, which can be divided by a^i, i.e.

$f(ay) = a^i g(ay)$, where $g(ay)$ is a polynomial of degree $d - i > 0$ in a, whose constant term is nonzero. By Lemma 9.1.5 again, there exist $a_1, a_2 \in \mathbb{C}$ with $g(a_1 y) = 0$ and $g(a_2 y) \neq 0$. Notice that $a_1 \neq 0$, since the constant term of $g(ay)$ does not vanish. The claim now holds by taking $P = a_1 y$ and $\alpha = a_2/a_1$. □

The previous Proposition shows that the vanishing of a polynomial in $\mathbb{C}[x_0, \ldots, x_n]$ is not defined over a cone, hence over a subset of $\mathbb{P}^n$, unless the polynomial is homogeneous. Conversely, if $f \in \mathbb{C}[x_0, \ldots, x_n]$ is homogeneous, then the vanishing of f at a set of projective coordinates $[p_0 : \cdots : p_n]$ of $P \in \mathbb{P}^n$ implies the vanishing of f at *any* set of homogeneous coordinates of P, because the vanishing set of f is a cone.

Consequently, we give the following, basic:

Definition 9.1.7 We call *projective variety* of $\mathbb{P}^n$ every subset of $\mathbb{P}^n_K$ defined by the vanishing of a family $J = \{f_j\}$ of homogeneous polynomials $f_j \in \mathbb{C}[x_0, \ldots, x_n]$.

In other words, projective varieties are subsets of $\mathbb{P}^n$ whose equivalence classes are the solutions of a system of *homogeneous* polynomial equations.

When V is any linear space of dimension d, we define the projective varieties in $\mathbb{P}(V)$ by taking an identification $V \sim \mathbb{C}^d$ (hence by fixing a basis of V).

We will denote with $X(J)$ the projective variety defined by the family J of homogeneous polynomials.

Example 9.1.8 Let $\{f_1, \ldots, f_m\}$ be a family of *linear* homogeneous polynomials in $\mathbb{C}[x_0, \ldots, x_n]$. The projective variety X defined by $\{f_1, \ldots, f_m\}$ is called a *linear* projective variety.

The polynomials $f_1, \ldots, f_m$ define also a linear subspace $W \subset \mathbb{C}^{n+1}$. It is easy to prove that there is a canonical identification of X with the projective space $\mathbb{P}(W)$ (see Exercise 19).

Remark 9.1.9 Let X be a projective variety defined by a set J of homogeneous polynomials and take a subset $J' \subset J$. Then the projective variety $X' = X(J')$ defined by J' *contains* X.

One can easily find examples (even of linear varieties) with $X' = X$ even if J' is properly contained in J (see Exercise 20).

Remark 9.1.10 Projective varieties provide a system of *closed* sets for a topology, called the *Zariski topology* on $\mathbb{P}^n$.

Namely, $\emptyset$ and $\mathbb{P}^n$ are both projective varieties, defined respectively by the families of polynomials $\{1\}$ and $\{0\}$. If $\{W_i\}$ is a family of projective varieties, with $W_i = X(J_i)$, then $\bigcap\{W_i\}$ is the projective variety defined by the family $J = \bigcup\{J_i\}$ of homogeneous polynomials. Finally, if $W_1 = X(J_1)$ and $W_2 = X(J_2)$ are projective varieties, then $W_1 \cup W_2$ is the projective variety defined by the family of homogeneous polynomials:

$$J_1 J_2 = \{fg : f \in J_1, g \in J_2\}.$$

Example 9.1.11 Every singleton in $\mathbb{P}^n$ is a projective variety.

Namely, if $[p_0 : \cdots : p_n]$ are homogeneous coordinates of a point P, with $p_i \neq 0$, then the set of linear homogeneous polynomials

$$I = \{p_0 x_i - p_i x_0, \ldots, p_n x_i - p_i x_n\}$$

defines $\{P\} \subset \mathbb{P}^n$.

In particular, the Zariski topology satisfies the first separation axiom T_1.

Example 9.1.12 Every Zariski closed subset of $\mathbb{P}^1$ is finite, except $\mathbb{P}^1$ itself.

Namely let f be any nonzero homogeneous polynomial of degree d in $\mathbb{C}[x_0, x_1]$. Then setting $x_1 = 1$, we get a polynomial $\bar{f} \in \mathbb{C}[x_0]$ which, by the Fundamental Theorem of Algebra, can be uniquely decomposed in a products $\bar{f} = e(x_0 - \alpha_0)^{m_0} \cdots (x_0 - \alpha_k)^{m_k}$, where $\alpha_1, \ldots, \alpha_k$ are the roots of $\bar{f}$ and $e \in \mathbb{C}$.

Going back to f, we see that there exists a power x_1^β (maybe $\beta = 0$) such that

$$f = e x_1^\beta (x_0 - \alpha_0 x_1)^{m_0} \cdots (x_0 - \alpha_k x_1)^{m_k}.$$

It follows immediately that f vanishes only at the points $[\alpha_0 : 1], \ldots, [\alpha_k : 1]$, with the addition of $[1 : 0]$ if $\beta > 0$.

Thus, the open sets in the Zariski topology on $\mathbb{P}^1$ are $\emptyset$ and the *cofinite* sets, i.e. sets whose complement is finite. In other words, the Zariski topology on $\mathbb{P}^1$ coincides with the *cofinite* topology.

Example 9.1.13 In higher projective spaces there are nontrivial closed subsets which are infinite. Thus the Zariski topology on $\mathbb{P}^n$, $n > 1$, is not the cofinite topology.

Indeed, let $f \neq 0$ be a homogeneous polynomial in $\mathbb{C}[x_0, \ldots, x_n]$, of degree bigger than 0, and assume $n > 1$. We prove the variety $X(f)$, which is not $\mathbb{P}^n$ by Lemma 9.1.5, has infinitely many points.

To see this, notice that if all points $Q = [q_0 : q_1 : \cdots : q_n]$ with $q_0 \neq 0$ belong to $X(f)$, then we are done. So we can assume that there exists $Q = [1 : q_1 : \cdots : q_n] \notin X(f)$. For any choice of $m = (m_2, \ldots, m_n) \in \mathbb{C}^{n-1}$ consider the line L_m, passing through Q, defined by the vanishing of the linear polynomials

$$x_2 - m_2(x_1 - q_1 x_0) - q_2 x_0, \ldots, x_n - m_n(x_1 - q_1 x_0) - q_n x_0.$$

Define the polynomial

$$f_m = f(x_0, x_1, m_2(x_1 - q_1 x_0) + q_2 x_0, \ldots, m_n(x_1 - q_1 x_0) + q_n x_0).$$

If (α_0, α_1) is a solution of the equation $f_m = 0$, then the intersection $X(f) \cap L_m$ contains the point

$$[\alpha_0 : \alpha_1 : m_2(\alpha_1 - q_1 \alpha_0) + q_2 \alpha_0, \ldots, m_n(\alpha_1 - q_1 \alpha_0) + q_n \alpha_0].$$

Since the polynomial f_m is homogeneous of the same degree than f, then it vanishes at some point, so that $X(f) \cap L_m \neq \emptyset$. Since two different lines L_m, $L_{m'}$ meet only at $Q \notin X(f)$, the claim follows.

9.1.1 Associated Ideals

Definition 9.1.14 Let I be an ideal of a polynomial ring $R = \mathbb{C}[x_0, \ldots, x_n]$. We say that I *is generated by* $J \subset R$, and write $I = \langle J \rangle$, when

$$I = \{h_1 f_1 + \cdots + h_m f_m : h_1, \ldots, h_m \in R, \quad f_1, \ldots, f_m \in J\}.$$

We say that I is a *homogeneous* ideal if there is a set of homogeneous elements $J \subset R$ such that $I = \langle J \rangle$.

Notice that not every element of a homogeneous ideal is homogeneous. for instance, in $\mathbb{C}[x]$ the homogeneous ideal $I = \langle x \rangle$ contains the nonhomogeneous element $x + x^2$.

Proposition 9.1.15 *The ideal I is homogeneous if and only if for any polynomial $f \in I$, whose homogeneous components are $f_d, \ldots, f_0$, then every f_i belongs to I.*

Proof Assume that I is generated by a set of homogeneous elements J and take $f \in I$. Consider the decomposition in homogeneous components $f = f_d + \cdots + f_0$. There are homogeneous polynomials $q_1, \ldots, q_m \in J$ such that $f = h_1 q_1 + \cdots + h_m q_m$, for some polynomials $h_j \in R$. Denote by d_j the degree of q_j and denote by $h_{i,j}$ the homogeneous component of degree i in h_j (with $h_{ij} = 0$ whenever $i < 0$). Then, by comparing the homogeneous components, one gets for every degree i

$$f_i = h_{1,i-d_1} q_1 + \cdots + h_{m,i-d_m} q_m$$

and thus $f_i \in \langle J \rangle = I$ for every i.

Conversely, I is always contained in the ideal generated by the homogeneous components of its elements. Thus, when these components are also in I for all $f \in I$, then I is generated by homogeneous polynomials. □

Remark 9.1.16 If W is a projective variety defined by the vanishing of a set J of homogeneous polynomials, then W is also defined by the vanishing of all the polynomials in the ideal $I = \langle J \rangle$.

Indeed if P is a point of W, then for all $f \in I$, write $f = h_1 f_1 + \cdots + h_m f_m$ for some f_i's in J. We have

$$f(P) = h_1(P) f_1(P) + \cdots + h_m(P) f_m(P) = 0.$$

It follows that every projective variety can be defined as the vanishing locus of a homogeneous ideal.

Before stating the basic result in the correspondence between projective varieties and homogeneous ideals (i.e. the homogeneous version of the celebrated Hilbert's Nullstellensatz), we need some more piece of notation.

Definition 9.1.17 For any ideal $I \subset R$, define the *radical* of I as the set

$$\sqrt{I} = \{f : f^m \in I \text{ for some exponent } m\}.$$

$\sqrt{I}$ is an ideal of R and contains I.

When I is a homogeneous ideal, then also $\sqrt{I}$ is homogeneous, and the projective varieties $X(I)$ and $X(\sqrt{I})$ are equal (see Exercise 21).

We say that an ideal I in R is *radical* if $I = \sqrt{I}$. For any ideal I, $\sqrt{I}$ is a radical ideal, since $\sqrt{\sqrt{I}} = \sqrt{I}$.

We call *irrelevant ideal* the ideal of $R = \mathbb{C}[x_0, \ldots, x_n]$ generated by the indeterminates $x_0, \ldots, x_n$.

The irrelevant ideal is a radical ideal that defines the empty set in $\mathbb{P}^n$. Indeed, no points of $\mathbb{P}^n$ can annihilate all the variables, as no points in $\mathbb{P}^n$ have all the homogeneous coordinates equal to 0.

Example 9.1.18 In $\mathbb{C}[x, y]$ consider the homogeneous element x^2. The radical of the ideal $I = \langle x^2 \rangle$ is the ideal generated by x. Indeed x belongs to $\sqrt{\langle x^2 \rangle}$, moreover if $f^n \in I$ for some polynomial f, then f cannot have a vanishing constant term, thus $f \in \langle x \rangle$.

The three sets $\{x^2\}$, $\langle x^2 \rangle$, $\sqrt{\langle x^2 \rangle} = \langle x \rangle$ all define the same projective subvariety of $\mathbb{P}^1$: the point of homogeneous coordinates $[0 : 1]$.

Now we are ready to state the famous Hilbert's Nullstellensatz, which clarifies the relations between different sets of polynomials that define the same projective variety.

Theorem 9.1.19 (Homogeneous Nullstellensatz) *Two homogeneous ideals I_1, I_2 in the polynomial ring $R = \mathbb{C}[x_0, \ldots, x_n]$ define the same projective variety X if and only if*

$$\sqrt{I_1} = \sqrt{I_2},$$

with the unique exception $I_1 = R$, $I_2 =$ the irrelevant ideal.

Thus, if two radical homogeneous ideals I_1, I_2 define the same projective variety X, and none of them is the whole ring R, then $I_1 = I_2$.

Moreover, two sets J_1, J_2 of homogeneous polynomials define the same projective variety X if and only if $\sqrt{\langle J_1 \rangle} = \sqrt{\langle J_2 \rangle}$.

A proof of the homogeneous Nullstellensatz can be found in the book [1].

We should notice that the Theorem works because our varieties are defined over $\mathbb{C}$, which is algebraically closed. The statement is definitely not true over a nonalgebraically closed field, as the real field $\mathbb{R}$. This is itself a good reason to define projective varieties on an algebraically closed field, as $\mathbb{C}$.

We list below other consequences of the Nullstellensatz.

Theorem 9.1.20 *Let $J \in \mathbb{C}[x_0 \ldots, x_n]$ be a set of homogeneous polynomials which define a projective variety $X \neq \emptyset$. Then the set*

$$J(X) = \{f \in \mathbb{C}[x_0 \ldots, x_n] : f \text{ is homogeneous and } f(P) = 0 \text{ for all } P \in X\}$$

coincides with the radical of the ideal generated by J.

We will call $J(X)$ the *homogeneous ideal* associated with X.

Corollary 9.1.21 *Let $I \in R = \mathbb{C}[x_0 \ldots, x_n]$ be a homogeneous ideal. I defines the empty set in $\mathbb{P}^n$ if and only if, for some m, all the powers x_i^m belong to I.*

Proof Indeed we get from the homogeneous Nullstellensatz that $\sqrt{I}$ is either R or the irrelevant ideal. In the former case, the claim is obvious. In the latter, for every i there exists m_i such that $x_i^{m_i} \in I$, and one can take m as the maximum of the m_i's. □

Another fundamental result in the study of projective varieties, still due to Hilbert, is encoded in the following algebraic result:

Theorem 9.1.22 (Basis Theorem) *Let J be a set of polynomials and let I be the ideal generated by J. Then there exists a* finite *subset $J' \subset J$ that generates the ideal I.*

In particular, any projective variety can be defined by the vanishing of a finite *set of homogeneous polynomials.*

A proof of a weaker version of this theorem will be given in Chap. 13 (or see, also, Sect. 4 of [1]). Let us list some consequences of the Basis Theorem.

Definition 9.1.23 We call *hypersuperface* any projective variety defined by the vanishing of a single homogeneous polynomial. By abuse, often we will write $X(f)$ instead of $X(\{f\})$.

When f has degree 1, then $X(f)$ is called a *hyperplane*.

Corollary 9.1.24 *Every projective variety is the intersection of a finite number of hypersurfaces. Equivalently, every open set in the Zariski topology is a finite union of complements of hypersurfaces.*

Proof Let $X = X(J)$ be a projective variety, defined by the set J of homogeneous polynomials. Find a finite subset $J' \subset J$ such that $\langle J \rangle = \langle J' \rangle$. Then:

$$X = X(J) = X(\langle J \rangle) = X(\langle J' \rangle) = X(J').$$

If $J = \{f_1, \ldots, f_m\}$ then, by Remark 9.1.10:

$$X = X(f_1) \cap \cdots \cap X(f_m).$$

□

Example 9.1.25 If $L \subset \mathbb{P}^n$ is a linear variety which corresponds to a linear subspace of dimension $m+1$ in $\mathbb{C}^{n+1}$, then L can be defined by $n-m$ linear homogeneous polynomials, i.e. L is the intersection of $n-m$ hyperplanes.

Remark 9.1.26 One could think that the homogeneous ideal of every projective variety in $\mathbb{P}^n$ can be generated by a finite set of homogeneous polynomials of cardinality bounded by a function of n.

F.S. Macaulay proved that this guess is false.

Indeed, in [2] he showed that for every integer m there exists a subset (curve) in $\mathbb{P}^3$ whose homogeneous ideal cannot be generated by a set of less than m homogeneous polynomials.

9.1.2 Topological Properties of Projective Varieties

The Basis Theorem provides a tool for the study of some aspects of the Zariski topology.

Definition 9.1.27 A topological space Y is *irreducible* when any pair of non-empty open subsets have a non-empty intersection.

Equivalently, Y is irreducible if it is not the union of two proper closed subsets.

Equivalently, Y is irreducible if the closure of every non-empty open subset A is Y itself, i.e. every non-empty open subset is dense in Y.

The following Proposition is easy and we will leave it as an exercise (see Exercise 22).

Proposition 9.1.28 *(i) Every singleton is irreducible.*
(ii) If Y is an irreducible subset, then the closure of Y is irreducible.
(iii) If an irreducible subset Y is contained in a finite union of closed subsets $X_1 \cup \cdots \cup X_m$, then Y is contained in some X_i.
(iv) If $Y_1 \subset \ldots Y_i \subset \ldots$ is an ascending chain of irreducible subsets, then $\bigcup Y_i$ is irreducible.

Corollary 9.1.29 *Any projective space $\mathbb{P}^n$ is irreducible and compact in the Zariski topology.*

Proof Let A_1, A_2 be non-empty open subsets, in the Zariski topology, and assume that A_i is the complement of the projective variety $X_i = X(J_i)$, where J_1, J_2 are two subsets of homogeneous polynomials in $\mathbb{C}[x_0, \ldots, x_n]$. We may assume, by the Basis Theorem, that both J_1, J_2 are finite. Notice that none of $X(J_1), X(J_2)$ can coincide with $\mathbb{P}^n$, thus both J_1, J_2 contain a nonzero element.

To prove that $\mathbb{P}^n$ is irreducible, we must show that $A_1 \cap A_2$ cannot be empty, i.e. that $X_1 \cup X_2$ cannot coincide with $\mathbb{P}^n$. By Remark 9.1.10, $X_1 \cup X_2$ is the projective variety defined by the set of products $J_1 J_2$. If we take $f_1 \neq 0$ in J_1 and $f_2 \neq 0$

in J_2, then $f = f_1 f_2$ is a nonzero element in $J_1 J_2$. By Lemma 9.1.5 there exist points $P \in \mathbb{P}^n$ such that $f(P) \neq 0$. Thus P does not belong to $X_1 \cup X_2$, and the irreducibility of $\mathbb{P}^n$ is settled.

For the compactness, we prove that $\mathbb{P}^n$ enjoys the Finite Intersection Property: if the intersection of a family of closed subset is empty, then there exists a finite subfamily with empty intersection.

Let $\{X_i\}$ be any family of closed subsets such that $\bigcap X_i = \emptyset$. Assume $X_i = X(J_i)$ and define $J = \bigcup J_i$. By Remark 9.1.10, $\bigcap X_i = X(J)$, thus also $\bigcap X_i = X(\langle J \rangle)$. By the Basis Theorem, there exists a finite subset J' of J such that $\langle J' \rangle = \langle J \rangle$. Thus there exists a finite subfamily $J_{i_1}, \ldots, J_{i_k}$ such that $\langle J_{i_1} \cup \cdots \cup J_{i_k} \rangle = \langle J \rangle$. Thus

$$\emptyset = X(\langle J \rangle) = X(\langle J_{i_1} \cup \cdots \cup J_{i_k} \rangle) = X(J_{i_1} \cup \cdots \cup J_{i_k}) = X_{i_1} \cap \cdots \cap X_{i_k}$$

and the Finite Intersection Property holds. □

Closed subsets in a compact space are compact. Thus any projective variety $X \subset \mathbb{P}^n$ is compact in the topology induced by the Zariski topology of $\mathbb{P}^n$.

Notice that irreducible topological spaces are far from being Hausdorff spaces. Thus no nontrivial projective space satisfies the Hausdorff separation axiom T_2.

Another important consequence of the Basis Theorem is the following.

Theorem 9.1.30 *Any non-empty family $\mathscr{F}$ of closed subsets of $\mathbb{P}^n$ (i.e. of projective varieties), partially ordered by inclusion, has a minimal element.*

Proof Let the claim fail. Then one can find an infinite chain of elements of $\mathscr{F}$,

$$X_0 \supset X_1 \supset \cdots \supset X_i \supset \ldots$$

where all the inclusions are strict. Consider for all i the ideal $I(X_i)$ generated by the homogeneous polynomials which vanish at X_i. Then one gets an *ascending* chain of ideals

$$I(X_0) \subset I(X_1) \subset \cdots \subset I(X_i) \subset \ldots$$

where again all the inclusions are strict. Let $I = \bigcup I(X_i)$. It is immediate to see that I is a homogeneous ideal. By the Basis Theorem, there exists a finite set of homogeneous generators $g_1, \ldots, g_k$ for I. Since every g_j belongs to $\bigcup I(X_i)$, for i_0 sufficiently large we have $g_j \in I(X_{i_0})$ for all j. Thus $I = I(X_{i_0})$, so that $I(X_{i_0}) = I(X_{i_0+1})$, a contradiction. □

Definition 9.1.31 For any projective variety X, a subset X' of X is an *irreducible component* of X if it is closed (in the Zariski topology, thus X' is a projective variety itself), irreducible and X' maximal with respect to these two properties.

It is clear that X is irreducible if and only if X itself is the unique irreducible component of X.

Theorem 9.1.32 *Let X be any projective variety. Then the irreducible components of X exist and their number is finite.*

Moreover there exists a unique decomposition of X as the union

$$X = X_1 \cup \cdots \cup X_k$$

where $X_1, \ldots, X_k$ are precisely the irreducible components of X.

Proof First, let us prove that irreducible components exist. To do that, consider the family $\mathscr{F}_p$ of closed irreducible subsets containing a point P. $\mathscr{F}_P$ is not empty, since it contains $\{P\}$. If $X_1 \subset \ldots X_i \subset \ldots$ is an ascending chain of elements of $\mathscr{F}_p$, then the union $Y = \bigcup X_i$ is irreducible by 9.1.28 (iv), thus the closure of Y sits in $\mathscr{F}_p$ (by 9.1.28 (ii)) and it is an upper bound for the chain. Then the family $\mathscr{F}_p$ has maximal elements, by the Zorn's Lemma. These elements are irreducible components of X.

Notice that we also proved that every point of X sits in some irreducible component, i.e. X is the union of its irreducible components. If Y is an irreducible component, by 9.1.28 (ii) also the closure of Y is irreducible. Thus, by maximality, Y must be closed.

Next, we prove that X is a finite union of irreducible closed subsets. For, assume this is false. Call $\mathscr{F}$ the family of closed subsets of X which are *not* a finite union of irreducible subsets. $\mathscr{F}$ is non-empty, since it contains X. By Theorem 9.1.30, $\mathscr{F}$ has some minimal element X'. As $X' \in \mathscr{F}$, then X' cannot be irreducible. Thus there are two closed subsets X_1, X_2, properly contained in X', whose union is X'. Since X' is minimal in $\mathscr{F}$, none of X_1, X_2 is in $\mathscr{F}$, thus both X_1, X_2 are union of a finite number of irreducible closed subsets. But then also X' would be a finite union of closed irreducible subsets. As $X' \in \mathscr{F}$, this is a contradiction.

Thus, there are irreducible closed subsets $X_1, \ldots, X_k$, whose union is X. Then, if Y is any irreducible component of X, we have $Y \subset X = X_1 \cup \cdots \cup X_k$. By 9.1.28 (iii), Y is contained in some X_i. By maximality, we get that Y coincides with some X_i. This proves that the number of irreducible components of X is finite.

We just proved that X decomposes in the union of its irreducible components $Y_1, \ldots, Y_m$. By 9.1.28 (iii), none of the Y_i can be contained in the union of the remaining components. Thus the decomposition is unique. □

Example 9.1.33 Let X be the variety in $\mathbb{P}^2$ defined by the vanishing of the homogeneous polynomial $g = x_0x_2 - x_1^2$. Then X is irreducible.

Proving the irreducibility of a projective variety, in general, is not an easy task. We do that, in this case, introducing a method that we will refine later.

Assume that X is the union of two proper closed subsets X_1, X_2, where X_i is defined by the vanishing of homogeneous polynomials in the set J_i.

We consider the map $f : \mathbb{P}^1 \to \mathbb{P}^2$ defined by sending each point $P = [y_0 : y_1]$ to the point $f(P) = [y_0^2 : y_0y_1 : y_1^2]$ of $\mathbb{P}^2$. It is immediate to check, indeed, that the point $f(P)$ does not depend on the choice of a particular pair of homogeneous coordinates for P. Here f is simply a set-theoretic map. We will see, later, that f has relevant geometric properties.

The image of f is contained in X, for any point with homogeneous coordinates $[x_0 : x_1 : x_2] = [y_0^2 : y_0y_1 : y_1^2]$ annihilates g. Moreover the image of f is exactly X. Indeed let $Q = [q_0 : q_1 : q_2]$ be a point of X. Fix elements $b, c \in \mathbb{C}$ such that $b^2 = q_0$ and $c^2 = q_2$. Then we cannot have both b, c equal to 0, for in this case $q_0 = q_2 = 0$ and also $q_1 = 0$, because $g(Q) = 0$, a contradiction. Moreover $(bc)^2 = q_0q_2 = q_1^2$. Thus, after possibly the change of the sign of one between b and c, we may also assume $bc = q_1$. Then:

$$f([b : c]) = [b^2 : bc : c^2] = [q_0 : q_1 : q_2] = Q.$$

The map f is one-to-one. To see this, assume $f([b : c]) = f([b' : c'])$. Then $(b'^2, b'c', c'^2)$ is equal to (b^2, bc, c^2) multiplied by some nonzero scalar $z \in \mathbb{C}$. Taking a suitable square root w of z, we may assume $b' = wb$. We have $c' = \pm wc$, but if $c' \neq wc$ then $b'c' = -zbc \neq zbc$, a contradiction. Thus also $c' = wc$ and (b', c'), (b, c) define the same point in $\mathbb{P}^1$.

In conclusion, f is a bijective map $f : \mathbb{P}^1 \to X$.

Next, we prove that $Z_1 = f^{-1}(X_1)$ is closed in $\mathbb{P}^1$. Indeed for any polynomial $p = p(y_0, y_1, y_2) \in J_1$ consider the polynomial $q = p(x_0^2, x_0x_1, x_1^2) \in \mathbb{C}[x_0, x_1]$. It is immediate to check that any $P \in \mathbb{P}^1$ satisfies $q(P) = 0$ if and only if $f(P)$ satisfies $p(f(P)) = 0$. Thus Z_1 is the projective variety in $\mathbb{P}^1$ associated to the set of homogeneous polynomials

$$J' = \{p(x_0^2, x_0x_1, x_1^2) : p \in J_1\} \subset \mathbb{C}[x_0, x_1].$$

Similarly $Z_2 = f^{-1}(X_2)$ is closed in $\mathbb{P}^1$.

Since f is bijective, then Z_1, Z_2 are proper closed subset of $\mathbb{P}^1$, whose union is $\mathbb{P}^1$. This contradicts the irreducibility of $\mathbb{P}^1$.

We will see (Exercise 28) that any linear variety is irreducible.

Example 9.1.34 Let X be the variety in $\mathbb{P}^2$ defined by the set of homogeneous polynomials $J = \{x_0x_1, x_0(x_0 - x_2)\}$. Then X is the union of the sets $L_1 = \{[x_0 : x_1 : x_2] : x_0 = 0\}$ and $L_2 = \{[x_0 : x_1 : x_2] : x_1 = 0, x_0 = x_2\}$. These are both linear varieties, hence they are irreducible (L_2 is indeed a singleton). Moreover $L_1 \cap L_2 = \emptyset$. It follows that X is not irreducible: L_1, L_2 are its irreducible components.

Definition 9.1.35 We say that a polynomial $f \in \mathbb{C}[x_0, \ldots, x_n]$ is *irreducible* when $f = g_1g_2$ implies that either g_1 or g_2 are constant.

Irreducible polynomials are the basic elements that determine a factorization of every polynomial. We refer to [1] I.14 for a proof of the following statement, which establishes a link between irreducible hypersurfaces and irreducible polynomials.

Theorem 9.1.36 (Unique factorization) *Any polynomial f can be written as a product $f = f_1f_2 \cdots f_h$ where the f_i's are irreducible polynomials. The f_i's are called* irreducible factors *of f and are unique up to scalar, in the sense that if $f = g_1 \cdots g_s$,*

with each g_j irreducible, then $h = s$ and, after a possible permutation, there are scalars $c_1, \ldots, c_h \in \mathbb{C}$ with $g_i = c_i f_i$ for all i.

If f is homogeneous, also the irreducible factors of f are homogeneous.

Notice that the irreducible factors of f need not be distinct. In any event, the irreducible factors of a product fg are the union of the irreducible factors of f and the irreducible factors of g.

Example 9.1.37 Let $X = X(f)$ be a hypersurface and take a decomposition $f = f_1 \cdots f_h$ of f into irreducible factors. Then the $X(f_i)$'s are precisely the irreducible components of X.

To prove this, first notice that when f is irreducible, then X is irreducible. Indeed assume that $X = X_1 \cup X_2$, where X_1, X_2 are closed and none of them contains X. Then take f_1 (respectively f_2) in the radical ideal of X_1 (respectively X_2) and such that X is not contained in $X(f_1)$ (respectively in $X(f_2)$). We have $X_1 \subset X(f_1)$ and $X_2 \subset X(f_2)$, thus:

$$X(f) \subset X(f_1) \cup X(f_2) = X(f_1 f_2).$$

It follows that $f_1 f_2$ belongs to the radical of the ideal generated by f, thus some power of $f_1 f_2$ belongs to the ideal generated by f, i.e. there is an equality

$$(f_1 f_2)^n = fh$$

for some exponent n and some polynomial h. It follows that f is either an irreducible factor of either f_1 or f_2. In the former case $f_1 = fh_1$ hence $X(f_1)$ contains X. In the latter, $X(f_2)$ contains X.

In particular, X is irreducible if and only if f has a unique irreducible factor. This clearly happens when f is irreducible, but also when f is a power of an irreducible polynomial.

9.2 Multiprojective Varieties

Let us move to consider *products* of projective spaces, which we will call also *multiprojective spaces*.

The nonexpert reader would be surprised, at first, by knowing that a product of projective spaces is *not* trivially a projective space itself.

For instance, consider the product $\mathbb{P}^1 \times \mathbb{P}^1$, whose points have a pair of homogeneous coordinates $([x_0 : x_1], [y_0 : y_1])$. These pairs can be multiplied separately by two different scalars. Thus, $([1 : 1], [1 : 2])$ and $([2 : 2], [1 : 2])$ represent the same point of the product. On the other hand, the most naïve association with a point in a projective space yields to relate $([x_0 : x_1], [y_0 : y_1])$ with $[x_0 : x_1 : y_0 : y_1]$ (which, by the way, sits in $\mathbb{P}^3$), but $([1 : 1 : 1 : 2])$ and $([2 : 2 : 1 : 2])$ are different points in $\mathbb{P}^3$.

We will see in the next chapters how a product can be identified with a subset (indeed, with a projective variety) of a higher dimensional projective space.

By now, we develop independently a theory for products of projective spaces and their relevant subsets: multiprojective varieties.

Remark 9.2.1 Consider a product $\mathbb{P} = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$. A point $P \in \mathbb{P}$ corresponds to an equivalence class whose elements are n-tuples

$$((p_{1,0}, \ldots, p_{1,a_1}), \ldots, (p_{n,0}, \ldots, p_{n,a_n}))$$

where, for all i, $(p_{i,0}, \ldots, p_{i,a_i}) \neq 0$. Two such elements

$$P = ((p_{1,0}, \ldots, p_{1,a_1}), \ldots, (p_{n,0}, \ldots, p_{n,a_n}))$$
$$Q = ((q_{1,0}, \ldots, q_{1,a_1}), \ldots, (q_{n,0}, \ldots, q_{n,a_n}))$$

belong to the same class when there are scalars $k_1, \ldots k_n \in \mathbb{C}$ (all of them necessarily nonzero) such that, for all i, j, $q_{ij} = k_i p_{ij}$.

We will denote the elements of the equivalence class that define P as sets of multihomogeneous coordinates for P, writing

$$P = ([p_{1,0} : \cdots : p_{1,a_1}], \ldots, [p_{n,0} : \cdots : p_{n,a_n}]).$$

Since we want to construct a projective geometry for multiprojective spaces, we need to define the vanishing of a polynomial

$$f \in \mathbb{C}[x_{1,0}, \ldots, x_{1,a_1}, \ldots, x_{n,0}, \ldots, x_{n,a_n}]$$

at a point P of the product above. This time, it is not sufficient that f is homogeneous, because subsets of coordinates referring to factors of the product can be scaled independently.

Definition 9.2.2 A polynomial $f \in \mathbb{C}[x_{1,0}, \ldots, x_{1,a_1}, \ldots, x_{n,0}, \ldots, x_{n,a_n}]$ is *multihomogeneous* of multidegree $(d_1, \ldots, d_n)$ if f, considered as a polynomial in the variables $x_{i,0}, \ldots, x_{i,a_i}$, is homogeneous of degree d_i, for every i.

Strictly speaking, the definition of a multihomogeneous polynomial in a polynomial ring $\mathbb{C}[x_0, \ldots, x_N]$ makes sense only after we defined a partition in the set of variables. Moreover, if we change the partition, the notion of multihomogeneous polynomial also changes.

Notice, however, that a partition is canonically determined when we consider the polynomial ring $\mathbb{C}[x_{1,0}, \ldots, x_{1,a_1}, \ldots, x_{n,0}, \ldots, x_{n,a_n}]$ associated to the multiprojective space $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$.

Multihomogeneous polynomials are homogeneous, but the converse is false.

Example 9.2.3 Consider the polynomial ring $\mathbb{C}[x_0, x_1, y_0, y_1]$, with the partition $\{x_0, x_1\}, \{y_0, y_1\}$, and consider the two homogenous polynomials

$$f_1 = x_0^2 y_0 + 2x_0 x_1 y_1 - 3x_1^2 y_0 \qquad f_2 = x_0^3 - 2x_1 y_0 y_1 + x_0 y_1^2.$$

Then f_1 is multihomogeneous (of multidegree (2, 1)) while f_2 is not multihomogeneous.

Example 9.2.4 In $\mathbb{C}[x_{1,0}, \ldots, x_{1,a_1}, \ldots, x_{n,0}, \ldots, x_{n,a_n}]$ the homogeneous linear polynomial $x_{1,0} + \cdots + x_{1,a_1} + \cdots + x_{n,0} + \cdots + x_{n,a_n}$ is never multihomogeneous, except for the trivial partition.

For the trivial partition, homogeneous and multihomogeneous polynomials coincide.

If for any i one takes a homogeneous polynomial $f_i \in \mathbb{C}[x_{i,0}, \ldots, x_{i,a_i}]$ of degree d_i, then the product $f_1 \cdots f_n$ is multihomogeneous of multidegree $(d_1, \ldots, d_n)$, in the ring $\mathbb{C}[x_{1,0}, \ldots, x_{1,a_1}, \ldots, x_{n,0}, \ldots, x_{n,a_n}]$ with the natural partition.

It is immediate to verify that given two representatives of the same class in $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$:

$$P = ((p_{1,0}, \ldots, p_{1,a_1}), \ldots, (p_{n,0}, \ldots, p_{n,a_n}))$$
$$Q = ((q_{1,0}, \ldots, q_{1,a_1}), \ldots, (q_{n,0}, \ldots, q_{n,a_n})),$$

when $f \in \mathbb{C}[x_{1,0}, \ldots, x_{1,a_1}, \ldots, x_{n,0}, \ldots, x_{n,a_n}]$ is multihomogeneous of any multidegree, then $f(P) = 0$ if and only if $f(Q) = 0$.

As a consequence, one can define the vanishing of f at a point $P = ([p_{1,0} : \cdots : p_{1,a_1}], \ldots, [p_{n,0} : \cdots : p_{n,a_n}])$ of the product, as the vanishing of f at any set of multihomogeneous coordinates.

Definition 9.2.5 We call *multiprojective variety* every subset $X \subset \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$ defined by the vanishing of a family J of multihomogeneous polynomials

$$J \subset \mathbb{C}[x_{1,0}, \ldots, x_{1,a_1}, \ldots, x_{n,0}, \ldots, x_{n,a_n}].$$

We will, as usual, write $X(J)$ to denote the multiprojective variety defined by J.

Example 9.2.6 Consider the product $\mathbb{P} = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$ and consider, for all i, a projective variety X_i in $\mathbb{P}^{a_i}$. Then the product $X_1 \times \cdots \times X_n$ is a multiprojective variety in $\mathbb{P}$.

Indeed, assume that X_i is defined by a finite subset J_i of homogeneous polynomials in the variables $x_{i,0}, \ldots, x_{i,a_i}$. For all i extend J_i to a finite set K_i which defines the empty set in $\mathbb{P}^{a_i}$ (e.g. just add to J_i all the coordinates in $\mathbb{P}^{a_i}$). Then $X_1 \times \cdots \times X_n$ is defined by the (finite) set of products of homogeneous polynomials:

$$J = \{f_1 \cdots f_n : f_j \in K_j \ \forall j \text{ and } \exists i \text{ with } f_i \in J_i\}.$$

Namely, take $P = (P_1, \ldots, P_n) \in \mathbb{P}$. Clearly if $P \in X_1 \times \cdots \times X_n$ then P annihilates all the elements in J. Conversely, if $P \notin X_1 \times \cdots \times X_n$, say $P_j \notin X_j$, then for every $i \neq j$ take g_i any homogeneous polynomial in K_i such that $g_i(P_i) \neq 0$, and

take $g_j \in J_j$ such that $g_j(P_j) \neq 0$. Clearly the product $g_1 \cdots g_n$ belongs to J and it does not vanish at P.

Example 9.2.7 There are multiprojective varieties that are not a product of projective varieties.

For instance, consider the multiprojective variety X defined by $x_0y_1 - x_1y_0$ in the product $\mathbb{P}^1 \times \mathbb{P}^1$. It is easy to see that X does not coincide with $\mathbb{P}^1 \times \mathbb{P}^1$ (Exercise 29), but for each point $P \in \mathbb{P}^1$ the point (P, P) of the product sits in X. Thus X cannot be the product of two subsets of $\mathbb{P}^1$, one of which is a proper subset.

Most properties introduced in the previous section for projective varieties also hold for multiprojective varieties. We give here a short survey (the proofs are left as an exercise).

Remark 9.2.8 Let $X(J)$ be a multiprojective variety, defined by a set J of multihomogeneous polynomials. Then for any $J' \subset J$, the variety $X(J')$ contains $X(J)$.

One can have $X(J') = X(J)$ even if J' is properly contained in J.

For instance, $X(J)$ is also defined by the ideal $\langle J \rangle$ and by its radical $\sqrt{\langle J \rangle}$.

Remark 9.2.9 Multiprojective varieties define a family of closed subset for a topology on a product $\mathbb{P} = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$. We call this topology the *Zariski topology* on $\mathbb{P}$.

$\mathbb{P}$ is irreducible and compact, in the Zariski topology. Thus every multiprojective variety is itself compact, in the induced topology.

Remark 9.2.10 The Basis Theorem 9.1.22 guarantees that every multihomogeneous variety is the zero locus of a *finite* set of multihomogeneous polynomials.

Every multihomogeneous variety is indeed the intersection of a finite number of *hypersurfaces* in $\mathbb{P} = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$, where a hypersurface is defined as a multiprojective variety $X(J)$, where J is a singleton.

Theorem 9.2.11 *Let X be a multiprojective variety. Then there exists a unique decomposition of X as the union*

$$X = X_1 \cup \cdots \cup X_k$$

where $X_1, \ldots, X_k$ are irreducible multiprojective varieties: the irreducible components of X.

9.3 Projective and Multiprojective Maps

A theory of algebraic objects in Mathematics cannot be considered complete unless one introduces also the notion of *good* maps between the objects.

We define in this section a class of maps between projective and multiprojective varieties, that are good for our purposes. We will call them *projective or multiprojective maps*.

In principle, a projective map is a map which is described by polynomials. Unfortunately, one cannot take this as a global definition, because it is too restrictive and would introduce some undesirable phenomenon.

Instead, we define projective maps in terms of a *local* description by polynomials.

Definition 9.3.1 Let $X \subset \mathbb{P}^n$ and $Y \subset \mathbb{P}^m$ be projective varieties. We say that a map $f : X \to Y$ is *projective* if the following property holds:

for any $P \in X$ there exist an open set U of X (in the Zariski topology), containing P, and $m+1$ polynomials $f_0, \ldots, f_m \in \mathbb{C}[x_0, \ldots, x_n]$, homogeneous *of the same degree*, such that for all $Q = [q_0 : \cdots : q_n] \in U$:

$$f(Q) = [f_0(q_0, \ldots, q_n) : \cdots : f_m(q_0, \ldots, q_n)].$$

We will also write that, over U, the map f is given parametrically by the system:

$$\begin{cases} y_0 & = f_0(x_0, \ldots, x_n) \\ \cdots & \quad \cdots \\ y_m & = f_m(x_0, \ldots, x_m) \end{cases}$$

Thus, X is covered by open subsets on which f is defined by polynomials. Since X is compact, we can always assume that the cover is finite.

Notice that if we take another set of homogeneous coordinates for the point $Q \in U$, i.e. we write $Q = [cq_0 : \cdots : cq_n]$, where c is a nonzero scalar, then since the polynomials are homogeneous of the same degree, say degree d, we get $f_i(cq_0, \ldots, cq_n) = c^d f_i(q_0, \ldots, q_n)$ for all i. Thus $f(Q)$ is independent on the choice of a specific set of homogeneous coordinates for Q.

We may always consider a projective map $f : X \to Y \subset \mathbb{P}^m$ as a map from X to the projective space $\mathbb{P}^m$. The following proposition shows that when the domain X of f is a projective space itself, then the localization to open sets, in the definition of projective maps, is useless.

Proposition 9.3.2 *Let $f : \mathbb{P}^n \to \mathbb{P}^m$ be a projective map. Then there exists a set of $m+1$ homogeneous polynomials $f_0, \ldots, f_m \in \mathbb{C}[x_0, \ldots, x_n]$ of the same degree, such that $f(Q)$ is defined by $f_0, \ldots, f_m$ for all $Q \in \mathbb{P}^n$.*

In other words, in the definition we can always take just one open subset $U = \mathbb{P}^n$.

Proof Take two open subsets U_1, U_2 where f is defined, respectively, by homogeneous polynomials $g_0, \ldots, g_m$ and $h_0, \ldots, h_m$ of the same degree. Since $\mathbb{P}^n$ is irreducible, then $U = U_1 \cap U_2$ is a non-empty, dense open subset. For any point $P \in U$ there exists a scalar $\alpha_P \in \mathbb{C} - \{0\}$ such that, if $P = [p_0 : \cdots : p_n]$, then

$$(g_0(p_0, \ldots, p_n), \ldots, g_m(p_0, \ldots, p_n)) = \alpha_P (h_0(p_0, \ldots, p_n), \ldots, h_m(p_0, \ldots, p_n)).$$

It follows that the homogeneous polynomials

$$g_j h_i - g_i h_j$$

vanish at all the points of U. Thus they must vanish at all the points of $\mathbb{P}^n$, since U is dense. In particular they vanish in all the points of $U_1 \cup U_2$. It follows immediately that for any $P \in U_1 \cup U_2$, $P = [p_0 : \cdots : p_n]$, the sets of coordinates $[g_0(p_0, \ldots, p_n) : \cdots : g_m(p_0, \ldots, p_n)]$ and $[h_0(p_0, \ldots, p_n) : \cdots : h_m(p_0, \ldots, p_n)]$ determine the same, well defined point of $\mathbb{P}^m$. The claim follows. □

After Proposition 9.3.2 one may wonder if the local definition of projective maps is really necessary.

Well, *it is*, as illustrated in the following Example 9.3.4.

The fundamental point is that necessarily the polynomials $f_0, \ldots, f_m$ that define the projective map f over U, cannot have a common zero $Q \in U$, otherwise, the map would not be defined in Q. Sometimes this property cannot be obtained globally by a unique set of polynomials. It is necessary to use an open cover and vary the polynomials, passing from one open subset to another one.

Example 9.3.3 Assume $n \leq m$ and consider the map between projective spaces $f : \mathbb{P}^n \to \mathbb{P}^m$, defined globally by polynomials $f_0(x_0, \ldots, x_n)$, ..., $f_m(x_0, \ldots, x_n)$, where:

$$f_i(x_0, \ldots, x_n) = \begin{cases} x_i & \text{if } i \leq n \\ 0 & \text{otherwise.} \end{cases}$$

Then f is a projective map. It is obvious that f is injective.

Be careful that the map f would not exist for $n > m$.

Indeed, if for instance $n = m + 1$, then the image of the point $P \in \mathbb{P}^n$, with coordinates $[0 : \cdots : 0 : 1]$, would be the point of coordinates $[0 : \cdots : 0]$, which does not exist in $\mathbb{P}^m$.

Example 9.3.4 Let X be the hypersurface of the projective plane $\mathbb{P}^2$, defined by $g(x_0, x_1, x_2) = x_0^2 + x_1^2 - x_2^2$. People can immediately realize that X corresponds to the usual circle of analytic geometry.

We define a projective map (the *stereographic projection*) $f : X \to \mathbb{P}^1$ (see Figure 9.1).

Consider the two hypersurfaces $X(h_1), X(h_2)$, where h_1, h_2 are respectively equal to $x_1 - x_2$ and $x_1 + x_2$. Notice that $X(h_1) \cap X(h_2)$ is just the point of coordinates $[1 : 0 : 0]$, which does not belong to X. Thus the open subsets of the plane $(X(h_1))^c$, $(X(h_2))^c$ cover X. Define two open subsets of X by $U_1 = X \cap (X(h_1))^c$, $U_2 = X \cap (X(h_2))^c$.

Then define the map f as follows:

$$\text{on } U_1,\ f = \begin{cases} y_0 &= x_0 \\ y_1 &= x_1 - x_2 \end{cases}, \qquad \text{on } U_2,\ f = \begin{cases} y_0 &= x_1 + x_2 \\ y_1 &= -x_0 \end{cases}.$$

We need to prove that the definition is consistent in the intersection $U_1 \cap U_2$. Notice that if $Q = [0 : q_1 : q_2]$ belongs to X, then $q_1^2 - q_2^2 = 0$, so that Q belongs either to U_1 or to U_2, but not to both, since one cannot have $q_1 = q_2 = 0$. Thus any point

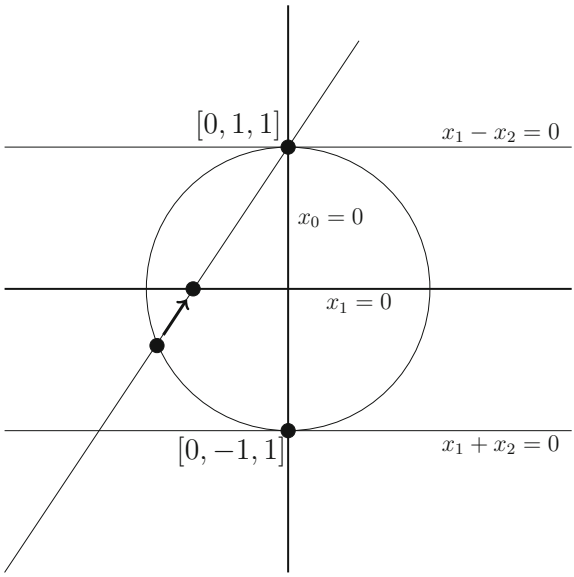


Fig. 9.1 Stereographic projection

$Q = [q_0 : q_1 : q_2] \in U_1 \cap U_2$ satisfies $q_0 \neq 0$. Since clearly Q also satisfies $q_1 + q_2 \neq 0$, then:

$$[q_0 : q_1 - q_2] = [q_0(q_1 + q_2) : (q_1 - q_2)(q_1 + q_2)] = \\ = [q_0(q_1 + q_2) : q_1^2 - q_2^2] = [q_0(q_1 + q_2) : -q_0^2] = [q_1 + q_2 : -q_0].$$

Thus f is a well defined projective map.

Notice that the two polynomials that define the map on U_1 cannot define the map globally, because $X \setminus U_1$ contains the point $[0 : 1 : 1]$, where both x_0 and $x_1 - x_2$ vanish.

The map f is one-to-one and onto. Indeed if $B = [b_0 : b_1] \in \mathbb{P}^1$, then $f^{-1}(B)$ consists of the unique point $[2b_0b_1 : -b_0^2 + b_1^2 : -b_0^2 - b_1^2]$, as one can easily compute.

Proposition 9.3.5 *Projective maps are continuous in the Zariski topology.*

Proof Consider $X \subset \mathbb{P}^n$ and a projective map $f : X \to Y \subset \mathbb{P}^m$. We may assume indeed $Y = \mathbb{P}^m$, to prove the continuity. Let U be an open subset of $\mathbb{P}^m$, which is the complement of a hypersurface $X(g)$, for some $g \in \mathbb{C}[y_0, \ldots, y_m]$. Let Ψ be an open subset of X where f is defined by the polynomials $f_0, \ldots, f_m$. Then $f^{-1}(U) \cap \Psi$ is the intersection of Ψ with the complement of the hypersurface defined by the homogeneous polynomial $g(f_0, \ldots, f_m) \in \mathbb{C}[x_0, \ldots, x_n]$. It follows that $f^{-1}(U)$ is a union of open sets, hence it is open.

Since every open subset of $\mathbb{P}^m$ is a (finite) union of complements of hypersurfaces (see Exercise 27), the claim follows. □

Definition 9.3.6 We will say that a projective map $f : X \to Y$ is an *isomorphism* if there is a projective map $g : Y \to X$ such that $g \circ f =$ identity on X and $f \circ g =$ identity on Y.

Equivalently, a projective map f is an isomorphism if it is one-to-one and onto, and the set-theoretic inverse g is itself a projective map.

Example 9.3.7 Let us prove that the map f defined in Example 9.3.4, from the hypersurface $X \subset \mathbb{P}^2$ defined by the polynomial $x_0^2 + x_1^2 - x_2^2$ to $\mathbb{P}^1$ is an isomorphism.

We already know, indeed, what is the inverse of f: it is the map $g : \mathbb{P}^1 \to X$ defined parametrically by

$$\begin{cases} x_0 &= 2y_0y_1 \\ x_1 &= -y_0^2 + y_1^2 \\ x_2 &= -y_0^2 - y_1^2 \end{cases}$$

It is immediate to check, indeed, that both $g \circ f$ and $f \circ g$ are the identity on the respective spaces.

Remark 9.3.8 We are now able to prove that the map f of Example 9.3.4 cannot be defined globally by a pair of polynomials:

$$\begin{cases} y_0 &= p_0(x_0, x_1, x_2) \\ y_1 &= p_1(x_0, x_1, x_2) \end{cases}$$

Otherwise, since the map g defined in the previous example is the inverse of f, we would have that for any choice of $Q = (b_0, b_1) \neq (0, 0)$, the homogeneous polynomials

$$h_{b_0,b_1} = b_1 p_0(y_0y_1, y_1^2 - y_2^2, -y_1^2 - y_2^2) - b_0 p_1(y_0y_1, y_1^2 - y_2^2, -y_1^2 - y_2^2),$$

whose vanishing defines $f(g(Q))$, vanishes at a single point of $\mathbb{P}^1$. Notice that the degree d of any h_{b_0,b_1} is at least 2.

Since $\mathbb{C}$ is algebraically closed, a homogeneous polynomial in two variables that vanishes at a single point is a power of linear form. Thus any polynomial h_{b_0,b_1} is a d-th power of a linear form. In particular there are scalars a_0, a_1, c_0, c_1 such that:

$$h_{1,0} = p_0(y_0y_1, y_1^2 - y_2^2, -y_1^2 - y_2^2) = (a_0y_0 - a_1y_1)^d$$
$$h_{0,1} = p_1(y_0y_1, y_1^2 - y_2^2, -y_1^2 - y_2^2) = (c_0y_0 - c_1y_1)^d$$

Notice that the point $Q' = [a_1 : a_0]$ cannot be equal to $[c_1 : c_0]$, otherwise both p_0, p_1 would vanish at $g(Q') \in X$. Then $h_{1,-1} = (a_0y_0 - a_1y_1)^d - (c_0y_0 - c_1y_1)^d$ vanishes at two different points, namely $[a_1 + c_1 : a_0 + c_0]$ and $[ea_1 + c_1 : ea_0 + c_0]$, where e is any d-root of unit, different from 1.

In the case of multiprojective varieties, most definitions and properties above can be rephrased and proved straightforwardly.

Definition 9.3.9 Let $X \subset \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$ be a multiprojective variety. A map $f : X \to \mathbb{P}^m$ is a *projective map* if there exists a open cover $\{U_i\}$ of the domain X such that f is defined over each U_i by multihomogeneous polynomials, all of the same multidegree.

In other words f is multiprojective if for any U_i of a cover there are multihomogeneous polynomials $f_0, \ldots, f_m$ in $\mathbb{C}[x_{0,1}, \ldots, x_{n,a_n}]$, of the same multidegree, such that for all $P \in X$, $P = ([p_{0,1} : \cdots : p_{0,a_1}], \ldots, [p_{n,1} : \cdots : p_{n,a_n}])$, then $f(P)$ has coordinates

$$f(P) = [f_0((p_{0,1}, \ldots, p_{0,a_1}), \ldots, (p_{n,1}, \ldots, p_{n,a_n})) : \ldots \\ \cdots : f_m((p_{0,1}, \ldots, p_{0,a_1}), \ldots, (p_{n,1}, \ldots, p_{n,a_n}))).$$

We will write in parametric form:

$$\begin{cases} y_0 &= f_0(x_{0,1}, \ldots, x_{n,a_n}) \\ \cdots & \quad \cdots \\ y_m &= f_m(x_{0,1}, \ldots, x_{n,a_n}) \end{cases}$$

We will say that $f : X \to \mathbb{P}^{b_1} \times \cdots \times \mathbb{P}^{b_m}$ is a *multiprojective map* if all of its components are.

Remark 9.3.10 The composition of two multiprojective maps is a multiprojective map.

The identity from a multiprojective variety to itself is a multiprojective map.

Multiprojective maps are continuous in the Zariski topology.

Proposition 9.3.11 *Let $f : \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n} \to \mathbb{P}^m$ be a multiprojective map. Then there exists a set of $m+1$ multihomogeneous polynomials $f_0, \ldots, f_m$ of the same multidegree, such that $f(Q)$ is defined by $f_0, \ldots, f_m$ for all $Q \in \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$.*

9.4 Exercises

Exercise 19 Prove the last assertion of Example 9.1.8. If $\{p_1, \ldots, p_m\}$ is a collection of *linear* homogeneous polynomials in $\mathbb{C}[t_0, \ldots, t_n]$, then the projective variety X defined by $\{p_1, \ldots, p_m\}$ can be canonically identified with the projective space $\mathbb{P}(W)$ over the linear subspace $W \subset \mathbb{C}^{n+1}$ defined by the p_i's.

Exercise 20 Prove that if X is the projective variety defined by a set J of homogeneous polynomials and $J' \subset J$, then $X' = X(J')$ contains X.

Find an examples of different subsets $J \neq J'$ of linear polynomials such that $X(J) = X(J')$.

Exercise 21 Prove that if I is a homogeneous ideal, then also $\sqrt{I}$ is a homogeneous ideal and $X(I) = X(\sqrt{I})$.

Exercise 22 Prove that if Y is an irreducible subset, then the closure of Y is irreducible.

Prove that if an irreducible subset Y is contained in a finite union of closed subsets $X_1 \cup \cdots \cup X_m$, then Y is contained in some X_i.

Prove that if $Y_1 \subset \ldots Y_i \subset \ldots$ is an ascending chain of irreducible subsets, then $\bigcup Y_i$ is irreducible.

Exercise 23 Prove that if X is an irreducible subset of a topological space, then X is not the union of a finite number of subsets $Y_i \subset X$ which are closed in the induced topology.

Exercise 24 Prove that if X_1 and X_2 are topological spaces, and X_1 has the separation property T_1, then for any $Q \in X_1$ the *fiber* $\{Q\} \times X_2$ is a closed subset of $X_1 \times X_2$ which is homeomorphic to X_2.

Prove that if X_1 and X_2 are irreducible, and one of them has the property T_1, then also the product $X_1 \times X_2$ is irreducible.

Exercise 25 Determine which Hausdorff topological space can be irreducible.

Exercise 26 Prove that if X is a finite projective variety, then the irreducible components of X are its singletons.

Exercise 27 Prove that any open subset of a projective variety X is covered by a finite union of open subsets which are the intersection of X with the complement of a hypersurface.

Exercise 28 Prove that if p is an irreducible polynomial and $c \in \mathbb{C}$, then also cp is irreducible.

Prove that every linear polynomial is irreducible.

Exercise 29 Prove that the multiprojective variety X defined by $x_0y_1 - x_1y_0$ in the product $Y = \mathbb{P}^1 \times \mathbb{P}^1$ does not coincide with Y.

Exercise 30 Prove that the composition of two projective maps is a projective map.

Prove that the identity from a projective variety to itself is a projective map.

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Chapter 10
Projective Maps and the Chow's Theorem



The chapter contains the proof of the Chow's Theorem, a fundamental result for algebraic varieties with an important consequence for the study of statistical models. It states that, over an algebraically closed field, like $\mathbb{C}$, the image of a projective (or multiprojective) variety X under a projective map is a Zariski closed subset of the target space, i.e., it is itself a projective variety.

The proof of Chow's Theorem requires an analysis of projective maps, which can be reduced to a composition of linear maps, Segre maps and Veronese maps.

The proof also will require the introduction of a basic concept of the elimination theory, i.e., the resultant of two polynomials.

10.1 Linear Maps and Change of Coordinates

We start by analyzing projective maps induced by linear maps of vector spaces.

The nontrivial case concerns linear maps which are surjective but not injective. After a change of coordinates, such maps induce maps between projective varieties that can be described as *projections*.

Despite the fact that the words *projective* and *projection* have a common origin (in the paintings of the Italian Renaissance, e.g.) projections not always give rise to projective maps.

The description of the image of a projective variety under projections relies indeed on nontrivial algebraic tools: the rudiments of the *elimination theory*.

Let us start with a generalization of Example 9.3.3.

Definition 10.1.1 Consider a linear map $\phi : \mathbb{C}^{n+1} \to \mathbb{C}^{m+1}$ *which is injective*.

Then ϕ defines a projective map (which, by abuse, we will still denote by ϕ) between the projective spaces $\mathbb{P}^n \to \mathbb{P}^m$, as follows:

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C. Bocci and L. Chiantini, *An Introduction to Algebraic Statistics with Tensors*, UNITEXT 118, https://doi.org/10.1007/978-3-030-24624-2_10

– for all $P \in \mathbb{P}^n$, consider a set of homogeneous coordinates $[x_0 : \cdots : x_n]$ and send P to the point $\phi(P) \in \mathbb{P}^m$ with homogeneous coordinates $[\phi(x_0, \ldots, x_n)]$.

Such maps are called *linear projective maps*.

It is clear that the point $\phi(P)$ does not depend on the choice of a set of coordinates for P, since ϕ is linear.

Notice that we cannot define a projective map in the same way when ϕ is not injective. Indeed, in this case, the image of a point P whose coordinates lie in the Kernel of ϕ would be indeterminate.

Since any linear map $\mathbb{C}^{n+1} \to \mathbb{C}^{m+1}$ is defined by linear homogeneous polynomials, then it is clear that the induced map between projective spaces is indeed a projective map.

Example 10.1.2 Assume that the linear map ϕ is an isomorphism of $\mathbb{C}^{n+1}$. Then the corresponding linear projective map is called a *change of coordinates*.

Indeed ϕ corresponds to a change of basis inside $\mathbb{C}^{n+1}$.

The associated map $\phi : \mathbb{P}^n \to \mathbb{P}^n$ is an isomorphism, since the inverse isomorphism ϕ^{-1} determines a projective map which is the inverse of ϕ.

Remark 10.1.3 By construction, any change of coordinates in a projective space is a homeomorphism of the corresponding topological space, in the Zariski topology.

So, the image of a projective variety under a change of coordinates is still a projective variety.

From now on, when dealing with projective varieties, we will freely act with the change of coordinates on them.

The previous remark generalizes to any linear projective map.

Proposition 10.1.4 *For every injective map $\phi : \mathbb{C}^{n+1} \to \mathbb{C}^{m+1}$, $m \geq n$, the associated linear projective map $\phi : \mathbb{P}^n \to \mathbb{P}^m$ sends projective subvarieties of $\mathbb{P}^n$ to projective subvarieties of $\mathbb{P}^m$.*

In topological terms, any linear projective map is closed *in the Zariski topology, i.e., it sends closed sets to closed sets.*

Proof The linear map ϕ factorizes in a composition $\phi = \psi \circ \phi'$ where ϕ' is the inclusion which sends $[x_0 : \cdots : x_n]$ to $[x_0 : \cdots : x_n : 0 : \cdots : 0]$, $m - n$ zeroes, and ψ is a change of coordinates (notice that we are identifying the coordinates in $\mathbb{P}^n$ with the first $n + 1$ coordinates in $\mathbb{P}^m$).

Thus, up to a change of coordinates, any linear projective map can be reduced to the map that embeds $\mathbb{P}^n$ into $\mathbb{P}^m$ as the linear space defined by equations $x_{n+1} = \cdots = x_m = 0$.

It follows that if $X \subset \mathbb{P}^n$ is the subvariety defined by homogeneous polynomials $f_1, \ldots, f_s$ then, up to a change of coordinates, the image of X is the projective subvariety defined in $\mathbb{P}^m$ by the polynomials $f_1, \ldots, f_s, x_{n+1}, \ldots, x_m$. □

The definition of linear projective maps, which requires that ϕ is injective, becomes much more complicated if we drop the injectivity assumption.

Let $\phi : \mathbb{C}^{n+1} \to \mathbb{C}^{m+1}$ be a *non* injective linear map. In this case, we cannot define through ϕ a projective map $\mathbb{P}^n \to \mathbb{P}^m$ as above, since for any vector $(p_0, \ldots, p_n)$ in the kernel of ϕ, the image of the point $[p_0 : \cdots : p_n]$ is undefined, because $\phi(p_0, \ldots, p_n)$ vanishes.

On the other hand, the kernel of ϕ defines a projective linear subspace of $\mathbb{P}^n$, the *projective kernel*, which will be denoted by K_ϕ.

If $X \subset \mathbb{P}^n$ is a subvariety *which does not meet* K_ϕ, then the restriction of ϕ to the coordinates of the points of X determines a well-defined map from X to $\mathbb{P}^m$.

Example 10.1.5 Consider the point $P_0 \in \mathbb{P}^m$ of projective coordinates $[1 : 0 : 0 : \cdots : 0]$ and let M be the linear subspace of $\mathbb{C}^{m+1}$ of the points with first coordinate equal to 0, i.e., $M = X(x_0)$. Let $\phi_0 : \mathbb{C}^{m+1} \to M$ be the linear surjective (but not injective) map which sends a vector $(x_0, x_1, \ldots, x_m)$ to $(0, x_1, \ldots, x_m)$.

Notice that M defines a linear projective subspace $\mathbb{P}(M) \subset \mathbb{P}^m$, of projective dimension $m - 1$ (i.e., a hyperplane), and $P_0 \notin \mathbb{P}(M)$. Moreover P_0 is exactly the projective kernel of ϕ_0. Let Q be any point of $\mathbb{P}^m$ different from P_0. If $Q = [q_0 : q_1 : \cdots : q_m]$ then $\phi_0(Q) = (0, q_1, \ldots, q_m)$ determines a well defined projective point, which corresponds to the intersection of $\mathbb{P}(M)$ with the line P_0Q. This is the reason why we call ϕ_0 the *projection from* P_0 *to* $\mathbb{P}(M)$. Notice that we cannot define a global projection $\mathbb{P}^m \to \mathbb{P}(M)$, since it would not be defined in P_0. What we get is a set-theoretic map $\mathbb{P}^m \setminus \{P_0\} \to \mathbb{P}(M)$. For any other choice of a point $P \in \mathbb{P}^m$ and a hyperplane H, not containing P, there exists a change of coordinates which sends P to P_0 and H to $\mathbb{P}(M)$. Thus the geometric projection $\mathbb{P}^m \setminus \{P\} \to H$ from P to H is equal to the map described above, up to a change of coordinates.

We can generalize the construction to projections from positive dimensional linear subspaces.

Namely, for a fixed $n < m$ consider the subspace $N \subset \mathbb{C}^{m+1}$, of dimension $m - n < m + 1$, formed by the $(m + 1)$-tuples of type $(0, \ldots, 0, x_{n+1}, \ldots, x_m)$ and let M be the $(n + 1)$-dimensional linear subspace of $(m + 1)$-tuples of type $(x_0, \ldots, x_n, 0 \ldots, 0)$. Let $\phi_0 : \mathbb{C}^{m+1} \to M$ be the linear surjective (but not injective) map which sends any $(x_0, \ldots, x_m)$ to $(x_1, \ldots, x_n, 0, \ldots, 0)$.

Notice that N and M define *disjoint* linear projective subspaces, respectively $\mathbb{P}(N)$, of projective dimension $m - n - 1$, and $\mathbb{P}(M)$, of projective dimension n. Let Q be a point of $\mathbb{P}^m \setminus \mathbb{P}(N)$ (this means exactly that Q has coordinates $[q_0 : \cdots : q_n]$, with $q_i \neq 0$ for some index i between 0 and n). Then the image of Q under ϕ_0 is a well defined projective point, which corresponds to the intersection of $\mathbb{P}(M)$ with the projective linear subspace spanned by $\mathbb{P}(N)$ and Q. This is why we get from ϕ_0 a set-theoretic map $\mathbb{P}^m \setminus \mathbb{P}(N) \to \mathbb{P}(M)$, which we call *the projection from* $\mathbb{P}(N)$ *to* $\mathbb{P}(M)$.

For any choice of two disjoint linear subspaces L_1, of dimension $m - n - 1$, and L_2, of dimension n,there exists a change of coordinates which sends L_1 to $\mathbb{P}(N)$ and L_2 to $\mathbb{P}(M)$. Thus the geometric projection $\mathbb{P}^m \setminus L_1 \to L_2$ from L_1 to L_2 is equal to the map described above, up to a change of coordinates.

Example 10.1.6 Let $\phi : \mathbb{C}^{m+1} \to \mathbb{C}^{n+1}$ be any surjective map, with kernel L_1 (of dimension $m - n$). We can always assume, up to a change of coordinates, that L_1 coincides with the subspace N defined in Example 10.1.5. Then considering the linear subspace $M \subset \mathbb{C}^{m+1}$ defined in Example 10.1.5, we can find an isomorphism of vector spaces ψ from M to $\mathbb{C}^{n+1}$ such that $\phi = \psi \circ \phi_0$, where ϕ_0 is the map introduced in Example 10.1.5. Thus, after an isomorphism and a change of coordinates, ϕ acts on points of $\mathbb{P}^m \setminus K_\phi$ as a geometric projection.

Example 10.1.6 suggests the following definition.

Definition 10.1.7 Given a linear surjective map $\phi : \mathbb{C}^{m+1} \to \mathbb{C}^{n+1}$ and a subvariety $X \subset \mathbb{P}^m$ which does not meet K_ϕ, the restriction map $\phi_{|X} : X \to \mathbb{P}^n$ is a well defined projective map, which will be denoted as a *projection of* X *from* K_ϕ. The subspace K_ϕ is also called the *center* of the projection.

Notice that $\phi_{|X}$ is a projective map, since it is defined, up to isomorphisms and change of coordinates, by (simple) homogeneous polynomials (see Exercise 31).

Thus, linear surjective maps define projections from suitable subvarieties of $\mathbb{P}^m$ to $\mathbb{P}^n$. Next section is devoted to prove that projections are closed, in the Zariski topology.

10.2 Elimination Theory

In this section, we introduce the basic concept of the elimination theory: the resultant of two polynomials.

The resultant provides an answer to the following problem:
– assume we are given two (not necessarily homogeneous) polynomials $f, g \in \mathbb{C}[x]$. Clearly both f and g factorize in a product of linear factors. Which algebraic condition must f, g satisfy to share a common factor, hence a common root?

Definition 10.2.1 Let f, g be nonconstant (nonhomogeneous) polynomials in one variable x, with coefficients in $\mathbb{C}$. Write $f = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ and $g = b_0 + b_1 x + b_2 x^2 + \cdots + b_m x^m$. The *resultant* $R(f, g)$ of f, g is the determinant of the *Sylvester matrix* $S(f, g)$, which in turn is the $(m + n) \times (m + n)$ matrix defined as follows:

$$S(f,g) = \det \begin{pmatrix} a_0 & a_1 & a_2 & \dots & a_n & 0 & 0 & 0 & \dots & 0 \\ 0 & a_0 & a_1 & a_2 & \dots & a_n & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & 0 & 0 & a_0 & a_1 & a_2 & \dots & a_n \\ b_0 & b_1 & \dots & b_m & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & b_0 & b_1 & \dots & b_m & 0 & 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & \dots & 0 & 0 & 0 & 0 & b_0 & b_1 & \dots & b_m \end{pmatrix}$$

where the a's are repeated m times and the b's are repeated n times.

Notice that when f is constant and g has degree $d > 0$, then by definition $R(f, g) = f^d$.

When both f, g are constant, the previous definition of resultant makes no sense. In this case we set:

$$R(f, g) = \begin{cases} 0 & \text{if } f = g = 0 \\ 1 & \text{otherwise.} \end{cases}$$

Example 10.2.2 Just to give an instance, if $f = x^2 - 3x + 2$ and $g = x - 1$ (both vanishing at $x = 1$), the resultant $R(f, g)$ is

$$R(f, g) = \det S(f, g) = \det \begin{pmatrix} 1 & -3 & 2 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix} = 0$$

Proposition 10.2.3 *With the previous notation, f and g have a common root if and only if $R(f, g) = 0$.*

Proof The proof is immediate when either f or g are constant (Exercise 33).

Otherwise write $\mathbb{C}[x]_i$ for the vector space of polynomials of degree $\leq i$ in $\mathbb{C}[x]$. Then the transpose of $S(f, g)$ is the matrix of the linear map:

$$\phi : \mathbb{C}[x]_{m-1} \times \mathbb{C}[x]_{n-1} \to \mathbb{C}[x]_{m+n-1}$$

which sends (p, q) to $pf + qg$ (matrix computed with respect to the natural basis defined by the monomials). Thus $R(f, g) = 0$ if and only if the map has a nontrivial kernel.

Let (p_0, q_0) be a nontrivial element of the kernel, i.e., $p_0 f + q_0 g = 0$. Consider the factors $(x - \alpha_i)$ of f, where the α_i's are the roots of f (possibly some factor is repeated). Then, all these factors must divide $q_0 g$. Since $\deg q_0 < \deg f$, at least one factor $x - \alpha_i$ must divide g. Thus α_i is a common root of f and g.

Conversely, if α is a common root of f and g, then $x - \alpha$ divides both f and g. Hence setting $p_0 = g/(x - \alpha)$, $q_0 = -f/(x - \alpha)$, one finds a nontrivial element (p_0, q_0) of the kernel of ϕ, so that $\det S(f, g) = 0$. □

We have the analogue construction if f, g are homogeneous polynomials in two or more variables.

Definition 10.2.4 If f, g are *homogeneous* polynomials in $\mathbb{C}[x_0, x_1, \ldots, x_r]$ one can define the 0th resultant $R_0(f, g)$ of f, g just by considering f and g as polynomials in x_0, with coefficients in $\mathbb{C}[x_1, \ldots, x_r]$, and taking the determinant of the corresponding Sylvester matrix $S_0(f, g)$.

$R_0(f, g)$ is thus a polynomial in $x_1, \ldots, x_r$.

For instance, if $f = x_0^2 x_1 + x_0 x_1 x_2 + x_2^3$ and $g = 2x_0 x_1^2 + x_0 x_2^2 + 3x_1^2 x_2$, then the 0th resultant is:

$$R_0(f,g) = \det S_0(f,g) = \det \begin{pmatrix} x_1 & x_1x_2 & x_2^3 \\ 2x_1^2 + x_2^2 & 3x_1^2x_2 & 0 \\ 0 & 2x_1^2 + x_2^2 & 3x_1^2x_2 \end{pmatrix} = \\ = 3x_1^5x_2^2 + 4x_1^4x_2^3 - 3x_1^3x_2^4 + 4x_1^2x_2^5 + x_2^7.$$

From Proposition 10.2.3, with an easy induction on the number of variables, one finds that the resultant of homogeneous polynomials in several variables has the following property:

Proposition 10.2.5 *Let f, g be homogeneous polynomials in $\mathbb{C}[x_0, x_1, \ldots, x_r]$. Then $R_0(f,g)$ vanishes at $(\alpha_1, \ldots, \alpha_r)$ if and only if there exists $\alpha_0 \in \mathbb{C}$ with:*

$$f(\alpha_0, \alpha_1, \ldots, \alpha_r) = g(\alpha_0, \alpha_1, \ldots, \alpha_r) = 0.$$

Less obvious, but useful, is the following remark on the resultant of two homogeneous polynomials.

Proposition 10.2.6 *Let f, g be homogeneous polynomials in $\mathbb{C}[x_0, x_1, \ldots, x_r]$. Then $R_0(f,g)$ is homogeneous.*

Proof The entries s_{ij} of the Sylvester matrix $S_0(f,g)$ are homogeneous and their degrees decrease by 1 passing from one element s_{ij} to the next element s_{ij+1} in the same row (unless some of them is 0). Thus for any nonzero entry s_{ij} of the matrix, the number $\deg s_{ij} - j$ depends only on the row i. Call it u_i. Then the summands given by any permutation, in the computation of the determinant, are homogeneous of same degree:

$$d = \frac{1}{2}(n+m+1)(n+m) + \sum_i u_i,$$

so that $R_0(f,g)$ is homogeneous, and its degree is equal to d. □

Next, a fundamental property of the resultant $R_0(f,g)$ is that it belongs to the ideal generated by f and g. We will not give a full proof of this property, and refer to the book [1] for it.

Instead, we just prove that $R_0(f,g)$ belongs to the radical of the ideal generated by f and g, which is sufficient for our aims.

Proposition 10.2.7 *Let f, g be homogeneous polynomials in $\mathbb{C}[x_0, x_1, \ldots, x_r]$. Then $R_0(f,g)$ belongs to the radical of the ideal generated by f and g.*

Proof In view of the Nullstellensatz, it is sufficient to prove that $R_0(f,g)$ vanishes at all points of the variety defined by f and g. But this is obvious from Proposition 10.2.5: if $[\alpha_0 : \alpha_1 : \cdots : \alpha_r]$ are homogeneous coordinates of $P \in V(f,g)$, then

$$R_0(f,g)(\alpha_0, \alpha_1, \ldots, \alpha_r) = R_0(f,g)(\alpha_1, \ldots, \alpha_r) = 0,$$

since $f(x_0, \alpha_1, \ldots, \alpha_r)$ and $g(x_0, \alpha_1, \ldots, \alpha_r)$, polynomials in $\mathbb{C}[x_0]$, have a common root α_0. □

10.3 Forgetting a Variable

In Sect. 10.1 we introduced the projection maps as projectifications of surjective linear maps $\phi : \mathbb{C}^{m+1} \to \mathbb{C}^{n+1}$. It is important to recall that when ϕ has nontrivial kernel (i.e., when $n < m$) the projection is *not* defined as a map between the two projective spaces $\mathbb{P}^m$ and $\mathbb{P}^n$. On the other hand, for any subvariety $X \subset \mathbb{P}^m$ which does not intersect the projectification of $Ker(\phi)$, the map ϕ corresponds to a well defined projective map $X \to \mathbb{P}^n$.

In this section, we describe the image of a variety in a projection π from a point, i.e., when the center of projections has dimension 0. It turns out, in particular, that $\pi(X)$ is itself an algebraic variety.

Through this section, consider the surjective linear map $\phi : \mathbb{C}^{m+1} \to \mathbb{C}^m$ which sends $(\alpha_0, \alpha_1, \ldots, \alpha_m)$ to $(\alpha_1, \ldots, \alpha_m)$. The kernel of the map is generated by $(1, 0, \ldots, 0)$. Thus, if X is a projective variety in $\mathbb{P}^m$ which misses the point $P_0 = [1 : 0 : \cdots : 0]$, then the map induces a well defined projective map $\pi : X \to \mathbb{P}^{m-1}$: the projection from P_0 (see Definition 10.1.7).

For any point $Q \in \pi(X)$, $Q = [q_1 : \cdots : q_n]$, the inverse image of Q in X is the the intersection of X with the line joining P_0 and Q. Thus $\pi^{-1}(Q)$ is the set of points in X with coordinates $[q_0 : q_1 : \cdots : q_n]$, for some $q_0 \in \mathbb{C}$.

Remark 10.3.1 For all $Q \in \pi(X)$, the inverse image $\pi^{-1}(Q)$ is finite.

Indeed $\pi^{-1}(Q)$ is a Zariski closed set in the line P_0Q, and it does not contain P_0, since $P_0 \notin X$. The claim follows since the Zariski topology on a line is the cofinite topology.

Let $J \subset \mathbb{C}[x_0, \ldots, x_n]$ be the homogeneous ideal associated to X. Define:

$$J_0 = J \cap \mathbb{C}[x_1, \ldots, x_n].$$

In other words, J_0 is the set of elements in J which are constant with respect to the variable x_0. In Chap. 13 we will talk again about Elimination Theory, but from the point of view of Groebner Basis; there, the ideal J_0 will be called the *first elimination ideal* of J (Definition 13.5.1).

Remark 10.3.2 J_0 is a homogeneous radical ideal in $\mathbb{C}[x_1, \ldots, x_n]$.

Indeed J_0 is obviously an ideal. Moreover for any $g \in J_0$, any homogeneous component g_d of g belongs to J, because J is homogeneous, and does not contain x_0. Thus $g_d \in J_0$, and this is sufficient to conclude that J_0 is homogeneous (see Proposition 9.1.15).

If $g^d \in J_0$ for some $g \in \mathbb{C}[x_1, \ldots, x_n]$, then $g \in J$, because J is radical, moreover g does not contain x_0. Thus also $g \in J_0$.

We prove that $\pi(X)$ is the projective variety defined by J_0. We will need the following refinement of Lemma 9.1.5:

Lemma 10.3.3 *Let $Q_1, \dots, Q_k$ be a finite set of points in $\mathbb{P}^n$. Then there exists a linear form $\ell \in \mathbb{C}[x_0, \dots, x_n]$ such that $\ell(Q_i) \neq 0$ for all i.*

If none of the Q_i's belong to the variety defined by a homogeneous ideal J, then there exists $g \in J$ such that $g(Q_i) \neq 0$ for all i.

Proof Fix a set of homogeneous generators $g_1, \dots g_s$ of J.

First assume that all the g_i's are linear. Then the g_i's define a subspace L of the space of linear homogeneous polynomials in $\mathbb{C}[x_0, \dots, x_n]$. For each Q_i, the set L_i of linear forms in L that vanish at Q_i is a linear subspace of L, which is properly contained in L, because some g_j does not vanish at Q_i. Since a nontrivial complex linear space cannot be the union of a finite number of proper subspaces, we get that for a general choice of $b_1, \dots, b_s \in \mathbb{C}$, the linear form $\ell = b_1 g_1 + \cdots + b_s g_s$ does not belong to any L_i, thus $\ell(Q_i) \neq 0$ for all i. This proves the second claim for ideals generated by linear forms.

The first claim now follows soon, since the (irrelevant) ideal J generated by *all* the linear forms defines the empty set.

For general $g_1, \dots, g_s$, call d_i the degree of g_i and $d = \max\{d_i\}$. If ℓ is a linear form that does not vanish at any Q_i, then $h_j = \ell^{d-d_j} g_j$ is a form of degree d that vanishes at Q_i precisely when g_j vanishes. The forms $h_1, \dots, h_s$ define a subspace L' of the space of forms of degree d. For all i, the set of forms in L' that vanish at Q_i is a proper subspace of L'. Thus, as before, for a general choice of $b_1, \dots, b_s \in \mathbb{C}$, the form $g = b_1 h_1 + \cdots + b_s h_s$ is an element of J which does not vanish at any Q_i. □

Theorem 10.3.4 *The variety defined in $\mathbb{P}^{n-1}$ by the ideal J_0 coincides with $\pi(X)$.*

Proof Let $Q \in \pi(X)$, $Q = [q_1 : \cdots : q_n]$. Then there exists $q_0 \in \mathbb{C}$ such that the point $P = \pi^{-1}(Q) = [q_0 : q_1 : \cdots : q_n] \in \mathbb{P}^n$ belongs to X. Thus $g(P) = 0$ for all $g \in J$. In particular, this is true for all $g \in J_0$. On the other hand, if $g \in J_0$ then g does not contain x_0, thus:

$$0 = g(P) = g(q_0, q_1, \dots, q_n) = g(q_1, \dots, q_n) = g(Q).$$

This proves that the variety defined by J_0 contains $\pi(X)$.

Conversely, identify $\mathbb{P}^{n-1}$ with the hyperplane $x_0 = 0$, and fix a point $Q = [q_1 : \cdots : q_n] \notin \pi(X)$. Consider an element $f \in J$ that does not vanish at P_0 and let W be the variety defined by f. The intersection of W with the line $P_0 Q$ is a finite set $Q_1, \dots, Q_k$. Moreover no Q_i can belong to X, since $\pi^{-1}(Q)$ is empty. Thus, by Lemma 10.3.3, there exists $g \in J$ that does not vanish at any Q_i. Consider the resultant $h = R_0(f, g)$. By Proposition 10.2.7, h belongs to the radical of the ideal generated by f, g, thus it belongs to J, which is a radical ideal. Moreover h does not contain the variable x_0. Thus $h \in J_0$. Finally, from Proposition 10.2.5 it follows that $h(Q) \neq 0$. Then Q does not belong to the variety defined by J_0. □

Remark 10.3.5 A direct consequence of Theorem 10.3.4 is that the projection π is a closed map, in the Zariski topology.

Indeed any closed subset Y of a projective variety X is itself a projective variety, thus by Theorem 10.3.4 the image of Y in π is Zariski closed.

We can repeat all the constructions of this section by selecting any variable x_i instead of x_0 and performing the elimination of x_i. Thus we can define the *i-th resultant* $R_i(f, g)$ and use it to prove that projections with center any coordinate point $[0 : \cdots : 0 : 1 : 0 : \cdots : 0]$ are closed maps.

10.4 Linear Projective and Multiprojective Maps

In this section, we prove that projective maps defined by linear maps of projective spaces are closed in the Zariski topology.

Remark 10.4.1 Let V, W be linear space, respectively, of dimension $n + 1$, $m + 1$.

The choice of a basis for V corresponds to fixing an isomorphism between V and $\mathbb{C}^{n+1}$. Thus we can identify, after a choice of the basis, the projective space $\mathbb{P}(V)$ with $\mathbb{P}^n$. We will use this identification to introduce all the concepts of Projective Geometry into $\mathbb{P}(V)$. Notice that two such identifications differ by a change of basis in $\mathbb{P}^n$, thus they are equivalent, up to an isomorphism of $\mathbb{P}^n$.

Similarly, the choice of a basis for W corresponds to fixing an isomorphism between W and $\mathbb{C}^{m+1}$.

A linear map $W \to V$ corresponds, under the choice of a basis, to a linear map $\mathbb{C}^{m+1} \to \mathbb{C}^{n+1}$. Thus, the study of projective maps in $\mathbb{P}^m$ and $\mathbb{P}^n$ induced by linear maps $\mathbb{C}^{m+1} \to \mathbb{C}^{n+1}$ corresponds to the study of projective maps in $\mathbb{P}(W)$, $\mathbb{P}(V)$ induced by linear maps $W \to V$.

Proposition 10.4.2 *Let $\phi : \mathbb{C}^{m+1} \to \mathbb{C}^{n+1}$ be a linear map. Let K_ϕ be the projective kernel of ϕ and let $X \subset \mathbb{P}^m$ be a projective subvariety such that $X \cap K_\phi = \emptyset$. Then ϕ induces a projective map $X \to \mathbb{P}^n$ (that we will denote again with ϕ) which is a closed map in the Zariski topology.*

Proof The map ϕ factors through a linear surjection $\phi_1 : \mathbb{C}^{m+1} \to \mathbb{C}^{m+1}/Ker(\phi)$ followed by a linear injection ϕ_2. After the choice of a basis, the space $\mathbb{C}^{m+1}/Ker(\phi)$ can be identified with $\mathbb{C}^{N+1}$, where $N = m - \dim(Ker(\phi))$, so that ϕ_1 can be considered as a map $\mathbb{C}^{m+1} \to \mathbb{C}^{N+1}$ and ϕ_2 as a map $\mathbb{C}^{N+1} \to \mathbb{C}^{n+1}$. Since X does not meet the kernel of ϕ_1, by Definition 10.1.7 ϕ_1 induces a projection $X \to \mathbb{P}^N$. The injective map ϕ_2 defines a projective map $\mathbb{P}^N \to \mathbb{P}^n$, by Definition 10.1.1. The composition of these two maps is the projective map $\phi : X \to \mathbb{P}^n$ of the claim. It is closed since it is the composition of two closed maps. □

Notice that the projective map ϕ is only defined up to a change of coordinate, since it relies on the choice of a basis in $\mathbb{C}^{m+1}/Ker(\phi)$.

The previous result can be extended to maps from multiprojective varieties to multiprojective spaces.

Example 10.4.3 Consider a multiprojective product $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$ and an injective linear map $\phi : \mathbb{C}^{a_1+1} \to \mathbb{C}^{m+1}$. Then the induced linear map:

$$\mathbb{P}^{a_1} \times \mathbb{P}^{a_2} \times \cdots \times \mathbb{P}^{a_s} \to \mathbb{P}^{m} \times \mathbb{P}^{a_2} \times \cdots \times \mathbb{P}^{a_s}$$

is multiprojective and closed, because the product of closed maps is closed.

Of course, the same statement holds if we replace i with any index or if we mix up the indices. Moreover we can apply it repeatedly.

Similarly, consider a linear map $\phi : \mathbb{C}^{a_1+1} \to \mathbb{C}^{m+1}$ and a multiprojective subvariety $X \subset \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$ such that X is disjoint from $\mathbb{P}(\ker \phi) \times \mathbb{P}^{a_2} \times \cdots \times \mathbb{P}^{a_s}$. Then there is an induced linear map: $X \to \mathbb{P}^{m} \times \mathbb{P}^{a_2} \times \cdots \times \mathbb{P}^{a_s}$ which is multiprojective and closed.

Proposition 10.4.4 *Any projection π_i from a multiprojective space $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$ to any of its factor $\mathbb{P}^{a_i}$ is a closed projective map.*

Proof The map π_i is defined by sending $P = ([p_{10} : \cdots : p_{1a_1}], \ldots, [p_{s0} : \cdots : p_{sa_s}])$ to $[p_{i0} : \ldots, p_{ia_i}]$. Thus the map is defined by multihomogeneous polynomials (of multidegree 1 in the ith set of variables and 0 in the other sets).

To prove that π_i is closed, we show that the image in π_i of any multiprojective variety is a projective subvariety of $\mathbb{P}^{a_i}$. Let $X \subset \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$ be a multiprojective subvariety and let $Y = \pi_i(X)$. If $Y = \mathbb{P}^{a_i}$, there is nothing to prove. Thus the claim holds if $n_i = 0$, i.e., $\mathbb{P}^{a_i}$ is a point. We will proceed then by induction on a_i, assuming that $Y \neq \mathbb{P}^{a_i}$.

Let Q be a point of $\mathbb{P}^{a_i} \setminus Y$. Then no points of type $(P_1, \ldots, P_s)$ with $P_i = Q$ can belong to X. Thus X does not contain points $(P_1, \ldots, P_s)$, with P_i in the projective kernel of the projection from Q.

The claim now follows from Example 10.4.3. □

Corollary 10.4.5 *Any projection π_i from a multiprojective space $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$ to a product of some of its factors is a closed projective map.*

10.5 The Veronese Map and the Segre Map

We introduce now two fundamental projective and multiprojective maps, which are the cornerstone, together with linear maps, of the construction of projective maps. The first map, the *Veronese map*, is indeed a generalization of the map built in Example 9.1.33.

We recall that a monomial is *monic* if its coefficient is 1.

Definition 10.5.1 Fix n, d and set $N = \binom{n+d}{d} - 1$. There are exactly $N + 1$ monic monomials of degree d in $n + 1$ variables $x_0, \ldots, x_n$. Let us call $M_0, \ldots, M_N$ these monomials, for which we fixed an order.

The *Veronese map* of degree d in $\mathbb{P}^n$ is the map $v_{n,d} : \mathbb{P}^n \to \mathbb{P}^N$ which sends a point $[p_0 : \cdots : p_n]$ to $[M_0(p_0, \ldots, p_n) : \cdots : M_N(p_0, \ldots, p_n)]$.

Notice that a change in the choice of the order of the monic monomials produces simply the composition of the Veronese map with a change of coordinates. After choosing an order of the variables, e.g., $x_0 < x_1 < \cdots < x_n$, a very popular order of the monic monomials is the order in which $x_0^{d_0} \cdots x_n^{d_n}$ preceeds $x_0^{e_0} \cdots x_n^{e_n}$ if in the smallest index i for which $d_i \neq e_i$ we have $d_i > e_i$. This order is called *lexicographic order*, because it reproduces the way in which words are listed in a dictionary. In Sect. 13.1 we will discuss different types of monomial orderings.

Notice that we can define an analogue of a Veronese map by choosing arbitrary (nonzero) coefficients for the monomials M_j's. This is equivalent to choose a *weight* for the monomials. The resulting map has the same fundamental property of our Veronese map, for which we choose to take all the coefficients equal to 1.

Remark 10.5.2 The Veronese maps are well defined, since for any $P = [p_0 : \cdots : p_n] \in \mathbb{P}^n$ there exists an index i with $p_i \neq 0$, and among the monomials there exists the monomial $M = x_i^d$, which satisfies $M(p_0, \ldots, p_n) = p_i^d \neq 0$.

The Veronese map is injective. Indeed if $P = [p_0 : \cdots : p_n]$ and $Q = [q_0 : \cdots : q_n]$, have the same image, then the powers of the p_i's and the q_i's are equal, up to a scalar multiplication. Thus, up to a scalar multiplication, one may assume $p_i^d = q_i^d$ for all i, so that $q_i = e_i p_i$, for some choice of a d-root of unit e_i. If the e_i's are not all equal to 1, then there exists a monic monomial M such that $M(e_0, e_1, \ldots, e_n) \neq 1$, thus $M(p_0, \ldots, p_n) \neq M(q_0, \ldots, q_n)$, which contradicts $v_{n,d}(P) = v_{n,d}(Q)$.

Because of its injectivity, sometimes we will refer to a Veronese map as a *Veronese embedding*.

The images of Veronese embeddings will be denoted as *Veronese varieties*.

Example 10.5.3 The Veronese map $v_{1,3}$ sends the point $[x_0 : x_1]$ of $\mathbb{P}^1$ to the point $[x_0^3 : x_0^2 x_1 : x_0 x_1^2 : x_1^3] \in \mathbb{P}^3$.

The Veronese map $v_{2,2}$ sends the point $[x_0 : x_1 : x_2] \in \mathbb{P}^2$ to the point $[x_0^2 : x_0 x_1 : x_0 x_2 : x_1^2 : x_1 x_2 : x_2^2]$ (notice the lexicographic order).

Proposition 10.5.4 *The image of a Veronese map is a projective subvariety of* $\mathbb{P}^N$.

Proof We define equations for $Y = v_{n,d}(\mathbb{P}^n)$.

Consider $(n+1)$-tuples of nonnegative integers $A = (a_0, \ldots, a_n)$, $B = (b_0, \ldots, b_n)$ and $C = (c_0, \ldots, c_n)$, with the following property:

(*) $\sum a_i = \sum b_i = \sum c_i = d$ and $a_i + b_i \geq c_i$ for all i. Define $D = (d_0, \ldots, d_n)$, where $d_i = a_i + b_i - c_i$, Clearly $\sum d_i = d$. For any choice of $A = (a_0, \ldots, a_n)$, the monic monomial $x_0^{a_0} \cdots x_n^{a_n}$ corresponds to a coordinate in $\mathbb{P}^N$. Call M^A the coordinate corresponding to A. Define in the same way M^B, M^C and then also M^D. The polynomial:

$$f_{ABC} = M^A M^B - M^C M^D$$

is homogeneous of degree 2 and vanishes at the points of Y. Indeed for any $Q = v_{n,d}([\alpha_0 : \cdots : \alpha_n])$ it is easy to see that:

$$M^A M^B = M^C M^D = \alpha_0^{a_0+b_0} \cdots \alpha_n^{a_n+b_n},$$

since any x_k appears in $M^C M^D$ with exponent $a_k + b_k$. It follows that Y is contained in the projective subvariety W defined by the forms f_{ABC}, when A, B, C vary in the set of $(n+1)$-tuples with the property (*) above.

To see that $Y = W$, take $Q = [m_0 : \cdots : m_N] \in W$. Each m_i corresponds to a monic monomial in the x_j's, and we assume they are ordered in the lexicographic order.

First, we claim that at least one coordinate m_i, corresponding to a power x_i^d, must be nonzero. Indeed, on the contrary, assume that all the coordinates corresponding to powers vanish, and consider a minimal q such that coordinate m_q of Q is nonzero. Let m_q correspond to the monomial $x_0^{a_0} \cdots x_n^{a_n}$. Since m_q is not a power, there are at least two indices $j > i$ such that $a_i, a_j > 0$. Put $A = (a_0, \ldots, a_n) = B$ and $C = (c_0, \ldots, c_n)$ where $c_i = a_i - 1$, $c_j = a_j + 1$ and $c_k = a_k$ for $k \neq i, j$. The $(n+1)$-tuples A, B, C satisfy condition (*). One computes that $D = (d_0, \ldots, d_n)$ with $d_i = a_i + 1$, $d_j = a_j - 1$ and $d_k = a_k$ for $k \neq i, j$. Moreover, since $Q \in W$, then:

$$M^A(Q)M^B(Q) = M^C(Q)M^D(Q).$$

But we have $M^A(Q) = M^B(Q) = m_q$, while $M^C(Q) = 0$, since $x_0^{c_0} \cdots x_n^{c_n}$ preceeds $x_0^{a_0} \cdots x_n^{a_n}$ in the lexicographic order. It follows $m_q^2 = 0$, a contradiction.

Then at least one coordinate corresponding to a power is nonzero. Just to fix the ideas, assume that m_0, which corresponds to x_0^d in the lexicographic order, is different from 0. After multiplying the coordinates of Q by $1/m_0$, we may assume $m_0 = 1$. Then consider the coordinates corresponding to the monomials $x_0^{d-1}x_1, \ldots x_0^{d-1}x_n$. In the lexicographic order, they turn out to be $m_1, \ldots, m_n$, respectively. Put $P = [1 : m_1 : \cdots : m_n] \in \mathbb{P}^n$. We claim that Q is exactly $v_{n,d}(P)$.

The claim means that for any coordinate m of Q, corresponding to the monomial $x_0^{a_0} \cdots x_n^{a_n}$ we have $m = m_1^{a_1} \cdots m_n^{a_n}$. We prove the claim by descending induction on a_0. The cases $a_0 = d$ and $a_0 = d - 1$ are clear by construction. Assume that the claim holds when $a_0 > d - s$ and take m such that $a_0 = d - s$. In this case there exists some index $j > 0$ such that $a_j > 0$. Put $A = (a_0, \ldots, a_n)$, $B = (d, 0, \ldots, 0)$ and $C = (c_0, \ldots, c_n)$ where $c_0 = a_0 + 1 = d - s + 1$, $c_j = a_j - 1$, and $c_k = a_k$ for $k \neq 0, j$. The $(n+1)$-tuples A, B, C satisfy condition (*). Thus

$$M^A(Q)M^B(Q) = M^C(Q)M^D(Q),$$

where $D = (d_0, \ldots, d_n)$ and $d_0 = d - 1$, $d_j = 1$, and $d_k = 0$ for $k \neq 0, j$. It follows by induction that $M^B(Q) = 1$, $M^D(Q) = m_j$ and

$$M^C(Q) = m_1^{c_1} \cdots m_n^{c_n} = m_1^{a_1} \cdots m_n^{a_n}/m_j.$$

Then $m = M^A(Q) = M^C(Q)M^D(Q) = m_1^{a_1} \cdots m_n^{a_n}$, and the claim follows. □

We observe that all the forms $M^A M^B - M^C M^D$ are quadratic forms in the variables M_i's of $\mathbb{P}^N$. Thus the Veronese varieties are defined in $\mathbb{P}^N$ by quadratic equations.

Example 10.5.5 Consider the Veronese map $v_{1,3} : \mathbb{P}^1 \to \mathbb{P}^3$. The monic monomials of degree three in 2 variable are (in lexicographic order):

$$M_0 = x_0^3, \quad M_1 = x_0^2 x_1, \quad M_2 = x_0 x_1^2, \quad m_3 = x_1^3.$$

The equations for the image are obtained by couples A, B, C satisfying condition (*) above. Up to trivialities, these couples are:

$$A = (3,0)\ B = (0,3)\ C = (1,2),\ \text{so } D = (2,1) \text{ and we get } M_0M_3 - M_1M_2 = 0,$$
$$A = (3,0)\ B = (1,2)\ C = (2,1),\ \text{so } D = (2,1) \text{ and we get } M_0M_2 - M_1^2 = 0,$$
$$A = (2,1)\ B = (0,3)\ C = (1,2),\ \text{so } D = (1,2) \text{ and we get } M_1M_3 - M_2^2 = 0$$

Consider the point $Q = [3 : 6 : 12 : 24]$, which satisfies the previous equations. Since the coordinate m_0 corresponding to x_0^3 is equal to $3 \neq 0$, we divide the coordinates of Q by 3 and obtain $Q = [1 : 2 : 4 : 8]$. Then $Q = v_{1,3}([1 : 2])$, as one can check directly.

Example 10.5.6 Equations for the image of $v_{2,2} \subset \mathbb{P}^5$ (the classical Veronese surface) are given by:

$$\begin{cases} M_0M_4 - M_1M_2 = 0 \\ M_3M_2 - M_1M_4 = 0 \\ M_5M_1 - M_2M_4 = 0 \\ M_3M_5 - M_4^2 = 0 \\ M_0M_5 - M_2^2 = 0 \\ M_0M_3 - M_1^2 = 0 \end{cases}$$

The point $Q = [0 : 0 : 0 : 1 : -2 : 4]$ satisfies the equations, and indeed it is equal to $v_{2,2}([0 : 1 : -2])$. Notice that in this case, to recover the preimage of Q, one needs to replace x_0 with x_1 in the procedure of the proof of Proposition 10.5.4, since the coordinate corresponding to x_0^2 is 0.

As a consequence of Proposition 10.5.4, one gets the following result.

Theorem 10.5.7 *All the Veronese maps are closed in the Zariski topology.*

Proof We need to prove that the image in $v_{n,d}$ of a projective subvariety of $\mathbb{P}^n$ is a projective subvariety of $\mathbb{P}^N$.

First notice that if F is a monomial of degree kd in the variables $x_0, \ldots, x_n$ of $\mathbb{P}^n$, then it can be written (usually in several ways) as a product of k monomials of degree d in the x_i's, which corresponds to a monomial of degree k in the coordinates $M_0, \ldots, M_N$ of $\mathbb{P}^N$. Thus, any form f of degree kd in the x_j's can be rewritten as a form of degree k in the coordinates M_j's.

Take now a projective variety $X \subset \mathbb{P}^n$ and let $f_1, \ldots, f_s$ be homogeneous generators for the homogeneous ideal of X. Call e_i the degree of f_i and let $k_i d$ be the

smallest multiple of d bigger or equal to e_i. Then consider all the products $x_j^{k_i d - e_i} f_i$, $j = 0, \ldots, n$. These products are homogeneous forms of degree $k_i d$ in the x_j's. Moreover a point $P \in \mathbb{P}^n$ satisfies all the equations $x_j^{k_i d - e_i} f_i = 0$ if and only if it satisfies $f_i = 0$, since at least one coordinate x_j of P is nonzero.

With the procedure introduced above, transform arbitrarily each form $x_j^{k_i d - e_i} f_i = 0$ in a form F_{ij} of degree k in the variables M_j's. Then we claim that $v_{n,d}(X)$ is the subvariety of $v_{n,d}(\mathbb{P}^n)$ defined by the equations $F_{ij} = 0$. Since $v_{n,d}(\mathbb{P}^n)$ is closed in $\mathbb{P}^N$, this will complete the proof.

Indeed let Q be a point of $v_{n,d}(X)$. The coordinates of Q are obtained by the coordinates of its preimage $P = [p_0 : \cdots : p_n] \in X \subset \mathbb{P}^n$ by computing in P all the monomials of degree d in the x_j's. Thus $F_{ij}(Q) = 0$ for all i, j if and only if $x_j^{k_i d - e_i} f_i(P) = 0$ for all i, j, i.e., if and only if $f_i(P) = 0$ for all i. The claim follows. □

Example 10.5.8 Consider the map $v_{2,2}$ and let X be the line in $\mathbb{P}^2$ defined by the equation $f = (x_0 + x_1 + x_2) = 0$. Since f has degree 1, consider the products:

$$\begin{aligned} x_0 f &= x_0^2 + x_0 x_1 + x_0 x_2, \\ x_1 f &= x_0 x_1 + x_1^2 + x_1 x_2, \\ x_2 f &= x_0 x_2 + x_1 x_2 + x_2^2. \end{aligned}$$

they can be transformed, respectively, in the monomials

$$M_0 + M_1 + M_2, \quad M_1 + M_3 + M_4, \quad M_2 + M_4 + M_5.$$

Thus the image of X is the variety defined in $\mathbb{P}^5$ by the previous three linear forms and the six quadratic forms of Example 10.5.6, that define $v_{2,2}(\mathbb{P}^2)$.

Next, let us turn to the Segre embeddings.

Definition 10.5.9 Fix $a_1, \ldots, a_n$ and $N = (a_1 + 1) \cdot (a_2 + 1) \cdots (a_n + 1) - 1$. There are exactly $N + 1$ monic monomials of multidegree $(1, \ldots, 1)$ (i.e., multilinear forms) in the variables $x_{1,0}, \ldots, x_{1,a_1}, x_{2,0}, \ldots, x_{2,a_2}, \ldots, x_{n,0} \ldots x_{n,a_n}$. Let us choose an order and denote with $M_0, \ldots, M_N$ these monomials.

The *Segre map* of $a_1, \ldots, a_n$ is the map $s_{a_1,\ldots,a_n} : \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n} \to \mathbb{P}^N$ which sends a point $P = ([p_{10} : \cdots : p_{1n_1}], \ldots, [p_{n0} : \cdots : p_{na_n}])$ to $[M_0(P) : \cdots : M_N(P)]$.

The map is well defined, since for any $i = 1, \ldots, n$ there exists $p_{ij_i} \neq 0$, and among the monomials there is $M = x_{1,j_1} \cdots x_{n,j_n}$, which satisfies $M(P) = p_{1j_1} \cdots p_{nj_n} \neq 0$.

Notice that when $n = 1$, then the Segre map is the identity.

Proposition 10.5.10 *The Segre maps are injective.*

Proof Make induction on n, the case $n = 1$ being trivial.

For the general case, assume that

$$P = ([p_{10} : \cdots : p_{1a_1}], \ldots, [p_{n,0} : \cdots : p_{n,a_n}]),$$
$$Q = ([q_{10} : \cdots : q_{1a_1}], \ldots, [q_{n,0} : \cdots : q_{n,a_n}])$$

have the same image. Fix indices such that $p_{1j_1}, \ldots, p_{nj_n} \neq 0$. The monomial $M = x_{1,j_1} \cdots x_{n,j_n}$ does not vanish at P, hence also $q_{1j_1}, \ldots, q_{nj_n} \neq 0$.

Call $\alpha = q_{1j_1}/p_{1j_1}$. Our first task is to show that $\alpha = q_{1i}/p_{1i}$ for $i = 1, \ldots, n_1$, so that $[p_{11} : \cdots : p_{1a_1}] = [q_{11} : \cdots : q_{1a_1}]$. Define $\beta = (q_{2j_2} \cdots q_{nj_n})/(p_{2j_1} \cdots p_{nj_n})$. Then $\beta \neq 0$ and:

$$\alpha\beta = (q_{1j_1} \cdots q_{nj_n})/(p_{1j_1} \cdots p_{nj_n}).$$

Since P, Q have the same image in the Segre map, then for all $i = 1, \ldots, a_1$, the monomials $M_i = x_{1,i}x_{2,j_2} \cdots x_{n,j_n}$ satisfy:

$$\alpha\beta M_i(P) = M_i(Q).$$

It follows immediately $\alpha\beta(p_{1i} \cdots p_{nj_n}) = (q_{1i} \cdots q_{nj_n})$ so that $\alpha\beta p_{1i} = q_{1i}$ for all i. Thus $[p_{10} : \cdots : p_{1a_1}] = [q_{10} : \cdots : q_{1a_1}]$.

We can repeat the argument for the remaining factors $[p_{i0} : \cdots : p_{ia_i}] = [q_{i0} : \cdots : q_{ia_i}]$ of P, Q $(i = 2, \ldots, n)$, obtaining $P = Q$. □

Because of its injectivity, sometimes we will refer to a Segre map as a *Segre embedding*.

The images of Segre embeddings will be denoted as *Segre varieties*.

Example 10.5.11 The Segre embedding $s_{1,1}$ of $\mathbb{P}^1 \times \mathbb{P}^1$ to $\mathbb{P}^3$ sends the point $([x_0 : x_1], [y_0 : y_1])$ to $[x_0y_0 : x_0y_1 : x_1y_0 : x_1y_1]$.

The Segre embedding $s_{1,2}$ of $\mathbb{P}^1 \times \mathbb{P}^2$ to $\mathbb{P}^5$ sends the point $([x_{10} : x_{11}], [x_{20} : x_{21} : x_{22}])$ to the point:

$$[x_{10}x_{20} : x_{10}x_{21} : x_{10}x_{22} : x_{11}x_{20} : x_{11}x_{21} : x_{11}x_{22}].$$

The Segre embedding $s_{1,1,1} : \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^7$ sends the point $P = ([x_{10} : x_{11}], [x_{20} : x_{21}], [x_{30} : x_{31}])$ to the point:

$$[x_{10}x_{20}x_{30} : x_{10}x_{20}x_{31} : x_{10}x_{21}x_{30} : x_{10}x_{21}x_{31} : x_{11}x_{20}x_{30} : x_{11}x_{20}x_{31} : x_{11}x_{21}x_{30} : x_{11}x_{21}x_{31}].$$

Recall the general notation that with $[n]$ we denote the set $\{1, \ldots, n\}$.

Proposition 10.5.12 *The image of a Segre map is a projective subvariety of* $\mathbb{P}^N$.

Since the set of tensors of rank one corresponds to the image of a Segre map, the proof of the proposition is essentially the same as the proof of Theorem 6.4.13. We give the proof here, in the terminology of maps, for the sake of completeness.

Proof We define equations for $Y = s_{a_1,\ldots,a_n}(\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n})$.

For any n-tuple $A = (\alpha_0, \ldots, \alpha_m)$ define the form M^A of multidegree $(1, \ldots, 1)$ as follows:

$$M^A = x_{0\alpha_0} \cdots x_{n\alpha_n}.$$

Then consider any subset $J \subset [n]$ and two n-tuples of nonnegative integers $A = (\alpha_0, \ldots, \alpha_n)$ and $B = (\beta_0, \ldots, \beta_n)$. Define $C = C^J_{AB}$ as the n-tuple $(\gamma_1, \ldots, \gamma_n)$ such that:

$$\gamma_i = \begin{cases} \alpha_i & \text{if } i \in J, \\ \beta_i & \text{if } i \notin J \end{cases}.$$

Define D as $D = C^{J^c}_{AB}$, where J^c is the complement of J in $[n]$. Thus $D = (\delta_1, \ldots, \delta_n)$, where:

$$\delta_i = \begin{cases} \beta_i & \text{if } i \in J, \\ \alpha_i & \text{if } i \notin J \end{cases}.$$

Consider the polynomials

$$f^J_{AB} = M^A M^B - M^C M^D.$$

Every f^J_{AB} is homogeneous of degree 2 in the coordinates of $\mathbb{P}^N$. We claim that the projective variety defined by the forms f^J_{AB}, for all possible choices of A, B, J as above, is exactly equal to Y.

One direction is simple. If:

$$Q = s_{a_1,\ldots,a_n}([q_{10} : \cdots : q_{1a_1}], \ldots, [q_{n0} : \cdots : q_{na_n}])$$

then it is easy to see that both $M^A M^B(Q)$ and $M^C M^D(Q)$ are equal to the product

$$q_{1\alpha_1} q_{1\beta_1} q_{2\alpha_2} q_{2\beta_2} \cdots q_{n\alpha_n} q_{n\beta_n}.$$

It follows that Y is contained in the projective subvariety W defined by the forms f^J_{AB}.

To see the converse, we make induction on the number n of factors. The claim is obvious if $n = 1$, for in this case the equations f^J_{AB} are trivial and the Segre map is the identity on $\mathbb{P}^{a_1}$.

Assume that the claim holds for $n - 1$ factors. Take $Q = [m_0 : \cdots : m_N] \in W$. Each m_i corresponds to a monic monomial $x_{1i_1} \cdots x_{ni_n}$ of multidegree $(1, \ldots, 1)$ in the x_{ij}'s. Fix a coordinate m of Q different from 0. Just to fix the ideas, we assume that m corresponds to $x_{10} \cdots x_{n-1\,0} x_{n0}$. If m corresponds to another multilinear form, the argument remains valid, it just requires heavier notation.

Consider the point Q' obtained from Q by deleting all the coordinates corresponding to multilinear forms in which the last factor is not x_{n0}. If we consider $N' = \Pi_2^n(a_i + 1) - 1$, then Q' can be considered as a point in $\mathbb{P}^{N'}$, moreover the coordinates of Q' satisfy all the equation $f^{J'}_{A'B'} = 0$, where A', B' are $(n-1)$-tuples $(\alpha_1, \ldots, \alpha_{n-1})$, $(\beta_1, \ldots, \beta_{n-1})$ and $J' \subset [n-1]$. It follows by induction that Q'

corresponds to the image of some $P' \in \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_{n-1}}$ in the Segre embedding in $\mathbb{P}^{N'}$.

Write $P' = ([p_{10} : \cdots : p_{1a_1}], \ldots, [p_{n-1\,0} : \cdots : p_{n-1\,a_{n-1}}])$. Since $m \neq 0$, i.e., the coordinate of Q' corresponding to m is nonzero, then we must have $p_{j0} \neq 0$ for all $j = 1, \ldots n$. Let m_i, $i = 1, \ldots, a_n$, be the coordinate of Q corresponding to the multilinear form $x_{10} \cdots x_{ni_n}$. Then we prove that the coordinate m' corresponding to $x_{1i_1} \cdots x_{ni_n}$ satisfies

$$m' = \frac{m_{i_n}}{m} p_{1i_1} \cdots p_{n-1\,i_{n-1}},$$

This will prove that Q is the image of the point:

$$P = ([p_{10} : \cdots : p_{1a_1}], \ldots, [p_{n-1\,0} : \cdots : p_{n-1\,a_{n-1}}], [p_{n0} : \cdots : p_{na_n}]),$$

where $p_{ni} = m_i/m$ for all $i = 0, \ldots, a_n$.

To prove the claim, take $A = (i_1, \ldots, i_n)$, $B = (0, \ldots, 0)$ and $J = \{n\}$. Then we have $\gamma_k = \alpha_k$ and $\delta_k = 0$ for $k = 1, \ldots, n-1$, while $\gamma_n = 0$ and $\delta_n = i_n$. Thus $M^A(Q) = m'$, $M^B(Q) = m$, $M^D(Q) = m_{i_n}$ and, by induction $M^C(Q) = p_{1i_1} \cdots p_{n-1\,i_{n-1}}$. Since $M^A(Q)M^B(Q) = M^C(Q)M^D(Q)$, the claim follows. □

We observe that all the forms $M^A M^B - M^C M^D$ are quadratic forms in the variables M_i's of $\mathbb{P}^N$. Thus the Segre varieties are defined in $\mathbb{P}^N$ by quadratic equations.

Example 10.5.13 Consider the Segre embedding $s_{1,1} : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$. The 4 variables M_0, M_1, M_2, M_3 in $\mathbb{P}^3$ correspond, respectively, to the multilinear forms

$$M_0 = x_{10}x_{20},\ M_1 = x_{10}x_{21},\ M_2 = x_{11}x_{20},\ M_3 = x_{11}x_{21}.$$

If we take $A = (0, 0)$, $B = (1, 1)$ and $J = \{1\}$, we get that $C = (0, 1)$, $D = (1, 0)$. Thus M^A corresponds to $x_{10}x_{20} = M_0$, M^B corresponds to $x_{11}x_{21} = M_3$, M^C corresponds to $x_{10}x_{21} = M_1$ and M^D corresponds to $x_{11}x_{20} = M_2$. We get thus the equation:

$$M_0M_3 - M_1M_2 = 0.$$

The other choices for A, B, J yield either trivialities or the same equation.

Hence the image of $s_{1,1}$ is the variety defined in $\mathbb{P}^3$ by the equation $M_0M_3 - M_1M_2 = 0$. It is a quadric surface (see Fig. 10.1).

Example 10.5.14 Equations for the image of $s_{1,2} \subset \mathbb{P}^5$ (up to trivialities) are given by:

$$\begin{cases} M_0M_4 - M_1M_3 = 0 \text{ for } A = (0, 0),\ B = (1, 1),\ J = \{1\} \\ M_0M_5 - M_2M_3 = 0 \text{ for } A = (0, 0),\ B = (1, 2),\ J = \{1\} \\ M_5M_1 - M_2M_4 = 0 \text{ for } A = (0, 1),\ B = (1, 2),\ J = \{1\}, \end{cases}$$

where $M_0 = x_{10}x_{20}$, $M_1 = x_{10}x_{21}$, $M_2 = x_{10}x_{22}$, $M_3 = x_{11}x_{20}$, $M_4 = x_{11}x_{21}$, $M_5 = x_{11}x_{22}$.

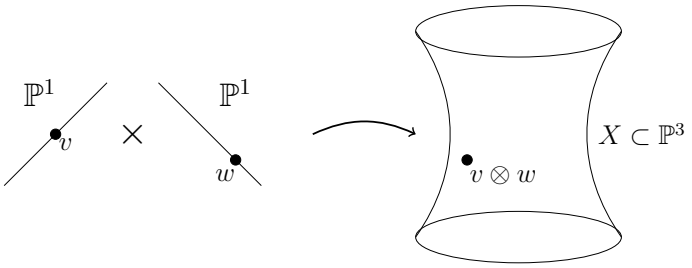


Fig. 10.1 Segre embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ in $\mathbb{P}^3$

Example 10.5.15 We can give a more direct representation of the equations defining the Segre embedding of the product of two projective spaces $\mathbb{P}^{a_1} \times \mathbb{P}^{a_2}$.

Namely, we can plot the coordinates of $Q \in P^N$ in a $(a_1 + 1) \times (a_2 + 1)$ matrix, putting in the entry ij the coordinate corresponding to $x_{1\,i-1}x_{2\,j-1}$.

Conversely, any matrix $(a_1 + 1) \times (a_2 + 1)$ (except for the null matrix) corresponds uniquely to a set of coordinates for a point $Q \in \mathbb{P}^N$. Thus we can identify $\mathbb{P}^N$ with the projective space over the linear space of matrices of type $(a_1 + 1) \times (a_2 + 1)$ over $\mathbb{C}$.

In this identification, the choice of $A = (i, j)$, $B = (k, l)$ and $J = \{1\}$ (choosing $J = \{2\}$ we get the same equation, up to the sign) produces a form equivalent to the 2×2 minor:

$$m_{ij}m_{kl} - m_{il}m_{kj}$$

of the matrix.

Thus, the image of a Segre embedding of two projective space can be identified with the set of matrices of rank 1 (up to scalar multiplication) in a projective space of matrices.

As a consequence of Proposition 10.5.12, one gets the following result.

Theorem 10.5.16 *All the Segre maps are closed in the Zariski topology.*

Proof We need to prove that the image in $s_{a_1,\ldots,a_n}$ of a multiprojective subvariety X of $V = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$ is a projective subvariety of $\mathbb{P}^N$.

First notice that if F is a monomial of multidegree $(d, \ldots, d)$ in the variables x_{ij} of V, then it can be written (usually in several ways) as a product of k multilinear forms in the x_{ij}'s, which corresponds to a monomial of degree d in the coordinates $M_0, \ldots, M_N$ of $\mathbb{P}^N$. Thus, any form f of multidegree $(d, \ldots, d)$ in the x_{ij}'s can be rewritten as a form of degree d in the coordinates M_j's.

Take now a projective variety $X \subset V$ and let $f_1, \ldots, f_s$ be multihomogeneous generators for the ideal of X. Call $(d_{k1}, \ldots, d_{kn})$ the multidegree of f_k and let $d_k = \max\{d_{k1}, \ldots, d_{kn}\}$. Consider all the products $x_{1j_1}^{d_k - d_{k1}} \cdots x_{nj_n}^{d_k - d_{kn}} f_k$. These products are multihomogeneous forms of multidegree $(d_k, \ldots, d_k)$ in the x_{ij}'s. Moreover a point

$P \in V$ satisfies all the equations $x_{1j_1}^{d_k-d_{k1}} \cdots x_{nj_n}^{d_k-d_{kn}} f_k = 0$ if and only if it satisfies $f_k = 0$, since for all i at least one coordinate x_{ij} of P is nonzero.

With the procedure introduced above, transform arbitrarily each form $x_{1j_1}^{d_k-d_{k1}} \cdots x_{nj_n}^{d_k-d_{kn}} f_k$ in a form $F_{kj_1,\ldots,j_n}$ of multidegree $(d_k, \ldots, d_k)$ in the variables of $\mathbb{P}^N$. Then we claim the $s_{a_1,\ldots,a_n}(X)$ is the subvariety of $s_{a_1,\ldots,a_n}(V)$ defined by the equations $F_{kj_1,\ldots,j_n} = 0$. Since $s_{a_1,\ldots,a_n}(V)$ is closed in $\mathbb{P}^N$, this will complete the proof.

To prove the claim, let Q be a point of $s_{a_1,\ldots,a_n}(X)$. The coordinates of Q are obtained by the coordinates of its preimage $P = ([p_{10} : \cdots : p_{1n_1}], \ldots, [p_{n0} : \cdots : p_{na_n}] \in X \subset V$ by computing all the multilinear forms in the x_{ij}'s at P. Thus $F_{kj_1,\ldots,j_n}(Q) = 0$ for all $k, j_1, \ldots, j_n$ if and only if $f_k(P) = 0$ for all k. The claim follows. □

Example 10.5.17 Consider the variety X in $\mathbb{P}^1 \times \mathbb{P}^1$ defined by the multihomogeneous form $f_1 = x_{10}x_{21}^2 + x_{11}x_{20}^2$ of multidegree $(1, 2)$. Then we have:

$$x_{10}f = x_{10}^2x_{21}^2 + x_{10}x_{11}x_{20}^2 = (x_{10}x_{21})^2 + (x_{10}x_{21}\, x_{01}x_{21}) = M_1^2 + M_1M_3,$$
$$x_{11}f = x_{11}x_{10}x_{21}^2 + x_{11}^2x_{20}^2 = (x_{10}x_{21}\, x_{11}x_{21}) + (x_{11}x_{21})^2 = M_1M_3 + M_3^2.$$

These two forms, together with the form $M_0M_3 - M_1M_2$ that defines $s_{1,1}(\mathbb{P}^1 \times \mathbb{P}^1)$ in $\mathbb{P}^3$, define the image of X in the Segre embedding.

Remark 10.5.18 Even if we take a minimal set of forms f_k's that define $X \subset \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$, with the procedure of Theorem 10.5.16 we *do not* find, in general, a *minimal* set of forms that define $s_{a_1,\ldots,a_n}(X)$.

Indeed the ideal generated by the forms $F_{kj_1,\ldots,j_n}$ constructed in the proof of Theorem 10.5.16 needs not, in general, to be radical or even saturated.

We end this section by pointing out a relation between the Segre and the Veronese embeddings of projective and multiprojective spaces.

Definition 10.5.19 A multiprojective space $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$ is *cubic* if $a_i = a$ for all i.

We can embed $\mathbb{P}^a$ into the cubic multiprojective space $\mathbb{P}^a \times \cdots \times \mathbb{P}^a$ (n times) by sending each point P to $(P, \ldots, P)$. We will refer to this map as the *diagonal* embedding. It is easy to see that the diagonal embedding is an injective multiprojective map.

Example 10.5.20 Consider the cubic product $\mathbb{P}^1 \times \mathbb{P}^1$ and the diagonal embedding $\delta : \mathbb{P}^1 \to \mathbb{P}^1 \times \mathbb{P}^1$.

The point $P = [p_0 : p_1]$ of $\mathbb{P}^1$ is mapped to $([p_0 : p_1], [p_0 : p_1]) \in \mathbb{P}^1 \times \mathbb{P}^1$. Thus the Segre embedding of $\mathbb{P}^1 \times \mathbb{P}^1$, composed with δ, sends P to the point $[p_0^2 : p_0p_1 : p_1p_0 : p_1^2] \in \mathbb{P}^3$.

We see that the coordinates of the image have a repetition: the second and the third coordinates are equal, due to the commutativity of the product of complex numbers. In other words the image $s_{1,1} \circ \delta(\mathbb{P}^1)$ satisfies the linear equation $M_1 - M_2 = 0$ in $\mathbb{P}^3$.

We can get rid of the repetition if we project $\mathbb{P}^3 \to \mathbb{P}^2$ by forgetting the third coordinate, i.e., by taking the map $\mathbb{C}^4 \to \mathbb{C}^3$ that maps (M_0, M_1, M_2, M_3) to (M_0, M_1, M_3). The projective kernel of this map is the point $[0:0:1:0]$, which does not belong to $\delta(\mathbb{P}^1)$, since $P = [p_0 : p_1] \in \mathbb{P}^1$ cannot have $p_0^2 = p_1^2 = 0$. Thus we obtain a well defined projection $\pi : s_{1,1} \circ \delta(\mathbb{P}^1) \to \mathbb{P}^2$.

The composition $\pi \circ s_{1,1} \circ \delta : \mathbb{P}^1 \to \mathbb{P}^2$ corresponds to the map which sends $[p_0 : p_1]$ to $[p_0^2 : p_0 p_1 : p_1^2]$. In other words $\pi \circ s_{1,1} \circ \delta$ is the Veronese embedding $v_{1,2}$ of $\mathbb{P}^1$ in $\mathbb{P}^2$.

The previous example generalizes to any cubic Segre product.

Theorem 10.5.21 *Consider a cubic multiprojective space $\mathbb{P}^n \times \cdots \times \mathbb{P}^n$, with $r > 1$ factors. Then the Veronese embedding $v_{n,r}$ of degree r corresponds to the composition of the diagonal embedding δ, the Segre embedding $s_{n,\dots,n}$ and one projection.*

Proof For any $P = [p_0 : \cdots : p_n] \in \mathbb{P}^n$ the point $s_{n,\dots,n} \circ \delta(P)$ has repeated coordinates. Indeed for any permutation σ on $[r]$ the coordinate corresponding to $x_{1i_1} \cdots x_{ri_r}$ of $s_{n,\dots,n} \circ \delta(P)$ is equal to $p_{i_1} \cdots p_{i_r}$, hence its equal to the coordinate corresponding to $x_{1i_{\sigma(1)}} \cdots x_{ri_{\sigma(r)}}$. To get rid of these repetition, we can consider coordinates corresponding to multilinear forms $x_{1i_1} \cdots x_{ri_r}$ that satisfy:

(**) $\quad i_1 \le i_2 \le \cdots \le i_r$.

By easy combinatorial computations, the number of these forms is equal to $\binom{n+d}{d}$. Forgetting the variables corresponding to multilinear forms that do not satisfy condition (**) is equivalent to take a projection $\phi : C^{N+1} \to \mathbb{C}^{N'+1}$, where $N' + 1 = \binom{n+d}{d}$. The kernel of this projections is the set of $(N+1)$-tuples in which the coordinates corresponding to linear forms that satisfy (**) are all zero. Among these coordinates there are those for which $i_1 = i_2 = \cdots = i_r = i$, $i = 0, \dots, n$. So $s_{n,\dots,n} \circ \delta(P)$ cannot meet the projective kernel of ϕ, because that would imply $p_0^d = \cdots = p_n^d = 0$.

Thus $\phi \circ s_{n,\dots,n} \circ \delta(P)$ is well defined for all $P \in \mathbb{P}^n$. The coordinate of $s_{n,\dots,n} \circ \delta(P)$ corresponding to $x_{1i_1} \cdots x_{ri_r}$ is equal to $p_0^{d_0} \cdots p_n^{d_n}$, where, for $i = 0, \dots, n$, d_i d_i, is the number in which i appears among $i_1, \dots, i_r$. Then $d_0 + \cdots + d_n = r$.

It is clear then that computing $\phi \circ s_{n,\dots,n} \circ \delta(P)$ corresponds to computing (once) in P all the monomials of degree r in $x_0, \dots, x_n$. □

10.6 The Chow's Theorem

We prove in this section the Chow's theorem: every projective or multiprojective map is closed in the Zariski topology.

Proposition 10.6.1 *Every projective map $f : \mathbb{P}^n \to \mathbb{P}^m$ factors through a Veronese map, a change of coordinates and a projection.*

Proof By Proposition 9.3.2, there are homogeneous polynomials $f_0, \dots, f_m \in \mathbb{C}[x_0, \dots, x_n]$ of the same degree d, which do not vanish simultaneously at any point $P \in \mathbb{P}^n$, and such that f is defined by the f_j's. Each f_j is a linear combination of monic monomials of degree d. Hence, there exists a change of coordinates g in the target space $\mathbb{P}^N$ of $v_{n,d}$ such that f is equal to $v_{n,d}$ followed by g and by the projection to the first $m+1$ coordinates. Notice that since $(f_0(P), \dots, f_m(P)) \neq 0$ for all $P \in \mathbb{P}^n$, then the projection is well defined on the image of $g \circ v_{n,d}$. □

A similar procedure holds to describe a canonical decomposition of multiprojective maps.

Proposition 10.6.2 *Every multiprojective map $f : \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n} \to \mathbb{P}^N$ factors through Veronese maps, a Segre map, a change of coordinates and a projection.*

Proof By Proposition 9.3.11, there are multihomogeneous polynomials $f_1, \dots, f_s$ in the ring $\mathbb{C}[x_{1,0}, \dots, x_{1,a_1}, \dots, x_{n,0} \dots x_{n,a_n}]$ of the same multidegrees $(d_1, \dots, d_n)$, which do not vanish simultaneously at any point $P \in \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n}$, and such that f is defined by the f_j's. Each f_j is a linear combination of products of monic monomials, of degrees $d_1, \dots, d_n$, in the set of coordinates $(x_{1,0}, \dots, x_{1,a_1}), \dots, (x_{n,0} \dots x_{n,a_n})$, respectively. If v_{a_i,d_i} denotes the Veronese embedding of degree d_i of $\mathbb{P}^{a_i}$ into the corresponding space $\mathbb{P}^{N_i}$, then f factors through $v_{a_1,d_1} \times \cdots \times v_{a_n,d_n}$ followed by a multilinear map $F : \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n} \to \mathbb{P}^N$, which in turn is defined by multihomogeneous polynomials $F_1, \dots, F_s$ of multidegree $(1, \dots, 1)$ (multilinear forms). Each F_j is a linear combination of products of n coordinates in the sets $(x_{1,0}, \dots, x_{1,a_1}), \dots, (x_{n,0} \dots x_{n,a_n})$, respectively. Hence F factors through a Segre map $s_{N_1,\dots,N_r}$, followed by a change of coordinates in $\mathbb{P}^M$, $M = (N_1+1)\cdots(N_r+1) - 1$ which sends the linear polynomial associated to the F_j's to the first $N+1$ coordinates of $\mathbb{P}^N$, and then followed by a projection to the first s coordinates. □

Now we are ready to state and prove the Chow's Theorem.

Theorem 10.6.3 (Chow's Theorem) *Every projective map $f : \mathbb{P}^n \to \mathbb{P}^N$ is Zariski closed, i.e., the image of a projective subvariety is a projective subvariety.*

Every multiprojective map $f : \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_n} \to \mathbb{P}^M$ is Zariski closed.

Proof In view of the two previous propositions, this is just an obvious consequence of Theorems 10.5.7 and 10.5.16. □

We will see, indeed, in Corollary 11.3.7, that the conclusion of Chow's Theorem holds for any projective map $f : X \to Y$ between any projective varieties.

Example 10.6.4 Let us consider the projective map $f : \mathbb{P}^1 \to \mathbb{P}^2$ defined by

$$f(x_1, x_2) = (x_1^3, x_1^2 x_2 - x_1 x_2^2, x_2^3).$$

We can decompose f as the Veronese map $v_{1,3}$, followed by the linear isomorphism $g(a, b, c) = (a, b - c, c - d, d)$ and then followed by the projection π to the first, second and fourth coordinate.

Namely:

$$(\pi \circ g \circ v_{1,3})(x_1, x_2) = (\pi \circ g)(x_1^3, x_1^2 x_2, x_1 x_2^2, x_2^3) = \\ = \pi(x_1^3, x_1^2 x_2 - x_1 x_2^2, x_1 x_2^2 - x_2^3, x_2^3) = (x_1^3, x_1^2 x_2 - x_1 x_2^2, x_2^3).$$

The image of π is a projective curve in $\mathbb{P}^3$, whose equation can be obtained by elimination theory. One can see that, in the coordinates z_0, z_1, z_2 of $\mathbb{P}^2$, $f(\mathbb{P}^1)$ is the zero locus of

$$z_1^3 - z_0 z_2 (z_0 - 3z_1 - z_2).$$

Example 10.6.5 Let us consider the subvariety Y of $\mathbb{P}^1 \times \mathbb{P}^1$, defined by the multihomogeneous polynomial $f = x_0 - x_1$, of multidegree $(1, 0)$ in the coordinates $(x_0, x_1), (y_0, y_1)$ of $\mathbb{P}^1 \times \mathbb{P}^1$. Y corresponds to $[1 : 1] \times \mathbb{P}^1$.

Take the Segre embedding $s : \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$,

$$(x_0, x_1), (y_0, y_1) = (x_0 y_0, x_0 y_1, x_1 y_0, x_1 y_1).$$

Then the image $s(\mathbb{P}^1 \times \mathbb{P}^1)$ corresponds to the quadric Q in $\mathbb{P}^3$ defined by the vanishing of the homogeneous polynomial $g = z_0 z_3 - z_1 z_2$.

The image of Y is a projective subvariety of $\mathbb{P}^3$, which is contained in Q, but it is no longer defined by g and another polynomial: we need two polynomials, other than g.

Namely, Y is defined also by the two multihomogeneous polynomials, of multidegree $(1, 1)$, $f_0 = f y_0 = x_0 y_0 - x_1 y_0$ and $f_1 = f y_1 = x_0 y_1 - x_1 y_1$. Thus $s(Y)$ is defined in $\mathbb{P}^3$ by g, $g_0 = z_0 - z_1$, $g_1 = z_2 - z_3$. (Indeed, in this case, g_0, g_1 alone are sufficient to determine $s(Y)$, which is a line).

Other examples and applications are contained in the exercise section.

10.7 Exercises

Exercise 31 Recall that a map between topological spaces is closed if it sends closed sets to closed sets.

Prove that the composition of closed maps is a closed map, the product of closed maps is a closed map, and the restriction of a closed map to a closed set is itself a closed map.

Exercise 32 Given a linear surjective map $\phi : \mathbb{C}^{m+1} \to \mathbb{C}^{n+1}$ and a subvariety $X \subset \mathbb{P}^m$ which does not meet K_ϕ, find the polynomials that define the projection of X from K_ϕ, in terms of the matrix associated to ϕ.

Exercise 33 Let f, g be nonzero polynomials in $\mathbb{C}[x]$, with f constant. Prove that the resultant $R(f, g)$ is nonzero.

Exercise 34 Consider the Veronese map $\mathbb{P}^1 \to \mathbb{P}^3$ defined in Example 10.5.17 and call X the image. Show that the three equations for X found in Example 10.5.17 are nonredundant: for any choice of two of them, there exists a point Q which satisfies the two equations but does not belong to X.

Reference

1. Walker, R.J.: Algebraic Curves. Princeton University Press, Princeton (1950)

Chapter 11
Dimension Theory



The concept of *dimension* of a projective variety is a fairly intuitive but surprisingly delicate invariant, from an algebraic point of view.

If one considers projective varieties over $\mathbb{C}$ with their natural structure of complex or holomorphic varieties, then the algebraic definition of dimension coincides with the usual (complex) dimension.

On the other hand, for many purposes, it is necessary to deal with the concept from a completely algebraic point of view, so the definition of dimension that we give below is fundamental for our analysis.

The point of view that we take is mainly concerned with a geometric, projective definition, though at a certain point, for the sake of completeness, we cannot avoid to invoke some deeper algebraic result.

11.1 Complements on Irreducible Varieties

The first step is rather technical: we need some algebraic properties of irreducible varieties. We recall that the definition of irreducible topological spaces, together with examples, can be found in Definition 9.1.27 of Chap. 9.

So, from now on, dealing with projective varieties, we will always refer to reducibility or irreducibility with respect to the induced Zariski topology.

Let us start with a characterization of irreducible varieties, in terms of the associated homogeneous ideal (see Corollary 9.1.15).

Definition 11.1.1 An ideal J of a polynomial ring $R = \mathbb{C}[x_0, \ldots, x_N]$ is a *prime ideal* if $f_1 f_2 \in J$ implies that either $f_1 \in J$ or $f_2 \in J$.

Equivalently, J is prime if and only if the quotient ring R/J is a domain, i.e., $a, b \in R/J$, $ab = 0$ implies that either $a = 0$ or $b = 0$.

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Proposition 11.1.2 *Let $Y \subset \mathbb{P}^N$ be a projective variety and call J the homogeneous ideal defined by Y. Then Y is irreducible if and only if J is a prime ideal.*

Proof Assume $Y = Y_1 \cup Y_2$, where the Y_i's are proper closed subsets. Then there exist polynomials f_1, f_2 such that f_i vanishes on Y_i but not on Y. Thus $f_1, f_2 \notin J$, while $f_1 f_2$ vanishes at any point of Y, i.e., $f_1 f_2 \in J$.

The previous argument can be inverted to show that the existence of $f_1, f_2 \notin J$ such that $f_1 f_2 \in J$ implies that Y is reducible. □

Definition 11.1.3 Let $Y \subset \mathbb{P}^N$ be an *irreducible* projective variety and let $J \subset \mathbb{C}[x_0, \ldots, x_N]$ be its homogeneous ideal. Then J is a prime ideal and $R_Y = \mathbb{C}[x_0, \ldots, x_N]/J$ is a domain. So, one can construct the *quotient field* $k(R_Y)$ as the field of all quotients $\{\frac{a}{b} : a, b \in R_Y, b \neq 0\}$, where $\frac{a}{b} = \frac{a'}{b'}$ if and only if $ab' = a'b$.

We call $k(R_Y)$ the *projective function field* of Y and we will indicate this field with K_Y.

Example 11.1.4 The space $\mathbb{P}^N$ itself is defined by the ideal $J = (0)$, thus the projective function field of Y is the field of fractions $\mathbb{C}(x_0, \ldots, x_N)$.

11.2 Dimension

There are several definitions of *dimension* of an irreducible variety. All of them have some difficult aspect. In some cases, it is laborious even to prove that the definition makes sense. For most approaches, it is not immediate to see that the geometric naïve notion of dimension corresponds to the algebraic notion.

Our choice is to make use, as far as possible, of the geometric approach, entering deeply in the algebraic background only to justify some computational aspect.

The final target is the theorem on the dimension of general fibers (Theorem 11.3.5), which allows to manage the computation of the dimension for most applications.

Definition 11.2.1 Given a projective map $f : X \to Y$ we call *fiber* of f over the point $P \in Y$ the inverse image $f^{-1}(P)$.

Remark 11.2.2 Since projective maps are continuous in the Zariski topology, and singletons are projective varieties, then the fiber over any point $P \in Y$ is closed in the Zariski topology, hence it is a projective variety.

Proposition 11.2.3 *For every projective variety $X \subsetneq \mathbb{P}^N$ there exists a linear subspace $L \subset \mathbb{P}^N$, not intersecting X, such that the projection of X from L to a linear subspace L' is surjective, with finite fibers.*

Proof We make induction on N. If $N = 1$ the claim follows immediately, since in this case X is a finite set.

For $N > 1$, fix a point $P \notin X$. A change of coordinates will not make any difference for our claim, so we assume $P = [1 : 0 : \cdots : 0]$. The projection π with center P

maps X to a subvariety of the hyperplane $L' = \mathbb{P}^{N-1}$ defined by $x_0 = 0$ (see Example 10.1.5). Since the fibers of π are closed subvarieties of a line and do not contain the point P of the line, then the fibers of π are finite. By induction we know the existence of a surjective projection ϕ of $\pi(X)$ from a linear subspace $L_0 \subset \mathbb{P}^{N-1} \setminus \pi(X)$ to a linear subspace, and such that ϕ has finite fibers. The composition $\phi \circ \pi$ is surjective, has finite fibers and corresponds to the projection of X from the span L of L_0 and P. Notice that L cannot intersect X, since L_0 does not intersect $\pi(X)$. □

Now we are ready for the definition of *dimension*. Notice that we already have the notion of projective dimension r of a linear subspace of $\mathbb{P}^N$, which corresponds to the projectivization of a linear subspace (of dimension $r + 1$) of the vector space $\mathbb{C}^{N+1}$.

Definition 11.2.4 We say that an irreducible projective variety $X \subset \mathbb{P}^N$ has *dimension n* if there exists a linear subspace $L \subset \mathbb{P}^N$ of (projective) dimension $N - n - 1$, which does not meet X, such that the projection with center L, which maps X to a linear subspace L' of dimension n, is surjective, with finite fibers.

We assign dimension -1 to the emptyset.

Also, when $X = \mathbb{P}^N$, we consider valid to take $L = \emptyset$ and the projection equal to the identity. This implies that $\mathbb{P}^N$ has dimension N.

In the rest of the section, since by elementary linear algebra any linear subspace L' of $\mathbb{P}^N$ is isomorphic to $\mathbb{P}^n$, by abuse we will consider the projection $X \to L'$ as a map $X \to \mathbb{P}^n$.

It is clear that two projective varieties which are isomorphic under a change of coordinates share the same dimension.

Example 11.2.5 Since $\mathbb{P}^0$ has just one point, clearly singletons have a surjective projection to $\mathbb{P}^0$, with finite fibers. So singletons have dimension 0.

Finite projective varieties are reducible, unless they are singleton (see Exercise 26). Thus, by definition, singletons are the only projective irreducible varieties of dimension 0.

Example 11.2.6 The linear subspace L in $\mathbb{P}^N$ defined by $x_{n+1} = \cdots = x_N = 0$ is isomorphic to $\mathbb{P}^n$ and a projection from the subspace L' defined by $x_1 = \cdots = x_n = 0$ to L is an isomorphism of L to itself. Thus L has dimension n.

Base changes provide isomorphisms between L and any linear variety of projective dimension n. Thus, for linear subspaces, the projective dimension is a value for the dimension, as defined above.

Example 11.2.7 Let C be a projective variety in $\mathbb{P}^2$, defined by the vanishing of one irreducible homogeneous polynomial $g \neq 0$. Then C is an irreducible variety (Example 9.1.37) and it contains infinitely many points (Example 9.1.13).

By Lemma 9.1.5, there exists a point $P_0 \in \mathbb{P}^2$ which does not belong to C. The projection π from P_0 maps C to $\mathbb{P}^1$. Every fiber of π is a proper projective subvariety of a line, since it cannot contain P_0. Since the Zariski topology on a line is the cofinite topology (Example 9.1.12), then every fiber of π is finite. The image of π, which is

a projective subvariety of $\mathbb{P}^1$ (by the Chow's Theorem), cannot be finite (since C is infinite), so it coincides with $\mathbb{P}^1$. Hence π is surjective.

We just proved that C has dimension 1.

The following remark, on the structure of projections, will be useful to produce inductive arguments for the dimension of a variety.

Proposition 11.2.8 *Fix a linear space $L \subset \mathbb{P}^N$, which does not meet a variety $X \subset \mathbb{P}^N$, and fix a linear subspace L', disjoint from L, such that* $\dim(L') + \dim(L) = N - 1$. *Fix a point $P \in L$, a linear subspace $M \subset L$ of dimension* $\dim(L) - 1$ *and disjoint from P, and a hyperplane H containing L' and disjoint from P.*

Then the projection ϕ of X from L to L' is equal to the projection ϕ_P of X from P to H, followed by the projection ϕ' (in $H = \mathbb{P}^{N-1}$) of $\phi_P(X)$ from $\phi_P(M)$ to L'.

Proof A point $Q \in X$ is sent by ϕ_P to the intersection of the line QP with H. Similarly, since the span of M and P is L, then M is sent by ϕ_P to the intersection $L \cap H$. In turn, $\phi_P(Q)$ is sent by ϕ' to the intersection of L' with the span of $\phi_P(Q)$ and $\phi_P(M)$ in H.

Since, by elementary linear algebra, the span of the line QP and $L \cap H$ is equal to the intersection of H with the span of L and Q, then the claim follows. □

By now, we do not know yet that the dimension of an irreducible projective variety is uniquely defined, since we did not exclude the existence of two different surjective projections of X to two linear subspaces of different dimensions, both with finite fibers.

It is not easy to face the problem directly. Instead, we show that the existence of a map $X \to \mathbb{P}^n$ with finite fibers is related with a numerical invariant of the irreducible variety X.

The invariant which defines the dimension is connected with a notion in the algebraic theory of field extensions: the *transcendence degree*. We recall some basic facts in the following remark. For the proofs, we refer to section II of the book [1] (see also some exercises, at the end of the chapter).

Remark 11.2.9 (Field extensions) Let K_1, K_2 be fields, with a nonzero homomorphism $\phi : K_1 \to K_2$. Then ϕ is injective, since the kernel is an ideal of K_1, hence it is trivial. So we can consider ϕ as a inclusion which realizes K_2 as an extension of K_1.

The extension is *algebraic*, when K_2 is finitely generated as a vector space over K_1. Otherwise the extension is *transcendent*. If K_2 is an algebraic extension of K_1, then for any $e \in K_2$ the powers $e, e^2, e^3, \ldots, e^h, \ldots$ become eventually linearly dependent over K_1. This means that there exists a polynomial $p(x)$, with coefficient in K_1, such that $p(e) = 0$.

Conversely, given any extension K_2 of K_1, for any $e \in K_2$ define $K_1(e)$ as the minimal subfield of K_2 which contains K_1 and e. We say that e is an *algebraic element* over K_1 if $K_1(e)$ is an algebraic extension of K_1, otherwise e is a *transcendental element*.

The set of all the elements of K_2 which are algebraic over K_1 is a field $K' \supset K_1$. We call K_1 the *algebraic closure* of K_1 in K_2. If $K' = K_1$, we say that K_1 is algebraically closed in K_2. In this case any element of $K_2 \setminus K_1$ is trascendent over K_1. A field K_1 is *algebraically closed* if any non-trivial extension K_2 of K_1 contains only transcendental elements, i.e., if K_1 is algebraically closed in any extension. $\mathbb{C}$ is the most popular example of an algebraically closed field.

If $K_2 = K_1(x)$ is the field of fractions of the polynomial ring $K[x]$, then K_2 is a trascendent extension. Conversely, if e is any transcendental element over K_1, then $K_1(e)$ is isomorphic to $K_1(x)$.

A set of elements $e_1, \ldots, e_n \in K_2$ such that for all i e_i is trascendent over $K_1(e_1, \ldots, e_{i-1})$ and K_2 is an algebraic extension of $K_1(e_1, \ldots, e_n)$ is a *transcendence* basis of the extension. All the transcendence basis have the same number of elements, which is called the *transcendence degree* of the extension.

If K_2 has transcendence degree d over K_1 and K_3 is an algebraic extension of K_2, then K_3 has transcendence degree d over K_1.

Proposition 11.2.10 *A surjective projection $\phi : X \to \mathbb{P}^n$ determines an inclusion κ_ϕ of the field $K_{\mathbb{P}^n} = \mathbb{C}(x_0, \ldots, x_n)$ into the projective function field K_X (see Definition 11.1.3).*

Proof Assume in the proof that $X \subset \mathbb{P}^m$.

The map ϕ is defined by n homogeneous polynomials $F_0, \ldots, F_n$ of the same degree d in the coordinates of $\mathbb{P}^m$, by Proposition 9.3.2. For any element $f/g \in \mathbb{C}(x_0, \ldots, x_n)$ define $F = f(F_0, \ldots, F_n)$ and $G = g(F_0, \ldots, F_n)$. Notice that we cannot have $G(P) = 0$ for all $P \in X$. Indeed this will imply $g(\phi(P)) = 0$ for all $P \in X$, hence $g(Q) = 0$ for all $Q \in \mathbb{P}^n$, since ϕ is surjective. By Lemma 9.1.5 this implies $g = 0$, a contradiction.

It follows that the quotient q of the equivalence classes of F and G is a well defined element of K_X. We define $\kappa_\phi(f/g) = q$.

Notice that q is 0 only if F vanishes on every point of X. As above this implies that f vanishes on every point of $\mathbb{P}^n$, i.e., $f = 0$. Thus there are elements $f/g \in \mathbb{C}(x_0, \ldots, x_n)$ such that $\kappa_\phi(f/g) \neq 0$, i.e., κ_ϕ is not the zero map. Thus κ_ϕ is injective. □

From now, when there exists a surjective projective map $\phi : X \to \mathbb{P}^n$ we will identify $\mathbb{C}(x_1, \ldots, x_n)$ with the subfield $\kappa_\phi(\mathbb{C}(x_0, \ldots, x_n))$ of K_X.

Theorem 11.2.11 *Assume there exists a surjective projection $\phi : X \to \mathbb{P}^n$ from the irreducible variety $X \subset \mathbb{P}^N$ to a linear space $\mathbb{P}^n$, with finite fibers. Then the quotient field K_X is an algebraic extension of $K_{\mathbb{P}^n} = \mathbb{C}(x_0, \ldots, x_n)$.*

Proof Assume that the map has finite fibers. We will indeed prove that the class of any variable x_i in the quotient ring R_X is algebraic over $\mathbb{C}(x_0, \ldots, x_n)$. Since these classes generate the quotient field of R_X, the claim will follow.

First assume that $N = n + 1$, so that ϕ is the projection from a point. We may also assume, after a change of coordinates, that ϕ is the projection from $P = [0 : \cdots :$

$0 : 1] \notin X$. Consider the homogeneous ideal J of X and an element $g \in J$ such that $g(P) \neq 0$. Write g as a polynomial in $x_N = x_{n+1}$ with coefficients in $\mathbb{C}(x_0, \ldots, x_n)$:

$$g = x_{n+1}^d a_d + x_{n+1}^{d-1} a_{d-1} + \cdots + a_0,$$

where each $a_i = a_i(x_0, \ldots, x_n)$ is a polynomial in $\mathbb{C}(x_0, \ldots, x_n)$. We cannot have $d = 0$, otherwise $g \in \mathbb{C}(x_0, \ldots, x_n)$ and g vanishes at P. Since the class of g vanishes in the quotient ring R_X, we get that the class of x_{n+1} is algebraic over $\mathbb{C}(x_0, \ldots, x_n)$. As the classes of $x_0, \ldots, x_n$ are clearly algebraic over $\mathbb{C}(x_0, \ldots, x_n)$, we are done in this case.

Then make induction on $N - n$. Let ϕ be the projection from L and fix a point $P \in L$. We may assume that $P = [0 : \cdots : 0 : 1] \notin X$. The projection ϕ factorizes though the projection ϕ_P from P to $\mathbb{P}^{N-1}$ followed by the projection from $\phi_P(L)$ (see Proposition 11.2.8). By induction we know that the classes of $x_0, \ldots, x_{N-1}$ are algebraic over $\mathbb{C}(x_0, \ldots, x_n)$. Arguing as above, we get that x_N is algebraic over $\mathbb{C}(x_0, \ldots, x_{N-1})$. This concludes the proof. □

We can now prove that our definition of *dimension* is unambiguous.

Corollary 11.2.12 *Let $X \subset \mathbb{P}^N$ be an irreducible variety. If there exists a surjective projection $\phi : X \to \mathbb{P}^n$ with finite fibers, then the transcendence degree of the quotient field of X is n.*

In particular, if $m \neq n$, then one cannot have a surjective projection $\phi' : X \to \mathbb{P}^m$ with finite fibers.

Proof The second statement follows since the transcendence degree of the quotient field of X does not depend on the projections ϕ, ϕ'. □

For reducible varieties, the definition of dimension is a straightforward extension of the definition for irreducible varieties.

Definition 11.2.13 Let $X_1, \ldots, X_m$ be the irreducible components of a variety X (we recall that, by Theorem 9.1.32, the number of irreducible varieties of X is finite). Then we define:

$$\dim(X) = \max\{\dim(X_i)\}.$$

From the definition of dimension and its characterization in terms of field extensions, one can prove the following properties.

Proposition 11.2.14 *Let $X \subset \mathbb{P}^N$ be any variety and let X' be a subvariety of X. Let ϕ be a projection of X to some linear subspace, whose fibers are finite. Then:*

(a) $\dim(X) = \dim(\phi(X))$.
(b) $\dim(X') \leq \dim(X)$.
(c) In particular $\dim(X) \leq N$.
(d) If $X \subset \mathbb{P}^N$ has dimension N, then $X = \mathbb{P}^N$.

Proof The first claim is clear: fix an irreducible component Y of X and take a projection ϕ' of $\phi(Y)$ to some linear space $\mathbb{P}^n$, whose fibers are finite (thus $n = \dim(\phi(Y))$). Then the composition $\phi' \circ \phi$ maps Y to $\mathbb{P}^n$ with finite fibers, so that $n = \dim(Y)$.

The proof of (c) is straightforward from the definition of dimension. Then (b) follows since a surjective projection ϕ with finite fibers from X to some $\mathbb{P}^n$ ($n = \dim(X)$) maps X', with finite fibers, to a subvariety of $\mathbb{P}^n$.

Finally, to see (d) assume $X \neq \mathbb{P}^N$ and fix a point $P \notin X$. Consider the projection ϕ from P. The fiber of ϕ which contains any point $Q \in X$ is a projective subvariety of the line QP and misses P, thus it is finite. Hence ϕ maps X to $\mathbb{P}^{N-1}$ with finite fibers. Thus $\dim(X) = \dim(\phi(X)) \leq N - 1$. □

Example 11.2.15 Hypersurfaces X in $\mathbb{P}^N$ have dimension $N - 1$.

Indeed take a points $P \notin X$ and consider the projection π of X from P to $\mathbb{P}^{N-1}$. The fibers of the projection are closed proper subvarieties of lines, hence they are finite. Moreover, the projection is surjective. Indeed let $p \in \mathbb{C}[x_0, \ldots, x_N]$ be a polynomial which defines X, i.e., $X = X(p)$. Take any point $Q \in \mathbb{P}^{N-1}$ and call L the line PQ. After a change of coordinates, we may assume $P = [1 : 0 : \cdots : 0]$ and $Q = [0 : 1 : 0 : \cdots : 0]$. Then the restriction of p to L is a polynomial $\bar{p}$ in the coordinates x_0, x_1, of the same degree of p, unless it is 0. In any case, $\bar{p}$ has some nontrivial solution, corresponding to points of X which are mapped to Q by the projection. The claim follows.

We generalize the computation of the dimension of hypersurfaces as follows.

Lemma 11.2.16 *Let X be an subvariety of $\mathbb{P}^N$, with infinitely many points. Then for any hyperplane H of $\mathbb{P}^N$ we have $X \cap H \neq \emptyset$.*

Proof Since the number of irreducible components of X is finite, there exists some component of X which contains infinitely many points. So, without loss of generality, we may assume that X is irreducible.

If $X \subset \mathbb{P}^1$, then $X = \mathbb{P}^1$ because X is closed in the Zariski topology, which is the cofinite topology, and the claim is trivial. Then assume $X \subset \mathbb{P}^N$, $N > 1$, and proceed by induction on N.

Fix a hyperplane H and assume that $H \cap X = \emptyset$. Fix a linear subspace $L \subset H$ of dimension $N - 2$ and consider the projection ϕ of X from L to a general line ℓ. By Chow's Theorem, the image $\phi(X)$ is a closed irreducible subvariety of ℓ, and it does not contain, by construction, the point $H \cap \ell$. Thus $\phi(X)$ is finite and irreducible, hence it is a point Q (see Exercise 26). Then X is contained in the hyperplane H' spanned by L and Q, which is a $\mathbb{P}^{N-1}$, and it does not meet the hyperplane $H' \cap H$ of H'. This contradicts the inductive assumption. □

Next result can be viewed as a generalization of Example 11.2.15.

Theorem 11.2.17 *Let $X \subset \mathbb{P}^N$ be an irreducible variety and consider a homogeneous polynomial $g \in \mathbb{C}[x_0, \ldots, x_N]$ which does not belong to the ideal of X. Then* $\dim(X \cap X(g)) = \dim(X) - 1$.

Proof Set $n = \dim(X)$. If $n = N$, then $X = \mathbb{P}^N$ (Proposition 11.2.14 part (d)) and $X \cap X(g) = X(g)$, so the claim holds in this case.

If $X \neq \mathbb{P}^N$, fix a surjective projection with finite fibers $\phi : X \to \mathbb{P}^n$ from some linear subspace $L \subset \mathbb{P}^N$, of dimension $N - n - 1$.

Assume first that g is a general linear form, so that $X(g)$ is a general hyperplane. Fix a general point $Q \in \mathbb{P}^n$ and consider the span L' of Q and L. The intersection of L' with X is a general fiber of ϕ, thus it is finite. Since $X(g)$ is a general hyperplane, then it will contain none of the points of $L' \cap X$. This means that Q is not in the image of $\phi(X \cap X(g))$. Thus ϕ maps (with finite fibers) $X \cap X(g)$ to a proper subvariety of $\mathbb{P}^n$. This proves that $\dim(X \cap X(g)) < n$.

Next, take a general hyperplane $H' = \mathbb{P}^{n-1}$ of $\mathbb{P}^n$ and consider the projection ϕ' of $\phi(X \cap X(g))$ from Q to H'. Notice that since $Q \notin \phi(X \cap X(g))$, then ϕ' has finite fibers, as in the proof of Proposition 11.2.14. The composition ϕ_0 of ϕ' and ϕ corresponds to the projection of $X \cap X(g)$ from L'. For all $Q' \in \mathbb{P}^{n-1}$, the fiber of ϕ_0 over Q' corresponds to the intersection of $X \cap X(g)$ with the span L'' of Q' and L', which is also the span of L, Q, Q', i.e., the span of L and the line QQ'. Since $\phi : X \to \mathbb{P}^n$ surjects, the inverse image of the line QQ' in ϕ is a subvariety $Y \subset X$ which contains infinitely many points. Thus, by Lemma 11.2.16, the intersection of Y with $H = X(g)$ is not empty. Any point $P \in Y \cap X(g)$ is a point of $X \cap X(g)$ mapped to Q' by the projection ϕ_0. It follows that ϕ_0 maps surjectively $X \cap X(g)$ to $\mathbb{P}^{n-1}$, with finite fibers. Thus $\dim(X \cap V(g)) = n - 1$.

Now we prove the general case, by induction on N. If $N = 1$, then X is either $\mathbb{P}^1$ or a point, and the claim is obvious. If $N > 1$, use the fact that for a general hyperplane H of $\mathbb{P}^N$, the intersection $X(g) \cap H$ is a hypersurface in $\mathbb{P}^{N-1}$ which does not contain $H \cap X$. Then, using the inductive hypothesis and the intersection with hyperplanes:

$$\begin{aligned}\dim(X) = \dim(X \cap H) + 1 &= \dim((X \cap H) \cap (X(g) \cap H)) + 2 \\ &= \dim((X \cap X(g)) \cap H) + 2 = \dim(X \cap X(g)) + 1.\end{aligned}$$

□

Corollary 11.2.18 *Let X be a variety of dimension n in $\mathbb{P}^N$ and let L be a linear subspace of dimension $d \geq N - n$. Then $L \cap X \neq \emptyset$.*

Proof If $n = 1$ then X is infinite and the claim follows from Lemma 11.2.16. Then make induction on N. The result is clear if $N = 1$. For $n, N > 1$, take a general hyperplane H containing L and identify H with $\mathbb{P}^{N-1}$. Fix an irreducible component X' of dimension n in X. Then either $X' \subset H$ or $\dim(H \cap X') = n - 1$ by Theorem 11.2.17. In any case, the claim follows by induction. □

The previous result has an important consequence, that will simplify the computation of the dimension of projective varieties.

Corollary 11.2.19 *Let L be a linear subspace which does not intersect a projective variety X and consider the projection ϕ of X from L to some linear space $\mathbb{P}^n$, disjoint from L.*

Then ϕ has finite fibers. Thus, ϕ is surjective if and only if $n = \dim(X)$.

Proof Assume that there exists a fiber $\phi^{-1}(Q)$ which is infinite, for some $Q \in \mathbb{P}^n$. Then $\phi^{-1}(Q)$ is an infinite subvariety of the span L' of L and Q. In particular $\dim(\phi^{-1}(Q)) \geq 1$. Since L' is a linear subspace of dimension $n+1$ which contains both L and $\phi^{-1}(Q)$, then by the previous corollary we have $L \cap \phi^{-1}(Q) \neq \emptyset$. This contradicts the assumption that L and X are disjoint. □

Corollary 11.2.20 *Let X be a variety of dimension n in $\mathbb{P}^N$ and let Y be the intersection of m hypersurfaces $Y = X(g_1) \cap \cdots \cap X(g_m)$. Then:*

$$\dim(X \cap Y) \geq n - m.$$

Proof Call $X_1, \ldots, X_k$ the irreducible components of X. First assume $m = 1$. For $i = 1, \ldots, k$ either Y contains X_i or $\dim(Y \cap X_i) = \dim(X_i) - 1$, by Theorem 11.2.17. It follows that either $\dim(Y \cap X) = \dim(X)$ or $\dim(Y \cap X) = \dim(X) - 1$. The general claim follows soon by induction. □

Example 11.2.21 By Corollary 11.2.20, a variety $Y = X(g_1) \cap \cdots \cap X(g_m)$ has dimension at least $N - m$, because $X(g_1)$ has dimension $N - 1$.

The inequality is indeed an equality if $m = 1$, by Example 11.2.15.

We observe that the dimension of an intersection $Y = X(g_1) \cap \cdots \cap X(g_m)$, $m > 1$, can be strictly bigger than $N - m$ (see Example 11.2.23).

A variety Y is called *complete intersection* if $\dim(Y) = N - m$.

Example 11.2.22 Let X be a Veronese variety in $\mathbb{P}^N$, image of the d-Veronese embedding $v_{n,d}$ of $\mathbb{P}^n$. Then $\dim(X) = n$.

Indeed we can produce a projection of X to $\mathbb{P}^n$, with finite fibers, as follows. Consider coordinates $x_0, \ldots, x_n$ for $\mathbb{P}^n$, so that the coordinates of $\mathbb{P}^N$ correspond to monomials of degree d in the x_i's. After a change of coordinates, we may arrange the monomials so that the first $n+1$ coordinates correspond to $x_0^d, \ldots, x_n^d$. Then take the projection ϕ of X to the linear space $\mathbb{P}^n$ spanned by the first $n+1$ coordinates. For each $Q = [q_0 : \cdots : q_n]$, the inverse image of Q in ϕ is the set of points in X which are of the form $v_d([p_0 : \cdots : p_n])$ with $p_i^d = q_i$ for $i = 0, \ldots, n$. Since we have only a finite number of choices for each of these numbers p_j's, the fibers of ϕ are finite.

Example 11.2.23 We know by Example 10.5.17 that the image X of the Veronese map of degree 3, $\mathbb{P}^1 \to \mathbb{P}^3$, is minimally defined in $\mathbb{C}[M_0, \ldots, M_3]$ by the three quadric equations

$$M_0M_3 - M_1M_2 = 0 \quad M_0M_2 - M_1^2 = 0 \quad M_1M_3 - M_2^2 = 0.$$

On the other hand, by Example 11.2.22, X has dimension 1. Thus X is not complete intersection.

Next, we have the following consequence of the description of projective maps $\mathbb{P}^n \to \mathbb{P}^N$, obtained in the proof of the Chow's Theorem.

Theorem 11.2.24 *Let $f : \mathbb{P}^n \to \mathbb{P}^N$ be a nonconstant projective map. Then the dimension of $f(\mathbb{P}^n)$ is n. Hence $N \geq n$.*

Proof We know that f is equivalent to a Veronese map $v_{n,d}$, followed by a change of coordinates and a projection π. Call Y the image of f and consider a surjective projection $\pi' : Y \to \mathbb{P}^m$ with finite fibers, with $m = \dim(Y)$. Then $\pi' \circ \pi$ is a surjective projection of $v_{n,d}(\mathbb{P}^n)$ to $\mathbb{P}^m$, with finite fibers, by Corollary 11.2.19. Thus $\dim(v_{n,d}(\mathbb{P}^n)) = m$, hence $m = n$ by the previous example. □

Example 11.2.25 Let X be a Segre variety in $\mathbb{P}^N$, image of the Segre embedding $s_{n,m}$ of $\mathbb{P}^n \times \mathbb{P}^m$. Then $\dim(X) = n + m$.

Indeed we can produce a projection of X to $\mathbb{P}^{n+m}$, with finite fibers, as follows. Consider coordinates $x_0, \ldots, x_n$ for $\mathbb{P}^n$ and coordinates $y_0, \ldots, y_m$ for $\mathbb{P}^m$, so that the coordinates z_{ij} of $\mathbb{P}^N$ correspond to the products $z_{ij} = x_i y_j$, with $i = 0, \ldots, n$ and $j = 0, \ldots, m$.

Thus the difference $i - j$ of the two indices of z_{ij} ranges between $-m$ and n. We define the map $\phi : \mathbb{P}^N \to \mathbb{P}^{n+m}$ by sending the point $[z_{ij}]$ to $[q_0 : \cdots : q_{n+m}]$ where

$$q_k = \sum_{i-j=k-m} z_{ij}.$$

Since ϕ is linear and surjective, it defines a projection of X, provided that X does not intersect the kernel of ϕ. This last fact can be proved by induction on n, m. It is clear if either m or n are 0. In the general case, if for some $P \in X$, $P = [x_i y_j]$ one has $q_k = 0$ for all $k = 0, \ldots, n + m$, then in particular $0 = z_{0m} = x_0 y_m$. If $x_0 = 0$, then the image of $\phi(P)$ corresponds to the image of the point $s(Q)$ of $Q = ([x_1 : \cdots : x_n], [y_0 : \cdots : y_m])$ in the Segre embedding of $\mathbb{P}^{n-1} \times \mathbb{P}^m$, so that we get a contradiction with the inductive assumption. A similar argument works when $y_m = 0$.

From Corollary 11.2.19, we obtain that the general fiber of the restriction $\phi_X = \phi_{|X}$ is finite. It remains to prove that ϕ_X is surjective. We prove it by induction on the number n. If $n = 1$, the claim is obvious. Assume that the claim holds for $n - 1, m$. Fix a *general* point $[1 : q_1 : \cdots : q_{n+m}] \in \mathbb{P}^{n+m}$. By induction the map $\phi_{X'}$ from the Segre embedding X' of $\mathbb{P}^{n-1} \times \mathbb{P}^m$ to $\mathbb{P}^{n+m-1}$ surjects. This implies that we can find a point $Q' = ([x_0 : \cdots : x_{n-1}], [1 : y_1 : \cdots : y_m])$ whose image in $\phi_{X'}$ is the point $[q'_1 : \cdots : q'_{n+m}]$, where $q'_i = q_i - y_i$. Then the image in ϕ_X of the point $Q = ([x_0 : \cdots : x_{n-1} : 1], [1 : y_1 : \cdots : y_m])$ is $[1 : q_1 : \cdots : q_{n+m}]$.

As in Theorem 11.2.24, one proves the following corollary on the dimension of the image of a multiprojective map.

Theorem 11.2.26 *Let $X = \mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_k}$ be a product of projective spaces and let $f : X \to \mathbb{P}^N$ be a multiprojective map, which is nonconstant on any factor. Then the dimension of $f(X)$ is $a_1 + \cdots + a_k$. Hence $N \geq a_1 + \cdots + a_k$.*

Remark 11.2.27 The definition of dimension given in this section can be used to provide a definition for the *degree* of a projective variety.

Namely, one can prove that given a surjective projection $\phi : X \to \mathbb{P}^n$, with finite fibers, the cardinality d of the general fiber is fixed, and it corresponds to the (linear) dimension of the residue field $k(X)$, considered as a vector field over $C[x_1, \ldots, x_n]$. Thus d does not depend on the choice of ϕ. The number d is called the degree of X.

In general, the computation of the degree of a projective variety is not straightforward. Since the results on which the very definition of degree is based, as well as the computation of the degree for elementary examples, requires a theory which goes beyond the aims of the book, we do not include it here. The interested reader can find an account of the theory in section VII of [1] and in the book [2].

11.3 General Maps

From Definition 9.3.1, only locally a general projective map between projective varieties $f : X \to Y$ is equivalent, up to changes of coordinates, to a Veronese map followed by a projection.

It is possible to construct examples of maps f for which an equivalent version of Theorem 11.2.24 does not hold. In particular, one can have $\dim(Y) < \dim(X)$.

We account in this section some relations between the dimension of projective varieties connected by projective maps and the dimension of general fibers of the map.

Lemma 11.3.1 *Let $X \in \mathbb{P}^n$ be a variety. Fix $P \in \mathbb{P}^n$ and let U be a Zariski open subset of X not containing P. Consider the projection $\pi : U \to \mathbb{P}^{n-1}$ and let U' be the image. Let $X' \subset U$ be the set of points Q such that P belongs to the closure of the fiber $\pi^{-1}(\pi(Q))$. Then X' is closed, in the Zariski topology of U. Moreover the set of points $Q \in U$ such that the fiber $\pi^{-1}(\pi(Q))$ is finite is a (possibly empty) open subset of U.*

Proof After a change of coordinates, we may assume that $P = [1 : 0 : \cdots : 0]$ and π maps a point $[p_0 : p_1 : \cdots : p_n]$ to $[p_1 : \cdots : p_n]$. Let f be a polynomial in the coordinates $x_0, \ldots, x_n$ of $\mathbb{P}^n$, which vanishes at the points of X, and write

$$f = g_0 + x_0 g_1 + \cdots + x_0^d g_d,$$

where each g_i is a polynomial in $x_1, \ldots, x_n$. Fix a point $Q = [q_0 : q_1 : \cdots : q_n] \in U$ and assume that $g_i(Q) = g_i(q_1, \ldots, q_n)$ does not vanish, for some i. Then the fiber $\pi^{-1}(\pi(Q))$ is contained in the subset of the line PQ defined by the polynomial $f_0 = f(x_0, q_1, \ldots, q_n) = 0$, which is finite, since f_0 is nontrivial. Thus the fiber $\pi^{-1}(\pi(Q))$ is finite, hence closed, and P cannot belong to the closure of $\pi^{-1}(\pi(Q))$. Conversely, assume that for all the generators f of the homogeneous ideal of X one has $g_i(Q) = 0$. Then any point $Q' = [b : q_1 : \cdots : q_n]$ of the line PQ belongs to X,

thus $\pi^{-1}(\pi(Q))$ contains an open subset in the line PQ which is non-empty, since Q belongs to it. Hence the closure of $\pi^{-1}(\pi(Q))$ coincides with the whole line PQ.

It follows that if we take a set of generators $f_1, \ldots, f_k$ for the homogeneous ideal of X and for each j we write $f_j = g_{0j} + x_0 g_{1j} + \cdots + x_0^d g_{dj}$, then the set of $Q \in U$ such that P belongs to the closure of $\pi^{-1}(\pi(Q))$, which is equal to the set such that $\pi^{-1}(\pi(Q))$ is not finite, is defined by the vanishing of all the polynomials g_{ij}'s. The claims follow. □

Corollary 11.3.2 *Let $f : X \to Y$ be a projective map. Then the set U of points $Q \in Y$ such that the fiber $f^{-1}(Q)$ of the map f over Q is finite is a (possibly empty) Zariski open set in X. Thus the set of points $P \in X$ such that the fiber $f^{-1}(f(P)))$ is finite is open in X.*

Proof There exists a finite open cover $\{U_i\}$ of X such that the restriction of f to each U_i coincides with a Veronese embedding followed by a change of coordinates and a projection π_i. We claim that $U \cap U_i$ is open for each i. Indeed π_i can be viewed as the composition of a finite chain of projections from points P_j's, and in each projection, by Lemma 11.3.1, the fibers are finite in an open subset. Thus $U = \bigcup(U \cap U_i)$ is open. □

Next, we give a formula for the dimension of projective varieties, which is the most useful and used formula in effective computations. It is based on the link between the dimensions of two varieties X, Y, when there exists a projective map $f : X \to Y$.

The first step toward the formula is the notion of *semicontinuous maps* on projective varieties.

Definition 11.3.3 Let X be a projective variety, and consider a map $g : X \to \mathbb{Z}$. We say that g is *upper semicontinuous* if for every $z \in \mathbb{Z}$ the set of points $P \in X$ such that $g(P) \geq z$ is closed in the Zariski topology.

The most important upper semicontinuous function that we will study in this section is constructed as follows. Take a projective map f between projective varieties $f : X \to Y$ and define μ_f by:

$$\mu_f(Q) = \text{ the dimension of the fiber } f^{-1}(Q).$$

Definition 11.3.4 A continuous map $f : X \to Y$ is *dominant* if the image is dense in Y.

Theorem 11.3.5 *Let X be irreducible and let $f : X \to Y$ be a dominant projective map. Then f is surjective, the function μ_f is upper semicontinuous and its minimum is* $\dim(X) - \dim(Y)$.

Proof Since the statement is trivial if $\dim(X) = 0$, we proceed by induction on $\dim(X)$.

First let us prove that $\dim(X) \geq \dim(Y)$. Let $Y \subset \mathbb{P}^m$. Notice that $f(X)$ cannot miss any component of Y, since it is dense in Y. Consider thus a hyperplane H not

containing a point of $f(X)$ chosen in any component of Y. Then $\dim(H \cap Y) = \dim(Y) - 1$, by Theorem 11.2.17. Consider an irreducible component Y' of $H \cap Y$, of dimension $\dim(Y) - 1$. The inverse image $f^{-1}(Y')$ is a proper subvariety of X, so it has finitely many irreducible components. Since Y' is irreducible, it cannot be the union of a finite number of proper closed subsets (see Exercise 23). Thus there exists a component X' of $f^{-1}(Y')$ such that $f_{|X'} : X' \to Y'$ is dominant. As X' is a proper closed subvariety of X, then $\dim(X') < \dim(X)$, by Theorem 11.2.17 again. Thus, by induction, $\dim(X) > \dim(X') \geq \dim(Y') = \dim(Y) - 1$, and the claim follows.

Next we prove that f is surjective. Assume there exists $P \in Y$ not belonging to $f(X)$. As above, fix a general hyperplane H of $\mathbb{P}^m$ containing P and missing some point of any component of Y in $f(X)$. Take a component Y' of $Y \cap H$, which contains P. As above, $f^{-1}(Y')$ is a proper closed subvariety of X and there exists a component X' of $f^{-1}(Y')$ which dominates Y'. By induction the map $f_{|X'} : X' \to Y'$ is dominant, which contradicts that $P \notin f(X)$.

Next, assume there exists a point $P \in X$ such that the fiber $f_P = f^{-1}(f(P))$ is finite. We know by Corollary 11.3.2 that the set of points $P' \in X$, such that the fiber $f_{P'} = f^{-1}(f(P'))$ is finite is open in X. We claim that $\dim(Y) = \dim(X)$. Indeed let $X \subset \mathbb{P}^n$ and fix a hyperplane H of $\mathbb{P}^n$ which misses all the points of f_P and set $X' = X \cap H$. Then there exists an irreducible component X' of $X \cap H$ of dimension $\dim(X') = \dim(X) - 1$. Moreover the restriction of f to X' maps X' to a closed subvariety Y' of Y. Since X is irreducible and f is surjective, then Y is irreducible, thus $\dim(Y') < \dim(Y)$. By induction, since $f_{|X'}$ has some finite fibers, then $\dim(X') = \dim(Y')$. One has

$$\dim(Y) \leq \dim(X) = \dim(X') + 1 = \dim(Y') + 1 \leq \dim(Y)$$

and the claim follows.

Finally, we prove the last statement by induction on the minimum of the dimension of the fibers of f. If the minimum is 0, the claim has been proved above. Assume that the minimum $e > 0$. Then by Theorem 11.2.17, the hyperplanes H of $\mathbb{P}^n$ intersect all the fibers of f. Fix a fiber F of f of dimension e and fix a general H which misses a point of each irreducible component of F. The restriction of f to $X \cap H$ still surjects onto Y, thus, being Y irreducible, there exists as above a component X' of $X \cap H$ such that $f_{|X'}$ still surjects onto Y. Thus X' contains some point $Q \in F$ and the fiber F' of $f_{|X'}$ which contains Q has dimension equal to $e - 1$. It follows by induction that:

$$\dim(X) - \dim(Y) = \dim(X') - \dim(Y) + 1 = (e - 1) + 1 = e.$$

The set of points $Q \in Y$ such that the dimension of $f_{|X'}^{-1}(Q)$ has dimension $e - 1$ is open in Y by the inductive assumption. For all these points the dimension of the fiber of f is e. Thus the set of points such that the dimension of the fiber is bigger than e is bounded to a closed subvariety Y''. The inverse image of Y'' is a proper closed subvariety X'' of X. Restricting f to X'' and using induction, we see that the set of points of Y whose fibers have dimension bigger than e is a closed subset Y_1 of

Y'', hence a closed subset of Y. It corresponds to the set of points $Q \in Y$ such that $\dim(f^{-1}(Q) > e$. Call e_1 the minimal dimension of a fiber of $F_{|X_1}$, where X_1 is the inverse image of Y_1. Then, arguing as above, we find a closed subset Y_2 of Y such that the fibers of f over the points of Y_2 have dimension greater than e_1. And so on. It follows that the map μ_f is semicontinuous. □

Remark 11.3.6 By Theorem 9.1.30, the chain of closed subsets

$$Z_0 = Y \supset Z_1 \supset Z_2 \cdots,$$

such that $Z_i = \{P \in Y : \mu_f(P) \geq e + i\}$, becomes constant after a finite number of steps. Thus the dimension of the fibers of f is bounded.

As a consequence of Theorem 11.3.5, we find the following extension of the Chow's Theorem.

Corollary 11.3.7 *The image of any projective map $f : X \to Y$ is closed in Y.*

Examples and applications are contained in the exercise section.

11.4 Exercises

Exercise 35 Prove the second characterization of prime ideals, given in Definition 11.1.1: the ideal J is prime if and only if the quotient ring R/J is a domain, i.e., for all $a, b \in R/J$, $ab = 0$ implies that either $a = 0$ or $b = 0$.

Exercise 36 Let P, Q be points of a projective space $\mathbb{P}^n$ and let H be a hyperplane in $\mathbb{P}^n$ not containing P. Prove that, for any linear subspace $L \subset \mathbb{P}^n$ containing P, the linear span of P, Q and the intersection $L \cap H$ is equal to the intersection of H with the linear span of L and Q.

Exercise 37 Prove that every homomorphism between two fields is either trivial of injective.

Exercise 38 Prove that if $K \subset K'$ is a field extension, then the set of elements of K' which are algebraic over K is a subfield of K' (containing K).

Exercise 39 Prove that if e is a transcendental element over K then the extension $K(e)$ is isomorphic to the field of fractions of the polynomial ring $K[x]$.

Exercise 40 Let $L \subset \mathbb{P}^N$ be a linear subspace of dimension m. Let M be another linear subspace of $\mathbb{P}^N$, which does not intersect L. Prove that the projection of M from L to some space $\mathbb{P}^{N-m-1}$ is a linear subspace of the same dimension of M.

Exercise 41 Prove that any semicontinuous map from a projective variety to $\mathbb{Z}$ is bounded.

Exercise 42 Find examples of maps $\mathbb{C} \to \mathbb{C}$ which are continuous and dominant (in the cofinite topology of $\mathbb{C}$, which is the restriction to $\mathbb{C}$ of the Zariski topology of $\mathbb{P}^1$) and not surjective.

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Chapter 12
Secant Varieties



The study of the rank of tensors has a natural geometric counterpart in the study of *secant varieties*. Secant varieties or, more generally, *joins* are a relevant object for several researches on the properties of projective varieties.

We introduce here the study of join varieties and secant varieties, mainly pointing out the most important aspects for applications to the theory of tensors.

12.1 Definitions

Consider k projective varieties $Y_1, \ldots, Y_k$ in some projective space $\mathbb{P}^n$. Roughly speaking, the join of $Y_1, \ldots, Y_k$ is the set obtained by taking points $P_1 \in Y_1, \ldots, P_k \in Y_k$ and taking the span of the points.

The formal definition, however, requires some caution, for otherwise one ends up with an object which is rather difficult to handle.

Definition 12.1.1 Let $Y_1, \ldots, Y_k$ be projective subvarieties of $\mathbb{P}^n$. Then $Y_1 \times \cdots \times Y_k$ is a multiprojective subvariety of $(\mathbb{P}^n)^k = \mathbb{P}^n \times \cdots \times \mathbb{P}^n$. The *total join* of $Y_1, \ldots, Y_k$ is the subset $TJ(Y_1, \ldots, Y_k) \subset Y_1 \times \cdots \times Y_k \times \mathbb{P}^n$ of all $(k+1)$-tuples $(P_1, \ldots, P_k, Q)$ such that the points $P_1, \ldots, P_k, Q$, as points in $\mathbb{P}^n$, are linearly dependent.

Proposition 12.1.2 *The total join of $Y_1, \ldots, Y_k$ is a multiprojective subvariety of $(\mathbb{P}^n)^{k+1}$.*

Proof A point $(P_1, \ldots, P_k, Q)$ belongs to the total join $TJ(Y_1, \ldots, Y_k)$ if and only if

(i) each P_i satisfies the equations of Y_i in the i-set of multihomogeneous coordinates in $(\mathbb{P}^n)^k$; and
(ii) taking homogeneous coordinates $(y_{i0}, \ldots, y_{in})$ for P_i and $(x_0, \ldots, x_n)$ for Q, then all the $(k+1) \times (k+1)$ minors of the matrix

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$$\begin{pmatrix} x_0 & x_1 & \dots & x_n \\ y_{10} & y_{11} & \dots & y_{1n} \\ \dots & \dots & \dots & \dots \\ y_{k0} & y_{k1} & \dots & y_{kn} \end{pmatrix}$$

vanish.

Since these last minors are multilinear polynomials in the multihomogeneous coordinates of $(\mathbb{P}^n)^{k+1}$, the claim follows. □

Notice that the total join coincides trivially with the product $Y_1 \times \cdots \times Y_k \times \mathbb{P}^n$ when $k > n + 1$. Thus, in order to avoid trivialities:

$$we\ assume\ from\ now\ on\ in\ this\ chapter\ that\ k \leq n + 1.$$

Definition 12.1.3 A set of varieties $Y_1, \dots, Y_k$ is *independent* if one can find linearly independent points $P_1, \dots, P_k$ such that $P_1 \in Y_1, \dots, P_k \in Y_k$.

Notice that, in the definition of total join, we are not excluding the case in which some of the Y_i's, or even all of them, coincide.

Remark 12.1.4 If we take the same variety Y k times, i.e., we take $Y_1 = \cdots = Y_k = Y$, then the set $Y_1, \dots, Y_k$ is independent exactly when Y is not contained in a linear subspace of (projective) dimension $k - 1$.

In particular, when Y is not contained in any hyperplane, i.e., when Y is *nondegenerate*, then for any $k \leq n + 1$ we obtain a set of k independent varieties by taking k copies of Y.

Notice also that, by definition, if the points $P_1 \in Y_1, \dots, P_k \in Y_k$ are linearly dependent (e.g., if some of them coincide), then for any $Q \in \mathbb{P}^n$ the point $(P_1, \dots, P_k, Q)$ belongs to the total join. This is a quite degenerate situation, and we would like to exclude the case of linearly dependent k-tuples $P_1, \dots, P_k$. We warn immediately that will not be able to exclude all of them. Yet, we will find a refined definition of joins, which excludes most of $(k + 1)$-tuples $(P_1, \dots, P_k, Q)$ in which the P_i's are dependent.

Proposition 12.1.5 *If $Y_1, \dots, Y_k$ are varieties, then the set of k-tuples of points $(P_1, \dots, P_k) \in Y_1 \times \cdots \times Y_k$ such that $P_1, \dots, P_k$ are dependent is a subvariety of the product.*

Proof Enough to observe that the set is defined by the (multilinear) $k \times k$ minors of the matrix obtained by taking as rows a set of coordinates of the P_i's. □

We will now consider the behavior of a total join with respect to the natural projections of $(\mathbb{P}^n)^k$ to its factors. Let us recall the following fact.

Proposition 12.1.6 *The product of a finite number of irreducible projective varieties is irreducible.*

Proof Follows immediately by a straightforward application of Exercise 24. □

Theorem 12.1.7 *Consider an independent set of irreducible varieties $Y_1, \ldots, Y_k$. Then there exists a unique irreducible component Z of the total join $TJ(Y_1, \ldots, Y_k)$ such that the restriction to Z of the projection π of $Y_1 \times \cdots \times Y_k \times \mathbb{P}^n$ to the first k factors surjects onto $Y_1 \times \cdots \times Y_k$.*

Proof Since, by construction, the projection of $TJ(Y_1, \ldots, Y_k)$ to $Y_1 \times \cdots \times Y_k$ surjects, then $Y_1 \times \cdots \times Y_k$ is contained in the finite union $\pi(Z_1) \cup \ldots \pi(Z_m)$, where $Z_1, \ldots Z_m$ are the irreducible components of $TJ(Y_1, \ldots, Y_k)$. Since each Z_i is closed and π is a closed map (see Proposition 10.4.4), then each $\pi(Z_j)$ is closed. Since by Proposition 12.1.6 the product $Y_1 \times \cdots \times Y_k$ is irreducible, it follows that there exists at least one component Z_j such that $\pi(Z_j) = Y_1 \times \cdots \times Y_k$.

Assume that there are two such components Z_i, Z_j. Then consider a set of points $P_1 \in Y_1, \ldots, P_k \in Y_k$, such that $P_1, \ldots, P_k$ are linearly independent. The set exists, because we are assuming that the set $Y_1, \ldots, Y_k$ is independent. It is easy to see that the fiber $\pi^{-1}(P_1, \ldots, P_k)$ is a subset of $\{(P_1, \ldots, P_k)\} \times \mathbb{P}^n$ which is naturally isomorphic to $\{P_1, \ldots, P_k\} \times L$, where L is the linear span of the P_i's. Thus L is a linear space of dimension $k-1$, so that $\pi^{-1}(P_1, \ldots, P_k)$ is irreducible. It follows that if the points $P_1 \in Y_1, \ldots, P_k \in Y_k$ are independent, then the fiber $\pi^{-1}(P_1, \ldots, P_k)$ is contained in one irreducible component of $TJ(Y_1, \ldots, Y_k)$.

We claim that for any irreducible component Z_i of the join, the set $W_i \subset Y_1 \times \cdots \times Y_k$ of k-tuples $(P_1, \ldots, P_k)$ such that $(P_1, \ldots, P_k, Q) \subset Z_i$ for all Q in the span of $P_1, \ldots, P_k$, is a subvariety.

To prove the claim, take any (multihomogeneous) equation f for Z_i as a subvariety of $(\mathbb{P}^n)^k \times \mathbb{P}^n$. Consider f as a polynomial in the variables of the last factor $\mathbb{P}^n$, with coefficients f_i's which are multihomogeneous polynomials in the coordinates of $(P_1 \ldots, P_k) \in (\mathbb{P}^n)^k$. Then $(P_1, \ldots, P_k, Q) \subset Z_i$ for all Q in the span of $P_1, \ldots, P_k$ if and only if $(P_1, \ldots, P_k)$ annihilates all the f_i's. This provides a set of multihomogeneous equations that defines W_i.

Now we show that the claim proves the statement. Assume that there are several irreducible components of the join, $Z_1, \ldots, Z_m$, which map onto $Y_1 \times \cdots \times Y_k$ in π and consider the sets $W_1, \ldots, W_m$ as above. We know that the fibers $\pi^{-1}(P_1, \ldots, P_k)$ belong to some Z_i, whenever the points $P_1, \ldots, P_k$ are independent. So, the union $\bigcup \pi(Z_i)$ contains the subset U of k-tuples $(P_1, \ldots, P_k)$ in $Y_1 \times \cdots \times Y_k$ such that the P_i's linearly independent. Since U is open, by Proposition 12.1.5, and non-empty, since the Y_i's are independent, then U is dense in $Y_1 \times \cdots \times Y_k$, which is irreducible, by Proposition 12.1.6. It follows that $Y_1 \times \cdots \times Y_k$, i.e., the closure of U, is contained in the union $\bigcup \pi(Z_i)$, hence it is contained in some $\pi(Z_i)$, say in $\pi(Z_1)$. In particular, $\pi(Z_1)$ contains an open, non-empty, hence dense, subset of $Y_1 \times \cdots \times Y_k$. Thus $\pi(Z_1) = Y_1 \times \cdots \times Y_k$. Assume that Z_2 in another component which satisfies $\pi(Z_2) = Y_1 \times \cdots \times Y_k$. We prove that $Z_2 \subset Z_1$, which contradicts the maximality of irreducible components. Namely $W = (Y_1 \times \cdots \times Y_k) \setminus U$ is closed in the product, so $Z_2 \cap (\pi^{-1}(W))$ is closed in Z_2 and it is a proper subset, since π restricted to Z_2 surjects. Hence $Z_2 \setminus (\pi^{-1}(W))$ is dense in Z_2, which is irreducible. On the other

hand if $(P_1, \dots, P_k, Q) \in Z_2 \setminus (\pi^{-1}(W))$, then $P_1, \dots, P_k$ are linearly independent, thus $(P_1, \dots, P_k, Q) \in Z_1$ because Z_1 contains the fiber of π over $(P_1, \dots, P_k)$. It follows that $Z_2 \setminus (\pi^{-1}(W)) \subset Z_1$, hence $Z_2 \subset Z_1$, a contradiction. □

Definition 12.1.8 Consider an independent set $Y_1, \dots, Y_k$ of irreducible varieties. The *abstract join* of the Y_i's is the unique irreducible component $AJ(Y_1, \dots, Y_k)$ of the total join $TJ(Y_1, \dots, Y_k)$ which maps onto $Y_1 \times \cdots \times Y_k$ in the natural projection.

The *(embedded) join* $J(Y_1, \dots, Y_k)$ is the image of $AJ(Y_1, \dots, Y_k)$ under the projection of $(\mathbb{P}^n)^k \times \mathbb{P}^n$ to the last copy of $\mathbb{P}^n$.

Since the image of an irreducible variety is irreducible, then $J(Y_1, \dots, Y_k)$ is irreducible.

We put the adjective *embedded* in parenthesis since we will (often) drop it and say simply that $J(Y_1, \dots, Y_k)$ is the *join* of $Y_1, \dots, Y_k$.

Notice that while the abstract join is an element of the product $(\mathbb{P}^n)^k \times \mathbb{P}^n$, the join is a subset of $\mathbb{P}^n$, which is Zariski closed, since the last projection is closed (see Proposition 10.4.4).

Definition 12.1.9 If we apply the previous definitions to the case $Y_1 = \cdots = Y_k = Y$, where Y is an irreducible variety in $\mathbb{P}^n$, we get the definitions of the *abstract k-th secant variety* $AS_k(Y)$ and the *(embedded) k-th secant variety* $S_k(Y)$ (which are both irreducible).

Example 12.1.10 Let Y be the d-th Veronese embedding of $\mathbb{P}^1$ in $\mathbb{P}^d$. The space $\mathbb{P}^d$ can be identified with the space of all forms of degree d in 2 variables x, y, by identifying the coordinates $z_0, \dots, z_d$ with monomials of degree d in two variables, e.g., $z_i = x^i y^{d-i}$. In this representation, Y can be identified with the set of powers, i.e., forms that can be written as $(ax + by)^d$, for some choice of the scalars a, b.

With this in mind, let us determine a representation of the secant variety $S_2(Y)$. A general point of the abstract secant variety $AS_2(Y)$ is a triplet (P, Q, T) where T lies on the line joining $P, Q \in Y$. Choose scalars a_P, b_P, a_Q, b_Q such that $P = (a_P x + b_P y)^d$ and $Q = (a_Q x + b_Q y)^d$. Then T belongs to the line P, Q if and only if there are scalars u, v such that $T = uP + vQ = u(a_P x + b_P y)^d + v(a_Q x + b_Q y)^d$. In particular, when $P = x^d$ and $Q = y^d$, then (P, Q, T) belongs to the abstract secant variety $AS_2(Y)$ if and only if T is a binomial of type $T = ux^d + vy^d$.

Notice that given two independent linear forms $L = (a_P x + b_P y)$, $M = (a_P x + b_P y)$, then after a change of coordinates in $\mathbb{P}^1$ we may always assume that $L = x$ and $M = y$.

Summarizing, it follows that the secant variety $S_2(Y)$ contains all the points $T \in \mathbb{P}^d$ which, after a change of coordinates, miss all the mixed monomials, i.e., are of the form $T = ux^d + vy^d$.

There are special points in $AS_2(Y)$ corresponding to triplets (P, P, T), i.e., with $P = Q$. They can arise as limit of families $(P, Q(t), T) \in AS_2(Y)$ for families of points $Q(t) \in Y$ which tend toward P for t going to 0, since the general point of the

irreducible variety $AS_2(Y)$ has $P \neq Q$ (notice that the limit of a family of points in $AS_2(Y)$ necessarily belongs to $AS_2(Y)$, since the abstract secant variety is closed).

For instance, consider the limit in the case $P = x^d$, $Q = (x + ty)^d$, $T = P - Q$. For $t \neq 0$, we have

$$T = (d-1)tx^{d-1}y + \binom{d-2}{2}t^2x^{d-2}y^2 + \cdots + t^d y^d.$$

Projectively, for $t \neq 0$, T is equivalent to $(d-1)x^{d-1}y + \binom{d-2}{2}tx^{d-2}y^2 + \cdots + t^{d-1}y^d$. Thus for $t \to 0$, T goes to $x^{d-1}y$. This implies that $(x^d, x^d, x^{d-1}y) \in AS_2(Y)$, i.e., $x^{d-1}y \in S_2(Y)$.

Notice that $x^{d-1}y$ cannot be written as $L^d + M^d$, for any choice of linear forms L, M. So $x^{d-1}y$ is a genuinely new point of $S_2(Y)$.

With a similar trick, one can prove that for any choice of two linear forms L, M in x, y, the form $L^{d-1}M$ belongs to $S_2(Y)$.

Proposition 12.1.11 *The abstract join $AJ(Y_1, \ldots, Y_k)$ has dimension $(k-1) + dim(Y_1) + \cdots + \dim(Y_k)$. In particular, the abstract secant variety $AS_k(Y)$ of a variety Y of dimension n has dimension $k - 1 + nk$.*

Proof The projection $AJ(Y_1, \ldots, Y_k) \to Y_1 \times \cdots \times Y_k$ has general fibers, over a point $(P_1, \ldots, P_k)$ of the product such that the P_i's are linearly independent, corresponding to the projective span of $P_1, \ldots, P_k$, which has dimension $k - 1$. The claim follows from Theorem 11.3.5. □

While the dimension of the abstract secant variety is always easy to compute, for the embedded secant variety the computation of the dimension can be complicated. To give an example, let us show first how the interpretation of secant varieties introduced in Example 12.1.10 can be extended.

Example 12.1.12 Consider the Veronese variety Y obtained by the Veronese embedding $v_{n,d} : \mathbb{P}^n \to \mathbb{P}^N$, where $N = \binom{d+n}{n} - 1$.

Y can be considered as the set of symmetric tensors of rank 1 and type $(n+1) \times \cdots \times (n+1)$, d times, i.e., the set of forms of degree d in $n+1$ variables, which are powers of linear forms.

Consider a form T of rank k. Then T can be written as a sum $T_1 + \cdots + T_k$ of forms of rank 1, with k minimal. The minimality of k implies that $T_1, \ldots, T_k$ are linearly independent, since otherwise we could barely forget one of them and write T as a sum of $k - 1$ powers. In particular $k \leq N + 1$. We get that $(T_1, \ldots, T_k, T)$ belongs to the k-th abstract secant variety of Y, so that T is a point of $S_k(Y)$.

The secant variety $S_k(Y)$ also contains forms whose rank is not k. For instance, if T has rank $k' < k$, then we can write $T = T_1 + \cdots + T_{k'}$, with $T_1, \ldots, T_{k'}$ linearly independent. Consider points $T_{k'+1}, \ldots, T_k \in Y$ such that $T_1, \ldots, T_k$ are linearly independent. Then consider the form

$$T(t) = T_1 + \cdots + T_{k'} + tT_{k'+1} + \cdots + tT_k,$$

where $t \in \mathbb{C}$ is a parameter. It is clear that for $t \neq 0$ the point $(T_1, \ldots, T_k, T(t))$ belongs to the abstract secant variety $AS_k(Y)$. Thus $T(t) \in S_k(Y)$ for $t \neq 0$. It follows that the limit of $T(t)$ for t going to 0, which is $T(0) = T$, also belongs to $S_k(Y)$, which is closed.

By arguing as in Example 12.1.10, one can prove that the form $T = x_0^{d-1}x_1 + x_2^d + \cdots + x_n^d$, whose rank is bigger than n, also belongs to $S_n(Y)$.

We can generalize the construction given in the previous example to show that

Proposition 12.1.13 *For every variety $Y \subset \mathbb{P}^n$ and for $k' < k \leq n+1$, we always have $S_{k'} \subseteq S_k$.*

Example 12.1.14 Let $Y = v_{2,2}(\mathbb{P}^2)$ be the Veronese variety of rank 1 forms of degree 2 in 3 variables, which is a subvariety of $\mathbb{P}^5$. The abstract secant variety $AS_2(Y)$, corresponding to triplets (L^2, M^2, T) such that L, M are linear forms and $T = aL^2 + bM^2$, has dimension $2 + 2 + 1 = 5$, by Proposition 12.1.11.

Let us prove that the dimension of the secant variety $S_2(Y)$ is 4. To do that, it is enough to prove that the general fiber of the projection $\pi : AS_2(Y) \to S_2(Y)$ is 1-dimensional. To this aim, consider $T = aL^2 + bM^2$, L, M general linear forms. L, M correspond to general points of the projective plane $\mathbb{P}^2$ of linear forms in 3 variables. Let $\ell \subset \mathbb{P}^2$ be the line joining L, M. After a change of coordinates, without loss of generality, we may assume that $L = x_0^2$, $M = x_1^2$, so that ℓ has equation $x_2 = 0$ and T becomes a form of degree 2 in the variables x_0, x_1, i.e., $T = x_0^2 + x_1^2$. It is easy to prove that $\pi^{-1}(T)$ has infinitely many points. Indeed for every point $P = ax_0 + bx_1 \in \ell$ there exists exactly one $Q = a'x_0 + b'x_1 \in \ell$ such that $(v_2(P), v_2(Q), T) \in AS_2(Y)$, i.e.,

$$T = (ax_0 + bx_1)^2 + (a'x_0 + b'x_1)^2,$$

as a projective point: enough to take $a' = b$ and $b' = -a$ (which is the unique choice, modulo scalar multiplication, for a, b general).

Thus we obtain a projective map f from ℓ to the fiber $\pi^{-1}(T)$ by sending $P = ax_0 + bx_1$ to $(v_2(P), v_2(Q), T)$, where $Q = bx_0 - ax_1$. The map f is clearly injective. We show that f is surjective, by proving that, when T is general, $(v_2(P), v_2(Q), T)$ cannot stay in $AS_2(Y)$, unless $P \in \ell$. Namely if $T = (ax_0 + bx_1 + cx_2)^2 + (a'x_0 + b'x_1 + c'x_2)^2$ with $c \neq 0$, then one computes $c = -ic'$, so that $a = -ia'$ and $b = -ib'$, thus $T = (-i)^2(a'x_0 + b'x_1 + c'x_2)^2 + (a'x_0 + b'x_1 + c'x_2)^2 = 0$, a contradiction. It follows that $\pi^{-1}(T)$ is isomorphic to ℓ, hence it has dimension 1.

Example 12.1.14 is a special case of the Alexander–Hirschowitz Theorem which determines the dimension of secant varieties of a Veronese variety Y (see Theorems 7.4.4 and 12.2.13). The example illustrates one of the few cases in which $AS_k(Y)$ and $S_k(Y)$ have different dimension.

Similar examples can be found by looking at projective spaces of matrices.

Example 12.1.15 Consider the Segre embedding $Y \subset \mathbb{P}^8$ of $\mathbb{P}^2 \times \mathbb{P}^2$. If we interpret $\mathbb{P}^8$ as the space of matrices of type 3×3, then $S_2(Y)$ contains matrices of rank 2. Namely matrices T of type $v \otimes w + v' \otimes w'$ have rank 2, for a general choice of $v, v', w, w' \in K^3$. Notice that the space of rows of a matrix $T = v \otimes w + v' \otimes w'$ is generated by v, v'. Thus a general element of $S_2(Y)$ has rank 2.

Conversely, as we saw in the proof of Proposition 6.3.2, if v, v' are generators for space of rows of a 3×3 matrix T of rank 2, then one can find vectors $w, w' \in K^3$ such that $T = v \otimes w + v' \otimes w'$.

It follows that while $AS_2(Y)$ has dimension $2\dim(Y) + 1 = 9$, yet $S_2(Y)$ has the dimension of the projective space of matrices of rank 2 in $\mathbb{P}^8$, which is the hypersurface of $\mathbb{P}^8$ defined by the vanishing of the determinant. Thus, $\dim(S_2(Y)) = 7$.

Definition 12.1.16 Let $Y \subset \mathbb{P}^N$ be any irreducible variety, which spans $\mathbb{P}^N$. We say that $T \in \mathbb{P}^N$ has *Y-border rank* k if k is the minimum such that $T \in S_k(Y)$.

Thus the set of points of Y-border rank k corresponds to the projective variety $S_k(Y)$.

The *generic Y-rank* is the minimum k such that $S_k(Y) = \mathbb{P}^N$.

The previous definition is particularly interesting when Y corresponds to the d-th Veronese embedding of some projective space $\mathbb{P}^n$, or to the Segre embedding of a product $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$. In the latter case, points $T \in \mathbb{P}^N$ can be identified with tensors of type $(a_1 + 1) \times \cdots \times (a_s + 1)$, while in the former case they can be identified with symmetric tensors of type $(n + 1) \times \cdots \times (n + 1)$, d times.

Points of Y correspond to tensors of rank 1. Every point corresponding to a tensor of rank k belongs to the secant variety $S_k(Y)$, so it has Y-border rank $\leq k$. Thus in general, for a tensor T one has

$$\text{border rank of } T \leq \text{ rank of } T$$

(we will drop the reference to Y, in the case of tensors of some given type, because by default we consider Y to be the variety of rank 1 tensors).

One can find examples in which the previous inequality is strict.

Example 12.1.17 Go back to Example 12.1.10, in which Y is the d-th Veronese embedding of $\mathbb{P}^1$ in $\mathbb{P}^d$, and $\mathbb{P}^d$ can be identified with the space of all forms of degree d in 2 variables x, y.

We proved there that the form $x^{d-1}y$ cannot be written as $L^d + M^d$, for any choice of linear forms L, M. So $x^{d-1}y$ has rank bigger than 2. On the other hand, we also proved that $x^{d-1}y$ belongs to $S_2(Y)$. Hence

$$\text{border rank of } x^{d-1}y < \text{ rank of } x^{d-1}y.$$

Some properties of the Y-border rank are listed in the following:

Remark 12.1.18 A general tensor of border rank k also has rank k.

Namely, by definition, the set of points $(P_1, \dots, P_k, P) \in AS_k(Y)$ such that $P_1, \dots, P_k$ are linearly independent (in particular, distinct) is open dense in $AS_k(Y)$. Thus a general tensor $T \in S_k(Y)$ has rank $\leq k$, hence it has rank k.

If r is the generic Y-rank, no points $P \in \mathbb{P}^N$ can have border rank bigger than r, because $P \in S_r(X) = \mathbb{P}^N$. It is, however, possible that some tensors T have rank bigger than the generic border rank.

In any event, if T is a tensor of border rank k, i.e., $T \in S_k(Y)$ for the variety Y of tensors of rank 1, then, since $S_k(Y)$ is irreducible, there exists a neighborhood of T in $S_k(Y)$ whose generic point is a tensor of rank k.

The latter property listed in the previous remark explains why the computation of the border rank k of a tensor T can be important for applications: we can realize T as a limit of tensors of rank k. In other words, T can be slightly modified to obtain a tensor of rank k.

Remark 12.1.19 Let $Y \subset \mathbb{P}^n$ be a projective variety and let $P \in \mathbb{P}^n$ be a point, not belonging to Y. Then the projection from P determines a well defined map $\pi : Y \to \mathbb{P}^{n-1}$. Call Y' the image of π. Assume that π is not injective. Then there are two points $P_1, P_2 \in Y$ which are sent to the same point of $\mathbb{P}^{n-1}$. This means that the points P, P_1, P_2 are aligned. Thus P belongs to the secant variety $S_2(Y)$.

It follows that the projection from a general point defines an injective map (i.e., a bijection from Y to Y'), provided that $S_2(Y) \neq \mathbb{P}^n$, i.e., provided that the generic Y-rank is bigger than 2.

In fact, one can prove (but the proof uses algebraic tools which go beyond the theory introduced in this book) that the projection from P determines an isomorphism between Y and Y' if and only if P does not belong to $S_2(Y)$. Thus, if $S_2(Y) = \mathbb{P}^n$, there is no chance of projecting isomorphically Y to a projective space of smaller dimension.

The study of projection of varieties to spaces $\mathbb{P}^n$ whose dimension n is only slightly bigger than the dimension of Y is a long-standing problem in classical Algebraic Geometry, which indeed stimulated the first studies on secant varieties. For instance, we know only a few examples of varieties of dimension $n - 3 > 1$ which can be isomorphically projected to $\mathbb{P}^{n-1}$.

Varieties Y for which the abstract secant variety $AS_k(Y)$ and the secant variety $S_k(Y)$ have the same dimension are of particular interest in the theory of secant spaces. For instance, if $Y \subset \mathbb{P}^N$ is a variety of tensors of rank 1, then $\dim(AS_k(Y)) = \dim(S_k(Y))$ implies that the general fiber of the projection $AS_k(Y) \to S_k(Y)$ is finite, which means that for a general tensor $T \in \mathbb{P}^N$ of rank k, there are (projectively) only a finite number of presentations $T = T_1 + \cdots + T_k$, with $T_i \in Y$ for all i. In particular, tensors for which the presentation is unique are called *identifiable tensors*.

Definition 12.1.20 For an irreducible variety $Y \subset \mathbb{P}^N$ and for any $T \in \mathbb{P}^N$, a *Y-decomposition* of T of length r is a set of linearly independent points $T_1, \dots, T_r \in Y$ such that $T = T_1 + \cdots + T_r$, i.e., $(T_1, \dots, T_r, T) \in AS_r(Y)$.

If $r = \text{rank}(T)$, we say that the decomposition *computes the rank*.

We say that T is *finitely r-identifiable* if there are only a finite number of decompositions of T of length r. We say that T is *r-identifiable* if there is only one decompositions of T of length r.

We say that Y is *generically finitely r-identifiable* if a general element of $S_r(Y)$ is finitely r-identifiable. Notice that Y is generically finitely r-identifiable if and only if $AS_r(Y)$ and $S_r(Y)$ have the same dimension.

We say that Y is *generically r-identifiable* if a general element of $S_r(Y)$ is r-identifiable.

We refer to the chapters on Multi-linear Algebra for a discussion on the importance of the identifiability properties of tensors.

12.2 Methods for Identifiability

Finding whether a given tensor is identifiable or not, or whether tensors of a given type are generically identifiable or not, is in general a difficult question. Results on these problems are not yet complete, and the matter is an important subject of ongoing investigations.

We briefly introduce in this section a couple of methods that are universally used to detect the identifiability, or the generic identifiability, of tensors of a given type.

12.2.1 Tangent Spaces and the Terracini's Lemma

The first method to compute the identifiability is based on the computation of the tangent space at a generic point.

Roughly speaking, the tangent space to a projective variety $X \subset \mathbb{P}^N$ at a general point $P \in X$ can be defined by considering that a sufficiently small Zariski open subset U of X around P is a differential subvariety of $\mathbb{P}^N$, for which the notion of tangent space is a well established, differential object. Yet, we will give an algebraic definition of tangent vectors, which are suitable for the computation of the dimension.

We will base the notion of tangent space first by giving the definition of embedded tangent space of a hypersurface at the origin, and then extending the notion to any (regular) point of any projective variety.

Definition 12.2.1 Let $X \subset \mathbb{P}^N$ by the hypersurface defined by the form

$$f = x_0^{d-1} g_1 + \cdots + g_d,$$

where each g_i is a form of degree i in $x_1, \ldots, x_N$. Clearly the point $P_0 = [1 : 0 : \cdots : 0]$ belongs to X. The *(embedded) tangent space to X at P_0* is the linear subspace $T_X(P_0)$ of $\mathbb{P}^N$ defined by the equation $g_1 = 0$.

It is clear that $P_0 \in T_X(P_0)$. Notice that the previous definition admits the case $g_1 = 0$: when this happens, then $T_X(P_0)$ coincides with $\mathbb{P}^N$. Otherwise $T_X(P_0)$ is a linear subspace of dimension $N - 1 = \dim(X)$.

We are ready for a rough definition of tangent space of any variety X.

Definition 12.2.2 Let X be any variety in $\mathbb{P}^N$, containing the point $P_0 = [1 : 0 : \cdots : 0]$. Each element f in the homogeneous ideal I_X of X defines a hypersurface $X(f)$ which contains P_0.

The *(embedded) tangent space to X at* P_0 is the intersection of the tangent spaces $T_{X(f)}(P_0)$, where f ranges among the elements of the ideal I_X.

If $P \in X$ is any point, then we define the tangent space to X at P as

$$T_X(P) = \phi^{-1}(T_{\phi(X)}(P_0)),$$

where ϕ is any change of coordinates which sends P to P_0.

We leave to the reader the proof that the definition of $T_X(P)$ does not depend on the choice of the change of coordinates ϕ.

Unfortunately, in a certain sense as it happens for Groebner basis (Chap. 13), it is not guaranteed that if $f_1, \ldots, f_s$ generate the ideal I_X, the intersection of the corresponding tangent spaces $T_{X(f_i)}(P_0)$ determines $T_X(P)$. The situation can however be controlled as follows.

Definition 12.2.3 We say that $P \in X$ is a *regular point* if $T_X(P)$ has dimension equal to $\dim(X)$.

Example 12.2.4 If X is a hypersurface defined by a form f as above, then P_0 is a regular point of X if and only if $g_1 \neq 0$.

We give without proof the following:

Theorem 12.2.5 *For every $P \in X$ it holds:*

$$\dim T_X(P) \geq \dim(X).$$

Moreover the equality holds in a Zariski open subset of X.

Example 12.2.6 The special case in which $X = \mathbb{P}^n$ is easy: the tangent space of X at any point coincides with $\mathbb{P}^n$. Thus any point of X is regular.

Next, we provide examples of tangent spaces to relevant varieties for tensor analysis.

Example 12.2.7 Consider the image X of the Veronese embedding of degree d of $\mathbb{P}^n$ into $\mathbb{P}^N$, $N = \binom{n+d}{n} - 1$. Let $Q_0 \in \mathbb{P}^n$ be the point of coordinates $[1 : 0 : \cdots : 0]$, which corresponds to the linear form x_0. Its image in $\mathbb{P}^N$ is the point $P_0 = [1 : 0 : \cdots : 0]$, which corresponds to the monomial $M_0 = x_0^d$.

Consider a quadratic equation $M_0M_i - M_jM_k = 0$ in the ideal of X. This equation exists unless M_i is divided by x_0^{d-1}, in which case the equation becomes null. Thus, the ideal of $T_X(P_0)$ contains all the equations of type $M_j = 0$, where M_j is a monomial in which the exponent of x_0 does not exceed $d-2$. It follows that $T_X(P_0)$ is contained in the space generated by the monomial corresponding to $x_0^d, x_0^{d-1}x_1, \ldots, x_0^{d-1}x_n$. Since this last space has dimension $n = \dim(X)$ and the dimension of $T_X(P_0)$ cannot be smaller than n, by Theorem 12.2.5, we get that $T_X(P_0)$ is the subspace generated by forms in $K[x_0, \ldots, x_n]_d$ which sit in the ideal spanned by $x_0^d, x_0^{d-1}x_1, \ldots, x_0^{d-1}x_n$.

A similar computation determines the tangent space to X at a general point $P \in X$ corresponding to a power L^d, where L is a linear form in $K[x_0, \ldots, x_n]_d$: $T_X(P)$ corresponds to the subspace generated by forms in $K[x_0, \ldots, x_n]_d$ which sit in the ideal spanned by $L^d, L^{d-1}x_1, \ldots L^{d-1}x_n$.

Example 12.2.8 Consider the image X of the Segre embedding of $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$ into $\mathbb{P}^N$, $N = \Pi(a_i + 1) - 1$. The image of the point $Q = ([1 : 0 : \cdots : 0], \ldots, [1 : 0 : \cdots : 0])$ is the point $P_0 \in \mathbb{P}^N$ of coordinates $[1 : 0 : \cdots : 0]$. If we call x_{ij} the coordinates in $\mathbb{P}^{a_i}$, then P_0 is the point corresponding to the multihomogeneous polynomial $M_0 = x_{10} \cdots x_{s0}$.

For every multihomogeneous polynomial $M = x_{1q_1} \cdots x_{sq_s}$, except those for which at least $s-1$ of the q_i's are 0, we can find an equation $M_0M - M_aM_b = 0$ for a suitable choice of M_a, M_b. Thus all the corresponding coordinates of $\mathbb{P}^N$ vanish on $T_X(P_0)$. It follows that $T_X(P_0)$ is the linear subspace of the coordinates representing multihomogeneous polynomial $M = x_{1q_1} \cdots x_{sq_s}$ for which at least $s-1$ of the q_i's are 0.

If we call Λ_i the coordinate space in $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$ of points $(Q_1, \ldots, Q_s)$ in which all the Q_j's, $j \neq i$, have coordinates $[1 : 0 : \cdots : 0]$, then it turns out that $T_X(P_0)$ is spanned by the images of the Λ_i's, $i = 1, \ldots, s$.

Similarly, for a general point $P \in X$ which is the image of the point $(Q_1, \ldots, Q_s)$ in $\mathbb{P}^{a_1} \times \cdots \times \mathbb{P}^{a_s}$, the tangent space $T_X(P)$ is spanned by the images of the spaces Λ_i of points $(R_1, \ldots, R_s)$ in which $R_j = Q_j$ for all $j \neq i$.

The tangent spaces to secant varieties are then described by the following:

Lemma 12.2.9 (Terracini's Lemma) *Let $U \in S_k(X)$ be a* general *point of the k-th secant variety of X. Assume that U belongs to the span of $P_1, \ldots, P_k \in X$. Then the tangent space to $S_k(X)$ at the point U is equal to the linear span of the tangent spaces $T_X(P_1), \ldots, T_X(P_k)$.*

The computation of tangent spaces allows us to study the dimension of secant varieties.

Remark 12.2.10 Thanks to the Terracini's Lemma, we can associate to each secant variety an *expected dimension*. Namely, we may assume that the tangent spaces at general points of $X \subset \mathbb{P}^N$ are as independent as they can. With this in mind, we define the expected dimension of $S_r(X)$ as

$$expdim_r(X) = \min\{N, r\dim(X) + r - 1\}.$$

Example 12.2.11 Let X be the image in $\mathbb{P}^5$ of the 2-Veronese embedding of $\mathbb{P}^2$. Since X is a surface, the dimension of the abstract secant variety $AS_2(X)$ is 5. On the other hand, $S_2(X)$ has dimension 4. Indeed, let $T \in S_2(X)$ belong to the span of $P_1 = v_2(L_1)$ and $P_2 = v_2(L_2)$. After a change of coordinates, we may always assume that L_1 corresponds to x and L_2 corresponds to y, in $\mathbb{P}(\mathbb{C}[x, y, z]_1) = \mathbb{P}^2$. Thus, the tangent space to X at T corresponds to the projectification of the linear subspace W of $\mathbb{C}[x, y, z]_2$, generated by x^2, xy, xz, y^2, yz. This last space has dimension 4.

Example 12.2.12 Let X be the image in $\mathbb{P}^8$ of the Segre embedding $s_{2,2}$ of $\mathbb{P}^2 \times \mathbb{P}^2$. Since X has dimension 4, the dimension of the abstract secant variety $AS_2(X)$ is 9. On the other hand, $S_2(X)$ has dimension 7. Indeed, let $T \in S_2(X)$ belong to the span of $P_1 = s(x_1, y_1)$ and $P_2 = s(x_2, y_2)$. The tangent spaces to X at P_1 and P_2 share the two points (x_1, y_2) and (x_2, y_1). Thus they share a line. Consequently their span has dimension 7.

Notice that if we identify $\mathbb{P}^8$ with the projective space of 3×3 matrices, then X corresponds to the variety of matrices of rank ≤ 2, which is the hypersurface whose equation is the determinant of a generic matrix.

In both the examples above, the dimension of the secant variety is smaller than the expected value. Such varieties are called *defective*.

Defective varieties, in fact, are quite rare in Algebraic Geometry. For instance, curves are never defective.

Moreover, reading Theorem 7.4.4 in this context, we have a complete list of defective Veronese varieties.

Theorem 12.2.13 (Alexander-Hirschowitz) *Let $X = v_{n,d}(\mathbb{P}^n)$ be a Veronese variety. Then the dimension of $S_r(X)$ equals the expected dimension, unless:*

- $n > 1$, $d = 2$, $2 \leq r \leq n$;
- $n = 2$, $d = 4$, $r = 5$;
- $n = 3$, $d = 4$, $r = 9$;
- $n = 4$, $d = 3$, $r = 7$;
- $n = 4$, $d = 4$, $r = 14$.

12.2.2 Inverse Systems

Inverse systems are a method to compute the rank and the identifiability of a given symmetric tensor T, identified as forms of given degree d in some polynomial ring $R = \mathbb{C}[x_0, \ldots, x_n]$.

The method is based on the remark that if

$$T = L_1^d + \cdots + L_r^d,$$

where the L_i's are linear forms in R, then every derivative of T is also spanned by $L_1, \ldots, L_r$.

Thus, the method starts by considering a *dual* ring of coordinates $S = \mathbb{C}[\partial_0, \ldots, \partial_n]$, where each ∂_i should be considered as the (divided) partial derivative with respect to x_i. In other words, ∂_i acts on R by the linear, associative action which satisfies:

$$\partial_i(x_j) = \begin{cases} 1 & \text{if } i = j; \\ 0 & \text{if } i \neq j, \end{cases}.$$

One can easily prove that, for all i, $\partial_i(x_i^m) = (\partial_i x_i)x_i^{m-1} = x_i^{m-1}$. It follows:

$$\partial_i^a x_i^b = \begin{cases} x_i^{b-a} & \text{if } a \leq b \\ 0 & \text{if } a > b \end{cases}.$$

We also have that if $L = a_0x_0 + \cdots + a_nx_n$ is a linear form, then by taking $D = \bar{a}_0\partial_0 + \cdots + \bar{a}_n\partial_n$ (where $\bar{a}_i$ is the conjugate of a_i, then $D(L) = 1$. More generally, $D^d(L^d) = 1$.

Definition 12.2.14 For every form $F \in R$, the *inverse system* of F is the set of all $D \in S$ such that $D(F) = 0$. The inverse system of F is indicated with the symbol $F^\perp$.

Proposition 12.2.15 *The inverse system of a form F is a homogeneous ideal in the ring S.*

Example 12.2.16 Let us consider the form $F = x_0^3 + x_0x_1x_2 + 2x_0x_2^2 - 2x_1x_2^2 - x_2^3$ in three variables. For any linear combination $\partial = a\partial_0 + b\partial_1 + c\partial_2$, one computes

$$\partial(F) = a(3x_0^2 + 2x_2^2) + b(-2x_2^2) + c(4x_0x_2 - 4x_1x_2 - 3x_2^2)$$

so that $\partial(F) = 0$ if and only if $a = b = c = 0$. Thus no linear form $\partial \neq 0$, in S, belongs to $F^\perp$. On the other hand, $\partial_1^2 \in F^\perp$.

It is clear that every homogeneous element in S of degree at least 4 belongs to $F^\perp$. Notice that $\partial_o^3(F) = 1$, so that $\partial_0^3 \notin F^\perp$.

Example 12.2.17 In the case $n = 1$, i.e., $R = \mathbb{C}[x_0, x_n]$ is a polynomial ring in two variables, consider $F = x_0^d + x_1^d$. Then $F^\perp$ is the ideal generated by

$$\partial_0\partial_1, \partial_0^d - \partial_1^d, \partial_0^{d+1}.$$

Example 12.2.18 If F is a form of rank 1, i.e., $F = L^d$ for some linear form L, then it is easy to compute $F^\perp$. Write $L = a_0x_0 + \cdots + a_nx_n$. Put $v_0 = (a_0, \ldots, a_n)$ and extend to a basis $v_0, v_1, \ldots, v_n$ of $\mathbb{C}^{n+1}$, where each v_i, $i > 0$, is orthogonal to v_0. Set $v_i = (a_{i0}, \ldots, a_{in})$. Define

$$D_0 = \bar{a}_0\partial_0 + \cdots + \bar{a}_n\partial_n, \quad D_i = a_{i0}\partial_0 + \cdots + a_{in}\partial_n.$$

Then $F^\perp$ is the ideal generated by $D_0^{d+1}, D_1, \ldots, D_n$.

Notice that if we consider S as a normal polynomial ring, then $F^\perp$ contains the ideal of all polynomials whose evaluation on $(a_0, \dots, a_n)$, which is a set of coordinates for the point $P \in \mathbb{P}^n$ associated with L, vanishes.

The previous example provides a way in which one can use $F^\perp$ to determine the rank of a symmetric tensor, i.e., of a polynomial form.

Namely, if $F = L_1^d + \cdots + L_r^d$, where L_i is a linear form associated to the point $P_i \in \mathbb{P}^n$, then clearly $F^\perp$ contains the intersection of the homogeneous ideals in S associated to the points P_i's.

It follows that

Proposition 12.2.19 *The rank r of F is the minimum such that there exist a finite set of points $\{P_1, \dots, P_r\} \subset \mathbb{P}^n$ such that the intersection of the ideals associated to the P_i's is contained in $F^\perp$.*

The linear forms associated to the points P_i's provide a decomposition for L.

Example 12.2.20 Consider the case of two variables x_0, x_1, i.e., $n = 1$, and let $F = x_0^2 x_1^2$. F is thus a monomial, but this *by no means* implies that it is easy to compute a decomposition of F in terms of (quartic) powers of linear forms.

In the specific case, one computes that $F^\perp$ is the ideal generated by $\partial_0^3, \partial_1^3$. Since the ideal of any point $[a : b] \in \mathbb{P}^n = \mathbb{P}^1$ is generated by one linear form $b\partial_1 - a\partial_0$, then the intersection of two ideals of points is simply the ideal generated by the product of the two generators. By induction, it follows that in order to determine a decomposition of F with r summands, we must find r *distinct* linear forms in S whose product is contained in the ideal generated by $\partial_0^3, \partial_1^3$. It is almost immediate to see that we cannot find such linear forms if $r = 2$, thus the rank of F is bigger than 2. It is possible to find a product of three linear forms which lies in $F^\perp$, namely:

$$(\partial_0 - \partial_1)(\partial_0 - (\frac{1 + i\sqrt{3}}{2})\partial_1)(\partial_0 + (\frac{3 + i\sqrt{3}}{2})\partial_1).$$

It follows from the theory that F has rank 3, and it is a linear combination of the 4-th powers of the linear forms $(x_0 - x_1)$, $(x_0 - (\frac{1+i\sqrt{3}}{2})x_1)$, $(x_0 + (\frac{3+i\sqrt{3}}{2})x_1)$.

It is laborious, but possible, to prove that indeed:

$$x_0^2 x_1^2 = a(x_0 - x_1)^4 + b(x_0 - (\frac{-1 + i\sqrt{3}}{2})x_1)^4 + c(x_0 + (\frac{1 + i\sqrt{3}}{2})x_1)^4,$$

where $a = \frac{1}{18}$, $b = \frac{-1+i\sqrt{3}}{36}$ and $c = -\frac{1+i\sqrt{3}}{36}$.

In general, it is not easy to determine which intersections of ideals of points are contained in a given ideal $F^\perp$. The solution is known, by now, only for a few special types of forms F.

12.3 Exercises

Exercise 43 Prove the first claim of Remark 12.1.4: if $Y_1 = \cdots = Y_k = Y$, then the set of varieties $Y_1, \ldots, Y_k$ is independent exactly when Y is not contained in a linear subspace of (projective) dimension $k - 1$.

Exercise 44 Prove the statement of Proposition 12.1.13: for every $Y \subset \mathbb{P}^n$ and for $k' < k \leq n + 1$, we always have $S_{k'} \subseteq S_k$.

Exercise 45 Prove that when $Y \subset \mathbb{P}^{14}$ is the image of the 4-veronese map of $\mathbb{P}^2$, then $\dim(AS_5(Y)) = 14$, but $\dim(S_5(Y)) < 14$.

Chapter 13
Groebner Bases



Groebner bases represent the most powerful tool for computational algebra, in particular for the study of polynomial ideals. In this chapter, based on [1, Chap. 2], we give a brief survey on the subject. For a deeper study of it, we suggest [1, 2].

Before treating Groebner bases, it is mandatory to recall many concepts, starting from the one of monomial ordering.

13.1 Monomial Orderings

In general, when we are working with polynomials in a single variable x, there is an underlined ordering among monomials in function of the degree of each monomial: $x^\alpha > x^\beta$ if $\alpha > \beta$. To be more precise we can say that we are working with an ordering, on the degree, on the monomials in one variable:

$$\cdots > x^{m+1} > x^m > \cdots > x^2 > x > 1.$$

When we are working in a polynomial ring with n variables $x_1, x_2, \ldots, x_n$ the situation is more complicated. As a matter of fact, it is not obvious to have an underlined ordering, since this would mean to fix, first of all, an order of "importance" on the variables.

We define now different orderings on $k[x_1, \ldots, x_n]$, which will be useful in different contexts. Before to do that is is important to observe that we can always recover a monomial $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ from its $n-$tuple of exponents $(\alpha_1, \ldots, \alpha_n) \in \mathbb{Z}^n_{\geq 0}$. This fact establishes an injective correspondence between the monomials in $k[x_1, \ldots, x_n]$ and $\mathbb{Z}^n_{\geq 0}$. In addition, each ordering between the vectors of $\mathbb{Z}^n_{\geq 0}$ defines an ordering between monomials: if $\alpha > \beta$, where $>$ is a given ordering on $\mathbb{Z}^n_{\geq 0}$, then we will say that $x^\alpha > x^\beta$.

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Since a polynomial is a sum of monomials, we want to be able to write its terms by ordering them in ascending or descending order (and, of course, in an unambiguous way). To do this we need:

(i) that it is possible to compare any two monomials. This requires that the ordering is a total ordering: given the monomials x^α and x^β, only one of the following statement must be true

$$x^\alpha > x^\beta, \quad x^\alpha = x^\beta, \quad x^\beta > x^\alpha.$$

(ii) to take into consideration the effects of the sum and product operations on the monomials. When we add polynomials, after collecting the terms, we can simply rearrange the terms. Multiplication could give problems if multiplying a polynomial for a monomial, the ordering of terms changed. In order to avoid this, we require that if $x^\alpha > x^\beta$ and x^γ are monomials, then $x^\alpha x^\gamma > x^\beta x^\gamma$.

Remark 13.1.1 If we consider an ordering on $\mathbb{Z}^n_{\geq 0}$, then property (ii) means that if $\alpha > \beta$ then, for any $\gamma \in \mathbb{Z}^n_{\geq 0}$, $\alpha + \gamma > \beta + \gamma$.

Definition 13.1.2 A *monomial ordering* on $k[x_1, \ldots, x_n]$ is any relation $>$ on $\mathbb{Z}^n_{\geq 0}$ or, equivalently, any relation on the set of monomials x^α, $\alpha \in \mathbb{Z}^n_{\geq 0}$ satisfying

(i) $>$ is a total ordering on $\mathbb{Z}^n_{\geq 0}$;
(ii) If $\alpha > \beta$ and $\gamma \in \mathbb{Z}^n_{\geq 0}$, then $\alpha + \gamma > \beta + \gamma$;
(iii) $x^\alpha > 1$ for every nonzero α.

Remark 13.1.3 Property (iii) is equivalent to the fact that $>$ is a well ordering on $\mathbb{Z}^n_{\geq 0}$, that is any non-empty subset of $\mathbb{Z}^n_{\geq 0}$ has a smallest element with respect to $>$. It is not difficult to prove that this implies that each sequence in $\mathbb{Z}^n_{\geq 0}$, strictly decreasing, at some point ends. This fact will be fundamental when we want to prove that some algorithm stops in a finite number of steps as some terms decrease strictly.

We pass now to introduce the most frequently used orderings.

Definition 13.1.4 Let $\alpha = (\alpha_1, \ldots, \alpha_n)$ and $\beta = (\beta_1, \ldots, \beta_n)$ be elements of $\mathbb{Z}^n_{\geq 0}$.

(lex) We say that $\alpha >_{lex} \beta$ if, in the vector $\alpha - \beta \in \mathbb{Z}^n_{\geq 0}$, the first nonzero entry, starting from left, is positive. We write $x^\alpha >_{lex} x^\beta$ if $\alpha >_{lex} \beta$ (*lexicographic ordering*).

(grlex) We say that $\alpha >_{grlex} \beta$ if,

$$|\alpha| = \sum_{i=1}^n \alpha_i > |\beta| = \sum_{i=1}^n \beta_i \text{ or } |\alpha| = |\beta| \text{ and } \alpha >_{lex} \beta.$$

We write $x^\alpha >_{grlex} x^\beta$ if $\alpha >_{grlex} \beta$ (*graded lexicographic ordering*).

(grevlex) We say that $\alpha >_{grevlex} \beta$ if,

$$|\alpha| = \sum_{i=1}^{n} \alpha_i > |\beta| = \sum_{i=1}^{n} \beta_i \text{ or } |\alpha| = |\beta|$$

and the first nonzero entry, starting from right, is negative. We write $x^{\alpha} >_{grevlex} x^{\beta}$ if $\alpha >_{grevlex} \beta$ (*reverse graded lexicographic ordering*).

Example 13.1.5

(1) $(1, 2, 5, 4) >_{lex} (0, 2, 4, 6)$ since $(1, 2, 5, 4) - (0, 2, 4, 6) = (1, 0, 1, -2)$;
(2) $(3, 2, 2, 4) >_{lex} (3, 2, 2, 3)$ since $(3, 2, 2, 4) - (3, 2, 2, 3) = (0, 0, 0, 1)$;
(3) $(1, 3, 1, 4) <_{lex} (2, 3, 2, 1)$ since the first entry, from left of $(1, 3, 1, 4) - (2, 3, 2, 1) = (-1, 0, 0, 3)$ is negative;
(4) $(1, 2, 3, 5) <_{grlex} (0, 1, 5, 6)$ since $|(1, 2, 3, 5)| = 11 < 12 = |(0, 1, 5, 6)|$;
(5) $(4, 2, 2, 4) >_{grlex} (4, 2, 1, 5)$ since $|(4, 2, 2, 4)| = |(4, 2, 1, 5)| = 12$ and $(4, 2, 2, 4) >_{lex} (4, 2, 1, 5))$ (in fact $(4, 2, 2, 4) - (4, 2, 1, 5)) = (0, 0, 1, -1)$;
(6) $(4, 3, 2, 1) <_{grevlex} (0, 2, 4, 6)$ since $|(4, 3, 2, 1)| = 10 < 12 = |(0, 2, 4, 6)|$;
(7) $(1, 3, 4, 4) >_{grevlex} (2, 3, 2, 5)$ since $|(3, 1, 2, 4)| = |(3, 1, 1, 5)| = 12$ and the first nonzero entry, from right, of $(1, 3, 4, 4) - (2, 3, 2, 5) = (-1, 0, 2, -1)$ is negative.

To each variable x_i is associated the vector of $\mathbb{Z}_{\geq 0}^n$ with 1 in the $i-$th position and zero elsewhere. It is easy to check that

$$(1, 0, \ldots, 0) >_{lex} (0, 1, \ldots, 0) >_{lex} \cdots >_{lex} (0, 0, \ldots, 0, 1)$$

from which we get $x_1 >_{lex} x_2 >_{lex} \cdots >_{lex} x_n$. In the practice, if we are working with variables $x, y, z, \ldots$ we assume that the alphabetic ordering among variables $x > y > z > \cdots$ is used to define the lexicographic ordering among monomials.

In the lexicographical ordering, each variable dominates any monomial composed only of smaller variables. For example, $x_1 >_{lex} x_2^4 x_3^5 x_4$ since $(1, 0, 0, 0) - (0, 4, 3, 1) = (1, -4, -3, -1)$. Roughly speaking, the lexicographical ordering does not take into account the total degree of monomial and, for this reason, we introduce the graded lexicographic ordering and the reverse graded lexicographic ordering. The two orderings behave in a different way: both use the total degree of the monomials, but $grlex$ uses the ordering lex and therefore "favors" the greater power of the first variable, while $grevlex$, looking at the first negative entry from right, "favors" the smaller power of the last variable. For example,

$$x_1^4 x_2 x_3^2 >_{grlex} x_1^3 x_2^3 x_3 \quad \text{and} \quad x_1^3 x_2^3 x_3 >_{grevlex} x_1^4 x_2 x_3^2.$$

It is important to notice that there are $n!$ orderings of type $grlex$ and $grevlex$ according to the ordering we give to the monomials of degree 1. For example, for two variables we can have $x_1 < x_2$ or $x_2 < x_1$.

Example 13.1.6 Let us show how monomial orderings are applied to polynomials. If $f \in k[x_1, \ldots, x_n]$ and we had chosen a monomial ordering $>$, then we can order, with respect to $>$, in a nonambiguous way the terms of f. Consider, for example, $f = 2x_1^2x_2^2 + 3x_2^4x_3 - 5x_1^4x_2x_3^2 + 7x_1^3x_2^3x_3$. With respect to the lexicographic ordering, f is written as

$$f = -5x_1^4x_2x_3^2 + 7x_1^3x_2^3x_3 + 2x_1^2x_2^2 + 3x_2^4x_3.$$

With respect to the degree lexicographic ordering, f is written as

$$f = -5x_1^4x_2x_3^2 + 7x_1^3x_2^3x_3 + 3x_2^4x_3 + 2x_1^2x_2^2.$$

With respect to the reverse degree lexicographic ordering, f is written as

$$f = 7x_1^3x_2^3x_3 - 5x_1^4x_2x_3^2 + 3x_2^4x_3 + 2x_1^2x_2^2.$$

Definition 13.1.7 Let $f = \sum_\alpha a_\alpha x^\alpha$ be a nonzero polynomial in $k[x_1, \ldots, x_n]$ and let $>$ be a monomial ordering

(i) the *multidegree* of f is

$$\text{multideg}(f) = \max\{\alpha \in \mathbb{Z}_{\geq 0}^n : a_\alpha \neq 0\}.$$

(ii) The *leading coefficient* of f is

$$LC(f) = a_{\text{multideg}(f)} \in k.$$

(iii) The *leading monomial* of f is

$$LM(f) = x^{\text{multideg}(f)}.$$

(iv) The *leading term* of f is

$$LT(f) = LC(f) \cdot LM(f).$$

If we consider the polynomial $f = 2x_1^2x_2^2 + 3x_2^4x_3 - 5x_1^4x_2x_3^2 + 7x_1^3x_2^3x_3$ of Example 13.1.6, once the reverse degree lexicographic ordering is chosen, one has:

$$\begin{gathered}\text{multideg}(f) = (3, 3, 1),\\ LC(f) = 7,\\ LM(f) = x_1^3x_2^3x_3,\\ LT(f) = 7x_1^3x_2^3x_3.\end{gathered}$$

Lemma 13.1.8 *Let $f, g \in k[x_1, \ldots, x_n]$ be nonzero polynomials. Then*

(i) $\text{multideg}(fg) = \text{multideg}(f) + \text{multideg}(g)$
(ii) If $f + g \neq 0$ *then* $\text{multideg}(f + g) \leq \max\{\text{multideg}(f), \text{multideg}(g)\}$. *If, moreover,* $\text{multideg}(f) \neq \text{multideg}(g)$, *then equality holds.*

From now on we will always assume that a particular monomial ordering has been chosen and therefore that $LC(f)$, $LM(f)$ and $LT(f)$ are calculated relative to that monomial ordering only.

The concepts introduced in Definition 13.1.7 permits to extend the classical division algorithm for polynomial in one variable, i.e., $f \in k[x]$, to the case of polynomials in more variable, $f \in k[x_1, \ldots, x_n]$. In the general case, this means to divide $f \in k[x_1, \ldots, x_n]$ by $f_1, \ldots, f_t \in k[x_1, \ldots, x_n]$, which is equivalent to write f as

$$f = a_1 f_1 + \cdots + a_t f_t + r,$$

where the a_i's and r are elements in $k[x_1, \ldots, x_n]$. The idea of this division algorithm is the same of the case of a single variable: we multiply f_1 for a suitable a_1 in such a way to cancel the leading term of f obtaining $f = a_1 f_1 + r_1$, Then we multiply f_2 for a suitable a_2 in such a way to cancel the leading term of r_1 obtaining $r_1 = a_2 f_2 + r_2$ and hence $f = a_1 f_1 + a_2 f_2 + r_2$. And we proceed in the same manner for the other polynomials $f_3, \ldots, f_t$. The following theorem assures the correctness of the algorithm.

Theorem 13.1.9 *Let $>$ be a fixed monomial ordering on $\mathbb{Z}^n_{\geq 0}$ and let $F = (f_1, \ldots, f_t)$ be an ordered $t-$uple of polynomials in $k[x_1, \ldots, x_n]$. Then any $f \in k[x_1, \ldots, x_n]$ can be written as*

$$f = a_1 f_1 + \cdots + a_t f_t + r$$

where $a_i, r \in k[x_1, \ldots, x_n]$ and $r = 0$ or r is a linear combination of monomials, with coefficients in k, none of them divisible by any of the leading terms $LT(f_1), \ldots, LT(f_t)$. We say that r is the remainder of the division of f by F. Moreover, if $a_i f_i \neq 0$, then

$$\text{multideg}(f) \geq \text{multideg}(a_i f_i).$$

The proof of Theorem 13.1.9, which we do not include here (see [1, Theorem 3, pag 61]), is based on the fact that the algorithm of division terminates after a finite number of steps, which is a consequence of the fact that $>$ is a well ordering (see Remark 13.1.3).

Example 13.1.10 We divide $f = x^2y + xy - x - 2y$ by $F = (f_1, f_2)$ where $f_1 = xy + y$ and $f_2 = x + y$ and using the lexicographic ordering. Both the leading terms $LT(f_1) = xy$ e $LT(f_2) = x$ divide the leading term of f, $LT(f) = x^2y$. Since F is ordered we start dividing by f_1:

$$a_1 = \frac{LT(f)}{LT(f_1)} = x.$$

Then we subtract $a_1 f_1$ from f

$$g = f - a_1 f_1 = x^3 y^2 + 3xy - 2y - x(xy + y) = -x - 2y.$$

The leading term of this polynomial, $LT(g) = -x$, is divisible for the one of f_2 and hence we compute

$$a_2 = \frac{LT(g)}{LT(f_2)} = -1, \qquad r = g - a_2 f_2 = -x - 2y + x + y = -y.$$

Hence one has

$$f = x \cdot (xy + y) - 1 \cdot (x + y) - y.$$

Unluckily, the division algorithm of Theorem 13.1.9 does not behave well as for the case of a single variable. This is shown in the following two examples.

Example 13.1.11 We divide $f = x^2 y + 4xy^2 - 2x$ by $F = (f_1, f_2)$ where $f_1 = xy + y$ and $f_2 = x + y$ and using again the lexicographic ordering. Proceeding as in the previous example we get

$$a_1 = \frac{LT(f)}{LT(f_1)} = \frac{x^2 y}{xy} = x, \qquad g = f - a_1 f_1 = 4xy^2 - xy - 2x$$

$$a_2 = \frac{LT(g)}{LT(f_2)} = \frac{4xy^2}{x} = 4y^2, \qquad r = g - a_2 f_2 = -xy - 2x - 4y^3.$$

Notice that the leading term of the remainder, $LT(r) = -xy$, is still divisible for the leading term of f_1. Hence we can again divide by f_1 getting

$$a_1' = \frac{LT(r)}{LT(f_1)} = \frac{-xy}{xy} = -1, \qquad r' = r - a_1' f_1 = -2x - 4y^3 + y.$$

Again the leading term of the new remainder, $LT(r') = -2x$, is still divisible for the leading term of f_2. Hence we can again divide by f_2 getting

$$a_2' = \frac{LT(r')}{LT(f_2)} = \frac{-2x}{x} = -2, \qquad r'' = r' - a_2' f_2 = -4y^3 + 3y.$$

Hence one has

$$\begin{aligned} f &= x(xy + y) + 4y^2(x + y) - 1(xy + y) - 2(x + y) - 4y^3 + 3y = \\ &= (x - 1)(xy + y) + (4y^2 - 2)(x^2 + 1) - 4y^3 + 3y. \end{aligned}$$

Example 13.1.12 Another problem of the division algorithm in $k[x_1, \ldots, x_n]$ concerns the fact that, changing the order of the f_i's, the values of a_i and r can change. In particular, the remainder r is not univocally determined. Consider, for example, the polynomial $f = x^2y + 4xy^2 - 2x$ of the previous example, dividing first by $f_2 = x + y$ and then by $f_1 = xy + y$.

$$a_2 = \frac{LT(f)}{LT(f_2)} = \frac{x^2y}{x} = xy, \qquad g = f - a_2 f_2 = 3xy^2 - 2x$$

$$a_1 = \frac{LT(g)}{LT(f_1)} = \frac{3xy^2}{xy} = 3y, \qquad g - a_1 f_1 = -2x - 3y^2.$$

Notice that the leading term of the remainder, $LT(r) = -2x$, is still divisible for the leading term of f_2. Hence we can again divide by f_2 getting

$$a_2' = \frac{LT(r)}{LT(f_2)} = \frac{2x}{x} = -2, \qquad r' = r - a_2' f_2 = -3y^2 + 2y$$

and giving a remainder, $-3y^2 + 2y$, which is different to the one of the Example 13.1.11.

From the previous examples, we can conclude that the division algorithm in $k[x_1, \ldots, x_n]$ is an imperfect generalization of the case of a single variable. To overcome these problems it will be necessary to introduce the Groebner basis. The basic idea is based on the fact that, when we work with a set of polynomials $f_1, \ldots, f_t$, this leads to working with the ideal generated by them $I = \langle f_1, \ldots, f_t \rangle$. This gives us the ability to switch from $f_1, \ldots, f_t$, to a different set of generators of I, but with better properties with respect to the division algorithm. Before introducing the Groebner basis we recall some concepts and results that will be useful to us.

13.2 Monomial Ideals

Definition 13.2.1 An ideal $I \subset k[x_1, \ldots, x_n]$ is a *monomial* ideal if there exists a subset $A \subset \mathbb{Z}^n_{\geq 0}$ (eventually infinite) such that I consists of all polynomials which are finite sums of the form $\sum_{\alpha \in A} h_\alpha x^\alpha$, where $h_\alpha \in k[x_1, \ldots, x_n]$. In such case, we write $I = \langle x^\alpha : \alpha \in A \rangle$.

An example of monomial ideal is given by $I = \langle x^5y^2, x^3y^3, x^2y^4 \rangle$.
It is possible to characterize all the monomials that are in a given monomial ideal.

Lemma 13.2.2 *Let $I = \langle x^\alpha : \alpha \in A \rangle$ be a monomial ideal. Then a monomial x^β is in I if and only if x^β is divisible for x^α, for some $\alpha \in A$.*

Proof Let x^β be a multiple of x^α for some $\alpha \in A$, then $x^\beta \in I$, by definition of ideal. On the other hand, if $x^\beta \in I$ then

$$x^\beta = \sum_{i=1}^{t} h_i x^{\alpha_i}, \tag{13.2.1}$$

where $h_i \in k[x_1, \ldots, x_n]$ and $\alpha_i \in A$. Writing each h_i as a combination of monomials, we can observe that each term in the right side of (13.2.1) is divisible for some α_i. Hence also the left side x^β of (13.2.1) must have the same property, i.e., it is divisible for some α_i. □

Observe that x^β is divisible by x^α when $x^\beta = x^\alpha \cdot x^\gamma$ for some $\gamma \in \mathbb{Z}^n_{\geq 0}$ which is equivalent to require $\beta = \alpha + \gamma$. Hence, the set

$$\alpha + \mathbb{Z}^n_{\geq 0} = \{\alpha + \gamma \,:\, \gamma \in \mathbb{Z}^n_{\geq 0}\}$$

consists of the exponents of monomials which are divisible by x^α. This fact, together with Lemma 13.2.2, permits us to give a graphical description of the monomials in a given monomial ideal. For example, if $I = \langle x^5y^2, x^3y^3, x^2y^4\rangle$, then the exponents of monomials in I form the set

$$\left((5,2) + \mathbb{Z}^n_{\geq 0}\right) \cup \left((3,3) + \mathbb{Z}^n_{\geq 0}\right) \cup \left((2,4) + \mathbb{Z}^n_{\geq 0}\right).$$

We can visualize this set as the union of the integer points in three translated copies of the first quadrant in the plane, as showed in Fig. 13.1.

The following lemma allows to say if a polynomial f is in a monomial ideal I, looking at the monomials of f.

Lemma 13.2.3 *Let I be a monomial ideal and consider $f \in k[x_1, \ldots, x_n]$. Then the following conditions are equivalent:*

(i) $f \in I$;
(ii) *every term of f is in I;*
(iii) *f is a linear combination of monomials in I.*

One of the main results on monomial ideals is the so-called Dickson's Lemma which assures us that every monomial ideal is generated by a finite number of monomials. For the proof, the interested reader can consult [1, Theorem 5, Chap. 2.4],

Lemma 13.2.4 (Dickson's Lemma) *A monomial ideal $I = \langle x^\alpha \,:\, \alpha \in A\rangle \subset k[x_1, \ldots, x_n]$ can be written as $I = \langle x^{\alpha_1}, x^{\alpha_2}, \ldots, x^{\alpha_t}\rangle$ where $\alpha_1, \alpha_2, \ldots, \alpha_t \in A$. In particular I has a finite basis.*

In practice, Dickson's Lemma follows immediately from the Basis Theorem 9.1.22, which has an independent proof. Since we did not provide a proof of Theorem 9.1.22, for the sake of completeness we show how, conversely, Dickson's Lemma can provide, as a corollary, a proof of a weak version of the Basis Theorem.

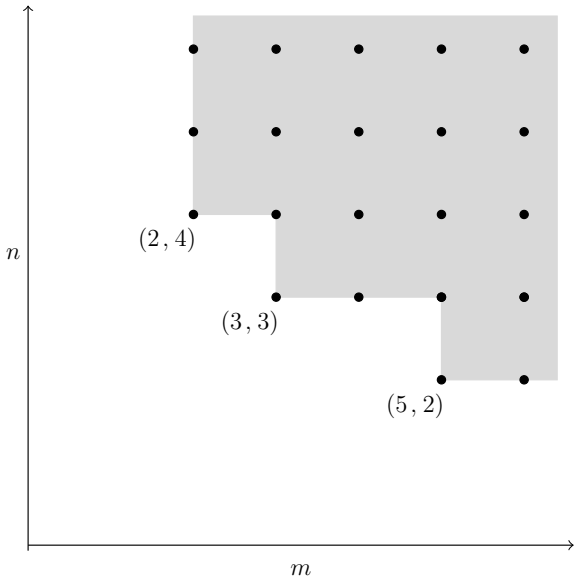


Fig. 13.1 $\left((5,2)+\mathbb{Z}^n_{\geq 0}\right) \cup \left((3,3)+\mathbb{Z}^n_{\geq 0}\right) \cup \left((2,4)+\mathbb{Z}^n_{\geq 0}\right)$

Theorem 13.2.5 (Basis Theorem, weak version) *Any ideal* $I \subset k[x_1, \ldots, x_n]$ *has a finite basis, that is* $I = \langle g_1, \ldots, g_t \rangle$ *for some* $g_1, \ldots, g_t \in I$.

Before proving the Hilbert Basis Theorem we introduce some concepts.

Definition 13.2.6 Let $I \subset k[x_1, \ldots, x_n]$ be an ideal different from the zero ideal $\{0\}$.

(i) We denote by $LT(I)$ the set of leading terms of I

$$LT(I) = \{cx^\alpha \ : \text{ there exists } f \in I \text{ with } LT(f) = cx^\alpha\}$$

(ii) We denote by $\langle LT(I) \rangle$ the ideal generated by the elements in $LT(I)$.

Given an ideal $I = \langle f_1, \ldots, f_t \rangle$, we observe that $\langle LT(f_1), \ldots, LT(f_t) \rangle$ is not necessarily equal to $\langle LT(I) \rangle$. It is true that $LT(f_i) \in LT(I) \subset \langle LT(I) \rangle$ from which it follows $\langle LT(f_1), \ldots, LT(f_t) \rangle \subset \langle LT(I) \rangle$. However $\langle LT(I) \rangle$ can contain strictly $\langle LT(f_1), \ldots, LT(f_t) \rangle$.

Example 13.2.7 Let $I = \langle f_1, f_2 \rangle$ with $f_1 = x^3y - x^2 + x$ e $f_2 = x^2y^2 - xy$. The ordering *grlex* is chosen. Since

$$y \cdot (x^3y - x^2 + x) - x \cdot (x^2y^2 - xy) = xy$$

one has $xy \in I$, from which $xy = LT(xy) \in \langle LT(I) \rangle$. However xy is not divisible by $LT(f_1) = x^3y$ and by $LT(f_1) = x^2y^2$, and hence, by Lemma 13.2.2 $xy \notin \langle LT(f_1), LT(f_2) \rangle$.

Proposition 13.2.8 *Let $I \subset k[x_1, \ldots, x_n]$ be an ideal.*

(i) $\langle LT(I)\rangle$ *is a monomial ideal;*
(ii) There exist $g_1, \ldots, g_t$ *such that* $\langle LT(I)\rangle = \langle LT(g_1), \ldots, LT(g_t)\rangle$.

Proof For (i), notice that the leading monomials $LM(g)$ of the elements $g \in I \setminus \{0\}$ generate the monomial ideal $J := \langle LM(g) : g \in I \setminus \{0\}\rangle$. Since $LM(g)$ and $LT(g)$ differ only by a nonzero constant, one has $J = \langle LT(g) : g \in I \setminus \{0\}\rangle = \langle LT(I)\rangle$. Hence $\langle LT(I)\rangle$ is a monomial ideal.
For (ii), since $\langle LT(I)\rangle$ is generated by the monomials $LM(g)$ with $g \in I \setminus \{0\}$, by Dickson's Lemma, we know that $\langle LT(I)\rangle = \langle LM(g_1), LM(g_2), \ldots, LM(g_t)\rangle$ for a finite number of polynomials $g_1, g_2, \ldots, g_t \in I$. Since $LM(g_i)$ and $LT(g_i)$ differ only by a nonzero constant, for $i = 1, \ldots, t$, one has $\langle LT(I)\rangle = \langle LT(g_1), LT(g_2), \ldots, LT(g_t)\rangle$. □

Using Proposition 13.2.8 and the division algorithm of Theorem 13.1.9 we can prove Theorem 13.2.5.

Proof of Hilbert Basis Theorem. If $I = \{0\}$ then, as a set of generators, we take $\{0\}$ which is clearly finite. If I contains some nonzero polynomials, then a set of generators $g_1, \ldots, g_t$ for I can be build in the following way. By Proposition 13.2.8 there exist $g_1, \ldots, g_t \in I$ such that $\langle LT(I)\rangle = \langle LT(g_1), LT(g_2), \ldots, LT(g_t)\rangle$. We prove that $I = \langle g_1, \ldots, g_t\rangle$.

Clearly $\langle g_1, \ldots, g_t\rangle \subset I$ since, for any $i = 1, \ldots, t$, $g_i \in I$. On the other hand, let $f \in I$ be a polynomial. We apply the division algorithm of Theorem 13.1.9 to divide f by $\langle g_1, \ldots, g_t\rangle$. We get

$$f = a_1 g_1 + \cdots + a_t g_t + r,$$

where the terms in r are not divisible for any of the leading terms $LT(g_i)$. We show that $r = 0$. To this aim, we observe, first of all, that

$$r = f - a_1 g_1 - \cdots - a_t g_t \in I.$$

If $r \neq 0$ then $LT(r) \in \langle LT(I)\rangle = \langle LT(g_1), LT(g_2), \ldots, LT(g_t)\rangle$ and, by Lemma 13.2.2, it follows that $LT(r)$ must be divisible by at least one leading term $LT(g_i)$. This contradicts the definition of remainder of division and hence r must be equal to zero, from which one has

$$f = a_1 g_1 + \cdots + a_t g_t + 0 \in \langle g_1, \ldots, g_t\rangle,$$

which proves $I \subset \langle g_1, \ldots, g_t\rangle$. □

13.3 Groebner Basis

Groebner basis is "good" basis for the division algorithm of Theorem 13.1.9. Here "good" means that the problems of Examples 13.1.11 and 13.1.12 do not happen.

Let us think about Theorem 13.2.5: the basis used in the proof has the particular property that $\langle LT(g_1), \ldots, LT(g_t)\rangle = \langle LT(I)\rangle$. It is not true that any basis of I has this property and so we give a specific name to the basis having this property.

Definition 13.3.1 Fix a monomial ordering. A finite subset $G = \{g_1, \ldots, g_t\}$ of an ideal I is a *Groebner basis* if

$$\langle LT(g_1), \ldots, LT(g_t)\rangle = \langle LT(I)\rangle.$$

The following result guarantees us that every ideal has a Groebner basis.

Corollary 13.3.2 *Fix a monomial ordering. Then every ideal $I \subset k[x_1, \ldots, x_n]$, different from $\{0\}$, admits a Groebner basis. Moreover, every Groebner basis for an ideal I is a basis of I.*

Proof Given an ideal I, different from the zero ideal, the set $G = \{g_1, \ldots, g_t\}$, built as in the proof of Theorem 13.2.5, is a Groebner basis by definition. To prove the second part of the statement it is enough to observe that, again, the proof of Theorem 13.2.5 assures us that $I = \langle g_1, \ldots, g_t\rangle$, that is G is a basis for I. □

Consider the ideal $I = \langle f_1, f_2\rangle$ of Example 13.2.7. According to Definition 13.3.1, $\{f_1, f_2\} = \{x^3y - x^2 + x, x^2y^2 - xy\}$ is not a Groebner basis.

Before to show how to find a Groebner basis for a given ideal, we point out some property of them.

Proposition 13.3.3 *Let $G = \{g_1, \ldots, g_t\}$ be a Groebner basis for an ideal $I \subset k[x_1, \ldots, x_n]$ and let $f \in k[x_1, \ldots, x_n]$ be a polynomial. Then there exists a unique $r \in k[x_1, \ldots, x_n]$ such that*

(i) no monomials in r are divisible by $LT(g_1), \ldots, LT(g_t)$;
(ii) there exists $g \in I$ such that $f = g + r$;

In particular, r is the remainder of the division of f by G, using the division algorithm, independently how the elements in G are listed.

Proof The division algorithm applied to f and G gives $f = a_1g_1 + \cdots + a_tg_t + r$ where r satisfies (i). In order that also (ii) be satisfied is sufficient to take $g = a_1g_1 + \cdots + a_tg_t \in I$. This proves the existence of r. To prove uniqueness, suppose that $f = g + r = \tilde{g} + \tilde{r}$ satisfy (i) and (ii). Then $\tilde{r} - r = \tilde{g} - g \in I$ and hence, if $\tilde{r} \neq r$, then $LT(\tilde{r} - r) \in \langle LT(I)\rangle = \langle LT(g_1), \ldots, LT(g_t)\rangle$. By Lemma 13.2.2 it follows that $LT(\tilde{r} - r)$ is divisible for some $LT(g_i)$. This is impossibile since no term of r and $\tilde{r}$ is divisible by $LT(g_1), \ldots, LT(g_t)$. Hence $\tilde{r} - r$ must be zero and uniqueness is proved. □

Remark 13.3.4 The remainder r of Proposition 13.3.3 is usually called *normal form* of f. The Proposition 13.3.3 tells us that Groebner basis can be characterized through uniqueness of the remainder. Observe that, even though the remainder is unique, independently of the order in which we divide f by the $LT(g_i)$'s, the coefficients a_i, in $f = a_1g_1 + \cdots + a_tg_t + r$, are not unique.

As a corollary of Proposition 13.3.3 we get the following criterion to establish if a polynomial is contained in a given ideal.

Corollary 13.3.5 *Let $G = \{g_1, \ldots, g_t\}$ be a Groebner basis for an ideal $I \subset k[x_1, \ldots, x_n]$ and let $f \in k[x_1, \ldots, x_n]$. Then $f \in I$ if and only if the remainder of the division of f by G is zero.*

Definition 13.3.6 We write $\overline{f}^F$ for the remainder of the division of f by an ordered $t-$uple $F = (f_1, \ldots, f_t)$. If F is a Groebner basis for $\langle f_1, \ldots, f_t\rangle$, then we can look at F as an unordered set, by Proposition 13.3.3.

Example 13.3.7 Consider the polynomial $f = x^2y + 4xy^2 - 2x$ and $F = \{f_1, f_2\}$ with $f_1 = xy + y$ and $f_2 = x + y$, From Example 13.1.11 we know that

$$\overline{f}^F = -4y^3 + 3y.$$

On the other side, if we consider $F' = \{f_2, f_1\}$, then, from Example 13.1.12, we get

$$\overline{f}^{F'} = -3y^2 + 2y.$$

Let's start now to explain how it is possible to build a Groebner basis for an ideal I from a set of generators $f_1, \ldots, f_t$ of I. As we saw before, one of the reasons for which $\{f_1, \ldots, f_t\}$ could not be a Groebner basis depends on the case that there is a combination of the f_i's whose leading term does not lie in the ideal generated by the $LT(f_i)$. This happens, for example, when the leading term of a given combination $ax^\alpha f_i - bx^\beta f_j$ are canceled, leaving only terms of a lower degree. On the other hand, $ax^\alpha f_i - bx^\beta f_j \in I$ and therefore its leading term belongs to $\langle LT(I)\rangle$. To study this cancelation phenomenon, we introduce the concept of S-polynomial.

Definition 13.3.8 Let $f, g \in k[x_1, \ldots, x_n]$ be two nonzero polynomials.

(i) If multideg$(f) = \alpha$ and multideg$(g) = \beta$, then, we define $\gamma = (\gamma_1, \ldots, \gamma_n)$ where $\gamma_i = \max\{\alpha_i, \beta_i\}$, and we call x^γ the lower common multiple of $LM(f)$ and $LM(g)$, by writing $x^\gamma = \text{lcm}(LM(f), LM(g))$.

(ii) The S-polynomial of f and g is the combination

$$S(f, g) = \frac{x^\gamma}{LT(f)} \cdot f - \frac{x^\gamma}{LT(g)} \cdot g.$$

Example 13.3.9 Consider the polynomials $f = 3x^3z^2 + x^2yz + xyz$ e $g = x^2y^3z + xy^3 + z^2$ in $k[x, y, z]$, If we choose the lexicographic ordering we get

$$\begin{aligned} \text{multideg}(f) &= (3, 0, 2), \\ \text{multideg}(g) &= (2, 3, 1), \end{aligned}$$

hence $\gamma = (3, 3, 2)$ and

$$S(f, g) = \frac{x^3y^3z^2}{3x^3z^2} \cdot f - \frac{x^3y^3z^2}{x^2y^3z} \cdot g = \frac{1}{3}x^2y^4z - x^2y^3z + \frac{1}{3}xy^4z - xz^3.$$

A S-polynomial $S(f, g)$ is used to produce the cancelation of the leading terms. In fact, any cancelation of leading terms between polynomials of the same multidegree is obtained from this type of polynomial combinations, as guaranteed by the following result.

Lemma 13.3.10 *Suppose to have a sum of polynomials* $\sum_{i=1}^{t} c_i f_i$ *where* $c_i \in k$ *and* $\text{multideg}(f_i) = \delta \in \mathbb{Z}^n_{\geq 0}$ *for all* i. *If* $\text{multideg}(\sum_{i=1}^{t} c_i f_i) < \delta$, *then* $\sum_{i=1}^{t} c_i f_i$ *is a linear combination, with coefficients in* k, *of the* S*-polynomials* $S(f_i, f_j)$, *for* $1 \leq i, j \leq t$. *Moreover, each* $S(f_i, f_j)$ *has multidegree* $< \delta$.

Using the concept of S-polynomial and the previous lemma we can prove the following criterion to establish whether a basis of an ideal is Groebner basis.

Theorem 13.3.11 (Buchberger S-pair criterion) *Let* I *be an ideal in* $k[x_1, \ldots, x_n]$. *Then a basis* $G = \{g_1, \ldots, g_t\}$ *for* I *is a Groebner basis for* I *if and only if, for any pair of indices* $i \neq j$, *the remainder of the division of* $S(g_i, g_j)$ *by* G *is zero.*

Proof The "only if" direction is simple because, if G is a Groebner basis, then since $S(g_i, g_j) \in I$, their remainder in the division by G is zero, by Corollary 13.3.5. It remains to prove the "if" direction.

Let $f \in I = \langle g_1, \ldots, g_t \rangle$ be a nonzero polynomial. Hence there exist polynomials $h_i \in k[x_1, \ldots, x_n]$ such that

$$f = \sum_{i=1}^{t} h_i g_i. \tag{13.3.1}$$

By Lemma 13.1.8 we know that

$$\text{multideg}(f) \leq \max\left(\text{multideg}(h_i g_i)\right). \tag{13.3.2}$$

Let $m_i = \text{multideg}(h_i g_i)$ and define $\delta = \max(m_1, \ldots, m_t)$. Hence the previous inequality can be written as $\text{multideg}(f) \leq \delta$. If we change, in (13.3.1), the way to write f in terms of G, we get a different value for δ. Since a monomial ordering is a well ordering, we can choose an expression for f, of the form (13.3.1), for which δ is minimal.

Let us show that, if δ is minimal, then multideg$(f) = \delta$.

Suppose that multideg $(f) < \delta$ and write f in order to isolate the terms of multidegree δ:

$$\begin{aligned} f &= \sum_{m_i=\delta} h_i g_i + \sum_{m_i<\delta} h_i g_i \\ &= \sum_{m_i=\delta} LT(h_i) g_i + \sum_{m_i=\delta} (h_i - LT(h_i)) g_i + \sum_{m_i<\delta} h_i g_i. \end{aligned} \tag{13.3.3}$$

The monomials appearing in the second and third addition of the second line, all have multidegree $< \delta$. So, our hypothesis multideg$(f) < \delta$ tells us that the first sum also has multidegree $< \delta$. Let $LT(h_i) = c_i x^{\alpha_i}$, then the first sum

$$\sum_{m_i=\delta} LT(h_i) g_i = \sum_{m_i=\delta} c_i x^{\alpha_i} g_i$$

has exactly the form described in Lemma 13.3.10 with $f_i = x^{\alpha_i} g_i$. Hence, again by Lemma 13.3.10, this sum is a linear combination of the S-polynomials $S(x^{\alpha_j} g_j, x^{\alpha_k} g_k)$. Moreover one has

$$\begin{aligned} S(x^{\alpha_j} g_j, x^{\alpha_k} g_k) &= \frac{x^{\delta}}{x^{\alpha_j} LT(g_j)} x^{\alpha_j} g_j - \frac{x^{\delta}}{x^{\alpha_k} LT(g_k)} x^{\alpha_k} g_k \\ &= x^{\delta - \gamma_{jk}} S(g_j g_k), \end{aligned}$$

where $x^{\gamma_{jk}}$ is the lower common multiple between $LM(g_j)$ and $LM(g_k)$. Hence there exist constants $c_{jk} \in k$ such that

$$\sum_{m_i=\delta} LT(h_i) g_i = \sum_{j,k} c_{jk} x^{\delta - \gamma_{jk}} S(g_j, g_k). \tag{13.3.4}$$

By hypothesis, we know that the remainder of the division of $S(g_j, g_k)$ by $g_1, \ldots, g_t$ is zero, hence, by the division algorithm, each S-polynomial can be written as

$$S(g_j, g_k) = \sum_{i=1}^{t} a_{ijk} g_i \tag{13.3.5}$$

where $a_{ijk} \in k[x_1, \ldots, x_n]$. The division algorithm tell us also that

$$\text{multideg}(a_{ijk} g_i) \leq \text{multideg}(S(g_j, g_k)) \tag{13.3.6}$$

for any choice of i, j and k. This means that, when the remainder is zero, we can find an expression for $S(g_j, g_k)$ in terms of G where not all the leading terms are canceled. As a matter of fact, we multiply the expression of $S(g_j, g_k)$ by $x^{\delta - \gamma_{jk}}$ having

$$x^{\delta-\gamma_{jk}} S(g_j, g_k) = \sum_{i=1}^{t} b_{ijk} g_i,$$

where $b_{ijk} = x^{\delta-\gamma_{jk}} a_{ijk}$. Hence, by (13.3.6) and Lemma 13.3.10 we get

$$\text{multideg}(b_{ijk} g_i) \leq \text{multideg}(x^{\delta-\gamma_{jk}} S(g_j, g_k)) < \delta. \tag{13.3.7}$$

If we replace the previous expression of $x^{\delta-\gamma_{jk}} S(g_j, g_k)$ in (13.3.4) we get the following equation

$$\sum_{m_i=\delta} LT(h_i) g_i = \sum_{j,k} c_{jk} x^{\delta-\gamma_{jk}} S(g_j, g_k) = \sum_{j,k} c_{jk} \left(\sum_{t} b_{ijk} g_i \right) = \sum_{i} \tilde{h}_i g_i$$

that, by (13.3.7), satisfies

$$\text{multideg}(\tilde{h}_i g_i) < \delta, \text{ for all } i.$$

Finally, we substitute $\sum_{m_i=\delta} LT(h_i) g_i = \sum_i \tilde{h}_i g_i$ in (13.3.3), obtaining an expression for f which is a linear combination of the g_i's where all terms have multidegree strictly less than δ. This contradicts the minimality of δ and hence $\text{multideg}(f) = \delta$.

Then $\text{multideg}(f) = \text{multideg}(h_i g_i)$ for some i, from which it follows that $LT(f)$ is divisible for $LT(g_i)$. Hence $LT(f) \in \langle LT(g_1), \ldots, LT(g_t) \rangle$ and the theorem is proved. □

Example 13.3.12 [from [1], page 84] Consider the ideal $I = \langle y - x^2, z - x^3 \rangle$ of the twisted cubic in $\mathbb{R}^3$. Fix the lexicographic ordering with $y > z > x$. We prove that $G = \{y - x^2, z - x^3\}$ is a Groebner basis for I. Compute the S-polynomial

$$S(y - x^2, z - x^3) = \frac{yz}{y}(y - x^2) - \frac{yz}{z}(z - x^3) = -zx^2 + yx^3.$$

By the division algorithm we get

$$-zx^2 + yx^3 = x^3(y - x^2) + (-x^2)(z - x^3) + 0,$$

hence $\overline{S(y - x^2, z - x^3)}_G = 0$ and, by Theorem 13.3.11 G is a Groebner basis for I. The reader check that, for the lexicographic ordering with $x > y > z$, G is not a Groebner basis for I.

13.4 Buchberger's Algorithm

We have seen, by Corollary 13.3.2, that every ideal admits a Groebner basis, but unfortunately, the corollary does not tell us how to build it. So let's see now how this problem can be solved via the Buchberger algorithm.

Theorem 13.4.1 *Let $I = \langle f_1, \ldots, f_s \rangle \neq \{0\}$ be an ideal in $k[x_1, \ldots, x_n]$. A Groebner basis for I can be built, in a finite number of steps, by the following algorithm.*

Input: $F = (f_1, \ldots, f_s)$

Output: a Groebner basis $G = (g_1, \ldots, g_t)$ for I, with $F \subset G$.

$G := F$
REPEAT
  $G' := G$
  For any pairs $\{p, q\}$, $p \neq q$ in G' DO
    $S := \overline{S(p,q)}^{G'}$
    IF $S \neq 0$ THEN $G := G \cup \{S\}$
UNTIL $G = G'$

For the proof, the reader can see [1].

Example 13.4.2 Consider again the ideal $I = \langle f_1, f_2 \rangle$ of Example 13.2.7. We already know that $\{f_1, f_2\} = \{x^3y - x^2 + x, x^2y^2 - xy\}$ is not a Groebner basis since $y \cdot (x^3y - x^2 + x) - x \cdot (x^2y^2 - xy) = xy = LT(xy) \notin \langle LT(f_1), LT(f_2) \rangle$. We fix $G' = G = \{f_1, f_2\}$ and compute

$$S(f_1, f_2) := \frac{x^3y^2}{x^3y} f_1 - \frac{x^3y^2}{x^2y^2} f_2 = xy.$$

Since $\overline{S(f_1, f_2)}^{G'} = xy$, we add $f_3 = xy$ to G. We repeat again the cycle with the new set of polynomials obtaining

$$S(f_1, f_2) = xy, \quad S(f_1, f_3) = -x^2 + x, \quad S(f_2, f_3) = -xy$$

and

$$\overline{S(f_1, f_2)}^{G'} = 0, \quad \overline{S(f_1, f_3)}^{G'} = -x^2 + x, \quad \overline{S(f_2, f_3)}^{G'} = 0.$$

Hence we add $f_4 = x^2 - x$ to G. Iterating again the cycle one has

$$\begin{array}{l} S(f_1, f_2) = xy, \quad S(f_1, f_3) = -x^2 + x, \quad S(f_1, f_4) = x^2y - x^2 + x \\ S(f_2, f_3) = -xy, \; S(f_2, f_4) = x^2y - xy, \; S(f_3, f_4) = xy \end{array}$$

from which we get

$$\begin{array}{l} \overline{S(f_1, f_2)}^{G'} = 0, \; \overline{S(f_1, f_3)}^{G'} = 0, \; \overline{S(f_1, f_4)}^{G'} = 0, \\ \overline{S(f_2, f_3)}^{G'} = 0, \; \overline{S(f_2, f_4)}^{G'} = 0, \; \overline{S(f_3, f_4)}^{G'} = 0. \end{array}$$

Thus we can exit from the cycle obtaining the Groebner basis

$$G = \{x^3y - x^2 + x, x^2y^2 - xy, xy, x^2 - x\}.$$

Remark 13.4.3 The algorithm of Theorem 13.4.1 is just a rudimentary version of the Buchberger algorithm, as it is not very practical from a computational point of view. In fact, once a remainder $\overline{S(p,q)}^{G'}$ is equal to zero, this will remain zero even if we add additional generators to G'. So there is no reason to recalculate those remainders that have already been analyzed in the main loop. In fact, if we add the new generators f_j, one at a time, the only remainders to be checked are those of the type $\overline{S(f_i, f_j)}^{G'}$, where $i \leq j - 1$. The interested reader can find a refined version of the Buchberger algorithm in [1, Chap. 2.9].

The Groebner basis obtained through Theorem 13.4.1 are often too large compared to the necessary. We can eliminate some generators using the following result.

Lemma 13.4.4 *Let G be a Groebner basis for an $I \subset k[x_1, \ldots, x_n]$ and let $p \in G$ be a polynomial such that $LT(p) \in \langle LT(G \setminus \{p\})\rangle$. Then $G \setminus \{p\}$ is still a Groebner basis for I.*

Proof We know that $\langle LT(G)\rangle = \langle LT(I)\rangle$. If $\langle LT(p)\rangle \in \langle LT(G \setminus \{p\})\rangle$, then $LT(G \setminus \{p\}) = LT(G)$. Hence, by definition, $G \setminus \{p\}$ is still a Groebner basis for I. □

If we multiply each polynomial in G by a suitable constant in such a way all leading coefficients are equal to 1, and then we remove from G, any p such that $LT(p) \in \langle LT(G \setminus \{p\})\rangle$, we obtain the so-called *minimal Groebner basis*.

Definition 13.4.5 A *minimal* Groebner basis for an ideal I is a Groebner basis G for I such that

(i) $LC(p) = 1$ for all $p \in G$.
(ii) For all $p \in G$, $LT(p) \notin \langle LT(G \setminus \{p\})\rangle$.

Example 13.4.6 Consider the Groebner basis $G = \{x^3y - x^2 + x, x^2y^2 - xy, xy, x^2 - x\}$ of Example 13.4.2 (with ordering $grlex$). The leading coefficients are all equal to 1, so condition i) is satisfied (otherwise it would be enough to multiply the polynomials of the basis for suitable constants). Observe that

$$\begin{aligned} &LT(x^3y - x^2 + x) = x^3y, \\ &LT(x^2y^2 - xy) = x^2y^2, \\ &LT(xy) = xy, \\ &LT(x^2 - x) = x^2. \end{aligned}$$

Thus the leading terms of $x^2y - x^2 + x$ e $xy^2 - xy$ are contained in the ideal $\langle xy, x^2\rangle = \langle LT(xy), LT(x^2 - x)\rangle$ and hence a minimal basis for the ideal $I = \langle x^2y - x^2 + x, xy^2 - xy\rangle$ is given by $\{xy, x^2 - x\}$.

An ideal can have many minimal Groebner bases. However, we can find one which is better than the others.

Definition 13.4.7 A *reduced Groebner basis* for an ideal $I \subset k[x_1, \ldots, x_n]$ is a Groebner basis G for I such that

(i) $LC(p) = 1$ for all $p \in G$.
(ii) For all $p \in G$, no monomial of p is in $\langle LT(G \setminus \{p\})\rangle$.

The reduced Groebner bases have the following important property.

Proposition 13.4.8 *Let $I \subset k[x_1, \ldots, x_n]$ be an ideal different from $\{0\}$. Then, given a monomial ordering, I has a unique reduced Groebner basis.*

13.5 Groebner Bases and Elimination Theory

The Elimination Theory, as shown in Sects. 10.2 and 10.3, is a systematic way to eliminate variables from a system of polynomial equations. The central part of this method is based on the so-called Elimination Theorem and Extension Theorem. We now define the concept of "eliminating variables" in terms of ideals and Groebner bases.

Definition 13.5.1 Given an ideal $I = \langle f_1, \ldots, f_t\rangle \subset k[x_1, \ldots, x_n]$, the $l-$th *elimination ideal* I_l of I is the ideal of $k[x_{l+1}, \ldots, x_n]$ defined as

$$I_l = I \cap k[x_{l+1}, \ldots, x_n].$$

It is easy to prove that I_l is an ideal of $k[x_{l+1}, \ldots, x_n]$. Obviously, the ideal I_0 coincides with I itself. It should also be noted that different monomial orderings give different elimination ideals.

Remark 13.5.2 The reader must notice that here the notation is different with respect to Sect. 10.3, where we define J_0 as the first elimination ideal in the fist variable x_0. The same ideal is called J_1 in Definition 13.5.1. For this reason, now our variables start from x_1, in such a way we have a more comfortable notation for indices.

It is clear, at this point, that eliminating $x_1, \ldots, x_l$ means to find the non-null polynomials contained in the $l-$th elimination ideal. This can easily be done through Groebner bases (once a proper monomial ordering has been fixed!).

Theorem 13.5.3 (of elimination) *Let $I \subset k[x_1, \ldots, x_n]$ be an ideal and G a Groebner basis for I with respect to the lexicographic ordering with $x_1 > x_2 > \cdots > x_n$. Then, for any $0 \leq l \leq n$, the set*

$$G_l = G \cap k[x_{l+1}, \ldots, x_n]$$

is a Groebner basis for the $l-$th elimination ideal I_l.

Proof Fix l with $0 \leq l \leq n$. By construction $G_l \subset I_l$, hence it is sufficient to prove that $\langle LT(I_l)\rangle = \langle LT(G_l)\rangle$. The inclusion $\langle LT(G_l)\rangle \subset \langle LT(I_l)\rangle$ is obvious. To prove the other inclusion, observe that if $f \in I_l$ than $f \in I$. Hence $LT(f)$ is divisible by $LT(g)$ for some $g \in G$. Since $f \in I_l$, then $LT(g)$ contains only the variables $x_{l+1}, \ldots, x_n$. Since we are using the lexicographic ordering with $x_1 > x_2 > \cdots > x_n$, any monomial formed by variables $x_1, \ldots, x_l$ is greater than all monomials in $k[x_{l+1}, \ldots, x_n]$ and hence $LT(g) \in k[x_{l+1}, \ldots, x_n]$ implies $g \in k[x_{l+1}, \ldots, x_n]$. This proves that $g \in G_l$, from which it follows $\langle LT(I_l)\rangle \subset \langle LT(G_l)\rangle$. □

The Elimination Theorem shows that a Groebner basis, in the lexicographic order, does not eliminate only the first variable, but also the first two variables, and the first three variables and so on. Often, however, we want to eliminate only certain variables, while we are not interested in others. In these cases, it may be difficult to calculate a Groebner basis using the lexicographical ordering, especially because this ordering can give some Groebner basis not particularly good. For different versions of the Elimination Theorem that are based on other orderings, refer to [1].

Now let us introduce the Extension Theorem. Suppose we have an ideal $I \subset k[x_1, \ldots, x_n]$ that defines the affine variety

$$V(I) = \{(a_1, \ldots, a_n) \in k^n \ : \ f(a_1, \ldots, a_n) = 0 \text{ for all } f \in I\}.$$

Consider the $l-$th elimination ideal. We denote by $(a_{l+1}, \ldots, a_n) \in V(I_l)$ a partial solution of the starting system of equations. To extend $(a_{l+1}, \ldots, a_n)$ to a complete solution of $V(I)$, first of all, we have to add a coordinate: this means to find a_l in such a way $(a_l, a_{l+1} \ldots, a_n) \in V(I_{l-1})$, that is $(a_l, a_{l+1} \ldots, a_n)$ is in the variety defined by the previous elimination ideal. More precisely, suppose that $I_{l-1} = \langle g_1, \ldots, g_s\rangle \subset k[x_l, \ldots, x_n]$. Hence we want to find solutions $x_l = a_l$ of the equations

$$g_1(x_l, a_{l+1}, \ldots a_n) = 0, \ldots, g_s(x_l, a_{l+1}, \ldots a_n) = 0.$$

The $g_i(x_l, a_{l+1}, \ldots a_n)$'s are polynomials in one variable hence their common solutions are the ones of the greater common divisors of these s polynomials. Obviously, it can happen that the $g_i(x_l, a_{l+1}, \ldots a_n)$'s do not have common solutions, depending on the choice of $a_{l+1}, \ldots a_n$. Hence, our aim, at the moment, is to try to determine, a priori, which partial solutions extend to complete solutions. We restrict to study the case in which we eliminated the first variable x_1 and hence we want to know if a partial solution $(a_2, \ldots, a_n) \in V(I_1)$ extends to a solution $(a_1, \ldots, a_n) \in V(I)$. The following theorem tells us when it is possible.

Theorem 13.5.4 (Extension Theorem) *Consider $I = \langle f_1, \ldots, f_t\rangle \subset \mathbb{C}[x_1, \ldots, x_n]$ and let I_1 the first elimination ideal of I. For each $1 \leq i \leq t$ we write f_i as*

$$f_i = g_i(x_2, \ldots, x_n)x_1^{N_i} + \textit{ terms in } x_1 \textit{ of degree } < N_i,$$

where $N_i \geq 0$ and $g_i \in \mathbb{C}[x_2, \ldots, x_n]$ is different from zero. Suppose there exists a partial solution $(a_2, \ldots, a_n) \in V(I_1)$. If $(a_2, \ldots, a_n) \notin V(g_1, \ldots g_t)$, then there exists $a_1 \in \mathbb{C}$ such that $(a_1, \ldots, a_n) \in V(I)$.

Notice that the Extension Theorem requires the complex field. Consider the equations

$$x^2 = y, x^2 = z.$$

If we eliminate x we get $y = z$ and, hence, all partial solutions (a, a) for all $a \in \mathbb{R}$. Since the leading coefficient of x in $x^2 = y$ and $x^2 = z$ never vanish, Theorem 13.5.4 guarantees that we can extend (a, a), under the condition that we are working on $\mathbb{C}$. As a matter of fact, on $\mathbb{R}$, $x^2 = a$ has not real solutions if a is negative, hence the only partial solutions (a, a) that we can extend, are the ones for all $a \in \mathbb{R}_{\geq 0}$.

Remark 13.5.5 Although the Extension Theorem gives a statement only in case the first variable is eliminated, it can still be used to eliminate any number of variables. The idea is to extend solutions to one variable at a time: first to x_{l-1}, then to x_{l-2} and so on up to x_1.

The Extension Theorem is particularly useful when one of the leading coefficients is constant.

Corollary 13.5.6 *Let $I = \langle f_1, \ldots, f_t \rangle \subset \mathbb{C}[x_1, \ldots, x_n]$ and assume that for some i, f_i can be written as*

$$f_i = c_i x_1^N + \text{ terms in } x_1 \text{ of degree } < N,$$

where $c \in \mathbb{C}$ is different from zero and $N > 0$. If I_1 is the first elimination ideal of I and $(a_2, \ldots, a_n) \in V(I_1)$, then there exists $a_1 \in \mathbb{C}$ such that $(a_1, \ldots, a_n) \in V(I)$.

We end this section by recalling again that the process of elimination corresponds to the projection of varieties in subspaces of lower dimension. For the rest of this section, we work over $\mathbb{C}$.

Let $V = V(f_1, \ldots, f_t) \subset \mathbb{C}^n$ be an affine variety. To eliminate the first l variables $x_1, \ldots, x_l$ we consider the projection map

$$\begin{array}{rccc} \pi_l : & \mathbb{C}^n & \to & \mathbb{C}^{n-l} \\ & (a_1, \ldots, a_n) & \mapsto & (a_{l+1}, \ldots a_n) \end{array}.$$

The following lemma explains the link between $\pi_l(V)$ and the $l-$th elimination ideal.

Lemma 13.5.7 *Let $I_l = \langle f_1, \ldots, f_t \rangle \cap \mathbb{C}[x_{l+1}, \ldots, x_n]$ be the $l-th$ elimination ideal of I. Then, in $\mathbb{C}^{n-l}$, one has*

$$\pi_l(V) \subset V(I_l).$$

Observe that we can write $\pi_l(V)$ as

$$\pi_l(V) = \left\{ \begin{array}{l} (a_{l+1}, \ldots, a_n) \in V(I_l) \,:\, \exists a_1, \ldots a_l \in \mathbb{C} \\ \text{with } (a_1, \ldots, a_l, a_{l+1}, \ldots, a_n) \in V \end{array} \right\}.$$

Hence, $\pi_l(V)$ consists exactly of the partial solutions that can be extended to complete solutions. Then, we can give a geometric version of the Extension Theorem.

Theorem 13.5.8 *Given $V = V(f_1, \ldots, f_t) \subset \mathbb{C}^n$, let g_i be as in Theorem 13.5.4. If I_1 is the first elimination ideal of $\langle f_1, \ldots, f_t \rangle$, then the following equality holds in $\mathbb{C}^{n-l}$:*

$$V(I_1) = \pi_1(V) \cup (V(g_1, \ldots, g_t) \cap V(I_1)),$$

where $\pi_1 : \mathbb{C}^n \to \mathbb{C}^{n-1}$ is the projection on the last $n-1$ components.

The previous theorem tell us that $\pi_1(V)$ covers the affine variety $V(I_1)$, with the exception, eventually, of a part in $V(g_1, \ldots, g_t)$. Unluckily we don't know how much this part is big and, there are cases where $V(g_1, \ldots, g_t)$ is enormous. However, the following result permits us to understand even better the link between $\pi_1(V)$ and $V(I_1)$.

Theorem 13.5.9 (Closure Theorem) *Given $V = V(f_1, \ldots, f_t) \subset \mathbb{C}^n$ let I_l be the $l-$th elimination ideal of $I = \langle f_1, \ldots, f_t \rangle$, then:*

(i) $V(I_l)$ is the smaller affine variety containing $\pi_l(V) \subset \mathbb{C}^{n-l}$.
(ii) When $V \neq \emptyset$, there exists an affine variety $W \subsetneq V(I_l)$ such that $V(I_l) \setminus W \subset \pi_l(V)$.

The Closure Theorem gives a partial description of $\pi_l(V)$ that covers $V(I_l)$ except for the points lying in a variety strictly smaller than $V(I_l)$.

Finally, we have also the geometric version of Corollary 13.5.6 that represents a good situation for elimination.

Corollary 13.5.10 *Let $V = V(f_1, \ldots, f_t) \subset \mathbb{C}^n$ and suppose that for some i, f_i can be written as*

$$f_i = c_i x_1^N + \text{terms in } x_1 \text{ of degree } < N,$$

where $c_i \in \mathbb{C}$ is different from zero and $N > 0$. If I_1 is the first elimination ideal of I, then, in $\mathbb{C}^{n-1}$,

$$\pi_1(V) = V(I_1),$$

where π_1 is the projection on the last $n-1$ components.

13.6 Exercises

Exercise 46 Prove Lemma 13.2.3.

Exercise 47 Compute the S-polynomials $S(f, g)$ where

(a) $f = x^2y + xy^2 + y^3$, $g = x^2y^3 + x^3y^2$, using $>_{lex}$;
(b) $f = y^2 + xy^2 + y^3$, $g = x^2 + y^3 + xy^2$, using $>_{grlex}$;
(c) $f = y^2 + x$, $g = x^2 + y$, using $>_{grevlex}$.

Exercise 48 Check if the following sets are Grobener basis. In case of negative answer compute the Groebner basis of the ideal they generate.

(a) $\{x_1 - x_2 + x_3 - x_4 + x_5, x_1 - 2x_2 + 3x_3 - x_4 + x_5, 4x_3 + 4x_4 + 5x_5\}$, using $>_{lex}$;
(b) $\{x_1 - x_2 + x_3 - x_4 + x_5, x_1 - 2x_2 + 3x_3 - x_4 + x_5, 4x_3 + 4x_4 + 5x_5\}$, using $>_{grlex}$;
(c) $\{x_1 + x_2 + x_3, x_2x_3, x_2^2 + x_3^2, x_3^3\}$, using $>_{grevlex}$.

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